SILENT SCALE SHIFT: VARIANCE CONTRACTION BREAKS CALIBRATED READOUTS

Anonymous authors
Paper under double-blind review

ABSTRACT

An encoder update can improve its current task while breaking a deployed readout. We study this separation between task information and readout compatibility. In matched HAR continuations, frozen accuracy falls from 0.946 to 0.453 while the current head reaches 0.948; unlabeled input rescaling restores 0.938. Exact affine crossings explain this scale-sensitive failure, and a signed update audit identifies contraction that task-gradient conflict alone cannot determine. Compatibility also depends on the output rule: preserving classes can leave fixed confidence thresholds invalid. We therefore evaluate the old consumer's decisions alongside the benefit and cost of an update. Retaining healthy computations supports both outputs efficiently. On RadioML, equal-distillation controls retain a 0.28-point auxiliary gain with unchanged old decisions. On the official CIFAR-100 test set, a ViT-B/32 student with selective current-tail computation gains 0.86 points over two-view supervision and 0.60 over doubled-epoch supervision while preserving all audited fine and coarse decisions. Complete FP32 CPU forwards cost $1.046/1.032\times$ the unrestricted current model at batches $1/128$.

1 INTRODUCTION

An encoder may keep training after its classifier or confidence gate has shipped. The deployed component keeps its weights and normalization statistics. A new encoder can then change the component's decisions on the same inputs, even as the current head improves. We call the deployed component a *consumer*, including the decision rule it implements. The problem arises in backward-compatible learning (Shen et al., 2020) and in activation monitoring (Duan, 2026).

Matched HAR continuations expose the separation. Starting from the same model and Adam state, joint training reduces frozen accuracy from 0.946 to 0.453 while the current head reaches 0.948. A global scale correction fitted without labels restores 0.938 (Figure 1). We call this component *silent scale shift*: information remains decodable, but the deployed readout no longer matches its input. Reconstruction training produces the same separation while improving its own objective.

For an affine readout, contraction changes the balance between features and bias. We derive its decision crossings and use input repairs to test how much damage scale and alignment explain. A signed audit connects this geometry to the applied encoder update. Task-gradient conflict alone, addressed by PCGrad (Yu et al., 2020) and CAGrad (Liu et al., 2021), does not determine whether that update contracts the representation.

Repair quality depends on the deployed rule. Accuracy can recover while individual decisions change, and classes can agree while confidence gates fail. We compare input repair, learned readouts and retention of the healthy function under the decisions each consumer actually makes. Shared healthy computations retain auxiliary gains on RadioML and CIFAR-100 at modest measured CPU cost. The question is whether an upgrade can keep its learning benefit and its existing interface together.

2 WHEN EXISTING READOUTS FAIL

We study a shared encoder h_θ, a current primary head p_ϕ, and an auxiliary head a_ψ. Ordinary joint training minimizes $L_p(p_\phi(h_\theta(x)), y) + \alpha L_a(a_\psi(h_\theta(x)), t)$. The nonnegative weight α sets

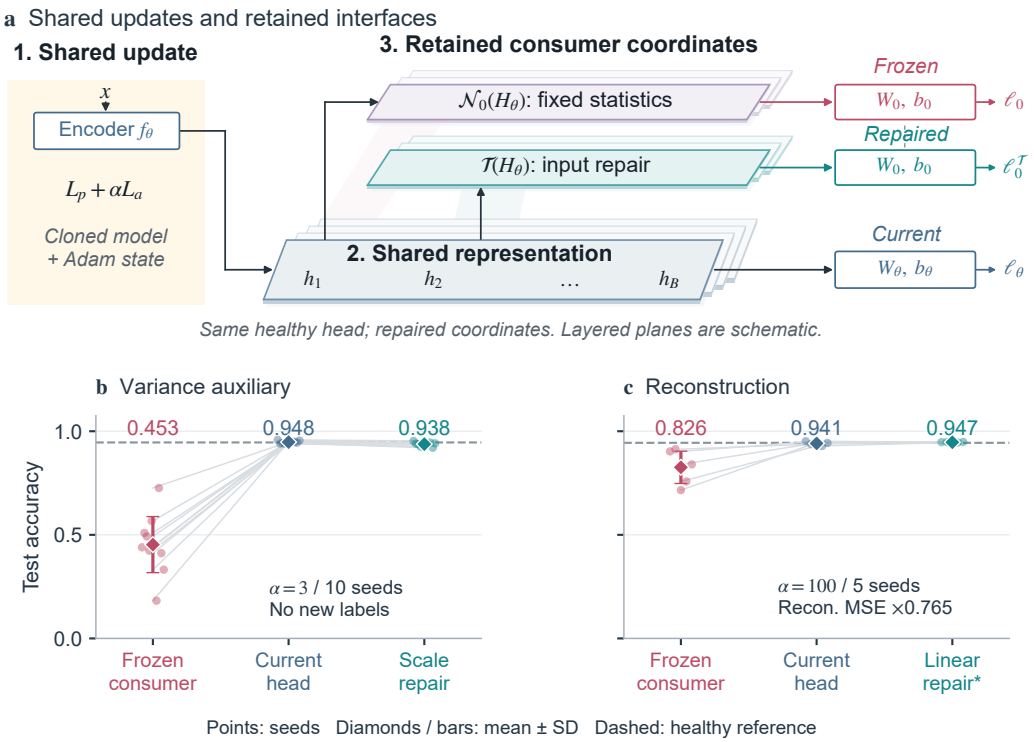


Figure 1: An updated representation can support the task while breaking an existing readout. (a) Three paths through shared features: the current head, the frozen readout, and the same readout after input repair. $\mathcal{N}_0$ is the healthy normalizer; $\mathcal{T}$ repairs normalization or alignment while retaining (W_0, b_0). (b) HAR variance continuation: global scale correction restores much of the lost frozen accuracy. (c) HAR reconstruction: auxiliary MSE falls to 0.765 of detachment; paired linear alignment restores readout accuracy. Repairs use training features without labels and leave both models fixed; *paired alignment also needs old features at the same anchors. Accuracy is on the full test set; bars show seed SD.

the auxiliary dose; t is an input, metadata or a second-view target, depending on the task. Joint training updates the encoder from both losses. Its matched detached control trains the auxiliary head on stop-gradient features, so only the primary loss updates the encoder. Gradient-combination baselines modify these two encoder gradients. A separate deployed readout C_0 keeps its weights, input statistics and decision thresholds fixed after the healthy checkpoint. The current head and a newly fitted probe can adapt; C_0 cannot.

We measure contraction by covariance trace, task information by a standardized-feature probe, and interface compatibility by the actual class or confidence decisions of C_0. A trace decrease alone establishes neither information loss nor decision failure. A practical intervention must also retain the learning benefit: our serving studies use positive paired gain bounds or their declared noninferiority margins, a fixed old-decision budget, and measured CPU latency. The per-study budgets and matched controls appear with the corresponding results.

Let $h_\theta(x) \in \mathbb{R}^m$ be the shared features. The old readout applies the fixed map $C_0(h) = W_0 D_0^{-1}(h - \mu_0) + b_0$, with healthy means μ_0, positive diagonal scales D_0, weights W_0, and bias b_0. A *consumer* is this map together with the output rule it feeds, such as a class decision or a fixed confidence threshold. Continuation holds the consumer fixed while its input $h_\theta(x)$ changes, so the complete function of x can change even though every stored parameter is unchanged. A refitted probe measures what the new features support; the consumer measures whether the old interface still works.

Pure contraction separates the two questions. For $H_c = cH$, an affine readout has logits $cWh + b$, which approach b as $c \to 0$. A unique bias winner then makes the prediction constant, although every $c > 0$ preserves linear decodability.

Proposition 1 (Two scale boundaries of a frozen affine readout)**.** *Write the logits after fixed input statistics as $z_i(c) = z_0 + c\, s_i$. Assume unique winners $t = \arg\max_j z_0[j]$ and $p_i = \arg\max_j z_i(1)[j]$. The bias-only class first loses as c increases from zero at*

$$c_i^{\text{bias}} = \min_{j:\, s_i[j] > s_i[t]} \frac{z_0[t] - z_0[j]}{s_i[j] - s_i[t]}, \tag{1}$$

with the minimum of an empty set taken as ∞. The first class change while shrinking from $c = 1$ instead occurs at

$$c_i^{\text{first}} = \max_{\substack{j:\, s_i[p_i] > s_i[j] \\ 0 < \xi_{ij} < 1}} \xi_{ij}, \qquad \xi_{ij} = \frac{z_0[j] - z_0[p_i]}{s_i[p_i] - s_i[j]}, \tag{2}$$

with zero denoting no positive-scale flip. The two boundaries differ in general. With nonzero feature variances and exact statistics, refitting the input statistics makes the readout invariant to every $c > 0$.

Proof. Each pairwise margin is affine in c. Increasing c from zero reaches the smallest positive crossing of the bias winner; decreasing c from one reaches the largest crossing of the healthy winner in $(0, 1)$. Standardising cH cancels c from numerator and denominator. For example, logits $(1, 10c, -1 + 12c)$ give $c^{\text{bias}} = 0.1$ but $c^{\text{first}} = 0.5$. □

The imposed-scale sweep in Figure 10 isolates this mechanism from the drift that training adds (Appendix H.6).

Matched HAR trajectories separate mean from scale. At $\alpha = 3$, global scale correction reaches 0.938 and adding a mean correction reaches 0.944, close to per-feature recalibration (0.947). At $\alpha = 100$, per-feature repair reaches 0.829 versus 0.935 for a supervised refit. Stronger and projected contraction require richer alignment (Appendix H.8).

All repairs act upstream of the frozen weights, through a versioned encoder output or a consumer-input adapter. Mean and scale repairs need cached healthy statistics and unlabeled new features; paired alignment also needs old features at the same anchors. A supervised refit reuses labels and is the stronger-access baseline. We report NLL and 15-bin ECE with accuracy.

The output rule decides whether a repair succeeds. A fixed confidence threshold can move even when classes and probabilities barely do. Let $p_0 = (1 - \epsilon/2, \epsilon/2)$, $q = (1 - 2\epsilon, 2\epsilon)$ and $\tau = 1 - \epsilon$ with $0 < \epsilon < 1/4$. Both predictors choose the same class, yet the healthy consumer accepts and the new one abstains on every example, while $\mathrm{KL}(p_0 \| q) = O(\epsilon) \to 0$. Classification agreement and a small distillation loss can therefore coexist with complete selective disagreement. A buffered log-odds metric avoids this saturation, but does not beat the standard controls in its planned comparison (Appendices I.6 and I.7). We audit consumer outputs rather than infer survival from one representation or loss statistic.

3 Auditing Shared Updates

Let $H(\theta) \in \mathbb{R}^{B \times m}$ be the outputs of a shared encoder on a fixed diagnostic batch; a linear primary head and an auxiliary head consume H. Training minimizes $L_p + \alpha L_a$, where L_p is the primary loss, L_a the auxiliary loss, and $\alpha \geq 0$ the dose; several auxiliaries can be folded into the weighted sum. Write $G_p = \nabla_\theta L_p$ and $G_a = \nabla_\theta L_a$. Define

$$V(\theta) = \frac{1}{2B} \| C_B H(\theta) \|_F^2, \qquad C_B = I - \mathbf{1}\mathbf{1}^\top / B, \qquad v = \nabla_\theta V. \tag{3}$$

Here $2V$ is the covariance trace with divisor B: a rotation-invariant measure of variation across examples. Under gradient flow,

$$\dot{V} = r - \alpha c, \qquad r = -v^\top G_p, \qquad c = v^\top G_a. \tag{4}$$

We call $c > 0$ *auxiliary contraction*, and $r > 0$ *primary restoration*. When both are positive, $\alpha c / r > 1$ defines the instantaneous contraction-dominated regime. The pair (r, c) also covers a

primary update that reduces variance on its own. Because the identity retains the encoder Jacobian, no reduced dynamical model is needed.

Proposition 2 (Conflict information does not identify contraction)**.** *The task-gradient Gram matrix does not determine the sign of $\dot{V}$. In particular, there are two auxiliary updates with identical task-gradient norms and conflict, but opposite effects on variance under the same primary update.*

Proof. Take $G_p = e_1$, $G_a^{\pm} = \pm e_2$, and $v = -e_1 + \beta e_2$ for $\beta > 0$. Both pairs have identity Gram matrix and zero conflict, whereas $r = 1$ and $c^{\pm} = \pm\beta$. For $\alpha\beta > 1$, $\dot{V}^+ < 0 < \dot{V}^-$. These gradients are locally realizable by a smooth encoder and losses (Appendix A). □

Local contraction and loss of task information are separate outcomes. In HAR, a $4{,}600\times$ trace contraction leaves standardized-probe accuracy unchanged (Appendix H.4). The spectral model in Theorem 3 distinguishes contraction of some modes from all modes. We measure variation, decodability and consumer utility separately. The cheaper radial feature-space proxy agrees with the exact derivative on 51% of 405 HAR checkpoints; its geometry and calibration controls are in Appendices A.1 and A.4. The one-shot unsigned STF-cal gate is an empirical intervention heuristic. The local sign condition requires its stated geometry and discrepancy terms; the PCA rank rule requires identifying the specified PSD operator. Neither supplies a general trajectory guarantee. Computing the full sign certificate is an offline audit, more expensive than measuring the signed flux directly (Appendix A).

4 Testing the Readout Interface

4.1 Matched continuation

We clone a healthy model with its optimizer state, freeze the deployed readout, and continue joint and detached branches on identical minibatches. A refitted probe measures retained decodability; the frozen consumer measures compatibility. Mean, scale and paired-alignment repairs use training features without new labels. We report held-out accuracy, NLL and 15-bin ECE with feature variation and the actual class and threshold decisions. The benchmark spans HAR (Anguita et al., 2012), RadioML (O'Shea et al., 2016), battery health (Saha & Goebel, 2007) and NYUv2; architectures, grids and the scatter criterion are in Appendices F–H and A.4. Error bars show population seed SD unless noted.

4.2 The update and architecture determine the failure

Real auxiliaries separate gradient loading from contraction. Across four heads, mean per-example task-gradient cosines stay below 0.06 in magnitude, while SNR radial loading reaches 88% (Figure 5a). The signed flow contribution is expanding at 74 of 75 historical epoch-end checkpoints, and flow and Adam signs disagree on $33/75$ sampled paired updates. We therefore audit the applied increment and a primary-only Adam counterfactual when attributing a change (Appendix H.7).

Encoder capacity changes the failure. At I/Q reconstruction dose 3000, the waveform CNN's frozen readout reaches 0.415 against 0.418 for detachment, while its current head reaches 0.437. The MLP is less compatible under the same auxiliary (Appendices H.11 and H.10). Failure is a property of the updated representation and its consumer, not of auxiliary loading or model size alone.

5 Preserving a Specified Consumer

5.1 Repairing the readout interface

Repairing the input can keep both the learning and the old readout, up to a point. At HAR reconstruction dose 100, the current head reaches 0.941 while frozen accuracy falls to 0.826. Paired linear alignment restores 0.947 and keeps the learned encoder with a reconstruction MSE of 0.765 relative to detachment (Figure 1c).

Recovered accuracy does not imply recovered decisions. A post-hoc chart improves accuracy, NLL and ECE yet changes 44.07% of decisions. Distilling a legacy head (Hinton et al., 2015; Li & Hoiem,

2016) gives a useful HAR tradeoff: reconstruction MSE is 0.780 of detachment, primary accuracy changes by $+0.30$ points $[-0.01, 0.61]$, and all ten class-disagreement bounds certify. Cheaper OLS and full-feature alignment preserve the class interface but fail the fixed confidence rules. The matched controls and vision counterexamples are in Appendices H.15–H.20 and H.21.

5.2 Retaining the healthy function

Retaining a healthy function gives exact outputs in real arithmetic. For an affine consumer A, the features $H_0 + (H_\theta - H_0)P_\perp$, with $P_\perp$ projecting onto $\ker A$, preserve its logits. On ten fresh SNR seeds this keeps a 0.89-point auxiliary gain, but evaluating both encoders costs $2.37\times$. Freezing the nonlinear trunk permits folding and loses the gain (Appendices H.12 and H.13). Sharing healthy computations offers a cheaper way to keep the function while adding learning capacity.

For a two-layer nonlinear trunk, let a_1^0 and t_2^0 denote the healthy first activation and second preactivation. If the first-map update changes rows S and the second-map update is UV, then

$$t_2 = t_2^0 + W_2^0[:, S](a_1 - a_1^0)[S] + UVa_1 + \delta b_2. \tag{5}$$

The activation difference stays on S through any coordinatewise nonlinearity, even when gates change. Shared dense products and correction terms therefore evaluate both healthy and current functions. Proposition 11 gives the construction and proof; its saved parameters retain the healthy path, while current features learn.

5.3 Learning capacity and measured cost

Sparse and dense replay retain old decisions and improve learning, but their small-network CPU cost exceeds the $1.20/1.10$ batch limits (Table 1). Their support/rank controls and negative development outcomes are in Appendices I.3–I.2.

On RadioML, we instead retain the healthy hidden units and distill the current output into 16 added nonlinear units. With equal architecture, initialization and distillation work, the joint student gains 0.28 points $[0.11, 0.45]$ over its detached counterpart across 16 fresh references. Every old class and confidence decision agrees. The folded graph costs $1.016\times$ at batch 1 and $1.070\times$ at batch 128 (Appendices J.20–J.21).

Pretrained vision models can share a frozen prefix while learning complete nonlinear suffixes. ResNet-18 and ViT-B/32 then improve supervised transfer over richer frozen-feature controls; smaller update scopes and the fixed reconstruction grid miss their gain targets (Appendices J.2–J.11).

On ViT-B/32, a cross-view contrastive auxiliary (Chen et al., 2020) improves training-only validation accuracy over both two-view and doubled-epoch supervision at matched suffix-forward budgets. A fixed-recipe AugReg replication also improves accuracy and contrastive loss. Directly applying the old coarse head to updated features changes 7.35% of its class decisions on average. Nine unlabeled repairs and the initial distilled-head screen fail their compatibility targets (Appendices J.6–J.7, J.12).

We can retain those visual gains with exact old decisions and FP32 execution. A 128-unit student reads the healthy final features and last MLP activation. Low-margin inputs also use the updated two-block suffix, sharing the frozen prefix (Figure 2). The cutoff is fixed on training-only validation before the official test. On all 10,000 CIFAR-100 test images, four references with three continuations each gain 0.86 points $[0.68, 1.05]$ over the paired two-view student and 0.60 $[0.41, 0.79]$ over its doubled-epoch counterpart. The direct supervised MLP uses the same features, labels and training schedule; the gain over it is 2.23 points. Every old fine and coarse decision agrees. The current tail handles 32.2% of inputs; the largest decision difference from the full joint teacher is 0.06%. Complete CPU forwards cost $1.046/1.032\times$ at batches $1/128$. The pure student loses 0.97 points from its teacher; selective computation closes that gap. In a retrospective equal-call comparison, fixed routing gains 0.63 points $[0.51, 0.75]$ over the exact random-routing expectation: the fixed gate captures 99.33% of student–teacher decision differences (Appendix J.22).

Table 1 places the retained-function programs alongside their learning gains and observed decision changes. The routed ViT result uses one pretrained checkpoint and one official test cohort; intervals

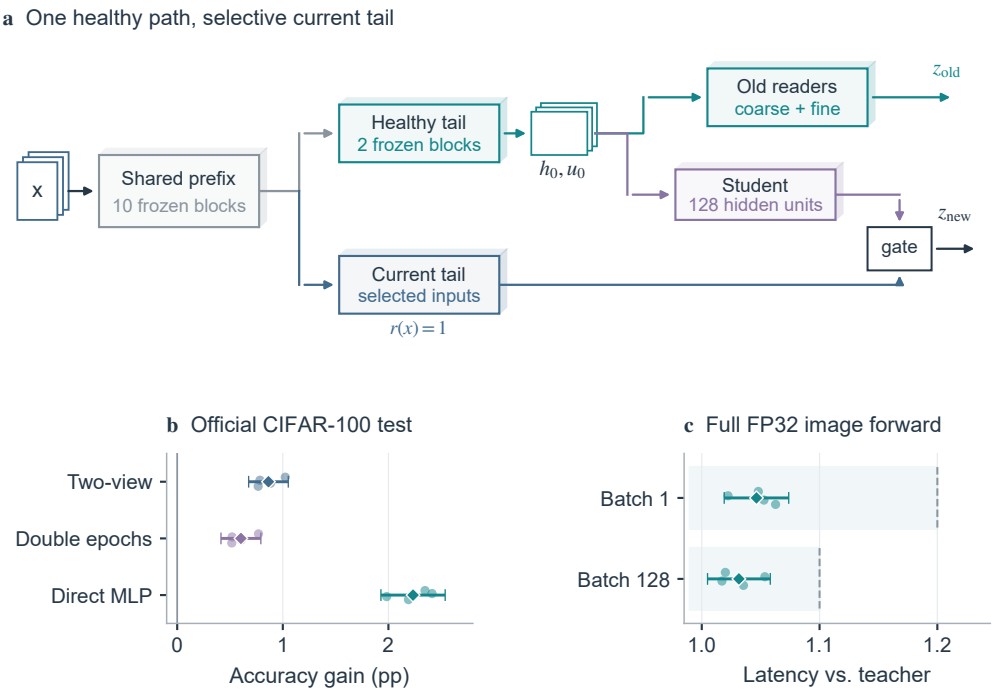


Figure 2: Keeping old vision interfaces while retaining auxiliary gains. (a) A healthy ViT path serves fixed fine and coarse readers. A student uses its features; inputs with low student margin also traverse the updated two-block tail. (b) Official CIFAR-100 accuracy gains over routed two-view and doubled-epoch supervised students, and a direct supervised MLP. (c) Complete FP32 image-forward latency relative to the unrestricted current teacher; dashed marks show the $1.20/1.10$ limits. Small points are four fitted-reference means, diamonds and bars paired 95% intervals. All old class and confidence decisions are unchanged.

Table 1: Learning, old decisions and CPU cost. Original replay and supervised-transfer rows use ten fresh continuations; cross-view uses four references with three continuations each. The exact student uses 16 fresh references and separate timing on the same architecture. HAR reports auxiliary MSE relative to detachment. SNR reports primary gain over detachment, with equal distillation work for the exact student. Vision gains are over H for transfer and D for cross-view training. The routed ViT row uses the official test and compares routed two-view students. Intervals are paired 95%. Old change is the largest observed class or confidence-rule disagreement. CPU batches are $1/1024$ for original replay and $1/128$ otherwise; cost targets are $1.20/1.10$.

Study	Learning outcome [95% interval]	Old change	CPU ratios
HAR	MSE 0.929 [0.917, 0.941]	0%	1.399/1.633
SNR dense replay	+1.257 pp [1.037, 1.477]	0%	1.286/2.195
SNR exact student	+0.28 pp [0.11, 0.45]	0%	1.016/1.070
ResNet-18	+0.49 pp [0.37, 0.60]	0.39%	1.126/1.118
ViT-B/32 transfer	+1.59 pp [1.49, 1.68]	0.14%	1.143/1.076
ViT-B/32 cross-view	+0.84 pp [0.66, 1.03]	0.18%	1.139/1.078
ViT-B/32 routed	+0.86 pp [0.68, 1.05]	0%	1.046/1.032

are across four fitted-reference means. ResNet-18 still exceeds the batch-128 cost limit. Reduced-precision reference serving offers a different accuracy–cost tradeoff, evaluated separately with the FP32 current function fixed (Appendix J.8).

6 Checking Compatibility Before Serving

Evaluation has three objects: the features, the current task, and the existing consumer. We measure variation and decodability for the features, primary and auxiliary utility for learning, and the specified class or threshold outputs for compatibility. A repair has to meet its consumer target without giving up the utility that motivated the update.

For a fixed corpus of N inputs, let f_0 be the healthy consumer, f a fixed candidate, and R_T the mean 0–1 error on subset T. For every assignment of labels,

$$R_T(f) - R_T(f_0) \le D_T(f, f_0), \qquad D_T = |T|^{-1} \sum_{i \in T} \mathbf{1}\{f(x_i) \neq f_0(x_i)\}. \tag{6}$$

The inequality is pointwise: error can increase only where predictions change, and it is sharp when the labels equal the healthy predictions. The bound is one-sided; large disagreement does not by itself imply harm.

When healthy predictions are available, this gives a label-free check. With $1{,}024$ monitor inputs we select the first member of a fixed repair hierarchy whose finite-corpus bound meets the budget, and otherwise abstain. Across 105 joint endpoints, a 10-point budget certifies 49, with 0 violations and 56 abstentions; a scatter heuristic makes 11 false-safe declarations (Appendix H.23). Where labels exist we also report negative flips (Yan et al., 2021), separating old-correct errors from useful corrections.

Early logits can also guide a pending upgrade. On CIFAR-10, an epoch-two monitor flags $15/17$ later budget exceedances and clears all seven safe updates. A fixed replay policy stays within 5% disagreement on all 24 held-out cases. RadioML trajectory forecasts reduce terminal-risk error from 2.87 to 2.25 points relative to a schedule-aware baseline; the corresponding advantage is uncertain on subject-disjoint HAR. These forecast studies and cheaper legacy-path experiments are reported in Appendices J.14–J.16 and J.17. They extend the interface audit to release decisions; the main serving results above use fixed, completed updates.

7 Related Work

Backward-compatible learning treats the readout as an interface (Shen et al., 2020; Hu et al., 2022b). Positive-congruent training targets negative flips (Yan et al., 2021). Weight interpolation changes the current function (Schumann et al., 2023), while BC-Aligner fits an encoder and backward map jointly. Unlabeled alignment also repairs stale language-model monitors (Duan, 2026). We test repairs against class and confidence contracts. Related tools include model stitching (Bansal et al., 2021; Mai et al., 2026; Smith et al., 2025), moment adaptation (Li et al., 2016; Gomez-Villa et al., 2024; Sun et al., 2016; Wang et al., 2021), calibration and unlabeled risk estimation (Guo et al., 2017; Ovadia et al., 2019; Garg et al., 2022; Zhao et al., 2023).

Multi-task learning and task grouping ask when shared training helps (Caruana, 1997; Standley et al., 2020). PCGrad and CAGrad address gradient conflict (Yu et al., 2020; Liu et al., 2021). Pareto, bargaining and progress-balancing methods choose different update directions (Sener & Koltun, 2018; Navon et al., 2022; Liu et al., 2023). The applied step's projection onto the variance gradient determines local contraction.

Self-supervised work studies collapse through stop-gradient mechanisms (Grill et al., 2020; Chen & He, 2021; Caron et al., 2021), variance regularization (Zbontar et al., 2021; Bardes et al., 2022; Ermolov et al., 2021) and spectral measures (Jing et al., 2022; Dong et al., 2021; Kim et al., 2025; Garrido et al., 2023). We measure variation across examples and use input repair to test decodability after contraction.

Our intervention controls build on auxiliary-gradient projection (Dery et al., 2021; Zhu et al., 2025), auxiliary selection (Navon et al., 2021; Jiang et al., 2023; Shin et al., 2024), and nullspace constraints (Pham et al., 2026; Fang et al., 2025). We test activity and modulation recognition (Anguita et al., 2012; O'Shea et al., 2016) and battery health (Saha & Goebel, 2007; Severson et al., 2019; Attia et al., 2020; Pradhan & Chakraborty, 2022).

DeltaCNN reuses sparse activation differences (Parger et al., 2022). LoRA and adapter serving share base parameters (Hu et al., 2022a; Chen et al., 2024; Sheng et al., 2024). Progressive networks and

side-tuning retain frozen paths while training new capacity (Rusu et al., 2016; Zhang et al., 2020; Sung et al., 2022). Confidence cascades allocate computation adaptively (Bolukbasi et al., 2017). We use these tools to separate two requirements: retaining the complete old consumer function and recovering the benefit of an updated model. The old path supplies the first; paired learning controls and measured serving costs test the second. Our replay shares computations across two versions on the same input. The frozen-feature controls also address distortion from fine-tuning (Kumar et al., 2022).

8 Conclusion

Task information can survive an update that invalidates an existing readout. Scale correction can recover the old interface, and exact replay can keep it in service while the encoder learns. The decisions an upgrade preserves and the cost of serving them deserve the same scrutiny as its task accuracy.

AI USE STATEMENT

Generative AI tools assisted with drafting, language editing, code, literature retrieval, proof development, and figure design. The new diagnostic identities were checked with executable numerical examples; assumptions are stated explicitly with each claim and in Appendix D. Figures are drawn from recorded experimental results. The authors remain responsible for verifying the final text, claims, code, and artifacts.

ETHICS STATEMENT

We use existing datasets: UCI-HAR, RadioML2016.10A, the NASA PCoE and MATR battery archives, public A-share market data, NYUv2, CelebA, CIFAR-10 and CIFAR-100, with ImageNet-pretrained vision weights. We collected no new human-subject data. The HAR identity auxiliary is evaluated as a training-head probe, not a held-out privacy attack. Battery prediction is safety-relevant; Appendix A.4.3 describes the observed limitations, and deployment requires task-specific validation.

REPRODUCIBILITY STATEMENT

The supplementary archive contains theoretical assumptions, source code, per-seed records, fixed development/confirmation plans, scale-boundary checks, actual-step audits and the corrected battery protocol. The separate numerical archive supplies the compact tensors used in result replay. Historical battery checkpoint selection and readout calibration are exploratory.

REFERENCES

Ahmed Allibhoy and Jorge Cortés. Control barrier function based design of gradient flows for constrained nonlinear programming. *IEEE Transactions on Automatic Control*, 69(6):3499–3514, 2024. doi: 10.1109/TAC.2023.3306492.

Anastasios N. Angelopoulos, Stephen Bates, Clara Fannjiang, Michael I. Jordan, and Tijana Zrnic. Prediction-powered inference. *Science*, 382:669–674, 2023. doi: 10.1126/science.adi6000. URL https://arxiv.org/abs/2301.09633.

Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge L. Reyes-Ortiz. Human activity recognition on smartphones using a multiclass hardware-friendly support vector machine. In *International Workshop on Ambient Assisted Living (IWAAL)*, volume 7657 of *Lecture Notes in Computer Science*, pp. 216–223. Springer, 2012.

Peter M. Attia, Aditya Grover, Norman Jin, Kristen A. Severson, Taylor M. Markov, Yu-Hsin Liao, Michael H. Chen, Bryan Cheong, Nicholas Perkins, Zi Yang, Patrick K. Herring, Muratahan Aykol, Stephen J. Harris, Richard D. Braatz, Stefano Ermon, and William C. Chueh. Closed-loop optimization of fast-charging protocols for batteries with machine learning. *Nature*, 578(7795): 397–402, 2020.

Dara Bahri, Heinrich Jiang, Yi Tay, and Donald Metzler. SCARF: Self-supervised contrastive learning using random feature corruption. In *International Conference on Learning Representations*, 2022.

Yamini Bansal, Preetum Nakkiran, and Boaz Barak. Revisiting model stitching to compare neural representations. In *Advances in Neural Information Processing Systems*, volume 34, pp. 225–236, 2021.

Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022.

Christopher M. Bishop. Mixture density networks. Technical Report NCRG/94/004, Aston University, 1994.

Tolga Bolukbasi, Joseph Wang, Ofer Dekel, and Venkatesh Saligrama. Adaptive neural networks for efficient inference. In *International Conference on Machine Learning*, pp. 527–536, 2017.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE International Conference on Computer Vision (ICCV)*, 2021.

Rich Caruana. Multitask learning. *Machine Learning*, 28(1):41–75, 1997.

Lequn Chen, Zihao Ye, Yongji Wu, Danyang Zhuo, Luis Ceze, and Arvind Krishnamurthy. Punica: Multi-tenant lora serving. In *Proceedings of Machine Learning and Systems*, 2024. URL https://arxiv.org/abs/2310.18547.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *International Conference on Machine Learning*, 2020. URL https://arxiv.org/abs/2002.05709.

Xinlei Chen and Kaiming He. Exploring simple Siamese representation learning. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.

Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning (ICML)*, 2018.

Lucio M. Dery, Yann Dauphin, and David Grangier. Auxiliary task update decomposition: The good, the bad and the neutral. In *International Conference on Learning Representations*, 2021.

Laurent Dinh, Razvan Pascanu, Samy Bengio, and Yoshua Bengio. Sharp minima can generalize for deep nets. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70, pp. 1019–1028, 2017. URL https://proceedings.mlr.press/v70/dinh17b.html.

Shibhansh Dohare, Juan Hernandez-Garcia, Qingfeng Lan, Parash Rahman, Ashique Rupam Mahmood, and Richard S. Sutton. Loss of plasticity in deep continual learning. *Nature*, 632(8026): 768–774, 2024.

Yihe Dong, Jean-Baptiste Cordonnier, and Andreas Loukas. Attention is not all you need: Pure attention loses rank doubly exponentially with depth. In *International Conference on Machine Learning (ICML)*, 2021.

Evan Duan. Do activation monitors survive model updates? benchmarking, predicting, and repairing activation-monitor staleness, 2026. URL https://arxiv.org/abs/2606.15980.

Vijay Ekambaram, Arindam Jati, Nam Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. TSMixer: Lightweight MLP-Mixer model for multivariate time series forecasting. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2023.

Aleksandr Ermolov, Aliaksandr Siarohin, Enver Sangineto, and Nicu Sebe. Whitening for self-supervised representation learning. In *International Conference on Machine Learning (ICML)*, 2021.

Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Jie Shi, Xiang Wang, Xiangnan He, and Tat-Seng Chua. AlphaEdit: Null-space constrained knowledge editing for language models. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/29c8c615b3187ee995029284702d3f43-Abstract-Conference.html.

Mehrdad Farajtabar, Navid Azizan, Alex Mott, and Ang Li. Orthogonal gradient descent for continual learning. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108, pp. 3762–3773, 2020.

Saurabh Garg, Sivaraman Balakrishnan, Zachary C. Lipton, Behnam Neyshabur, and Hanie Sedghi. Leveraging unlabeled data to predict out-of-distribution performance. In *International Conference on Learning Representations*, 2022.

Quentin Garrido, Randall Balestriero, Laurent Najman, and Yann LeCun. RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank. In *International Conference on Machine Learning (ICML)*, 2023.

Alex Gomez-Villa, Dipam Goswami, Kai Wang, Andrew D. Bagdanov, Bartlomiej Twardowski, and Joost van de Weijer. Exemplar-free continual representation learning via learnable drift compensation. In *European Conference on Computer Vision*, 2024. URL https://arxiv.org/abs/2407.08536.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. Bootstrap your own latent: A new approach to self-supervised learning. In *Advances in Neural Information Processing Systems*, 2020.

Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. In *First Conference on Language Modeling (COLM)*, 2024.

Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, pp. 1321–1330, 2017.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022a.

Weihua Hu, Rajas Bansal, Kaidi Cao, Nikhil Rao, Karthik Subbian, and Jure Leskovec. Learning backward compatible embeddings. In *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2022b. doi: 10.1145/3534678.3539194.

Junguang Jiang, Baixu Chen, Junwei Pan, Ximei Wang, Dapeng Liu, Jie Jiang, and Mingsheng Long. ForkMerge: Mitigating negative transfer in auxiliary-task learning. In *Advances in Neural Information Processing Systems*, 2023.

Li Jing, Pascal Vincent, Yann LeCun, and Yuandong Tian. Understanding dimensional collapse in contrastive self-supervised learning. In *International Conference on Learning Representations*, 2022.

Wittawat Jitkrittum, Neha Gupta, Aditya K. Menon, Harikrishna Narasimhan, Ankit Rawat, and Sanjiv Kumar. When does confidence-based cascade deferral suffice? In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://papers.nips.cc/paper_files/paper/2023/hash/1f09e1ee5035a4c3fe38a5681cae5815-Abstract-Conference.html.

Donghyeon Kim, Chae-Bong Sohn, Do-Yup Kim, and Dae-Yeol Kim. A taxonomy and theoretical analysis of collapse phenomena in unsupervised representation learning. *Mathematics*, 13(18): 2986, 2025.

Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations*, 2022. URL https://arxiv.org/abs/2202.10054.

Lubor Ladický, Jianbo Shi, and Marc Pollefeys. Pulling things out of perspective. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 89–96, 2014.

Yanghao Li, Naiyan Wang, Jianping Shi, Jiaying Liu, and Xiaodi Hou. Revisiting batch normalization for practical domain adaptation. *arXiv preprint arXiv:1603.04779*, 2016.

Zhizhong Li and Derek Hoiem. Learning without forgetting. In *European Conference on Computer Vision*, 2016.

Julian Lienen and Eyke Hüllermeier. From label smoothing to label relaxation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pp. 8583–8591, 2021. doi: 10.1609/aaai.v35i10.17041.

Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems*, 2021.

Bo Liu, Yihao Feng, Peter Stone, and Qiang Liu. FAMO: Fast adaptive multitask optimization. In *Advances in Neural Information Processing Systems*, pp. 57226–57243, 2023.

Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *IEEE International Conference on Computer Vision (ICCV)*, pp. 3730–3738, 2015.

Zheda Mai, Ke Zhang, Fu-En Wang, Zixiao Ken Wang, Albert Y. C. Chen, Lu Xia, Min Sun, Wei-Lun Chao, and Cheng-Hao Kuo. Revisiting model stitching in the foundation model era. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 41342–41351, 2026.

Aviv Navon, Idan Achituve, Haggai Maron, Gal Chechik, and Ethan Fetaya. Auxiliary learning by implicit differentiation. In *International Conference on Learning Representations*, 2021.

Aviv Navon, Aviv Shamsian, Idan Achituve, Haggai Maron, Kenji Kawaguchi, Gal Chechik, and Ethan Fetaya. Multi-task learning as a bargaining game. In *International Conference on Machine Learning (ICML)*, 2022.

Timothy J. O'Shea, Johnathan Corgan, and T. Charles Clancy. Convolutional radio modulation recognition networks. In *Engineering Applications of Neural Networks (EANN)*, volume 629 of *Communications in Computer and Information Science*, pp. 213–226. Springer, 2016.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019.

Mathias Parger, Chengcheng Tang, Christopher D. Twigg, Cem Keskin, Robert Wang, and Markus Steinberger. Deltacnn: End-to-end cnn inference of sparse frame differences in videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2203.03996.

Cuong Anh Pham, Praneeth Vepakomma, and Samuel Horváth. Learning in the null space: Small singular values for continual learning. In *Conference on Parsimony and Learning*, volume 328 of *Proceedings of Machine Learning Research*, pp. 612–628, 2026.

Sunil K. Pradhan and Basab Chakraborty. Battery management strategies: An essential review for battery state of health monitoring techniques. *Journal of Energy Storage*, 51:104427, 2022.

Andrei A. Rusu, Neil C. Rabinowitz, Guillaume Desjardins, Hubert Soyer, James Kirkpatrick, Koray Kavukcuoglu, Razvan Pascanu, and Raia Hadsell. Progressive neural networks. *arXiv preprint arXiv:1606.04671*, 2016. URL https://arxiv.org/abs/1606.04671.

Bhaskar Saha and Kai Goebel. Battery data set. NASA Ames Prognostics Data Repository, 2007. URL https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/.

Raphael Schumann, Elman Mansimov, Yi-An Lai, Nikolaos Pappas, Xibin Gao, and Yi Zhang. Backward compatibility during data updates by weight interpolation, 2023. URL https://arxiv.org/abs/2301.10546.

Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems*, 2018.

Kristen A. Severson, Peter M. Attia, Norman Jin, Nicholas Perkins, Benben Jiang, Zi Yang, Michael H. Chen, Muratahan Aykol, Patrick K. Herring, Dimitrios Fraggedakis, Martin Z. Bazant, Stephen J. Harris, William C. Chueh, and Richard D. Braatz. Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5):383–391, 2019.

Yantao Shen, Yuanjun Xiong, Wei Xia, and Stefano Soatto. Towards backward-compatible representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.

Ying Sheng, Shiyi Cao, Dacheng Li, Coleman Hooper, Nicholas Lee, Shuo Yang, Christopher Chou, Banghua Zhu, Lianmin Zheng, Kurt Keutzer, Joseph E. Gonzalez, and Ion Stoica. S-lora: Serving thousands of concurrent lora adapters. In *Proceedings of Machine Learning and Systems*, 2024. URL https://arxiv.org/abs/2311.03285.

Seokwon Shin, Hyungrok Do, and Youngdoo Son. Learning representation for multitask learning through self-supervised auxiliary learning. In *European Conference on Computer Vision (ECCV)*, 2024.

Nathan Silberman, Derek Hoiem, Pushmeet Kohli, and Rob Fergus. Indoor segmentation and support inference from RGBD images. In *European Conference on Computer Vision (ECCV)*, pp. 746–760, 2012.

Damian Smith, Harvey Mannering, and Antonia Marcu. Functional alignment can mislead: Examining model stitching. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267, pp. 55972–55998, 2025. URL https://proceedings.mlr.press/v267/smith25a.html.

Trevor Standley, Amir Zamir, Dong Chen, Leonidas Guibas, Jitendra Malik, and Silvio Savarese. Which tasks should be learned together in multi-task learning? In *International Conference on Machine Learning (ICML)*, 2020.

Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your ViT? data, augmentation, and regularization in vision transformers. *Transactions on Machine Learning Research*, 2022. URL https://arxiv.org/abs/2106.10270.

Steven H. Strogatz. *Nonlinear Dynamics and Chaos*. Addison-Wesley, Reading, MA, 1994.

Baochen Sun, Jiashi Feng, and Kate Saenko. Correlation alignment for unsupervised domain adaptation. *arXiv preprint arXiv:1612.01939*, 2016.

Yi-Lin Sung, Jaemin Cho, and Mohit Bansal. Lst: Ladder side-tuning for parameter and memory efficient transfer learning. In *Advances in Neural Information Processing Systems*, 2022. URL https://arxiv.org/abs/2206.06522.

James Tobin. Estimation of relationships for limited dependent variables. *Econometrica*, 26(1): 24–36, 1958.

Muhammad Usama and Muhammad Saad Qadir. Where did the update go? certifying declared change envelopes for model updates. *arXiv preprint arXiv:2611.04192*, 2026.

Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations*, 2021.

Zekun Wu, Swati Dhiman, and Adriano Koshiyama. Which decisions low-bit quantization breaks, and how to predict them, 2026. URL https://arxiv.org/abs/2608.06564.

Sijie Yan, Yuanjun Xiong, Kaustav Kundu, Shuo Yang, Siqi Deng, Meng Wang, Wei Xia, and Stefano Soatto. Positive-congruent training: Towards regression-free model updates. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 14299–14308, 2021. URL https://arxiv.org/abs/2011.09161.

Greg Yang and J. Edward Hu. Feature learning in infinite-width neural networks. In *International Conference on Machine Learning (ICML)*, 2021.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, 2020.

Jure Zbontar, Li Jing, Ishan Misra, Yann LeCun, and Stéphane Deny. Barlow Twins: Self-supervised learning via redundancy reduction. In *International Conference on Machine Learning (ICML)*, 2021.

Jeffrey O Zhang, Alexander Sax, Amir Zamir, Leonidas Guibas, and Jitendra Malik. Side-tuning: A baseline for network adaptation via additive side networks. In *European Conference on Computer Vision (ECCV)*, 2020.

Hao Zhao, Yuejiang Liu, Alexandre Alahi, and Tao Lin. On pitfalls of test-time adaptation. In *International Conference on Machine Learning*, volume 202, pp. 42058–42080, 2023.

Shijie Zhu, Hui Zhao, Tianshu Wu, Pengjie Wang, Hongbo Deng, Jian Xu, and Bo Zheng. Gradient deconfliction via orthogonal projections onto subspaces for multi-task learning. In *ACM International Conference on Web Search and Data Mining (WSDM)*, pp. 204–212, 2025.

Xiaojin Zhu, Zoubin Ghahramani, and John Lafferty. Semi-supervised learning using gaussian fields and harmonic functions. In *International Conference on Machine Learning*, 2003.

The supplementary material is organized by question. Readout crossings and update geometry are developed in Appendices A and D. Matched continuation, repairs and their failure cases are in Appendix H. Shared computation and serving controls begin at Appendix J; the matched RadioML and official-test vision programs are in Appendices J.20 and J.22. Complete development grids and recorded protocols accompany the source artifact.

A Proofs and Audit Details for the Diagnostic Boundary

A.1 Unsigned feature-space probes

Let h and u stack the rows of H and $C_B H$, respectively, and let $g_p = \nabla_h L_p$, $g_a = \nabla_h L_a$. A cheap feature-space probe uses the centered radial direction:

Definition 1 (Unsigned radial toxicity proxy)**.** *For $u \neq 0$ and $\|g_p\| > 0$, define*

$$\rho_b = \frac{\|\Pi_u g_a\|}{\|g_p\|}, \qquad \Pi_u g = \frac{u^\top g}{\|u\|^2} u. \tag{7}$$

The experimental estimator $\hat{\rho}$ instead averages regularized per-example ratios. The frozen decision uses a recorded value $q = \hat{\rho}_{t_0}$ and the nominal dose $\alpha_{\mathrm{probe}} = 1/q$.

The proxy drops the sign of the auxiliary direction and takes the whole primary-gradient norm as its reference. Expansion and contraction can then give the same ρ_b, and a large primary gradient need not restore centered variation. The two drivers also measure different objects: the controlled one probes the isotropic surrogate $\|C_B H\|_F^2/(Bm)$ while training with fixed projected-variance auxiliaries, and the real-head one probes the actual auxiliary.

The exact encoder metric and calibration drift further limit a frozen radial probe. Appendix A gives a local sign certificate and separates restoration alignment, geometry, discarded sign and drift. On 405 HAR checkpoints its canonical frozen certificate fires on only 2, both correct. The exact signed flux remains available where the proxy is unresolved.

A.2 A local validity boundary

The discrepancy can be exposed without assuming radial training dynamics. Let $J = \partial h/\partial\theta$, $K = JJ^\top$, $e = u/\|u\|$, and $n_p = \|g_p\| > 0$. At an audited checkpoint choose any $k > 0$ and write $K = k(I + E)$. Define

$$a = \frac{e^\top g_a}{n_p}, \quad d = -\frac{e^\top g_p}{n_p}, \quad \epsilon_a = \frac{e^\top E g_a}{n_p}, \quad \epsilon_p = -\frac{e^\top E g_p}{n_p}. \tag{8}$$

Here a is signed radial loading relative to the primary norm, d measures restoration alignment, and the ϵ terms measure the effect of encoder geometry. For the actual auxiliary, $|a| = \rho_b$. Products with K can be evaluated by Jacobian–vector and vector–Jacobian products, without constructing the full matrix.

Theorem 1 (Local validity boundary of a frozen probe)**.** *For any frozen probe value $q \geq 0$, the variance dynamics satisfy*

$$\frac{\dot{V}}{N} = 1 - \alpha q + \mathcal{E}, \qquad N = \frac{k\|u\| n_p}{B} > 0, \qquad \mathcal{E} = (d-1) + \epsilon_p - \alpha(a - q + \epsilon_a). \tag{9}$$

Consequently, with

$$\Delta = |d-1| + |\epsilon_p| + \alpha\big(|a-q| + |\epsilon_a|\big), \qquad |1 - \alpha q| > \Delta, \tag{10}$$

the probe and the exact variance derivative agree in sign. If $1 - \alpha q > \Delta$, variance increases; if $1 - \alpha q < -\Delta$, it decreases. The remaining region requires the signed audit.

Proof. The chain rule gives $\dot{V} = -u^\top K(g_p + \alpha g_a)/B$. Substitute $K = k(I + E)$, divide by N, and add and subtract $1 - \alpha q$. The triangle inequality bounds $|\mathcal{E}|$ by Δ. □

The checkpoint-local margin uses the exact flux to explain the cheap probe offline. On 405 HAR checkpoints, the frozen certificate fires on 2; the signed balance $\psi = d - \alpha a$ with a combined residual certifies 10 at the same scale. Per-checkpoint scale optimization raises the counts to 15 and 154 (Table 2). Every certified sign is correct. The corrected balance alone agrees in sign on 58% of checkpoints, leaving substantial encoder-geometry mismatch.

In the identity metric ($E = 0$), $\dot{V}/N = d - \alpha a$. If $a = q$ and $0 < d < \alpha q < 1$, the nominal gate stays closed while variance contracts: the primary norm overstates restoration. If $a = -q$, the same unsigned gate can open while the auxiliary expands variance. Drift after calibration contributes through $a - q$. These examples identify concrete sources of a wrong sign.

The mismatch separates into drift, sign and estimation components. If a_0 is signed loading at calibration for the actual auxiliary, then

$$|a_t - q| \le |a_t - a_0| + |a_0 - |a_0|| + \big||a_0| - q\big|. \tag{11}$$

The terms separate temporal drift, discarded sign, and surrogate or aggregation mismatch. A radial approximation is informative when the auxiliary contracts, the primary gradient restores centered variation, the encoder metric is compatible with the approximation, and calibration transfers to the audited checkpoint.

A scalar reduction, $\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma)$, yields a saddle-node threshold $\alpha_c = \lambda_\star/\kappa$ under the explicit shape and endpoint assumptions of Theorem 2. Equating $1/q$ with this threshold additionally requires calibration to $\kappa/\lambda_\star$. Empirically, we compare $1/q$ with the half-height crossing $\alpha_{1/2}$ and report the operational collapse threshold separately.

Let $\theta \in \mathbb{R}^2$, $v_0 = -e_1 + \beta e_2$, and $z(\theta) = v_0 + \theta$. Choose a two-example encoder $H(\theta) = (z^\top, -z^\top)^\top$. Then $V = \|z\|^2/2$ and $\nabla_\theta V = v_0$ at $\theta = 0$. The losses $L_p = \|z - (v_0 - e_1)\|^2/2$ and $L_a^\pm = \|z - (v_0 \mp e_2)\|^2/2$ are nonnegative smooth functions of the representation. Their gradients at zero are e_1 and $\pm e_2$. Thus Proposition 2 concerns realizable encoder/loss configurations, not arbitrary incompatible gradients. Both task pairs have the same Gram matrix, while the signed variance derivatives differ.

Vectorization of the centered batch gives $u = (C_B \otimes I_m)h$ and $\nabla_\theta V = J^\top u/B$. Hence $\dot{V} = -u^\top J J^\top (g_p + \alpha g_a)/B$. For any fixed positive scale k, write $JJ^\top = k(I + E)$. This is a decomposition, not an assumption of isotropy. With the definitions in (8), the normalized derivative is $d + \epsilon_p - \alpha(a + \epsilon_a)$, which proves (9). Taking absolute values term by term proves (10). The exact discrepancy $\mathcal{E}$ can be smaller than Δ because signed terms cancel. Failure of the sufficient margin condition therefore leaves the probe undecided; it does not prove the probe wrong. Choosing k must be reported, since it changes the decomposition and the tightness of this conservative envelope, although it cannot change $\dot{V}$.

Record the batch, checkpoint, losses, auxiliary dose, probe value q, and scale k. Compute g_p, g_a for the actual operating losses and u from the unfiltered representation. Evaluate Kg by first computing $J^\top g$, then a Jacobian–vector product with J; a dense K is unnecessary. This yields $a, d, \epsilon_a, \epsilon_p$, the exact signed flux, and the validity margin. When $u = 0$ or $\|g_p\| = 0$, the normalized decomposition is undefined and the unnormalized identity (4) remains the appropriate diagnostic. This audit costs more than a frozen feature-space probe: at $B = 256$ and $m = 64$ on CPU the exact flux costs 0.30 s while the exact $k = \mathrm{tr}(K)/(Bm)$ certificate costs 20.7 s, a factor of 69.6 (`results/CERTCOST/`). The trace enters only through the canonical scale: an envelope at any other k, including the per-checkpoint minimizer of Table 2, needs the exact flux and the head-level gradients but no trace of K. The certificate is an offline audit of the deployed probe q: it identifies which terms prevent transporting that cheap decision to the operating checkpoint. The archive contains cheap proxies and intervention outcomes for 405 HAR checkpoints. The frozen predictor agrees in sign with the exact variance derivative on 51%. Its sufficient certificate fires on 2 checkpoints at the trace-normalized scale $k = \mathrm{tr}(K)/(Bm)$ and 15 under per-checkpoint scale optimization, with no incorrect firing.

The signed balance $\psi = d - \alpha a$ replaces the unsigned loading and primary-norm reference with directional quantities. When $d \neq 0$, it equals $d(1 - \alpha q^\star)$ with $q^\star = a/d$. Sign agreement rises to 58% overall and 61% at positive dose. Computing the balance needs the same two head-level backward passes as the original probe; certifying it additionally needs the exact flux.

Table 2 varies the predictor, residual envelope, and scale separately. At canonical k, replacing the frozen predictor with ψ raises coverage from 2 to 3 with separate absolute geometry terms and to 10 when their signed cancellation is retained. Per-checkpoint scale optimization gives 15, 121, and 154. All rows have zero false firings, as required by the sufficient condition. For comparison, the frozen predictor with directly measured $|\mathcal{E}|$ certifies 128 at canonical k. Every arm uses the common normalization N, so these counts compare the full retrospective audits on the same prediction scale.

The optimized combined certificate covers $57/226$ contracting checkpoints (25%) and $97/179$ expanding checkpoints (54%). Restricting to $|d| \geq 0.1$ leaves 168 checkpoints, of which $47/97$ contracting and $52/71$ expanding points certify. The archive also contains 51 points with $|d| \leq 0.01$. Using ψ directly avoids division by weak restoration and remains defined at $d = 0$. These subsets describe the same audit archive.

Figure 3 ranks observed sign agreement using margins that already consume the exact derivative. AUC is 0.60 for the canonical corrected combined arm, 0.80 after scale optimization, and 0.63 for the canonical frozen arm. Scale optimization can itself improve these retrospective scores: whenever ψ and $r - \alpha c$ have matching nonzero signs, choosing $k = (r - \alpha c)/(\|u\| n_p \psi / B) > 0$ makes the combined residual zero. Unrestricted optimization would therefore certify all 236 matching-sign points; the recorded 154 uses a finite 241-point grid. Script `certificate_matched_audit.py` regenerates the arms from stored primitives.

Table 2: Matched retrospective certificate ablations on 405 HAR checkpoints. "Contracting" counts certified points among the 226 negative exact derivatives. All rows have zero false firings. Canonical $k = \mathrm{tr}(K)/(Bm)$ is fixed; optimized k is selected per checkpoint on the same 241-point grid. Triangle and combined geometry residuals are $|\epsilon_p| + \alpha|\epsilon_a|$ and $|\epsilon_p - \alpha\epsilon_a|$. Every envelope uses the exact flux.

Predictor	scale	residual	fires	contracting
frozen $1 - \alpha q$	canonical	original triangle	2	2
frozen $1 - \alpha q$	optimized	original triangle	15	10
corrected ψ	canonical	geometry triangle	3	3
corrected ψ	optimized	geometry triangle	121	41
corrected ψ	canonical	geometry combined	10	10
corrected ψ	optimized	geometry combined	154	57
frozen $1 - \alpha q$	canonical	exact $\|\mathcal{E}\|$	128	115

The operational gate and the rank rule rely on different assumptions. For the gate, Theorem 1 requires the actual operating losses and encoder Jacobian, a recorded calibration value q, and discrepancy smaller than the sign margin at the audited checkpoint. For the minimum-rank claim, Theorem 5 requires a specified PSD operator; an empirical PCA basis needs separate identification. Deployed STFCAL evaluates neither certificate online. Table 3 records the scope of each arm.

Table 3: Assumptions behind the gate, rank rule, and deployed filters. The sign certificate and minimum-rank theorem concern different mathematical objects; the empirical arms retain their own calibration and basis assumptions.

Arm	Claim	Required assumption
gate $\alpha\hat{\rho} \geq 1$	local sign of $\dot{V}$ (Theorem 1)	frozen q; actual operating losses and encoder Jacobian; drift inside Δ
rank rule $\mathrm{rank}_{\mathrm{sel}}$	minimum deletion rank (Theorem 5)	specified PSD operator; nested candidate subspaces
PCA basis	empirical surrogate	identified separately; not assumed to be the operator's eigenbasis
deployed STFCAL	local heuristic	no online certificate; calibration audit only

Corollary 1 (Transport to a later audited checkpoint)**.** *Let q_0 be the probe frozen at calibration, and let q_t and Δ_t be the live probe and the envelope of Theorem 1 at a later checkpoint. If*

$$|1 - \alpha q_0| > \Delta_t + \alpha|q_t - q_0|,$$

then the frozen decision has the same sign as the exact variance derivative at that checkpoint.

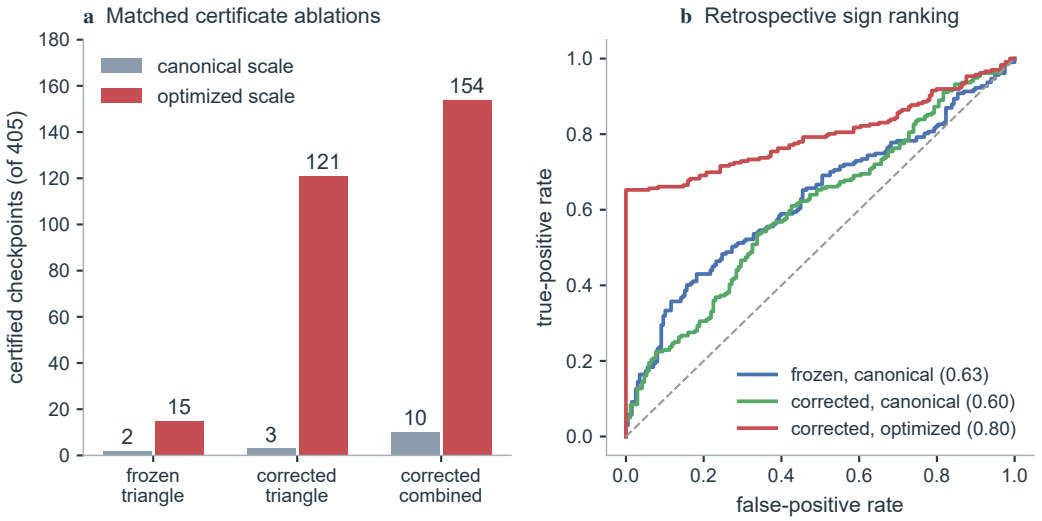


Figure 3: Certificate coverage depends on the predictor, residual, and scale. (a) Frozen and signed predictors under canonical and optimized scales, with separate and combined geometry residuals. (b) ROC curves for observed sign agreement. Every margin consumes the exact flux, so the curves describe retrospective audit discrimination.

Proof. $|(1-\alpha q_0)-(1-\alpha q_t)| = \alpha|q_t - q_0|$. If the displayed inequality holds, then $|1-\alpha q_t| \geq |1-\alpha q_0| - \alpha|q_t - q_0| > \Delta_t$, so Theorem 1 applies at t with the live terms. □

Corollary 2 (Signed directional balance)**.** *Let $\psi = d - \alpha a$. With the decomposition of Theorem 1,*

$$\frac{\dot{V}}{N} = \psi + \epsilon_p - \alpha\epsilon_a.$$

If $|\psi| > |\epsilon_p - \alpha\epsilon_a|$ at a specified $k > 0$, then $\dot{V}$ has the sign of ψ, including when $d = 0$. When $d \neq 0$, define $q^\star = a/d$ and $\Delta^\star = |\epsilon_p - \alpha\epsilon_a|/|d|$; the same condition is $|1-\alpha q^\star| > \Delta^\star$, with predicted sign $\operatorname{sign}[d(1-\alpha q^\star)]$. *The $|d-1|$ discrepancy is absent.*

Proof. Regroup (9) as $d - \alpha a + \epsilon_p - \alpha\epsilon_a$. A perturbation smaller in magnitude than ψ cannot change its sign. Substitution of $q^\star$ gives the ratio form when $d \neq 0$. □

The quantities $a \propto \sum_i (h_i - \bar{h})^\top g_{a,i}$ and $d \propto -\sum_i (h_i - \bar{h})^\top g_{p,i}$ are the centered auxiliary and primary radial derivatives. Their signed ratio $q^\star$ allows cancellation across examples. The archived probe instead averages per-example magnitudes divided by per-example primary norms. Replacing that primary reference by $d\, n_p$ removes the $|d-1|$ discrepancy.

Corollary 1 requires the frozen margin to exceed the current envelope plus probe drift. It certifies $2/405$ checkpoints at the canonical trace-normalized scale and $14/405$ under scale optimization, with no incorrect firing. The corresponding frozen certificate without the added drift term fires on $15/405$. Outside the certified checkpoints, calibration remains an empirical decision rule.

Figure 4 shows how closely the margin approaches the transport envelope. The 14 certified checkpoints in (a) lie just above it; the margin/envelope ratio in (b) has median 0.86 and upper quartile 0.996. Across the 27 width–dose cells in (c), sign agreement is 0.56 above median restoration alignment and 0.47 below it. Signed mismatch $|a - q|$ varies largely independently of d (log-scale correlation -0.05). Panel (d) compares $|\mathcal{E}|$ and the minimum envelope at the same per-checkpoint scale, with median ratio 0.885. Matching the normalization is essential to interpreting envelope tightness. These associations locate the informative region of the retrospective audit.

For $K = I$, the local condition is $d - \alpha a < 0$. Take $d = 0.1$, $a = q = 0.2$, and $\alpha = 1$: the nominal gate is closed while the exact derivative is negative. This is a false negative caused by treating the primary norm as radial restoration. Conversely, with $d = 1$, $a = -q = -2$, and $\alpha = 1$, the gate is

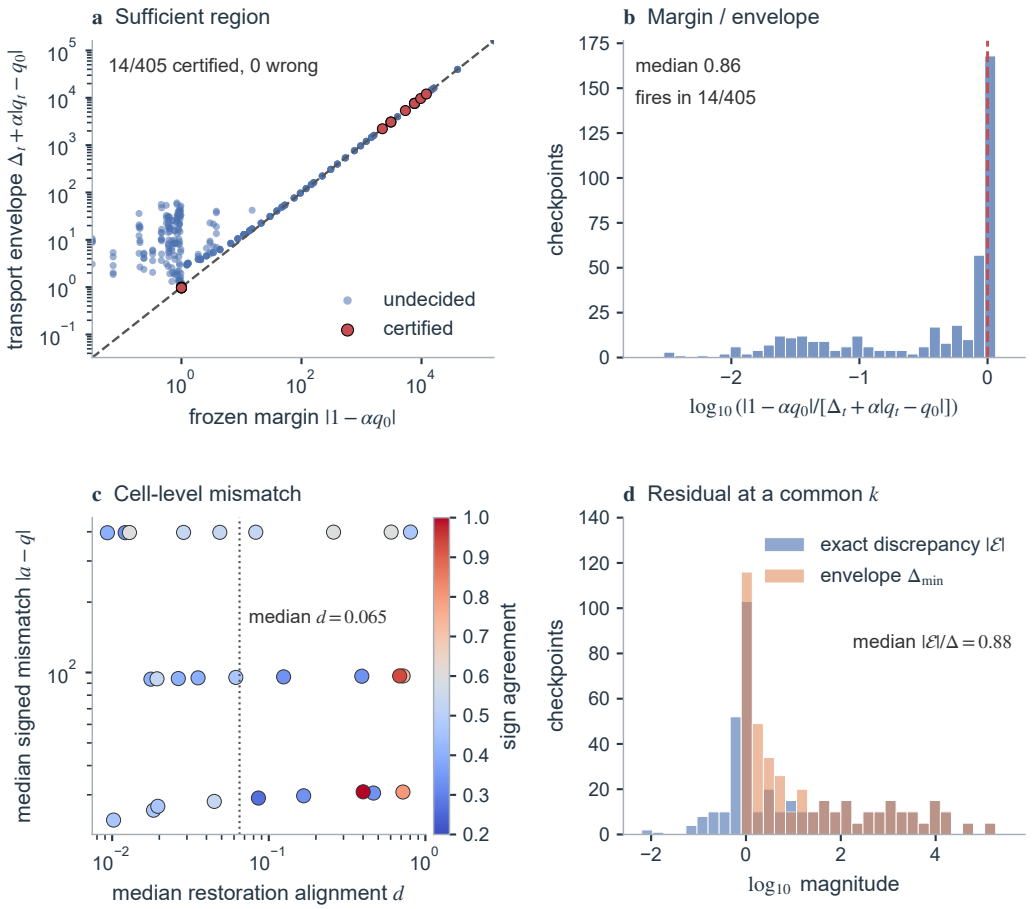


Figure 4: Most audited checkpoints fall outside the sufficient transport margin. Audit of 405 HAR checkpoints. (a) Frozen margin versus the optimized transport envelope; 14 checkpoints certify, all correctly. (b) Margin/envelope ratio: median 0.86, upper quartile 0.996. (c) Restoration alignment and signed mismatch across 27 width–dose cells. (d) Exact discrepancy and envelope at the same optimized scale, with median $|\mathcal{E}|/\Delta_{\min} = 0.885$.

open but the exact derivative is positive. For a fixed q, the theorem applies at each time with time-indexed components. If $\sup_{t\in I} \Delta_t < |1 - \alpha q|$, its sign conclusion holds throughout the interval I. Obtaining this uniform bound is an additional task; a single frozen observation does not supply it. Even a negative variance derivative on a finite interval does not by itself imply convergence to zero.

Let s_θ be the actual proposed parameter increment, including any optimizer preconditioning, momentum, and weight decay. If ∇V is L-Lipschitz on the segment from θ to $\theta + s_\theta$, then

$$\left|V(\theta + s_\theta) - V(\theta) - v^\top s_\theta\right| \le \tfrac{L}{2}\|s_\theta\|^2. \tag{12}$$

This follows by integrating the gradient along the segment. A first-order margin exceeding this remainder certifies the sign of the finite-step change. The gradient-flow decomposition applies directly to plain gradient updates; the actual-step audit is the corresponding observable for Adam. Separating primary and auxiliary contributions through a nonlinear optimizer requires specifying the counterfactual update and its state. A feature-gradient norm ratio is not an Adam-step certificate.

The controlled driver computes the probe in `rho_probe` within `cross_domain_mtl.py`:

$$\hat{\rho} = \frac{1}{B}\sum_{i=1}^{B} \frac{\|\Pi_{u_i}\widetilde{g}_{a,i}\|}{\|g_{p,i}\| + 10^{-8}}, \qquad \widetilde{g}_a = \nabla_H \frac{\|C_B H\|_F^2}{Bm}. \tag{13}$$

Training instead uses the model's fixed random projections. Thus a contracting surrogate, per-example averaging, and regularization all enter q. The real-head driver differentiates the specified reconstruction, identity, or SNR loss. The A1 alignment records are probes *after* joint training; their mean cosine is an average of examplewise cosines, not the cosine of the aggregate parameter gradients. Their unsigned loading cannot establish the sign of $v^\top G_a$. The complete main-figure audit is generated by `paper/revision/build_figures.py`, with source hashes and reconstructed statistics. Duplicate low-label dose logs are reconciled by $(\alpha, \mathrm{mode}, \mathrm{seed})$ and checked for equality before removal.

`paper/revision/check_boundary.py` checks the exact discrepancy, the sufficient sign certificate, the realizable conflict example, and the finite-step remainder on seeded linear encoders. These are algebraic checks of the new theory, separate from the archived neural-network experiments.

A.3 Conditional Dynamical Models

Definition 2 (Degenerate minimizer)**.** *A loss has a degenerate minimizer if a constant representation attains its minimum. For a differentiable interior minimum, stationarity is necessary but insufficient. Variance shrinkage has such minima; reconstruction with varying targets generally does not, because a constant code need not be stationary for a fixed decoder. Accordingly, we evaluate trained reconstruction heads empirically.*

Theorem 2 (Conditional scalar collapse transition)**.** *Assume an independently justified scalar reduction*

$$\dot{\sigma} = -\alpha\kappa\sigma + b(\sigma), \qquad \kappa > 0, \tag{14}$$

where $b(0) = 0$. Let $f(\sigma) = b(\sigma)/\sigma$ on $(0, \infty)$ tend to zero at both ends, increase strictly to a unique nondegenerate maximum $\lambda_\star = f(\sigma_\star) > 0$, and decrease strictly afterwards. Then $\alpha_c = \lambda_\star/\kappa$. For $0 < \alpha < \alpha_c$ there are two positive equilibria, the smaller unstable and the larger stable; initial values below the smaller one still collapse. At α_c the equilibria coalesce in a saddle-node; for $\alpha > \alpha_c$, every positive trajectory tends to zero.

Proof. The vector field is $\sigma(f(\sigma) - \alpha\kappa)$. The crossings and their stability follow from monotonicity; the nonzero second derivative gives the nondegenerate fold. Above the maximum the field is strictly negative for every $\sigma > 0$, whose only possible limiting equilibrium is zero. □

Lemma 1 (Departure from a primary optimum)**.** *At $G_p = 0$, any nonzero G_a makes the joint gradient nonzero when $\alpha > 0$. This implies departure from the primary optimum, but neither collapse nor irreversibility.*

Corollary 3 (Fixed-dose attenuation)**.** *In Theorem 2, a constant auxiliary multiplier $q \in (0, 1]$ replaces the effective dose by $q\alpha$ and the threshold by α_c/q. Positive attenuation can therefore preserve the positive equilibrium at a fixed dose. Exact deletion removes the contraction term but is not necessary for fixed-dose protection.*

Theorem 3 (First modal contraction versus full linear collapse)**.** *For the linear symmetric model $\dot{z} = (R - \alpha A)z$, assume $R \succ 0$ and $A \succeq 0$ in the same coordinates. Along an eigenvector of $R - \alpha A$ with eigenvalue ν_j, the modal variance obeys*

$$\dot{\varsigma}_j^2 = 2\nu_j\varsigma_j^2. \tag{15}$$

Let

$$T = R^{-1/2}AR^{-1/2}, \qquad \rho_\star = \lambda_{\max}(T). \tag{16}$$

Some mode contracts iff $\alpha\rho_\star > 1$. Every mode contracts exponentially iff $\alpha\lambda_{\min}(T) > 1$. Equality leaves a neutral mode.

Proof. $R - \alpha A = R^{1/2}(I - \alpha T)R^{1/2}$ has the same inertia as $I - \alpha T$ by congruence. Since the generator is symmetric, its eigenvalue signs determine modal growth or decay. □

The scalar model and the symmetric linear model answer different questions. Neither follows from measuring the proxy $\hat{\rho}$ alone. For example, with $R = I$, $A = \mathrm{diag}(2, 1/2)$, and $\alpha = 1$, one mode contracts and another expands. This permits task performance on a surviving direction without preserving the full representation. The NYUv2 result is compatible with this mechanism but does not identify its dynamical eigenvectors.

A.4 CALIBRATION AND PROTOCOL CONTROLS

A.4.1 PROTOCOLS AND OBSERVABLES

The controlled benchmark spans HAR (Anguita et al., 2012), RadioML (O'Shea et al., 2016), battery health (Saha & Goebel, 2007), and NYUv2. A three-layer MLP with a fixed projected-variance auxiliary isolates contraction; Transformer, CNN, real-head, CelebA, and financial studies test its scope. Appendices F–H give the protocols and optimizer grids. Error bars show population seed SD unless stated otherwise.

The operational label is relative scatter $s = \mathcal{S}_h/\mathcal{S}_{\text{single}} < 0.3$, using the same-seed single-task reference and

$$\mathcal{S}_h = \mathbb{E}_b\Big[\mathrm{std}_j\Big(h_j^{(b)} - \mathbb{E}_{b'} h_j^{(b')}\Big)\Big]. \tag{17}$$

The proxy is coordinate-dependent and blind to the all-ones direction. "Collapse" therefore refers to the declared 0.3 criterion; information is measured separately. The log-dose interpolated crossing at $s = 0.5$, $\alpha_{1/2}$, summarizes the dose curve.

A.4.2 CALIBRATION IS TASK- AND WIDTH-DEPENDENT

Across sixteen domain–width cells, the ratio of predicted dose to observed half-height crossing is 0.95–3.01 for battery and 30–116 for RadioML; HAR reverses bias with width (Figure 7a). All cells train projected auxiliaries and probe the isotropic surrogate.

A.4.3 CALIBRATION AND DRIFT CHANGE THE DECISION

On HAR at $m = 64$, detached warm-up moves $1/\bar{\rho}$ from 1.34 at $W = 1$ to 0.00907 at $W = 4$, against a half-height crossing of 0.00141. At dose 0.03, $W = 1, 2$ leave all gates closed and all ten seeds collapse; $W \geq 4$ opens every gate and prevents collapse. At dose 100, all tested warm-ups protect scatter (Figure 7b,d). An extreme dose can therefore hide a calibration error.

Recomputing the probe during training can produce false alarms. HAR ratios grow even at zero auxiliary dose, and running gates activate in every seed of ten configurations with final scatter 0.50–1.04. Battery has one activation among 35 seeds ending above half-height (Appendix H.2). A shrinking primary-gradient denominator can inflate the ratio. On 405 HAR checkpoints, the unsigned predictor agrees with the exact variance derivative on 51%, with median restoration alignment 0.065.

A.4.4 REAL AUXILIARIES AND STRONGER ENCODERS

Four real auxiliary heads separate loading from contraction (Figure 5a). Mean per-example task-gradient cosines are at most 0.06 in magnitude, while SNR radial loading is 88%. Yet the endpoint signed radial cosine is negative on SNR, and $c_t = v^\top G_a < 0$ at $74/75$ historical epoch-end checkpoints, so sampled flow contraction does not explain its collapse. Flow and Adam signs also disagree on $33/75$ sampled paired updates; we record the applied increment and a primary-only Adam counterfactual (Appendix H.7).

Protocol changes alter the observed failure. Historical SNR scatter onset lies between doses 3 and 10, whereas I/Q reconstruction stays above the criterion through 10000. Under healthy-checkpoint continuation, I/Q at 3000 gives frozen accuracy 0.301 versus 0.304 detached, with trace 1.061; SNR at 30 gives 0.243 versus 0.304, with scatter 0.870. Resetting auxiliary weights or their Adam state also fails to reproduce the historical severe collapse (Appendix H.11). Warm-up, split, architecture, and horizon define the operating regime.

A waveform CNN on the same I/Q split reaches 0.424 healthy-readout accuracy. At reconstruction dose 3000, frozen accuracy is 0.415 versus 0.418 detached, while the current head reaches 0.437. This stronger encoder leaves the consumer largely intact on the fixed real-auxiliary grid (Appendix H.10). A separate ResNet fine-to-coarse consumer screen fails its utility gate (Appendix H.22).

B Detailed Intervention Comparisons

B.1 Filtering the Auxiliary Update

Detachment blocks auxiliary gradients to the encoder and provides the reference when the auxiliary brings no primary-task value. The spectral toxicity filter (STF) instead removes a chosen component of the auxiliary channel. Its idealized backward form is

$$F_w[g_a] = g_a - w\Pi_u g_a, \qquad w \in [0,1], \tag{18}$$

The implemented selective arms transform the auxiliary input, $\widetilde{H} = H - wC_BHP$, using a frozen projector P or a centered radial direction. The transformation changes the forward input as well as its gradient, while the auxiliary head continues training (Appendix E).

The calibrated arm freezes the warm-up probe and opens the gate when $\alpha\hat{\rho} \geq 1$. The rank variant then selects

$$\mathrm{rank}_{\mathrm{sel}} = \min\{\mathrm{rank} \leq \mathrm{rank}_{\max} : \alpha\hat{\rho}_{\mathrm{res}}[\mathrm{rank}] < 1\}, \tag{19}$$

from nested candidate subspaces. If none passes, it uses the cap and records the unresolved residual. The radial arm is a separate intervention. Theorem 5 gives the minimum deletion rank for a specified PSD operator; using an empirical PCA basis adds a basis-identification question. Gate calibration and subspace choice therefore need separate evaluation.

The low-label SNR dose scan shows both the value and limits of filtering (Figure 8a,b). With 20% primary labels, joint training is healthy at low dose but collapses in every seed from $\alpha = 30$. Calibrated STF avoids collapse at every sampled dose and exceeds detachment by 0.5–1.0 percentage points over $\alpha \in [15, 100]$; the advantage disappears at extreme doses. This five-seed scan uses a different label budget from the fully labeled study in Appendix H.25.

The fully labeled SNR comparison shows why clearing the scatter threshold is insufficient (Figure 8c). Detachment reaches 0.307 accuracy with $0/10$ collapses. The radial arm reaches 0.199 with $1/10$, and the rank-3 arm 0.211 with $0/10$. Basis choice also changes how much auxiliary learning is removed: in HAR reconstruction, an auxiliary-gradient basis uses 30.75 directions versus PCA's 62.0 among the same four gate-open seeds, at similar accuracy (Table 7).

At the controlled high doses, calibrated filtering reduces collapse from $10/10$ to $0/10$ on HAR, battery, and RadioML, and from $5/5$ to $0/5$ on CelebA Smiling (Liu et al., 2015). With a Transformer at RadioML dose 3000, joint training collapses in $10/10$ seeds at accuracy 0.156; filtering gives $0/10$ collapses and relative scatter 0.970. Reliable gate activation at these extreme doses coexists with the calibration errors in Figure 7.

The battery application uses a temporal encoder with a mixer and gated state-space block, feeding SOH, RUL, reconstruction, and distributional heads. The corrected protocol keeps train, validation, and test cells disjoint, selects checkpoints on validation SOH, and separates observed-lifetime MAE from censored shortfall. Across ten initializations, detachment minus joint training is -3.47 SOH percentage points (95% t interval $[-4.20, -2.74]$) and -34.8 observed-RUL cycles ($[-57.8, -11.8]$). These intervals describe initialization variability on the fixed split.

On this MATR protocol (Severson et al., 2019), detachment improves SOH RMSE from 5.57 to 2.10 percentage points and observed-RUL MAE from 81.5 to 46.7 cycles (Table 5); selective filtering gives no further improvement. Joint training has relative covariance trace 3.55, so the gain is a negative-transfer result without representation contraction. The rank arm's large trace and low participation rank also illustrate why amplitude and diversity need separate measurement. Checkpoint selection and censoring-aware evaluation changed together, so their separate contributions to the new ranking are unknown.

Figure 6 collects the dose sweeps, width crossings, NYUv2 gate boundary, intervention counts, and financial outcomes. Width changes both empirical crossing locations and probe calibration (Yang & Hu, 2021); Appendix D examines their scaling separately.

Table 4: Controlled optimizer comparisons, ten seeds per method. Cells give collapse rate and mean task metric. The optimizer grid uses $(m, \alpha) = (128, 0.1)$ for HAR, $(32, 3000)$ for battery, and $(64, 10000)$ for RadioML. Battery uses the controlled variance auxiliary. Detachment's collapse counts come from separate $m = 64$ controls at doses $100, 100, 3000$; its task metrics belong to those protocols (Tables 5–6). Detachment removes the encoder's dose dependence, but these widths differ from the optimizer grid.

Combiner	HAR acc. ↑	Battery log-SOH RMSE ↓	RadioML acc. ↑
JOINT (plain summation)	1.00 (0.945)	1.00 (0.611)	1.00 (0.256)
conflict resolution			
PCGrad	1.00 (0.585)	1.00 (0.601)	1.00 (0.091)
CAGrad (c=0.5)	1.00 (0.952)	1.00 (0.607)	1.00 (0.091)
Pareto			
MGDA	1.00 (0.200)	1.00 (0.385)	1.00 (0.091)
gradient geometry			
Aligned-MTL	1.00 (0.865)	1.00 (0.609)	1.00 (0.091)
IMTL-G	1.00 (0.419)	1.00 (0.601)	1.00 (0.091)
Nash-MTL	1.00 (0.280)	1.00 (0.552)	1.00 (0.091)
loss weighting			
FAMO	1.00 (0.182)	1.00 (0.613)	1.00 (0.091)
UW (uncert.)	1.00 (0.940)	1.00 (0.611)	1.00 (0.091)
GradNorm	1.00 (0.958)	1.00 (0.611)	1.00 (0.170)
IMTL-L	1.00 (0.947)	1.00 (0.611)	1.00 (0.091)
Detachment (STF-0)	0.0	0.0	0.0

Table 5: Corrected MATR test results: ten initialization seeds on a fixed cell-disjoint split, with validation-based checkpoint selection. $S = \mathrm{tr}\,\mathrm{Cov}(H)$; relative S is the mean same-seed ratio to the SOH-only reference; PR is participation rank. Censored shortfall measures lower-bound violations in cycles. Table 12 reports uncertainty.

Mode	SOH RMSE% ↓	obs. RUL MAE ↓	shortfall ↓	S	relative S	PR
SOH-only	0.61	—	—	1.01	1.00	1.35
JOINT	5.57	81.5	17.60	2.86	3.55	2.49
STF-0	**2.10**	**46.7**	**2.06**	4.77	6.24	1.86
STFCAL	2.71	63.6	5.73	1.40	1.89	2.03
STFCALRANK	4.98	107.6	17.57	617.90	651.77	1.21

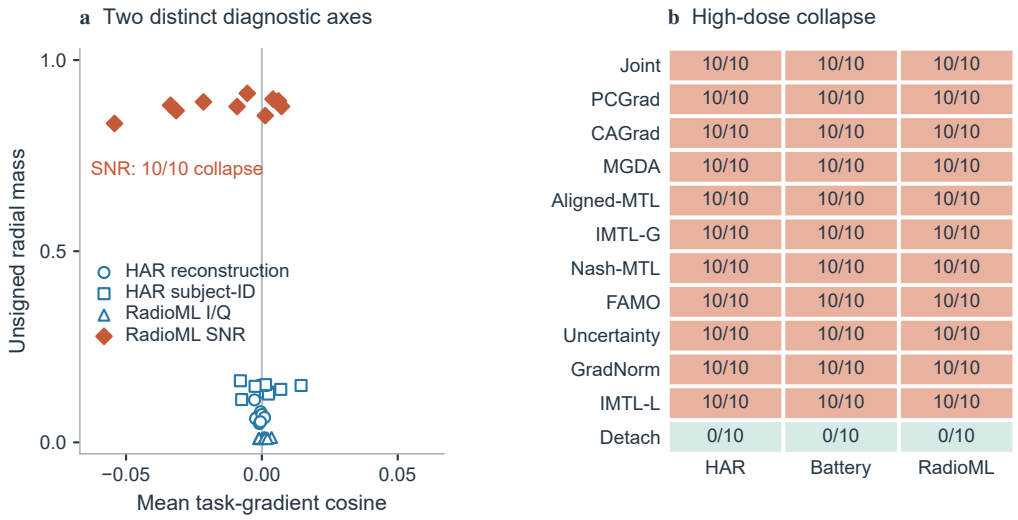


Figure 5: Task-gradient alignment leaves contraction unresolved. (a) Terminal probes of four real auxiliary heads, ten seeds each: mean per-example task-gradient cosine versus unsigned radial loading. SNR has high loading and $10/10$ collapsed endpoints; its signed flux is audited separately. (b) All ten optimizer baselines meet the scatter-collapse criterion in $10/10$ seeds at controlled doses 0.1 (HAR), 3000 (battery), and 10000 (RadioML). Detachment uses separate ten-seed controls. Table 4 gives widths and task metrics.

Table 6: Real auxiliary heads at width 64, ten seeds: mean primary accuracy and collapse rate in parentheses. The listed radial selective arms do not match detachment on high-dose SNR regression. The separate rank-3 arm also avoids collapse in all ten SNR seeds, but reaches only 0.211 accuracy (Figure 8c).

Auxiliary head	α	JOINT	STF-0	STF-HARD	STF-SOFT	STFCAL
HAR input-reconstruction	0.1	.942 (0.0)	.943 (0.0)	.942 (0.0)	.942 (0.0)	.942 (0.0)
HAR subject-ID probe	0.1	.941 (0.0)	.943 (0.0)	.944 (0.0)	.943 (0.0)	.942 (0.0)
HAR subject-ID probe	10	.830 (0.0)	.940 (0.0)	.940 (0.3)	.939 (0.3)	.940 (0.0)
RadioML I/Q-reconstruction	3000	.343 (0.0)	.307 (0.0)	.329 (0.0)	.327 (0.0)	.315 (0.0)
RadioML SNR regression	3000	.205 (1.0)	**.307 (0.0)**	.208 (1.0)	.200 (0.9)	.199 (0.1)

Table 7: Real HAR reconstruction ($m = R = 64$, ten seeds). Candidate ranks include closed-gate seeds; active ranks average only gate-open seeds. Accuracy and scatter average all ten seeds. PCA and auxiliary-gradient bases activate on identical seeds; these results establish neither superiority over joint training nor globally optimal deletion.

α	basis	gate open	candidate rank	active rank	rel. scatter	accuracy
12	PCA	$4/10$	57.0	62.00	0.842	0.9492
12	Aux. grad.	$4/10$	22.6	30.75	0.811	0.9497
30	PCA	$8/10$	62.8	63.62	0.754	0.9512
30	Aux. grad.	$8/10$	44.1	47.88	0.746	0.9531
100	PCA	$10/10$	64.0	64.00	0.630	0.9524
100	Aux. grad.	$10/10$	59.4	59.40	0.654	0.9520

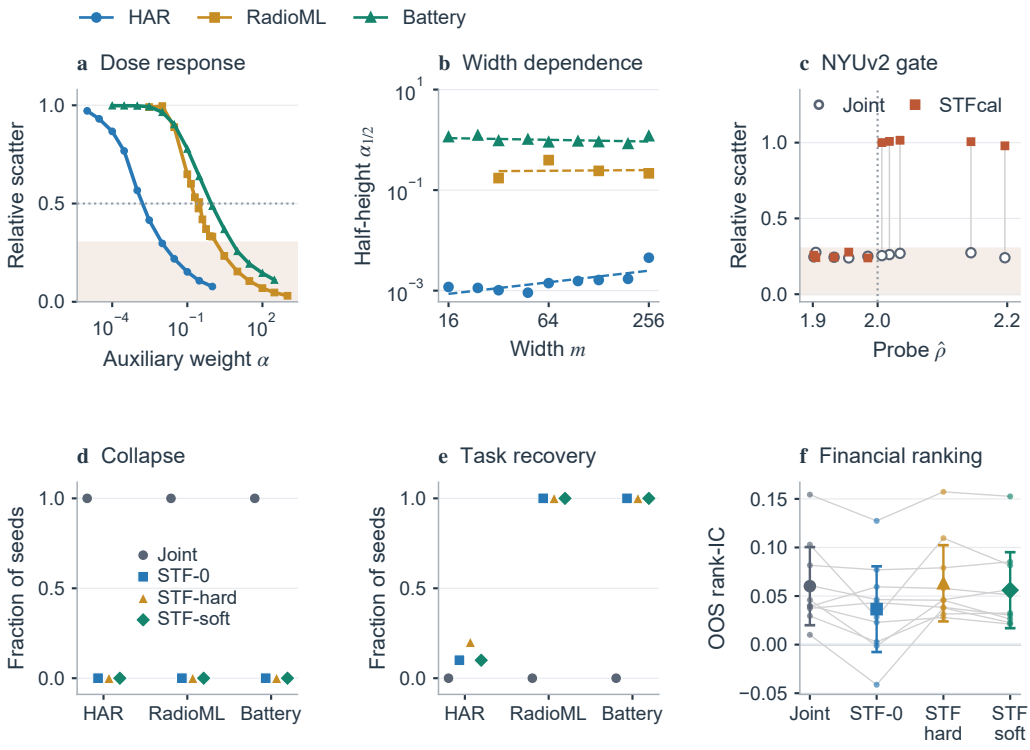


Figure 6: Additional diagnostics across domains. (a) Dose–scatter curves and empirical half-height crossings. (b) Width dependence. (c) Paired NYUv2 gate-boundary outcomes at $\alpha = 0.5$. (d,e) Collapse and task-recovery rates for the controlled interventions. (f) Financial rank-IC, paired by temporal fold. The corresponding tables specify protocols, seed counts, and the separate 0.3 collapse criterion.

C Financial Ranking Application

The motivating observation came from multi-task equity selection. A production model shared an encoder of A-share factors across pairwise ranking, distributional (MDN (Bishop, 1994)), reconstruction (AE), and view-contrastive (SCARF-style (Bahri et al., 2022)) heads. After deployment, predicted rankings became nearly constant under the unchanged training objective: cross-sectional prediction spreads fell by approximately 93% relative to validation while individual training losses stayed low. The observation prompted us to measure feature and prediction variation alongside loss; identifying its cause required the controlled studies.

The separate financial experiment uses walk-forward training with real distributional, reconstruction, and contrastive heads. Its R5fin records set `fixed_aux=0`, so Table 15 evaluates real auxiliaries. Across ten folds and ten initialization seeds, mean out-of-sample (OOS) rank-IC is 0.060 for joint training, 0.036 for detachment, 0.063 for the hard gate, and 0.056 for the soft gate. Averaging seeds within each fold gives paired differences from joint of -0.0237, $+0.0029$, and -0.0042. Two-sided Wilcoxon p values are 0.037, 0.432, and 0.695; none survives Holm correction across the three comparisons. Temporal folds are dependent, so these comparisons are sensitivity analyses. Detachment costs accuracy in this sample, and no retained subspace shows a benefit. Script `finance_fold_audit.py` reproduces the comparisons from all 40 source files.

In the separate synthetic testbed, initialization probes give $\rho = 0.74 \pm 0.19$ over 12 seeds and predicted critical weight $\alpha_c = \langle 1/\rho \rangle = 1.45 \pm 0.37$, consistent with the observed onset. Other domains use the projected-gradient diagnostic of Definition 1, but its numerical value depends on the task, encoder, and calibration time.

D Proofs and Additional Analysis

D.1 Statements Deferred from the Main Text

We formalize several statements discussed in the main text. Theorem numbering continues from the main text.

Proposition 3 (Sufficient conditions for gradient vanishing)**.** *For a differentiable prediction map with Jacobian J and output loss gradient ℓ, the parameter gradient vanishes iff $J^\top \ell = 0$. Sufficient conditions are $\ell = 0$, or identical Jacobian rows together with $\mathbf{1}^\top \ell = 0$. Equal prediction values alone imply neither condition.*

Proof. Apply the chain rule. For a counterexample to the stronger assertion, $\hat{y}_i = \theta x_i$ is constant at $\theta = 0$ but has Jacobian rows x_i. The pairwise logistic loss $\log(1 + e^{-(\hat{y}_1 - \hat{y}_2)})$ has output gradient $(-1/2, 1/2)$ at a tie, so its parameter gradient is $(x_2 - x_1)/2$, generally nonzero. □

Proposition 4 (Scope of positive attenuation)**.** *Under Theorem 2, a fixed positive multiplier q leaves a finite threshold α_c/q but can prevent the transition at a specified dose. A dose-dependent positive multiplier can keep the effective dose subcritical for every α: for $0 < \epsilon < 1$, take $q(\alpha) = (1 - \epsilon)/(1 + \alpha/\alpha_c)$.*

Proof. Substitute $q\alpha$ for α. The proposed choice gives $q\alpha/\alpha_c = (1-\epsilon)\alpha/(\alpha+\alpha_c) < 1$. A positive stable equilibrium remains, although its lower basin boundary persists. Preserving gradient direction therefore does not imply that positive task weighting must fail. □

Proposition 5 (Rank-dilution scaling)**.** *Assume the effective ratio scales as $\rho_\star \asymp (m_\Pi/m)^{\gamma_0}$, $\gamma_0 \geq 1$, and $m_\Pi \propto m^\beta$, $0 \leq \beta < 1$. Then $\alpha_c = 1/\rho_\star$ implies*

$$\gamma \;=\; \gamma_0\,(1-\beta), \qquad \gamma_0 \geq 1, \tag{20}$$

The separable scaling is assumed; rank alone does not imply it. More generally, restoration $\asymp m^a$ and contraction $\asymp m^b m_\Pi^c$ yield $\gamma = a - b - c\beta$, with the displayed form requiring $c = a - b = \gamma_0$. If the toxic norm scales as m_Π^p, then $c = 2p$ only when contraction depends quadratically on that norm and has no additional rank dependence. The equality $\gamma_0 = 2p$ further requires $a - b = 2p$. Neither exponent is identified by the toxic eigenvalue spectrum alone. A fixed effective rank gives $\beta = 0$; a fixed per-head output dimension need not.

Corollary 4 (Composition of rank-one PSD operators)**.** *Let $T = \sum_k \rho_k u_k u_k^\top$, with unit u_k and $\rho_k \geq 0$, in common restoration-normalized coordinates. If the directions are mutually orthogonal, $\lambda_{\max}(T) = \max_k \rho_k$; if aligned, it is $\sum_k \rho_k$. An additional orthogonal direction leaves the maximum unchanged only if its coefficient does not exceed the previous maximum.*

Proof. For orthogonal directions, the u_k are eigenvectors. For aligned directions, the sum is a single rank-one operator. Neither case establishes the geometry of the production model's heads. □

For an explicit scalar example, take $b(\sigma) = \lambda\sigma^2/(1+\sigma^2)$. Then

$$\sigma_\pm = \frac{\lambda \pm \sqrt{\lambda^2 - 4(\alpha\kappa)^2}}{2\alpha\kappa}, \qquad \alpha_c = \lambda/(2\kappa). \tag{21}$$

This verifies the fold and its basin dependence. Radial auxiliary behavior,

$$g_a = \kappa\delta h + o(\|\delta h\|), \tag{22}$$

requires isotropic curvature; stationarity at a constant representation is insufficient. For a general encoder, the Jacobian still determines its effect on variance through (4). In discrete time, linearizing a scalar stable equilibrium (Strogatz, 1994) with derivative $-\ell$ gives

$$e_{t+1} = (1 - \eta\ell)e_t, \qquad 0 < \eta\ell < 2, \tag{23}$$

where e_t is a deviation from that equilibrium, not total representation scatter.

Power-law spectra do not determine the capacity exponent. For eigenvalues $\lambda_i = Ci^{-(1+s)}$ with C independent of rank, $\|A_r\|_F^2 = C^2 \sum_{i=1}^r i^{-2(1+s)}$. Hence the Frobenius norm is bounded for $s > -1/2$, grows as $\sqrt{\log r}$ for $s = -1/2$, and as $r^{-s-1/2}$ for $s < -1/2$. A decaying spectrum ($s > -1$) therefore has power exponent at most $1/2$; for the commonly intended $s > 0$ it has exponent zero. The trace instead involves $\sum_i i^{-(1+s)}$, and the operator norm is C for a decreasing spectrum. Rank-dependent normalization $C(r, m)$ changes these conclusions. Thus neither a universal $\gamma_0(s)$ nor $\gamma_0 = 2p$ follows without specifying the norm, normalization, restoration and coupling to dynamics. Concretely, if $\kappa \asymp m^b \|A_r\|^q$, $\|A_r\| \asymp r^p$, and $r \asymp m^\beta$, then $\gamma = a - b - qp\beta$. The restricted form $\gamma_0(1-\beta)$ requires $a - b = qp = \gamma_0$. Hence $\gamma_0 = 2p$ holds only under the additional quadratic coupling $q = 2$ and matching restoration exponent $a - b = 2p$; these are not implied by the measured width slope.

The original two-head generator has $W_k \in \mathbb{R}^{m\times r}$ with independent entries of variance $1/r$. Its combined span has rank $\min(m, 2r)$ almost surely. For the averaged quadratic loss the sample-normalized Hessian is $A = 2\sum_k W_k W_k^\top/(2r)$, with $\mathbb{E}A = 2I/r$. At $r = 32$, widths $32, 64, 128, 256$ therefore have ranks $32, 64, 64, 64$. Fixed per-head rank is not fixed combined rank over this range, and changing r also changes curvature. The supplied orthonormal control instead constructs $A = Q\,\mathrm{diag}(\lambda)Q^\top$ at exactly specified rank and normalization.

The RS1 experiment fixes per-head rank $r \in \{4, 8, 16\}$ across widths $32, 48, 64, 96$, giving combined ranks $8, 16, 32$. At anchor 3×10^{-4}, descriptive half-height slopes are -0.31, 0.45, and 3.02; minimum relative scatter ranges are 0.040–0.068, 0.013–0.018, and 0.0056–0.0132. Using the unpenalized anchor changes the slopes to 0.65, 5.55, and 3.88. The fitted exponent is therefore sensitive to both rank and anchor. The analyzer retains all twelve curves and reports crossings and scatter floors separately.

The orthonormal control constructs the sample-normalized auxiliary Hessian $A = Q\,\mathrm{diag}(\lambda)Q^\top$ at exact rank r and verifies its amplitude. Three normalizations hold different quantities fixed: $\lambda = 1$ fixes curvature per direction, so the Frobenius norm grows as $\sqrt{r}$; $\lambda = 1/r$ fixes the trace; and $\lambda = 1/\sqrt{r}$ fixes the Frobenius norm. The training grid tests the first two at $m \in \{32, 64, 128\}$, $r \in \{4, 16\}$, seven doses, and three seeds, holding the loss, optimizer, and data generator otherwise fixed. Three observations follow. First, at fixed width and per-direction curvature, the $r = 16$ crossing is lower than the $r = 4$ crossing by factors of 4.8 at $m = 64$ and 228 at $m = 32$. Second, the fitted width slope for $r = 4$ changes from -0.29 under $\lambda = 1$ to $+0.91$ under $\lambda = 1/r$. Third, minimum scatter is 3.5–11% of the healthy reference for $r = 4$, versus 0.4–1.1% for $r = 16$. Thus normalization affects the estimated exponent even when rank is controlled exactly. Each three-width fit has only one residual degree of freedom and cannot identify a precise exponent. The commands

`controlled_rank_scan.py --self-test` and `run_controlled_rank_grid.py` verify the construction and reproduce the twelve configurations. The historical squared pairwise hinge has zero margin, so tied scores can attain zero primary loss. These controls concern that objective; they do not establish the same transition for informative ranking objectives. The separate `--primary-loss logistic` option removes this degeneracy, but no unfinished logistic sweep is counted as evidence.

The synthetic testbed uses random toxic projections of rank $\lfloor m/2 \rfloor$ and a low-rank primary task with 2–8 active directions across seeds (mean 5.3). It sweeps 17 logarithmically spaced doses over $[3 \times 10^{-4}, 10]$ at six widths $m \in \{8, \dots, 256\}$ under a pairwise-ranking loss. The toxic-rank schedule is proportional to width. The relative half-height rule gives $\gamma = 0.07$ across six widths and $\gamma = 0.01$ across the four widths used below. This nearly flat trend is consistent with the restricted law at $\beta = 1$, but does not independently establish that law.

Using a fixed absolute scatter threshold instead gives $\gamma = 1.53$ ($R^2 = 0.91$) on the same curves. At $\sigma_h = 1.5 \times 10^{-3}$, the crossing depends on the width-dependent healthy amplitude: the fixed-rank family has $\sigma_h(0) \propto m^{0.95}$. The absolute rule also gives nearly constant slopes $1.94, 2.02, 2.04, 1.96, 1.89$ across rank schedules $\beta = 0, \dots, 1$, varying by only 0.15 and largely masking schedule dependence. Using relative crossings, the fixed-per-head-rank family instead gives $\gamma_0 = 2.43$ ($R^2 = 0.90$, four widths). Its combined span can still grow as $\min(m, 2r)$, so this is not an exact fixed-effective-rank estimate.

Both synthetic families extend the dose grid to $\alpha = 0$ (21 points), using four widths $m \in \{32, 64, 128, 256\}$, five seeds, and 800 epochs. The fixed-rank family is the $\beta = 0$ case with constant per-head rank r. A corresponding ten-seed run yields crossings differing by $+3.1\%, +1.8\%, -5.8\%, +19.7\%$ across these widths. This comparison quantifies sensitivity to seed count, including variation in the healthy anchor at larger widths.

Two checks qualify the crossing estimator. First, a half-height crossing can occur without near-total collapse. For fixed per-head rank $r \leq 4$, scatter plateaus at 2–23% of its healthy value; only $r \geq 8$ reaches a floor of at most 0.9%. These floors must be distinguished from the operational 0.3 collapse threshold. Second, the synthetic fits use the first penalized dose, $\alpha = 3 \times 10^{-4}$, as the anchor. The unpenalized value is set by the ranking loss, is shared across toxic ranks, and lies 30–40% above the penalized trend. Varying the anchor over $\alpha \in [0, 3 \times 10^{-3}]$ preserves the decreasing ordering of $\gamma(\beta)$: $\gamma_0 \in [1.55, 3.23]$ at $\beta = 0$, versus $\gamma(1) \in [-0.06, +0.74]$. The ordering is more stable than the fitted coefficients.

The rank-schedule fits test a conditional scaling hypothesis. Given $\alpha_c \asymp (m/m_\Pi)^{\gamma_0}$ and an independently measured $m_\Pi \asymp m^\beta$, substitution gives $\gamma = \gamma_0(1 - \beta)$. Testing the premise requires measuring effective rank separately from the observed slope. The generator audit above specifies how per-head and combined ranks differ.

Crossing rules affect the fitted magnitudes. For HAR, an absolute threshold $T \in [0.20, 0.50]$ gives $\gamma \in [0.66, 0.82]$, compared with 0.66 under the four-width relative rule. Battery requires a grid that reaches a healthy branch: earlier sweeps started at $\alpha = 300$, where relative scatter was already 0.07–0.11, making thresholds left-censored. Extending to $\alpha \in [10^{-4}, 300]$ with five seeds gives a healthy branch at $\alpha \leq 10^{-3}$ (relative scatter 1.000 ± 0.005) and crossings $0.95, 0.89, 0.90, 1.18$ at $m = 32, \dots, 256$. The fitted slope is $\gamma_{\text{bat}} = 0.10$ ($R^2 = 0.43$), varying over $[-0.10, +0.12]$ across crossing rules. These four widths are consistent with weak width dependence but do not resolve a precise exponent.

Treating each width's crossing as one observation gives four-width OLS 95% intervals $[-0.07, +4.93]$ for the $\beta = 0$ schedule, $[-0.16, +1.49]$ for HAR, $[-0.24, +0.43]$ for battery, and $[-1.16, +1.20]$ for RadioML. None excludes zero. Interior schedules yield positive slopes: 2.57 (CI $[0.87, 4.27]$) at $\beta = 0.25$ and 1.86 ($[0.81, 2.91]$) at $\beta = 0.50$; $\beta = 1$ remains consistent with zero. At the reported anchor, all four remaining schedule intervals contain $\gamma_0(1 - \beta)$, whereas the $\beta = 0.75$ interval excludes it at anchor 10^{-4}. These results support schedule dependence without identifying an anchor-independent coefficient.

The denser SYN8 family supplies 26 completed (m, β) cells, ten seeds each. Since $r(64) = 32$ for every schedule, we fit $\log \alpha_c = c + \gamma_0(1 - \beta) \log(m/64)$. At anchor 3×10^{-4}, the constrained fit gives $\gamma_0 = 2.50$ (nominal OLS 95% CI $[1.94, 3.06]$). Across anchors $\{10^{-4}, 3 \times 10^{-4}\}$ and

inclusion/exclusion of $m = 128$, all four conditional intervals exclude zero (lowest bound 1.04). In the unconstrained model $c + ax + b\beta x$, $x = \log(m/64)$, every interval for $a + b$ contains zero and $-b/a = 0.78$–1.03: the linear form is not rejected, not established. These are cell-level fit intervals, not independent-seed uncertainty: the schedules share seeds and the $m = 64$ curves are identical. Joint/anchor-row point estimates range from 1.35 to 2.87. At $m \geq 96$ the sampled curve never resolves the collapsed floor; half-height crossings interpolate the observed decline, not a verified collapse threshold. Thus SYN8 sharpens a conditional trend without supplying a precise physical exponent. `synth_beta_joint.py` reproduces the fits.

Nine-width HAR and battery sweeps use $m \in \{16, 24, 32, 48, 64, 96, 128, 192, 256\}$ with the same driver, seeds, epochs, and dose grids. Shared configurations agree to floating-point precision (maximum difference 2×10^{-16}). HAR gives $\gamma_{\mathrm{HAR}} = 0.38$ (95% CI $[+0.10, +0.67]$, $R^2 = 0.60$, $n = 9$); excluding the two narrowest widths gives 0.60. Battery gives $\gamma_{\mathrm{bat}} = -0.06$ (CI $[-0.17, +0.06]$), consistent with weak width dependence. The denser grid narrows conditional fit intervals but does not eliminate crossing uncertainty. For example, HAR at $m = 32$, $\alpha = 10^{-3}$ has relative scatter 0.5026 ± 0.0524, placing the half-height crossing close to a sampled grid point and making interpolation sensitive to seed variation. We report both four- and nine-width estimates rather than treating either as an identified dynamical exponent.

We also examine probe scaling across encoders. Repeating calibration with a Transformer at $m \in \{32, 64, 128\}$ and ten seeds gives probe exponents -0.93 for HAR ($R^2 = 0.99$), $+0.39$ for battery ($R^2 = 0.87$), and -0.17 for RadioML ($R^2 = 0.79$). The corresponding MLP values are -1.63 ($R^2 = 0.96$), $+0.67$ ($R^2 = 0.86$), and $+0.28$ ($R^2 = 0.40$). Signs agree for HAR and battery; RadioML is consistent with zero for both encoders, and the 95% intervals overlap in all three domains. This is evidence of broadly similar probe trends in the tested encoders, not architecture invariance. The script `probe_exponent.py` reproduces the analysis.

Probe exponents and scatter-crossing exponents estimate different quantities. Their MLP point estimates agree in sign for RadioML ($+0.28$ versus $+0.02$), but disagree for HAR (-1.63 versus $+0.38$) and battery ($+0.67$ versus -0.06). A half-height crossing locates a dose at which trained scatter reaches half its reference value; the probe estimates $1/\hat{\rho}$ at the warm-up anchor. Agreement within a factor of a few at one width (a factor of 7 for NYUv2 at $m = 64$) does not imply agreement in width scaling. Negative probe slopes do not satisfy the restricted nonnegative exponent assumed in (20), but that assumption is not a general constraint on empirical slopes. The width analysis here therefore reports crossing exponents separately, while Appendix G reports probe values as calibration statistics.

Several sweeps show bimodal per-seed scatter near the empirical crossing. Such outcomes are compatible with a saddle-node model, but can also arise from initialization, finite training time, and optimization. We report the heterogeneity without assigning it a unique dynamical cause.

The original GPU experiments use fixed integer seeds, bf16 autocast with TF32 matrix multiplication, and Adam (lr $= 10^{-3}$, batch 256) on an NVIDIA A100 80 GB. Exceptions are specified with their protocols. Operational collapse means scatter below 0.3 times the same-seed single-task reference; the released recovery rate compares each seed with the mean joint-training metric. Code, seeds, hyperparameters, and per-fold results are included in the repository.

E Generalizations and Finite-Size Theory

Proposition 6 (Fixed attenuation in a specified model)**.** *For $T \succeq 0$ in restoration-normalized coordinates, uniform attenuation $T \mapsto (1-\lambda)T$, $0 \leq \lambda < 1$, multiplies the first-modal-contraction threshold by $1/(1-\lambda)$. At fixed dose the smallest λ meeting $\alpha(1-\lambda)\|T\| \leq 1$ is $\max(0, 1 - 1/(\alpha\|T\|))$ when $\alpha\|T\| > 0$.*

Proof. All eigenvalues scale by $1 - \lambda$; solve the displayed inequality. This statement concerns a fixed PSD model, not arbitrary signed or nonsymmetric filtered dynamics. □

Theorem 4 (Annihilation by orthogonal compression)**.** *Let $T \succeq 0$, $U = \mathrm{range}(T)$, and $Q = I - P_S$ be an orthogonal projector. Then $QTQ = 0$ iff $U \subseteq S$. Among such deletions the unique smallest deleted subspace is U, and $I - P_U$ preserves its orthogonal complement.*

Proof. $QTQ = (T^{1/2}Q)^\top (T^{1/2}Q) = 0$ iff $\operatorname{range}(Q) \subseteq \ker T$, equivalently $U \subseteq S$. A general one-sided backward operator QT need not describe symmetric neural-network dynamics; a signed filter can even reverse contraction, so its operator norm cannot supply the missing implication. □

Corollary 5 (Subspace estimation bound)**.** *If $\|P_U - P_{\hat{U}}\| \le \epsilon$ and $Q = I - P_{\hat{U}}$, then*

$$\|QT\| \le \epsilon \|T\|, \qquad \|QTQ\| \le \epsilon^2 \|T\|. \tag{24}$$

Proof. Use $T = P_U T P_U$ and $\|QP_U\| \le \epsilon$ on one or both sides. Whitening must precede this argument if T is restoration-normalized; an angle bound in other coordinates is insufficient. The bounds concern residual magnitude, not the existence of a residual. □

Theorem 5 (Minimum rank for a PSD compression)**.** *Fix a PSD operator T in specified Euclidean coordinates, with eigenvalues $\kappa_1 \ge \cdots \ge \kappa_d \ge 0$, and a threshold $\tau \ge 0$. Among orthogonal projectors $Q = I - P_S$ satisfying $\|QTQ\| \le \tau$, the minimum deletion rank is*

$$r^\star = \#\{j : \kappa_j > \tau\}.$$

Deleting $S_c = \operatorname{span}\{u_j : \kappa_j > \tau\}$ attains it. Within the class of spectral deletions, S_c also minimizes any inclusion-monotone deleted energy. Uniqueness of the energy optimum requires positive energy in every additionally deleted eigendirection.

Proof. The spectral choice leaves only eigenvalues at most τ. For any $\dim S < r^\star$, dimension counting gives a unit vector $v \in S^\perp \cap S_c$. Its Rayleigh quotient satisfies $v^\top QTQv = v^\top Tv > \tau$, violating feasibility. For spectral deletions feasibility requires containing S_c, giving the energy statement by inclusion. Rotated subspaces can also attain the same minimum rank; no global minimum-energy claim over rotations follows. □

Corollary 6 (Saturation and its limits)**.** *For $\tau = 1/\alpha$, the optimal rank changes in steps as α crosses inverse positive eigenvalues and eventually equals $\dim U$. The limiting projector is $I - P_U$, which equals full detachment only on U, or globally when U is the entire gradient space. Exhausting a finite probe depth does not prove saturation of U.*

The theorem concerns a specified operator, not a mean of gradient-norm ratios. For a linear reconstruction decoder Wh with squared loss, $g_a = W^\top \delta$ (up to loss normalization) and the Hessian is $W^\top W$. The uncentered second moment relevant to normalized residual energy is

$$C_g = \mathbb{E}\left[\frac{g_a g_a^\top}{(\|g_p\| + \varepsilon)^2}\right] = W^\top \mathbb{E}\left[\frac{\delta\delta^\top}{(\|g_p\| + \varepsilon)^2}\right] W.$$

For any rank-r projector P, the squared residual is $\operatorname{tr}((I - P)C_g)$, minimized by the top r eigenvectors of C_g (the Rayleigh–Ritz principle). Its square root upper-bounds the corresponding mean norm by Jensen's inequality. It does not generally minimize that mean itself. Hessian eigenvectors coincide with this basis under isotropic weighted residual covariance; representation PCA requires its leading subspace to coincide with the leading eigenspace of C_g. Commutation gives a common eigenbasis but does not ensure the same ordering. A nonlinear decoder additionally contributes residual-weighted second derivatives to its Hessian. A nearly flat PCA residual curve is therefore consistent with gradient energy outside leading representation components, without establishing orthogonality of the reconstruction residual to those components.

Table 9 gives the variance-shrinkage rank sweep. For real HAR reconstruction at $\alpha = 12$, auxiliary-gradient and PCA bases give mean candidate ranks 22.6 and 57.0. Both gates open in four of ten seeds, with active ranks 30.75 and 62.0 at similar aggregate accuracy (Table 7). This comparison isolates basis sensitivity. The artifact also includes the three-seed RMS-basis pilot and its negative accuracy result.

In the ten-seed battery cap32 experiment, `stfcalrank` reaches its cap in every seed and gives SOH RMSE 5.26, versus 5.24 for joint training. The larger cap therefore does not establish minimum-rank protection. A single-seed full-depth audit gives representation covariance traces 3.131, 4.065, 0.906, and 0.821 for joint training, detachment, the radial calibrated arm, and the rank arm, respectively, versus 1.147 for the SOH-only reference. Corresponding normalized SOH prediction variances are 0.00826, 0.01380, 0.01253, and 0.01019, versus 0.01379. These quantities separate variance from

prediction spread and establish neither representation recovery nor constant predictions. Historical checkpoints were selected on the evaluation split and remain exploratory. The corrected cell-disjoint protocol fits preprocessing on training cells and evaluates test data after validation-based checkpoint selection.

Forward and backward-only filters differ even when their Jacobians agree. For fixed U, let $\widetilde{H}_f = H - C_B HUU^\top$ and $\widetilde{H}_b = \mathrm{sg}(H) + D - C_B DUU^\top$, where $D = H - \mathrm{sg}(H)$ and sg stops gradients. Both induce the backward map $G \mapsto G - C_B GUU^\top$, but only $\widetilde{H}_b$ preserves the auxiliary input. Consequently, the loss gradients are evaluated at different forward values and need not produce the same update. Even at full rank, both retain a batch-mean gradient channel, unlike full detachment.

On HAR reconstruction at $\alpha = 100$, $m = R = 64$, paired forward and backward-only filters share warm-up, seed, basis, and selected rank. All ten gates open. Forward filtering gives primary accuracy 0.9507 ± 0.0033, reconstruction MSE 0.05112 ± 0.01083, and relative scatter 0.681 ± 0.039; backward-only filtering gives 0.9478 ± 0.0058, 0.02053 ± 0.00115, and 0.597 ± 0.061. Auxiliary errors use unfiltered features. The paired accuracy difference (backward minus forward) is -0.00285 (95% t interval $[-0.00815, 0.00245]$); reconstruction improves in every seed while scatter decreases. Holm-adjusted Wilcoxon p values across these outcomes are 0.242, 0.00586, and 0.0195. Thus the forward choice changes the auxiliary-fit/scatter tradeoff, with no resolved primary-accuracy advantage. Joint and detached accuracy in the same harness are 0.9461 and 0.9431. A three-seed $\alpha = 12$ pilot, including two closed-gate pairs, is also retained. Scripts `filter_semantics_audit.py` and `summarize_filter_semantics.py` provide the checks and paired records. Probe algebra and eigendecomposition run in fp32, with invariance to enclosing bf16 autocast checked separately.

A complementary experimental intervention uses $v = \nabla_\theta V$, proposed update $u_0 = G_p + \alpha G_a$, and slack $h = V - V_{\min}$. For $k > 0$ and $v \neq 0$, the closest update satisfying $v^\top u \leq kh$ is

$$u^\star = u_0 - \frac{(v^\top u_0 - kh)_+}{\|v\|^2} v.$$

The KKT conditions prove this Euclidean halfspace projection. Under a well-defined continuous flow, $\dot{h} \geq -kh$ preserves feasibility on this *fixed batch*. At $v = 0$ feasibility requires $kh \geq 0$. Discrete updates require checking the actual post-step variance: the supplied SGD implementation backtracks and restores parameters on failure. The argument covers neither unseen data nor Adam without modeling its applied update. The projection is the established constrained-gradient construction (Allibhoy & Cortés, 2024); its empirical benefit is untested here.

Table 8 compares deletion with positive attenuation on UCI-HAR ($m = 64$, sixty epochs). The calibrated multiplier uses the unprojected mean gradient ratio, so its $c = 1$ is a calibration setting rather than an identified dynamical threshold. The $c = 0.1$ setting is competitive with the tested fixed multipliers. This exploratory grid has three seeds per configuration and includes non-finite runs. At $\alpha = 300$, seed 0 diverges for every arm, including detachment, whose auxiliary heads still train during detached warm-up. We record numerical divergence separately from representation collapse.

Proposition 7 (Sampling fluctuation of a frozen statistic)**.** *For independent identically distributed scalar diagnostics X_i with finite variance σ_X^2, $\hat{\rho} = B^{-1}\sum_i X_i$ has variance σ_X^2/B and obeys a central limit theorem. Under a normal approximation the probability that the* gate $\alpha\hat{\rho} > 1$ *opens is* $\Phi(\sqrt{B}(\rho - 1/\alpha)/\sigma_X)$. *This is not the probability of training collapse. Batch-dependent centering or fitted bases require additional analysis or an independent frozen calibration sample.*

Proof. Use variance additivity and the central limit theorem. For a ratio A/P, the delta-method variance includes $-2\mu_A \operatorname{Cov}(A, P)/\mu_P^3$ as well as the two marginal variance terms. Mean per-example ratios and ratios of means are distinct estimands. □

Proposition 8 (Conditional calibration bound)**.** *Assume independent $X_i \in [0, M]$ with common mean ρ and population variance at most v_0. Bernstein's inequality gives*

$$\mathbb{P}(|\hat{\rho} - \rho| \geq \varepsilon) \leq 2\exp\left[-\frac{B\varepsilon^2}{2v_0 + 2M\varepsilon/3}\right]. \tag{25}$$

Thus an error probability at most δ follows if

$$B \geq (2v_0 + 2M\varepsilon/3)\varepsilon^{-2}\log(2/\delta). \tag{26}$$

Table 8: Matched attenuation on UCI-HAR ($m = 64$, sixty epochs). Accuracy and calibrated multiplier q are averaged over completed seeds; `div` indicates non-finite evaluation. There are three seeds per configuration, except at $\alpha = 300$, where seed 0 diverges for every arm and is excluded. Bold identifies the best fixed setting and the best calibrated setting when it exceeds the best fixed accuracy.

	accuracy			multiplier q		
arm	30	100	300	30	100	300
plain summation	.9293	`div`	`div`	1	1	1
detachment (STF-0)	.9276	.9276	`div`	0	0	0
fixed 10^{-4}	.9275	.9276	.9272	—	—	—
fixed $3{\cdot}10^{-3}$	.9278	**.9304**	.9286	—	—	—
fixed 10^{-2}	**.9301**	.9299	**.9293**	—	—	—
fixed $3{\cdot}10^{-2}$	**.9301**	.9294	.9265	—	—	—
fixed 10^{-1}	.9295	.9291	.9260	—	—	—
fixed $3{\cdot}10^{-1}$	.9242	.8842	`div`	—	—	—
calibrated $c = 0.1$	.9299	**.9306**	.9289	.076	.016	.004
calibrated $c = 0.3$	.9261	.9285	.9281	.228	.049	.012
calibrated $c = 1.0$	.9257	.9266	.9276	.759	.163	.039

Table 9: Selected rank versus auxiliary dose on UCI-HAR ($m = 64$, five seeds, probe depth $R = 64$). Candidate ranks are given for representation-PCA and auxiliary-gradient bases. A closed gate removes no directions; full detachment blocks all 64. Relative scatter is $\mathcal{S}_h/\mathcal{S}_{single}$ for `stfcalrank`. The PSD optimum in Theorem 5 additionally requires identifying the operator and basis.

α	gate	$r^\star$ (principal)	$r^\star$ (aux. basis)	rel. scatter (rank arm)	rel. scatter (JOINT)
0.003	0	–	–	0.404	0.402
0.01	0	–	–	0.287	0.280
0.03	1	1.6	1.6	1.007	0.200
0.1	1	3.8	3.8	1.026	0.138
0.3	1	4.2	4.2	1.018	0.101
1.0	1	11.6	11.6	0.978	0.070
3.0	1	25.8	25.8	0.987	0.051
10	1	39.8	40.0	0.986	0.035
30	1	47.4	47.8	0.968	0.025
100	1	61.6	61.0	0.950	0.016

A nonzero primary-gradient denominator alone does not provide a uniform bound M. Population bounds cannot be replaced by observed sample moments without a separate valid empirical-concentration result.

Proof. Apply the bounded-variable Bernstein inequality to the centered independent sum and solve for B. □

Corollary 7 (Certifying a frozen gate or nested rank rule)**.** *Under Proposition 8, a gate with population margin $d = |\rho - 1/\alpha| > 0$ has error probability at most $2\exp[-Bd^2/(2v_0 + 2Md/3)]$. For $R + 1$ fixed residual statistics with uniformly positive margins, a union bound controls all their threshold decisions and hence the selected nested rank. This does not identify the statistics with Hessian eigenvalues, certify an adaptively fitted basis without conditioning, or certify a subsequent training trajectory.*

Proof. A wrong threshold decision requires an estimation error at least its margin; apply the bound and then the union bound. Strict margins exclude equality cases. □

Proposition 9 (Local linear noise floor)**.** *For deviations $e_{t+1} = ae_t + \eta\xi_t$ from a fixed equilibrium, assume $|a| < 1$ and independent mean-zero innovations of variance D. The stationary variance is $\eta^2 D/(1 - a^2)$.*

Table 10: Historical battery results with evaluation-selected checkpoints, retained as exploratory comparisons. The legacy RUL score combines errors against observed lifetimes and censored lower bounds. Reference rows use four seeds; other rows use ten. SOH and legacy RUL are separate outcomes; single-task rows provide references.

	MATR		NASA	
Mode	SOH RMSE (%)	legacy RUL score	SOH RMSE (%)	legacy RUL score
single-task reference	0.51	694.6	9.17	18.5
JOINT (aux heads)	5.24 ± 0.54	209.0 ± 59.8	11.42 ± 0.85	17.5 ± 1.1
JOINT (toxifier)	1.23 ± 0.44	306.6 ± 48.2	9.66 ± 0.44	17.6 ± 2.4
STF-0	1.68 ± 0.41	574.7 ± 80.7	9.35 ± 0.55	18.0 ± 2.0
STF-0W (tri-weight)	1.54 ± 0.44	541.2 ± 91.3	9.40 ± 0.62	18.6 ± 1.4
STFCAL ($\hat{\rho}$=17.4/0.40)	1.77 ± 0.47	419.8 ± 33.6	11.40 ± 0.78	18.3 ± 1.7

All MATR rows: censoring-aware RUL supervision, all 185 cells retained; bf16 autocast training, fp32 evaluation.

Table 11: Frozen-anchor sampling experiment on HAR ($m = 64$, four detached warm-up epochs, five seeds). Each seed uses 2000 batches of size B, sampled with replacement. Standard deviation and CV describe variation across batches at a fixed anchor. "Meas." is the observed frequency of $|\hat{\rho} - \rho| > \eta\rho$; "bound" is the Bernstein plug-in envelope from (25), using estimated probe moments ($\kappa^2 = 2\rho^2 c_0^2 + \frac{2}{3} R_{\max}\varepsilon$). Values above one are uninformative probability bounds. The envelopes exceed observed frequencies but are not population confidence guarantees. The fitted CV slope is -0.500 ($R^2 = 0.96$), and the mean slope is $+0.006$. The norm of $\sum_i g_a^{(i)}$ is at most 4×10^{-8} of $\sum_i \|g_a^{(i)}\|$. Raising nominal confidence from 90% to 99.9% multiplies the sample-size expression by $\ln(2/10^{-3})/\ln(2/0.1) = 2.54$.

				$\eta = 0.2$		$\eta = 0.1$	
B	$\hat{\rho}$	sd($\hat{\rho}$)	CV($\hat{\rho}$)	meas.	bound	meas.	bound
16	74.30	69.52	0.947	0.820	1.94	0.911	1.98
32	76.69	50.40	0.664	0.740	1.90	0.866	1.97
64	77.03	36.00	0.471	0.657	1.83	0.836	1.94
128	77.00	25.34	0.332	0.526	1.71	0.756	1.90
256	77.74	18.24	0.237	0.378	1.49	0.657	1.83
512	77.71	12.92	0.168	0.221	1.15	0.538	1.69
1024	77.97	9.20	0.119	0.094	0.70	0.384	1.45
2048	78.00	6.34	0.082	0.022	0.29	0.218	1.07
4096	77.90	4.54	0.059	0.003	0.06	0.095	0.61

Proof. The second-moment recursion is $\mathbb{E}e_{t+1}^2 = a^2\mathbb{E}e_t^2 + \eta^2 D$; sum the convergent geometric series. This is variance of deviations, not total representation scatter. Divergence of a local linear approximation as $|a| \to 1$ does not prove a universal ordering of half-height and binary-collapse thresholds in a nonlinear experiment. □

At a frozen HAR anchor, 2000 resampled batches per size over five seeds give slopes -0.500 for probe CV, -0.494 for standard deviation, and $+0.006$ for the mean. Table 11 compares observed fluctuations with plug-in envelopes based on estimated variance and range. Shared batch centering and estimated moments limit these envelopes to diagnostic use. The records also distinguish mean per-example ratios from ratios of batch-mean gradients: the latter can vanish for centered variance losses.

E.1 Prediction Variance Normalization and Target z-Scoring

For $\hat{y} = w^\top h + b$, shrinking centered features by $1/c$ and expanding w by c preserves predictions. Shrinking both changes the regression loss, and constant predictions generally cannot fit varying targets. This scale ambiguity motivates reporting prediction variance and across-example feature covariance alongside task error, and testing prediction normalization and target scaling as controls.

Table 12: Corrected battery test results. Means ± population standard deviations over ten initialization seeds on one fixed cell split. RUL is observed-only MAE; S is across-example covariance trace, PR is participation rank, and V_{pred} is SOH prediction variance in squared percentage points. The SOH-only head does not train RUL.

Data	Mode	SOH RMSE (%)	RUL MAE	S	PR	V_{pred}
MATR	SOH-only	0.61 ± 0.19	—	1.01 ± 0.50	1.35 ± 0.14	141.67 ± 5.92
MATR	JOINT	5.57 ± 0.48	81.54 ± 26.84	2.86 ± 0.57	2.49 ± 0.65	66.10 ± 8.59
MATR	STF-0	2.10 ± 0.62	46.73 ± 11.32	4.77 ± 1.33	1.86 ± 0.22	109.50 ± 13.01
MATR	STFCAL	2.71 ± 0.54	63.59 ± 19.43	1.40 ± 0.44	2.03 ± 0.20	99.26 ± 9.69
MATR	STFCALRANK	4.98 ± 1.09	107.58 ± 31.10	617.90 ± 489.76	1.21 ± 0.25	70.11 ± 20.06
NASA	SOH-only	12.66 ± 0.99	—	3.39 ± 1.57	2.59 ± 0.74	183.51 ± 44.46
NASA	JOINT	14.43 ± 0.64	20.05 ± 1.59	5.41 ± 0.85	3.25 ± 0.49	127.49 ± 24.32
NASA	STF-0	12.74 ± 1.20	19.43 ± 2.66	4.38 ± 1.28	3.16 ± 0.58	135.70 ± 37.07
NASA	STFCAL	14.47 ± 1.22	19.20 ± 2.40	4.55 ± 1.42	2.75 ± 0.62	109.72 ± 24.59
NASA	STFCALRANK	14.47 ± 1.22	19.20 ± 2.40	4.55 ± 1.42	2.75 ± 0.62	109.72 ± 24.59

The historical regression recipe standardizes targets and divides predictions by a detached batch standard deviation. This changes optimization and can reduce sensitivity to scale, but detached normalization does not implement the derivative of a truly scale-invariant loss. Softmax also does not normalize logit scale: changing that scale changes probabilities. These operations therefore require empirical ablations, with scaling statistics fit only on training data.

F Additional Tables and Protocol

The corrected battery protocol uses cell-disjoint train/validation/test partitions: 130/27/28 MATR cells and 24/4/5 NASA cells. MATR test data contain 1069 observed and 1797 censored cycles; all 344 NASA test cycles are observed. Variation is over initializations on these fixed splits. Checkpoints minimize validation SOH, prioritizing that task. Participation rank is $(\operatorname{tr}\Sigma)^2/\operatorname{tr}(\Sigma^2)$; prediction variance uses squared SOH percentage points. Trace measures amplitude and participation rank measures spectral concentration; both depend on the encoder–head parameterization. Script `current_evidence_audit.py` checks split disjointness, seed pairing, and source hashes and regenerates the table. The remaining tables give datasets, historical censoring and head comparisons, auxiliary-weight sweeps, and the training protocol.

Algorithm 1 STF training protocol (per domain, per configuration)

1. **Input:** domain data $\mathcal{D}$, encoder width m, toxicity schedule $\{\alpha\}$, seeds
2. Phase 1 – health anchor: train the single-task encoder ($\alpha = 0$), record healthy scatter $\mathcal{S}_{single} = \mathcal{S}_h(\theta^*)$ and healthy spectral ratio ρ_0.
3. **Phase 2 – calibration:** sweep α with mode JOINT; record ρ, relative scatter $\mathcal{S}_h/\mathcal{S}_{single}$, collapse flag; locate α_c by the threshold crossing.
4. **Phase 3 – repair:** at a selected dose, train each arm on $L_p(H) + \alpha L_a(\widetilde{H})$, using its specified forward map $\widetilde{H}$ (or detached H for STF-0); backpropagate this loss and monitor the gradient diagnostic and relative scatter.
5. **Report:** collapse rate, recovery rate, ρ, $\mathcal{S}_h/\mathcal{S}_{single}$.

On NASA, the corrected protocol gives SOH RMSE 14.43 for joint and 12.74 for detachment, against the SOH-only reference 12.66. The paired difference is -1.69 percentage points (95% t interval $[-2.54, -0.83]$, nine of ten seeds improve). Observed-RUL MAE changes from 20.05 to 19.43 cycles, but the paired interval $[-3.37, 2.13]$ includes zero (exact two-sided Wilcoxon $p = 0.770$). The calibrated gate stays closed in all ten seeds: its two arms coincide with each other (14.47 SOH, 19.20 RUL), but not with joint because their encoder warm-up is detached. NASA has only five test cells; initialization intervals cannot establish population generalization.

In the historical ten-seed MATR experiment, the radial calibrated arm improves SOH over joint training (1.77 ± 0.47 versus 5.24 ± 0.54), while detachment reaches 1.68 ± 0.41. The cap32 rank arm remains near joint (5.26), so the radial result does not validate minimum-rank protection; Corollary 6

Table 13: The sixteen controlled toxifier cells behind Figure 7a. Every training auxiliary is a fixed projected-variance toxifier; the probe uses an isotropic variance surrogate. $\alpha_{\mathrm{probe}} = 1/\bar{\rho}$ is the inverse mean warm-up probe, while $\alpha_{1/2}$ interpolates the mean dose–scatter curve at relative scatter 0.5. Their ratio is a calibration-transfer diagnostic. Each probe value is a ten-seed mean of the archived warm-up records.

Domain	Head	Width m	Probe $\bar{\rho}$	α_{probe}	$\alpha_{1/2}$	Ratio
HAR	toxifier	32	30.2	3.32×10^{-2}	1.02×10^{-3}	32.5
HAR	toxifier	64	110.2	9.07×10^{-3}	1.41×10^{-3}	6.4
HAR	toxifier	128	552.0	1.81×10^{-3}	1.63×10^{-3}	1.1
HAR	toxifier	256	1504.9	6.65×10^{-4}	4.52×10^{-3}	0.15
Battery	toxifier	32	1.10	9.07×10^{-1}	9.51×10^{-1}	0.95
Battery	toxifier	64	0.71	1.42	8.95×10^{-1}	1.58
Battery	toxifier	128	0.85	1.18	9.03×10^{-1}	1.31
Battery	toxifier	256	0.28	3.56	1.18	3.01
RadioML	toxifier	32	0.14	7.13	1.73×10^{-1}	41.1
RadioML	toxifier	64	0.08	11.9	3.95×10^{-1}	30.3
RadioML	toxifier	128	0.05	19.8	2.41×10^{-1}	82.2
RadioML	toxifier	256	0.04	24.9	2.16×10^{-1}	115.6
NYUv2	toxifier	32	2.26	4.43×10^{-1}	1.14×10^{-1}	3.9
NYUv2	toxifier	64	2.01	4.98×10^{-1}	6.89×10^{-2}	7.2
NYUv2	toxifier	128	1.64	6.11×10^{-1}	3.56×10^{-2}	17.1
NYUv2	toxifier	256	1.29	7.78×10^{-1}	2.39×10^{-2}	32.6

locates the saturation limit at full-width deletion. Censored RUL requires separate observed-error and lower-bound-shortfall metrics; a lower legacy MAE need not indicate better lifetime prediction. NASA's closed gate and its discrepancy from detachment further limit the diagnostic. Because these historical checkpoints were evaluation-selected, the corrected validation protocol provides the primary battery evidence.

Algorithm 2 Released calibrated forward filters: radial STFCAL and capped STFCALRANK

1: **Input:** data $\mathcal{D}$, encoder width m, operating weight α, probe depth R, warm-up length W
2: **Warm-up:** train θ with the auxiliary branch detached ($\nabla_\theta L_a \equiv 0$) for W epochs; record the healthy anchor $\mathcal{S}^\star$ of (17)
3: **Probe:** obtain per-example gradients on an unfiltered frozen batch, radial projectors $\Pi_{c,i}$, and principal basis U. With $\varepsilon = 10^{-8}$, compute $\hat{\rho} = B^{-1} \sum_i \|\Pi_{c,i} g_a^{(i)}\| / (\|g_p^{(i)}\| + \varepsilon)$ and $\hat{\rho}_{\mathrm{res}}[r] = B^{-1} \sum_i \|(I - U_r U_r^\top) g_a^{(i)}\| / (\|g_p^{(i)}\| + \varepsilon)$ for $r = 0, \ldots, R$.
4: **Calibrate:** $g \leftarrow \mathbf{1}[\alpha\hat{\rho} \geq 1]$; $\mathcal{R} \leftarrow \{r \leq R : \alpha\hat{\rho}_{\mathrm{res}}[r] < 1\}$; choose $r \leftarrow \max(1, \min \mathcal{R})$ if nonempty, else $r \leftarrow R$. Record whether a feasible rank exists.
5: **Latch:** freeze g, r, U. If $g = 0$, pass H to the auxiliary head. Otherwise, STFCALRANK passes $H - C_B H U_r U_r^\top$; STFCAL passes $h_i - \langle h_i - \bar{h}, u_i \rangle u_i$, $u_i = (h_i - \bar{h}) / \max(\|h_i - \bar{h}\|, \varepsilon)$. Differentiate the resulting auxiliary loss through this map.
6: **Report:** gate, r, $\hat{\rho}$, $\mathcal{S}_h / \mathcal{S}^\star$

A four-seed MATR ablation removes each weighting factor in turn. Omitting the rate factor gives fast-charge SOH RMSE 0.95; omitting the temperature factor gives extreme-temperature SOH RMSE 2.35. These results support the intended association between factors and operating regimes. A sweep over $\omega \in \{0.5, 1, 2\}$ evaluates sensitivity around the chosen value $\omega = 1$.

Table 14: Four scientific domains and the financial discovery application. Sample counts describe the listed experimental datasets; individual protocols specify their partitions. Controlled scientific experiments use variance-shrinkage auxiliaries. The financial application uses real heads, and the battery and NYUv2 application studies also evaluate practical reconstruction heads.

Domain	Signal	Primary task	Samples	Metric
HAR (UCI-HAR)	motion (561-d)	6-class activity	7,352 / 2,947	accuracy
RadioML (2016.10A)	I/Q (256-d)	11-class modulation	110,000 / 110,000	accuracy
Battery (NASA+MATR)	charge/discharge	SOH + RUL	18,696 / 3,210	RMSE / MAE
Finance (A-share)	1,680-d factors	5-d return rank	800 stocks $\times$ 2,059 d	rank-IC
NYUv2	RGB-D (144$\times$192)	13-class seg + depth + normals	795 / 654 scenes	pxacc / mIoU / RMSE / angle

Table 15: Financial application: mean OOS rank-IC $\pm$ population s.d. over 100 fold-seed cells (ten temporal folds, ten initialization seeds). Cells sharing a fold are dependent. After averaging seeds within folds and correcting three paired comparisons, no method differs significantly from joint training. These records use real auxiliary heads without the fixed variance toxifier.

Mode	OOS rank-IC	vs. joint
JOINT	$+0.060 \pm 0.055$	—
STF-0 (`.detach()`)	$+0.036 \pm 0.056$	-39%
STF-HARD	$+0.063 \pm 0.052$	$+5\%$
STF-SOFT	$+0.056 \pm 0.053$	-7%

Table 24 reports the composition experiment. On HAR, accuracy changes from 0.943 with filtering alone to 0.933 with PCGrad and 0.940 with CAGrad. Relative scatter is at most 0.15 without filtering, compared with 0.97–1.14 after filtering. These measurements support compatibility for the tested filter-then-combiner order.

The synthetic testbed uses 17 log-spaced doses, six widths, and random projections of rank $\lfloor m/2 \rfloor$. Its near-flat relative-crossing slope ($\gamma = 0.07$) contrasts with the absolute-threshold estimate (1.53), as analyzed above. In RadioML, a five-seed sweep over $\alpha \in [3\times10^{-3}, 1]$, refined near the transition, gives crossings $\{0.17, 0.39, 0.24, 0.22\}$ at widths $\{32, 64, 128, 256\}$. The eightfold width range changes the crossing by at most a factor of 2.3, much less than the factor of eight implied by $\gamma = 1$. At $m = 64$, per-seed crossings span 0.29–0.54. Together with the near-zero fit R^2, this variation limits inference about the width slope.

The repository supplies the scripts and recorded results listed in its reproduction manifest; data-dependent reruns additionally require the stated datasets and preprocessing. New audits are identified separately from historical experiments. The drivers `cross_domain_mtl.py`, `battery_bmtl_v3.py`, and `nyu_bmtl.py` implement the STF family for the unified, battery, and dense-perception experiments. Algorithm 2 specifies the calibrated forward maps; each driver records application-specific details. PCGrad, CAGrad, and FAMO share these experiment pipelines, including the composition study. The remaining methods (MGDA, Aligned-MTL, Nash-MTL, IMTL-G, IMTL-L, GradNorm, and uncertainty weighting) use `moo_combiners.py`, which ports kernels from the authors' reference implementations. Comparisons use the recorded optimizer and schedule with published method defaults; batch-size exceptions are stated below.

JSON job manifests specify each experimental configuration by domain, width, mode, and seed, together with its command, output path, and log path. A deterministic runner skips existing outputs. The manifests cover collapse scans, width sweeps, filter composition, real auxiliary heads, battery weight sweeps, calibrated validation, NYUv2 experiments, and the three-domain multi-task baseline comparison. Experiments use fixed integer seeds, with counts reported for each study. Where available, JSON records retain individual seed outcomes; aggregate-only historical records are identified in the artifact audit.

The original unified runs use an NVIDIA A100 80 GB, Adam ($\mathrm{lr} = 10^{-3}$), bf16 autocast, and TF32 matrix multiplication. Calibration runs in fp32 outside autocast on detached features. Batch size is 256 for the unified protocol, real-head comparison, regularizer sweep, and MATR application; 128 for NASA conflict baselines versus 256 for its plain-summation runs; 32 for NYUv2; and 512 for financial walk-forward training. The NASA batch-size difference qualifies those historical

Table 16: Historical MATR censoring ablation: SOH RMSE (%) and legacy RUL MAE (cycles), with leave-cell-out evaluation, $d = 128$, bf16 training, and fp32 evaluation. Legacy rows use four seeds; the censoring-aware row uses ten. Legacy supervision treats censored bounds as exact targets; the EOL-only variant additionally excludes $96/185$ cells, leaving 14 test cells. Censoring-aware supervision applies a one-sided hinge to the 59.2% censored samples and retains all cells. Evaluation-selected checkpoints and legacy RUL scoring limit interpretation.

	SOH RMSE (%)		RUL MAE (cyc)		
Mode	JOINT	STF-0	JOINT	STF-0	cells used
legacy (RUL=0 labels)	5.56 ± 0.17	2.05 ± 0.38	94.1 ± 5.7	97.0 ± 15.3	185
legacy + `--eol-only`	1.53 ± 0.13	0.53 ± 0.03	77.5 ± 2.3	76.6 ± 4.1	89
censoring-aware	5.24 ± 0.54	1.68 ± 0.41	209.0 ± 59.8	574.7 ± 80.7	185

The legacy rows reproduce the `nocens` supervision, every other setting held fixed.

Table 17: Historical battery-head comparison at $\alpha = 10$ (five seeds): SOH RMSE (%), legacy all-sample RUL MAE (cycles), and "scatter" $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$ (sample correction $d - 1$, no centering across examples), not (17). PCGrad, FAMO, and CAGrad yield higher SOH error than detachment in both datasets. Scatter alone does not establish operational collapse; Table 4 reports separate controlled-toxifier experiments. These evaluation-selected results are exploratory.

	MATR			NASA		
Combiner	SOH RMSE (%)	RUL MAE	scatter	SOH RMSE (%)	RUL MAE	scatter
single-task reference	0.51	694.6	–	9.17	18.5	–
JOINT (sum, plain)	6.43 ± 0.19	208.8	0.126	13.33 ± 1.07	18.7	0.214
PCGrad	4.02 ± 0.98	172.9	0.094	11.89 ± 0.73	18.6	0.164
FAMO	10.10 ± 2.17	295.3	0.435	16.02 ± 0.50	32.4	0.474
CAGrad	6.33 ± 0.45	198.0	0.166	13.04 ± 0.75	17.7	0.198
STF-0 (detach)	1.83 ± 0.51	623.4	0.198	8.81 ± 0.72	17.1	0.225

comparisons. A unified-protocol configuration takes a few GPU-minutes; the battery application uses 150 epochs per configuration. Corrected battery runs use Adam at 2×10^{-3}, the parameter-constraint pilot uses SGD, and exact-rank controls use AdamW on CPU. Individual scripts and JSON configurations record these optimizer differences. Calibration records contain each seed's $\hat{\rho}$, residual statistics $\hat{\rho}_{\mathrm{res}}[0..R]$, gate, and selected rank. The script `calib_cert_b.py` repeats probes on a frozen healthy anchor at different batch sizes to produce Table 11.

Table 19 measures per-epoch training cost on UCI-HAR at four widths. We report the median of per-repetition median epoch times over ten repetitions, exclude epoch 0, and synchronize the device within each timed epoch. Runs begin after three consecutive idle-device readings, and method order rotates between repetitions to reduce timing drift. Contention materially affects the result: a non-idle run measured the reference at 0.180 s rather than 0.099 s and FAMO overhead at 0.0% rather than approximately 31%.

On the idle device, hard and soft filtering cost at most 21.8% on HAR. At $m = 256$, their overheads are 9.5% and 12.2%, respectively. For the tested radial implementation, projection costs $O(Bd)$ compared with the encoder's $O(Bd^2)$; this accounting does not include arbitrary full-rank eigendecompositions. Detachment and calibrated radial filtering reduce measured time by up to 65.0% and 47.8%. PCGrad, CAGrad, and FAMO cost 21.1–41.6%, 38.5–75.2%, and 19.6–31.1%, respectively. Across five domains and four widths, spectral-variant overhead is 6.3–26.0%, versus 5.8–98.2% for the comparison methods. Hard and soft filtering are cheaper than PCGrad and CAGrad in every measured configuration; FAMO is an exception to the broader ordering on CelebA and NYUv2 at $m = 256$. These are empirical timings for the specified implementations.

Table 18: Historical MATR auxiliary-weight sweep (seed 0): SOH RMSE (%) and legacy RUL MAE (cycles), with censoring-aware supervision and bf16 training/fp32 evaluation. Joint training uses reconstruction/distributional weight λ_{aux}; detachment makes encoder learning independent of this weight. SOH error generally increases with dose, although the $d = 32$ curve is not strictly monotone. Single-task rows are width-matched references; legacy RUL scores and evaluation-selected checkpoints are not independent lifetime-error estimates.

		SOH RMSE (%)↓			RUL MAE (cyc)↓		
Mode	λ_{aux}	d=32	d=128	d=256	d=32	d=128	d=256
JOINT	10^{-2}	2.72	1.36	1.52	478.0	386.3	377.5
	10^{-1}	2.69	2.23	2.87	230.4	399.7	411.0
	1	5.96	4.69	5.53	285.6	198.0	176.4
	10	6.52	6.58	6.01	178.5	197.0	182.0
STF-0 (dose-free)	—	2.36	1.18	1.51	607.2	627.4	540.2
single-task reference	—	0.60	0.33	0.57	951.8	761.7	325.8

Table 19: UCI-HAR per-epoch overhead relative to plain summation, batch 256. Entries are medians over ten repetitions of median epoch time, excluding epoch 0 and synchronizing each epoch. Method order rotates after three idle-device readings. A paired-ratio estimator differs by at most five percentage points. The first row gives reference times; negative overhead indicates speedup for the measured implementation.

Method	m=32	m=64	m=128	m=256
plain summation (s/epoch)	0.0813	0.0986	0.1283	0.1454
overhead relative to plain summation				
STF-0 (detach)	−35.8%	−48.1%	−59.5%	−65.0%
STF-HARD	+21.8%	+18.1%	+14.3%	+9.5%
STF-SOFT	+21.8%	+19.8%	+8.9%	+12.2%
STFCAL	−0.7%	−20.1%	−37.7%	−47.8%
PCGrad	+41.6%	+31.4%	+26.6%	+21.1%
CAGrad	+75.2%	+60.2%	+46.0%	+38.5%
FAMO	+31.0%	+31.1%	+23.9%	+19.6%

G NYUv2: Dense Perception Beyond Time Series

NYUv2 (Silberman et al., 2012; Ladický et al., 2014) contains 795 training and 654 test scenes with semantic segmentation (13 classes), depth, and surface normals. Images are resized to 144×192; segmentation pixels labeled -1 and zero-norm normal targets are masked. The convolutional encoder uses four stride-2 blocks with batch normalization and ReLU ($3 \to 32 \to 64 \to 128 \to 256$ channels), followed by a $d = 128$ representation and linear task heads. Losses are cross-entropy, L_1 depth error, and cosine normal error. We report final-epoch pixel accuracy, mIoU, depth RMSE/MAE, and mean angular error over ten seeds, without test-based checkpoint selection. The *proj* auxiliary is a fixed random variance-shrinkage projection to $d/4$, evaluated at $\alpha = 1$. The *ae* auxiliary is a trainable three-layer input-reconstruction head, evaluated at $\alpha = 250$, or 3.4 times its probe-based threshold. A separate unified-protocol experiment uses an MLP and a 9×12 coarse depth grid (108 outputs), with the optimizer, probe, and collapse criterion of Appendix A.4.1.

After four detached warm-up epochs, the unified MLP probe gives $\hat{\rho} = 2.26 \pm 0.16$, 2.01 ± 0.10, 1.64 ± 0.09, and 1.29 ± 0.05 at widths $32, 64, 128, 256$. Per-seed ranges are 2.06–2.65, 1.90–2.20, 1.51–1.77, and 1.22–1.38. The inverse-mean calibration values are 0.44, 0.50, 0.61, and 0.78, yielding a probe-based width slope of 0.27 ($R^2 = 0.98$, 95% CI $[+0.15, +0.40]$).

This is not the scatter-crossing exponent. Half-height crossings on the same widths yield $\gamma = -0.77$; varying the crossing level from 0.3 to 0.7 gives $[-0.94, -0.73]$ with $R^2 \geq 0.99$, whereas HAR remains positive ($+0.34$ to $+0.46$). At $m = 64$, the crossing and calibration values are 0.069 and 0.50, a factor of 7 apart. Extending the grid four decades lower restores relative scatter to 0.99–

Table 20: NYUv2 under the unified protocol (MLP, $m = 64$, ten seeds): depth RMSE in meters on a 9×12 grid and scatter relative to the same-seed single-task anchor. Calibration gives mean $\hat{\rho} = 2.01$ and inverse-mean threshold 0.50. Recorded candidate ranks are 1 at $\alpha \leq 0.5$ and 3 at $\alpha \geq 1$; the radial STFCAL arm does not use this rank. Full detachment yields RMSE 1.353 and relative scatter 1.000 at every dose.

α	0.03	0.1	0.3	0.5	1	3	10	30	100
JOINT RMSE	1.335	1.336	1.316	1.307	1.305	1.280	1.221	1.100	1.092
JOINT rel. scatter	0.62	0.45	0.30	0.26	0.22	0.16	0.10	0.04	0.04
JOINT collapsed	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10	10/10
STFCAL RMSE	1.339	1.327	1.314	1.328	1.354	1.353	1.353	1.354	1.354
STFCAL rel. scatter	0.63	0.45	0.31	0.63	1.01	1.01	1.01	1.00	1.01
STFCAL collapsed	0/10	0/10	4/10	5/10	0/10	0/10	0/10	0/10	0/10
calibrated gate open	0/10	0/10	0/10	5/10	10/10	10/10	10/10	10/10	10/10

1.01 at $\alpha = 10^{-6}$, compared with 0.64–0.83 at the previous lowest dose 0.01, without materially changing the crossing. The opposite width trends therefore cannot be explained solely by the original grid's lower endpoint.

We report the probe because it determines the implemented gate, while retaining the crossing estimate as a separate diagnostic. The probe slope varies over a range of 0.17 across calibration times on NYUv2, versus 2.55 on HAR (Appendix H.3); this relative stability does not make it a capacity exponent. For the convolutional encoder, the *proj* probe gives $\hat{\rho} = 3.27 \pm 0.54$ (range 2.45–3.92), predicting approximately 0.31, while the *ae* probe gives 0.0136 ± 0.0015, predicting approximately 73.

Table 20 shows joint collapse in $10/10$ seeds for every tested $\alpha \geq 0.5$. Calibrated filtering achieves $0/10$ only for $\alpha \geq 1$, when all gates open and relative scatter is approximately 1.00; the recorded candidate rank is 3. At $\alpha = 0.5$, the gate opens in the five seeds with $\hat{\rho} > 2$, and all five avoid collapse (candidate rank 1). This is an empirical threshold comparison, subject to Corollary 7's scope restrictions. At lower doses, closed-gate outcomes are close to joint training.

The task metric can conceal shrinkage: joint depth RMSE falls overall from 1.335 to 1.092 (approximately 18%) while relative scatter falls from 0.62 to 0.04. Thus the high-dose joint model has lower depth error than the filtered models despite meeting the representation-collapse criterion. At every tested $\alpha \geq 1$, paired Wilcoxon tests on relative scatter give $p \approx 2 \times 10^{-3}$, the smallest two-sided value attainable with ten nonzero, same-sign paired differences.

Table 21 compares dense tasks under a shared small-backbone protocol. Its recorded statistic is mean within-example feature variance, which differs from the across-example quantity in (17). With the projection auxiliary, the amplitude ratio is 0.22: segmentation mIoU decreases while depth RMSE improves, and radial filtering returns amplitude near the auxiliary-free reference. With reconstruction at $\alpha = 250$, filtered amplitude is 3.84, compared with 3.99 for joint training and 7.19 for detachment. These task and amplitude comparisons motivate an across-example covariance audit before assigning a collapse mechanism.

H Supplementary Experimental Results

Ten-seed calibrated experiments support Section 5. At high doses, joint versus STFCAL relative scatter is 0.016 versus 0.990 for HAR ($\alpha = 100$), 0.147 versus 1.001 for battery ($\alpha = 100$), and 0.014 versus 0.982 for RadioML ($\alpha = 3000$). In each case, collapse rates are $10/10$ versus $0/10$. At low doses with closed gates, outcomes are close: HAR at $\alpha = 0.001$ gives 0.525 versus 0.528, and battery at $\alpha = 0.01$ gives 0.963 versus 0.965.

With Transformer encoders at $\alpha = 100$, HAR scatter changes from 0.030 to 1.041 and battery scatter from 0.036 to 1.000, again reducing collapse from $10/10$ to $0/10$. Battery SOH RMSE improves

Table 21: NYUv2 three-task comparison, shared convolutional encoder ($d = 128$), split, and ten seeds; final-epoch evaluation. Blocks give single-task and auxiliary-free references, projection auxiliary (*proj*, $\alpha = 1$), and reconstruction (*ae*, $\alpha = 250$). Depth error is in meters and normal error in radians. Recorded "scatter" is within-example $\mathbb{E}_b[\mathrm{Var}_j(h_j^{(b)})]$, with correction $d - 1$; it measures a different quantity from across-example scatter in (17). Absolute task scores depend on this small-backbone protocol.

Configuration	seg pxacc	seg mIoU	depth RMSE	normal angle	scatter
single-seg	0.359	0.083	2.780	1.574	2.22
single-depth	0.076	0.032	1.159	1.573	5.08
single-normal	0.074	0.031	3.346	0.593	12.43
joint3 (no aux)	0.356	0.100	1.130	0.586	7.30
+ proj aux (α=1)	0.365	0.090	1.085	0.580	1.64
STF-0	0.356	0.100	1.131	0.585	7.30
STFCAL	0.357	0.100	1.134	0.586	7.37
STFCALRANK	0.357	0.095	1.115	0.583	2.65
+ ae aux (α=250)	0.350	0.092	1.168	0.580	3.99
STF-0	0.357	0.099	1.133	0.586	7.19
STFCAL	0.338	0.091	1.171	0.594	3.84

from 0.447 to 0.267. On CelebA at $\alpha = 100$ (five seeds), scatter increases from 0.003 to 0.934 and collapse falls from $5/5$ to $0/5$.

The real-head experiments use $m = 64$ and ten seeds (Table 6). Neither reconstruction head collapses at its tested dose; filtering has little effect or modestly reduces primary accuracy. The training-subject classifier presents a different trade-off: joint training achieves identity accuracy 0.646 against chance $1/21$, while filtering reduces it to 0.055 and improves primary accuracy. This is a training-head measurement, not a held-out privacy attack. For high-dose SNR regression, joint training collapses in every seed; hard and soft filtering do not restore accuracy, and detachment performs best among the tested interventions. Probe magnitudes differ substantially across these heads, but the false negatives discussed in Section 4.2 preclude a universal decision rule. Figure 9 shows the battery repair outcome, Table 23 the corresponding collapse and recovery rates, and Table 22 the intervention families.

The results identify several distinct regimes: task error can improve as scatter shrinks, real auxiliaries can remain benign over the tested grid, and calibration can miss contraction. Frozen-basis error and sampling variation provide additional diagnostic checks. Numerical instability and loss of plasticity (Dohare et al., 2024) remain outside the fixed-operator model.

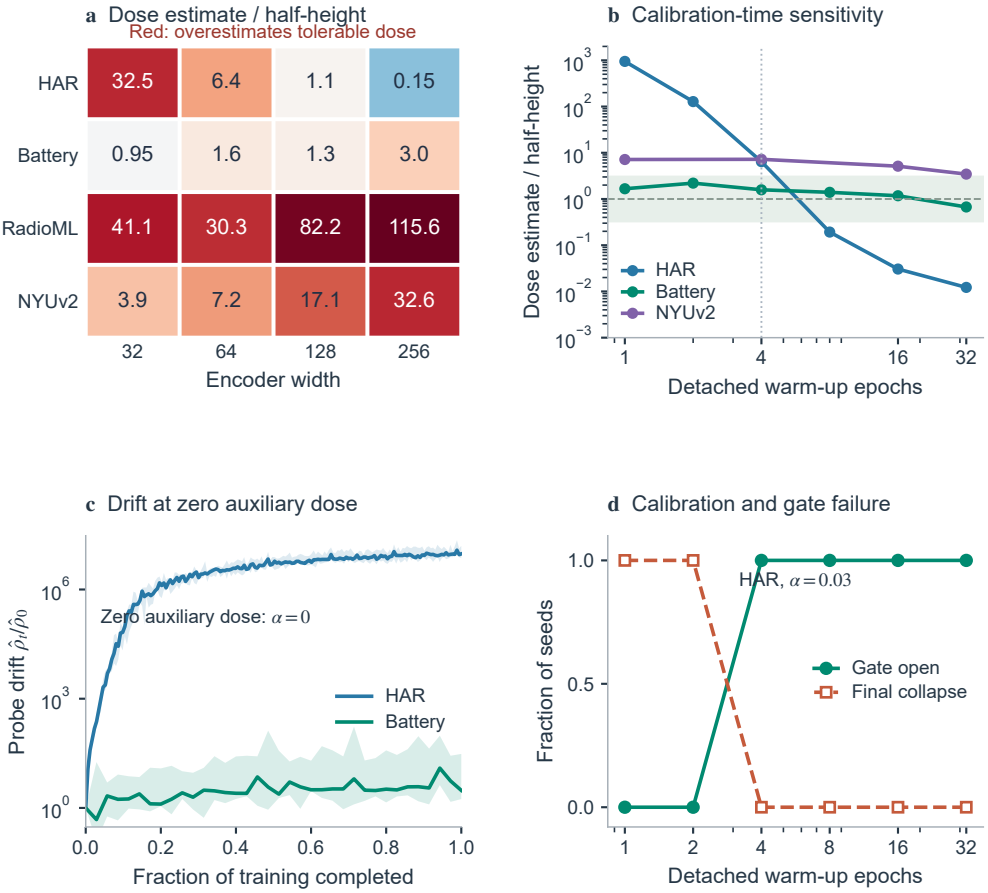


Figure 7: A frozen radial probe does not transfer uniformly. (a) $\alpha_{\text{probe}}/\alpha_{1/2}$ for 16 controlled domain–width cells. Red indicates an optimistic dose estimate; blue, a conservative one. (b) Calibration-time sensitivity at $m = 64$; shading marks a factor-3 reference band. Both panels use ten-seed mean probes. (c) Zero-dose probe drift: median and full range over five seeds, normalized to the initial value. (d) HAR at $\alpha = 0.03$: early calibration leaves all ten gates closed and all runs collapse; $W \geq 4$ opens every gate and prevents collapse. Values and crossing definitions are in Table 13 and Appendices H.2–H.3.

Table 22: Intervention families and their evaluation requirements. Performance comparisons under the stated implementations and protocols appear in Tables 4 and 25.

Family	Intervention	Relevant limitation
Task weighting (Chen et al., 2018; Liu et al., 2023)	scalar auxiliary strength	positive attenuation can help at a fixed dose
Gradient geometry (Yu et al., 2020; Liu et al., 2021; Navon et al., 2022)	task-gradient combination	alignment does not determine variance change
Variance regularization (Bardes et al., 2022)	representation penalty	penalty weight and primary-task trade-off
Full detachment	remove auxiliary encoder gradient	also removes useful auxiliary learning
Selective filtering	remove selected directions	basis identification and subsequent drift
Parameter variance constraint	constrain the actual parameter update	fixed guard batch; finite-step feasibility must be checked

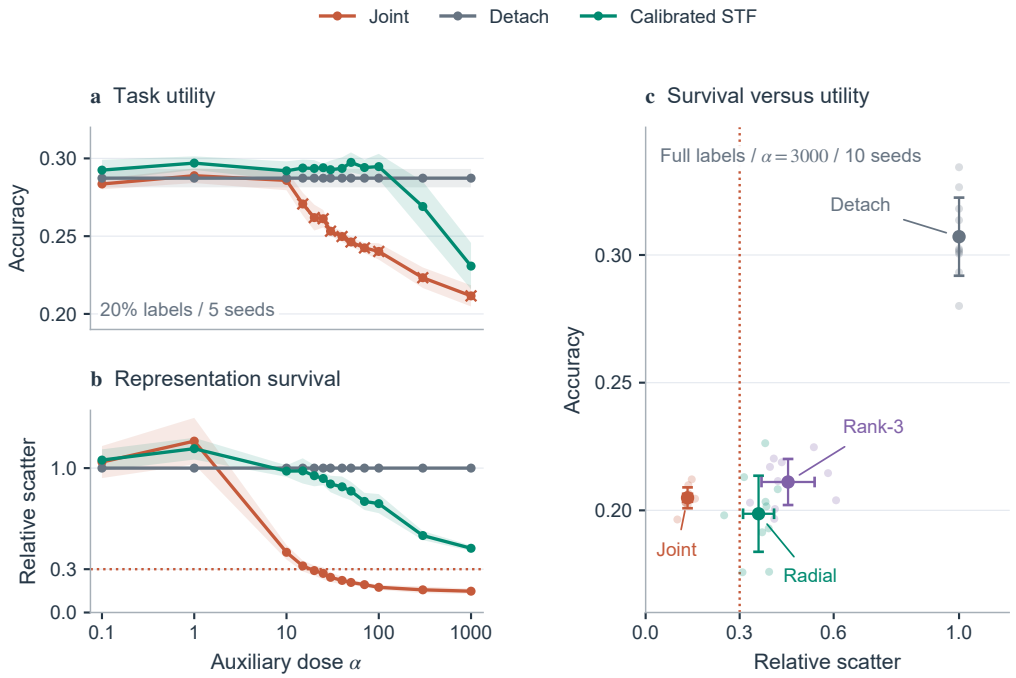


Figure 8: Filtering can retain an auxiliary gain, but scatter protection alone is insufficient. RadioML SNR regression, width 64. (a,b) With 20% primary labels, STF exceeds detachment at intermediate doses; the gain disappears at extreme doses despite scatter above 0.3. Five unique seeds per dose; bands show seed SD and crosses mark at least one collapsed joint run. (c) In the separate fully labeled dose-3000 protocol, rank-3 filtering avoids collapse in all ten seeds but underperforms detachment. Small points are seeds; large points and bars show means and SD. The dotted line marks the 0.3 criterion.

Table 23: Controlled high-dose collapse and task-recovery rates for three domains. Recovery is the fraction of seeds whose primary metric improves over the mean joint-training metric. HAR can retain high accuracy despite substantial shrinkage, so scatter recovery does not imply better accuracy. Battery uses 20 seeds per mode and RadioML ten; the historical HAR file provides aggregate rates. The joint recovery rate is set to zero by convention. All plotted rates are reproduced directly from the released records.

Domain	Metric	JOINT coll. / rec.	STF-0 coll. / rec.	STF-HARD coll. / rec.	STF-SOFT coll. / rec.
HAR	accuracy (flat)	1.0 / 0.0	0.0 / 0.1	0.0 / 0.2	0.0 / 0.1
RadioML	accuracy	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0
Battery	log-SOH RMSE	1.0 / 0.0	0.0 / 1.0	0.0 / 1.0	0.0 / 1.0

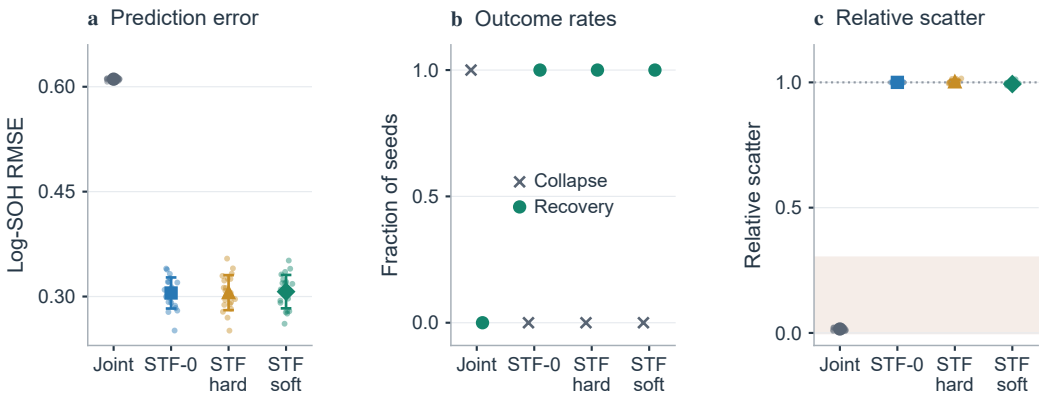


Figure 9: Controlled battery contraction can be prevented by filtering. NASA PCoE unified protocol, $m = 128$, $\alpha = 10^4$, 20 seeds per mode. (a) Mean log-SOH RMSE falls from 0.611 to about 0.305. (b) Collapse and task-recovery rates. (c) Scatter relative to the same-seed single-task reference; the dotted line marks 1.0 and shading the region below 0.3. Small points are seeds; large markers and whiskers show means and population SD. Log-SOH error and percentage-point SOH error in the separate application have different units.

The battery encoder maps a per-cycle window $x \in \mathbb{R}^{T \times D}$ of normalized voltage, current, temperature, and resistance to $h \in \mathbb{R}^d$. It combines a TSMIXER feature/temporal-mixing block (Ekambaram et al., 2023) with a gated state-space block (Gu et al., 2022; Gu & Dao, 2024). SOH (L_2), RUL (L_1), reconstruction, and distributional heads share this representation. The empirical real-head studies complement the degenerate-minimum model of Definition 2.

RUL targets require careful handling of right censoring. If a cell has not reached end of life when recording stops, its remaining lifetime is known only to exceed a lower bound. Treating that bound as an exact target can label a healthy cell with zero remaining life. In MATR, 59.2% of samples are censored and $96/185$ cells do not reach EOL within their records. Initialization zeros in summary arrays cause a further $48/185$ cells to receive an EOL index of zero under a naive threshold search. Restricting the search to each trajectory's physically valid suffix recovers these cells. We retain censored cells and use $|\hat{r} - r|$ for observed lifetimes and $\max(0, r_{\text{lb}} - \hat{r})$ for censored lower bounds. This one-sided loss is motivated by censored regression (Tobin, 1958), but is not a Tobit likelihood. Excluding cells without observed EOL would leave only 14 test cells in the historical split (Table 16).

To emphasize fast-charge, extreme-temperature, and rapid-degradation windows, BatteryMTL uses three multiplicative weighting factors:

$$\begin{aligned} w_i &= w_i^{(\text{rate})} \cdot w_i^{(\text{temp})} \cdot w_i^{(\text{degrad})}, \\ w_i^{(\text{rate})} &= 1 + \omega\, \mathbf{1}[\text{C-rate}_i > r_q], \\ w_i^{(\text{temp})} &= 1 + \omega\, \mathbf{1}[T_i < t_{lo} \vee T_i > t_{hi}], \\ w_i^{(\text{degrad})} &= 1 + \omega\, \frac{|\Delta\text{SOH}_i|}{\max_j |\Delta\text{SOH}_j|}, \end{aligned} \tag{27}$$

Equation (27) amplifies windows in which several factors are elevated, while also up-weighting windows with a single elevated factor. Thresholds are estimated from the training fold: r_q is the 70th C-rate percentile, and t_{lo}, t_{hi} are the 10th and 90th temperature percentiles. These relative thresholds distinguish regimes within MATR, where 99.8% of cycles exceed 1.5C and temperatures lie within 31–36°C. In contrast, fixed cuts at 1.5C and $0/40$°C would mark nearly all cycles for rate and none for temperature. Section B.1 reports regime-specific ablations; SOH-only and RUL-only models provide task-specific references, not guaranteed performance ceilings.

Table 24: Filter-then-combiner experiments at high auxiliary doses (five seeds, three domains). Unfiltered branches retain collapse rates of 1.0 and relative scatter at most 0.15. The hard filter, alone or followed by PCGrad/CAGrad, avoids the collapse criterion in every seed. Primary metrics change modestly after composition; the reverse order was not tested. Bold identifies the filter-only reference within each block.

		unfiltered (JOINT)			filtered (STF-HARD)		
	combiner	sum	PCGrad	CAGrad	sum	PCGrad	CAGrad
Battery	log-SOH RMSE↓	0.611	0.592	0.603	**0.294**	0.283	0.291
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.03	0.03	**1.00**	1.05	1.18
HAR	accuracy↑	0.941	0.667	0.952	0.943	0.933	0.940
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.15	0.00	0.03	**0.97**	1.14	1.03
RadioML	accuracy↑	0.293	0.091	0.109	**0.301**	0.296	0.295
	collapse rate	1.0	1.0	1.0	0.0	0.0	0.0
	relative scatter	0.02	0.00	0.00	**0.97**	0.99	1.04

H.1 Composition and Regime Weighting

The following measurements expand the composition and weighting results in Sections B.1 and B.1.

At the tested high doses, applying PCGrad or CAGrad to the unfiltered battery branch leaves collapse rate at 1.0 and relative scatter at $0.029/0.030$ (Table 24). Hard filtering alone reduces collapse to 0.0, restores relative scatter to 1.00, and improves log-SOH RMSE from 0.611 to 0.294. Adding PCGrad or CAGrad afterward gives RMSE $0.283/0.291$ and scatter up to 1.18. HAR and RadioML show the same qualitative preservation of scatter under composition. These results establish compatibility for the tested order; they neither show that sequencing is necessary nor establish why the unfiltered combiners fail.

BatteryMTL's multiplicative regime weights use training-data percentiles for C-rate and temperature, together with degradation rate. The aggregate SOH results change little, while factor ablations associate improvements in fast-charge and extreme-temperature windows with the rate and temperature terms, respectively (errors 0.95 and 2.35 when the corresponding terms are removed). Table 10 reports the historical aggregate results.

H.2 Probe Dynamics During Training

The frozen calibration in Section 5 is motivated by the behavior of a continuously recomputed probe. We repeat unfiltered joint training with `--rho-trace`, using five seeds per configuration over 1730 HAR steps and 350 battery steps. Monitoring does not alter the measured trajectories: all 28 shared BSCALE configurations match exactly in task metric, collapse rate, relative scatter, and $\hat{\rho}$ (maximum difference zero).

Three observations matter. First, HAR probe values rise sharply during training: from 0.017 to 1.9×10^5 at $m = 32$, 0.010 to 1.0×10^5 at $m = 64$, and 0.006 to 3.5×10^4 at $m = 256$. Second, similar terminal values occur when the toxifier is inactive ($\alpha = 0$) and at the lowest positive dose (10^{-4}), so a large probe value alone does not establish auxiliary-induced contraction. At $m = 32$, increasing the dose instead lowers the terminal probe, to 5.3×10^2 at 10^{-2} and 1.1×10^2 at 3×10^{-2}. Third, in ten HAR configurations whose final scatter lies between 0.50 and 1.04 of the reference, the gate $\alpha\hat{\rho} \geq 1$ opens in every seed between steps 110 and 590. A gate using this running statistic therefore activates on runs that do not meet the collapse criterion at the recorded endpoint.

Battery behaves differently: probe values remain $O(1)$, ranging from 0.29–0.49 initially to 0.3–2.9 at the final step. Among 35 seeds whose final scatter lies above the half-height level, the gate opens

Table 25: Anti-collapse regularizers at the doses and seeds of Table 4 (ten seeds per configuration). Penalties of weight λ are added to joint training. Entries report collapse rate, primary metric in parentheses, and relative scatter (1.0 is the single-task reference). Joint training, detachment, and hard filtering are evaluated in the same runs. Avoiding the scatter threshold need not improve the primary task.

	HAR		Battery		RadioML	
	coll. / acc.↑	rel. scat.	coll. / RMSE↓	rel. scat.	coll. / acc.↑	rel. scat.
JOINT	1.00 (0.945)	0.15	1.00 (0.610)	0.022	1.00 (0.256)	0.008
STF-0	0.00 (0.945)	1.00	0.00 (0.297)	1.00	0.00 (0.341)	1.00
STF-HARD	0.00 (0.946)	0.99	0.00 (0.299)	1.00	0.00 (0.339)	0.97
VICReg (variance + covariance)						
λ=1	1.00 (0.950)	0.22	1.00 (0.610)	0.023	1.00 (0.249)	0.007
λ=3	0.20 (0.946)	0.31	1.00 (0.610)	0.023	1.00 (0.253)	0.007
λ=10	0.00 (0.944)	0.36	1.00 (0.609)	0.024	1.00 (0.256)	0.008
λ=30	0.00 (0.926)	0.36	1.00 (0.604)	0.033	1.00 (0.260)	0.008
λ=100	0.00 (0.902)	0.36	1.00 (0.451)	0.14	1.00 (0.301)	0.011
λ=300	0.00 (0.873)	0.36	0.00 (0.339)	0.59	1.00 (0.362)	0.035
λ=10^3	0.00 (0.843)	0.36	0.00 (0.390)	1.95	1.00 (0.320)	0.15
λ=3×10^3	0.00 (0.831)	0.36	0.00 (0.423)	3.70	0.10 (0.207)	0.45
λ=10^4	0.00 (0.821)	0.36	0.00 (0.423)	4.54	0.00 (0.215)	1.29
Barlow Twins (redundancy reduction)						
λ=1	1.00 (0.935)	0.11	1.00 (0.607)	0.11	1.00 (0.367)	0.016
λ=3	1.00 (0.765)	0.022	1.00 (0.607)	0.13	1.00 (0.330)	0.024
λ=10	1.00 (0.435)	0.006	1.00 (0.606)	0.15	1.00 (0.121)	0.074
λ=30	1.00 (0.352)	0.006	1.00 (0.606)	0.16	1.00 (0.105)	0.096
λ=100	1.00 (0.338)	0.006	1.00 (0.605)	0.16	1.00 (0.126)	0.072
λ=300	1.00 (0.347)	0.006	1.00 (0.606)	0.16	1.00 (0.141)	0.043
λ=10^3	1.00 (0.343)	0.006	1.00 (0.606)	0.16	1.00 (0.140)	0.042
λ=3×10^3	1.00 (0.365)	0.006	1.00 (0.607)	0.16	1.00 (0.148)	0.053
λ=10^4	1.00 (0.339)	0.006	1.00 (0.607)	0.16	1.00 (0.153)	0.044

only once, at the last step. The contrast with HAR shows that calibration time must be specified by protocol. The script `repro/scripts/rho_trace_analyze.py` reproduces both analyses.

H.3 Sensitivity to Calibration Time

The default calibration measures $\hat{\rho}$ after four detached warm-up epochs and fixes the gate and rank. We vary only this measurement time: $W \in \{1, 2, 4, 8, 16, 32\}$ at five widths 32–512 for HAR and battery, and $W \in \{1, 4, 16, 32\}$ at four widths for NYUv2, with ten seeds per cell. The $W = 4$ configuration exactly reproduces all 100 shared HAR and battery probe values. NYUv2 gives $2.258/2.008/1.637/1.285$, matching the rounded values in Appendix G.

Probe values depend on the training state. Across the warm-up sweep, HAR at $m = 32$ increases from 0.63 to 6.9×10^4, and at $m = 512$ from 51.9 to 1.6×10^4. Battery remains $O(1)$: 0.96 to 1.92 at $m = 32$ and 0.058 to 0.67 at $m = 512$. NYUv2 at $m = 32$ changes from 1.94 to 4.19, a factor of 2.2. Accordingly, width slopes fitted to $1/\hat{\rho}$ vary from -2.00 to $+0.55$ for HAR, $+0.37$ to $+0.97$ for battery, and $+0.11$ to $+0.27$ for NYUv2. These are calibration-dependent probe slopes, not interchangeable estimates of capacity scaling. Differences in primary-gradient magnitude, which appears in the denominator, can contribute to this sensitivity.

Measurement time also changes gate activation. For HAR at $m = 64$, the inverse-mean probe value is 1.34 at $W = 1$, 0.181 at $W = 2$, 9.07×10^{-3} at $W = 4$, and 1.7×10^{-5} at $W = 32$, compared with the measured transition interval $[0.001, 0.01]$. In the sixty-epoch, ten-seed experiment at $\alpha = 0.03$,

gates remain closed for $W = 1$ and $W = 2$: both settings collapse in $10/10$ seeds, with accuracy 0.947 and relative scatter $0.19/0.20$, close to joint training (0.949, 0.19). For $W \geq 4$, all gates open, no seed collapses, and scatter returns to 0.99–1.00. At $\alpha = 100$, all six warm-up settings activate the gate and avoid collapse.

A short warm-up therefore fails at the marginal dose in this experiment. Detached warm-up also changes the training trajectory, so closed-gate runs need not equal joint training from initialization. Checking whether $\alpha\hat{\rho} \geq 1$ verifies activation at the chosen operating dose; it does not independently validate protection. The script `repro/scripts/warmup_sensitivity.py` reproduces these results.

H.4 Scale-Invariant Audit of the Scatter Criterion

The operational criterion in (17) is a coordinate-dependent, scale-sensitive proxy. To measure what a contraction preserves, we run the released HAR protocol at three doses and audit the frozen representation with covariance geometry and linear probes on raw, per-dimension-standardized, whitened, 10%-label, and new-label-space features. Thirty cells use $m = 64$, 60 epochs, Adam (10^{-3}, batch 256), and seeds 0–9; the driver is `repro/scripts/representation_audit.py`, and the records are in `results/REPAUDIT/`.

Table 26: Scale-invariant audit of the controlled HAR toxifier. Relative scatter uses the same-seed healthy reference. Covariance trace is absolute; participation rank (PR), effective (entropy) rank, and standardized probes are scale-invariant. At $\alpha = 100$ the scatter criterion records $10/10$ collapses, while the standardized probe and the new-label-space probe are unchanged and PR rises.

α	rel. scatter	cov. trace	PR	eff. rank	standardized probe	primary acc.
0 (healthy)	1.000	578.7	2.00	2.57	0.947	0.941
0.1	0.130	8.51	4.97	5.52	0.958	0.956
100	0.016	0.125	4.90	5.02	0.955	0.952

The trace falls by $4{,}600\times$ and every seed meets the scatter criterion, yet per-dimension standardisation preserves activity-probe accuracy. A static-versus-dynamic activity probe stays at 0.999. A one-hidden-layer MLP probe (64 units, 120 epochs) similarly reaches 0.945 at $\alpha = 0$ and 0.954 at $\alpha = 100$ over five seeds. These measurements show a large change in feature amplitude with retained access to the measured task information.

H.5 Small-Dose Operator Audit

The main-text comparison holds each operator at the domain's supercritical dose. The complementary question is whether the default scalarizations contract the representation when the auxiliary weight is far below the calibrated threshold. Table 27 reports five operators at three doses on HAR ($m = 64$, 60 epochs, three seeds); the driver is `repro/scripts/lowalpha_operator_audit.py`, and the records are in `results/LOWALPHA/`.

Table 27: Accuracy (scatter-collapse count over three seeds) at small auxiliary doses, HAR, $m = 64$. Plain summation is healthy at $\alpha = 0.001$, while the default scalarizations contract in every seed. This is operator conditioning rather than toxifier dose.

α	joint	MGDA	IMTL-G	Aligned	Nash	FAMO
0.001	0.943 (0/3)	0.551 (3/3)	0.242 (3/3)	0.927 (3/3)	0.425 (3/3)	0.176 (3/3)
0.01	0.950 (2/3)	0.409 (3/3)	0.301 (3/3)	0.884 (3/3)	0.461 (3/3)	0.176 (3/3)
0.1	0.953 (3/3)	0.182 (3/3)	0.183 (3/3)	0.832 (3/3)	0.565 (3/3)	0.181 (3/3)

At $\alpha = 0.001$, the tested operators contract features even though the auxiliary gradient is tiny and plain summation remains healthy. The low-dose behavior therefore depends on the default scalarization as well as the auxiliary dose. MGDA still collapses after lowering the learning rate to

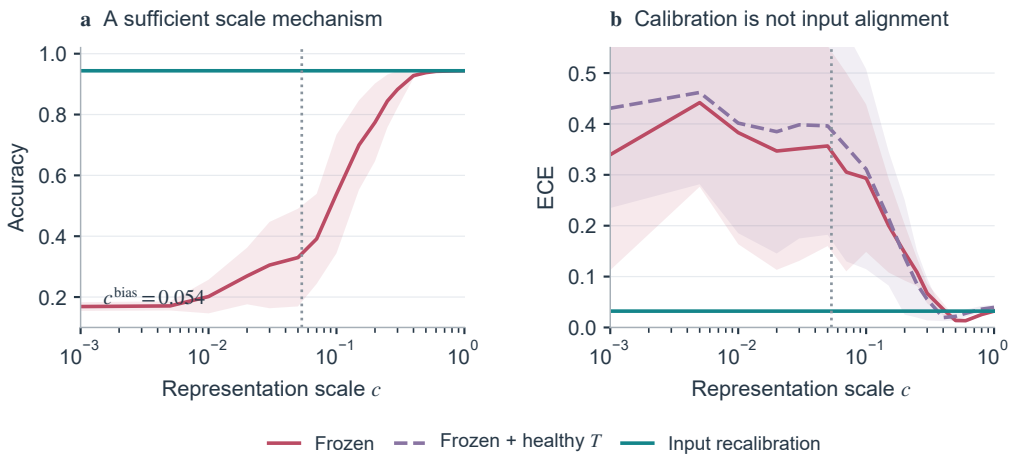


Figure 10: Pure scale change can invalidate a fixed affine readout. Healthy HAR features are multiplied by c before readout. (a) Frozen accuracy degrades; refitting input statistics cancels the imposed scale change. (b) ECE before and after applying the healthy temperature map. Five seeds, $m = 64$; bands show seed SD. The dashed line summarizes finite bias-exit thresholds: median of seed medians $c^{\text{bias}} = 0.054$, seed range 0.020–0.207. Bias exit and the first flip when shrinking from $c = 1$ are distinct boundaries.

10^{-4} (Table 28), which rules out the tested step-size reduction as a remedy. This conditioning effect is evaluated under the recorded default scalarizations and schedules.

Table 28 compares the default learning rate with a tenfold smaller one at $\alpha = 0.001$. The loss-weighting families (UW, GradNorm, IMTL-L) become healthy at this low dose, as does plain summation; the gradient-geometry operators (PCGrad, CAGrad, FAMO, IMTL-G, Nash, Aligned, MGDA) collapse in every seed at both learning rates, and several of their accuracies worsen at the lower rate. The low-dose contraction is therefore operator-specific, not a single step-size artifact. At the high doses of Table 4, all ten operators collapse.

Table 28: Small-dose operator audit at $\alpha = 0.001$, HAR, $m = 64$, three seeds. Accuracy (scatter-collapse count) at the default 10^{-3} and a tenfold smaller 10^{-4} learning rate. The weighting families become healthy at low dose; the gradient-geometry operators collapse at both rates.

Method	$lr = 10^{-3}$	$lr = 10^{-4}$
joint (sum)	0.943 (0/3)	0.934 (0/3)
PCGrad	0.660 (3/3)	0.656 (3/3)
CAGrad	0.954 (3/3)	0.953 (3/3)
FAMO	0.176 (3/3)	0.176 (3/3)
UW	0.941 (0/3)	0.935 (0/3)
GradNorm	0.954 (0/3)	0.934 (0/3)
IMTL-L	0.944 (0/3)	0.935 (0/3)
IMTL-G	0.242 (3/3)	0.187 (3/3)
Nash-MTL	0.425 (3/3)	0.338 (3/3)
Aligned-MTL	0.927 (3/3)	0.790 (3/3)
MGDA	0.551 (3/3)	0.186 (3/3)

H.6 Readout Transfer Under Contraction

The next experiment measures the compatibility of a readout fitted to healthy features. For five HAR seeds, we train healthy ($\alpha = 0$) and contracted ($\alpha = 100$) encoders and split training examples into readout-fitting and calibration subsets. Table 29 evaluates four maps on held-out test data: the

Table 29: Historical transfer between separately trained healthy and contracted HAR encoders (five seeds, means). "Frozen" applies the healthy head and input statistics unchanged; the reported ECE is the raw softmax ECE, and applying the healthy temperature map raises frozen ECE to 0.512. Input recalibration reuses the weights but refits the feature statistics; retraining fits a new head. The frozen readout degrades with large NLL/ECE, while cheap input recalibration recovers accuracy but not the NLL of a retrained head. The last two rows apply the same healthy readout to the STF-0 and STF-cal encoders.

Readout	accuracy	NLL	ECE
healthy reference	0.945	0.187	0.031
frozen (healthy statistics)	0.241	2.516	0.430
input recalibration	0.952	0.396	0.043
retrained on contracted features	0.955	0.190	0.034
STF-0 encoder (same readout)	0.946	0.188	0.031
STF-cal encoder (same readout)	0.943	0.201	0.033

Table 30: Historical real-head transfer between separately trained encoders. Frozen reuses the healthy head and statistics; recalibration refits statistics; retrained fits a new head; detached reuses the healthy readout on the STF-0 encoder. Relative S is the same-seed scatter ratio; five seeds for SNR and subject-ID, three for I/Q.

Head	α	rel. S	healthy	frozen	recal.	retrained	detached
RadioML SNR	3000	0.137	0.315	0.099	0.094	0.229	0.315
RadioML I/Q	3000	1.053	0.312	0.195	0.195	0.351	0.312
HAR subject-ID	10	0.932	0.944	0.384	0.466	0.901	0.944

healthy reference; the unchanged healthy head and statistics on contracted features; the same head with input statistics refitted without new labels; and a new head fitted to the contracted features.

Proposition 1 isolates the scale mechanism. A unique bias-only winner gives a constant predicted class at sufficiently small positive scale; ties need a specified convention. Fixed input centering contributes to the effective offset even with a bias-free head. Figure 10 summarizes finite c_i^{bias} by their within-seed median and then the median across seeds. This statistic describes bias exit, distinct from the first prediction flip when shrinking from $c = 1$. The standard-deviation floor is inactive over the plotted range; exact invariance for every $c > 0$ assumes unfloored statistics.

The historical transfer study measures compatibility between encoders trained separately from the same initialization. Its calibration subset is withheld from readout fitting but participates in supervised encoder training. Historical nominal-90% conformal coverage (0.802 healthy, 0 frozen, 0.806 recalibrated, 0.834 retrained) is therefore descriptive, not a split-conformal guarantee; its quantile also omits the finite-sample order-statistic correction. The paired continuation study below excludes calibration examples from encoder and head fitting and uses the corrected quantile. The official HAR subject split still changes the test population, so we report empirical coverage without assuming calibration–test exchangeability.

The unchanged readout models a fixed deployment interface behind a changing encoder. Its NLL and ECE rise sharply; input-statistic recalibration restores accuracy and most of ECE, while a new head also restores NLL. Applying the same healthy readout to STF-0 and STF-cal encoders gives 0.946 and 0.943 accuracy (`stf_readout_audit.py`). The gap also appears at $m = 32$ (frozen 0.236, recalibrated 0.887, refitted 0.909), $m = 128$ (0.166, 0.955, 0.954), and with a Transformer (0.162, 0.950, 0.952). These tested widths and encoders exhibit the same separation between decodability and compatibility. Driver: `repro/scripts/readout_transfer_audit.py`; records: `results/READOUT/`.

H.7 Paired Checkpoint Continuation

The paired experiment updates the encoder behind an already fitted consumer. We use the official HAR subject split and a prepared RadioML split with the same three-layer, width-64 MLP; the Ra-

dioML source-split audit appears below. A class-stratified 20% training reservation is made before any encoder update for calibration. The remaining 80% fits the encoder, primary head, and replacement probes. Reference readouts use per-feature standardization and the historical regularized LBFGS fit. Temperature fitting follows the fixed reference predictor; raw conformal scores use the corrected $\lceil (n+1)0.9 \rceil$ order statistic. Because HAR separates subjects between training and test, we report empirical coverage without assuming calibration–test exchangeability.

Each seed first trains a healthy encoder for 60 HAR or 25 RadioML epochs. A real SNR head is trained on detached features during this warm-up. We clone all model weights and Adam moments, retain the fitted consumer readout, and continue each branch for 30 HAR or 15 RadioML epochs. The learning rate is 10^{-3}, batch size 256, and every operation uses CPU fp32 except the double-precision linear readout fit and evaluation. Future minibatch permutations are generated once per seed and reused for joint training, detachment, STF-cal, and primary-only continuation. Initial parameter, optimizer-state, and minibatch-order hashes verify the pairing; all detached encoder and primary-head endpoints match primary-only continuation exactly. Evaluation never selects a checkpoint. The isotropic variance objective matches the historical readout audit; a fixed projected-variance objective tests the benchmark operator separately. The first five HAR variance seeds are followed by a predeclared replication on seeds 5–9, with the recipe unchanged and all outcomes retained.

Table 31 gives every sampled dose for both variance objectives. At isotropic $\alpha = 3$, frozen accuracy is 0.453, current-head accuracy 0.948, and recalibrated accuracy 0.947. Continuation thus causes a large compatibility loss while task decodability remains high. The fitted scalar residual shows additional feature drift, which matters at $\alpha = 100$: recalibration reaches 0.829 versus 0.935 after refitting. The imposed affine sweep and the trained continuations provide complementary tests of the scale component.

The SNR continuation provides a protocol boundary. At $\alpha = 3, 10, 30$, frozen accuracy is 0.293, 0.269, and 0.243, against 0.304 detached. At 30, recalibration reaches 0.253, refitting 0.285, and relative scatter 0.870. The severe collapse seen in historical from-initialization training is absent at these nominal doses after healthy warm-up. Calibration split, auxiliary state, warm-up, and total horizon also differ; the head-state factorial below tests one of these factors.

At the first minibatch of specified continuation epochs, we evaluate V and its gradient v on a fixed 256-example fitting batch. Loss gradients use the actual update minibatch. We record flow $r - \alpha c$, the applied increment s_θ, $v^\top s_\theta$, and the exact post-step change in V. A primary-only Adam counterfactual starts from identical pre-step weights and moments; its difference from the joint step measures the auxiliary's conditional effect at that optimizer state. Across SNR joint branches, flow and actual-step signs disagree on $33/75$ sampled updates. Table 32 includes negative-c checkpoints with a contracting auxiliary counterfactual, demonstrating why flow attribution and applied-step attribution need separate records.

Each new run uses one CPU thread, with at most four concurrent workers and explicit memory and time limits. The remote directories contain numerical code, inputs, checkpoints, and logs; analysis, writing, figures, and compilation run locally. Driver `paired_readout_continuation.py` writes the complete per-seed records to `repro/results/CPUCONT/`. Script `summarize_cpu_evidence.py` checks the paired controls and regenerates the numerical macros, tables, and figure. Seed intervals describe variation conditional on these datasets.

H.8 A Fixed Unlabeled Repair Hierarchy

To distinguish scale from richer drift, we apply a fixed repair hierarchy to every saved endpoint from the 10 isotropic HAR, 5 projected HAR, and 5 SNR continuation seeds. Deterministic replay reconstructs each healthy readout and reproduces its accuracy, NLL, ECE, and all archived frozen and recalibrated endpoints within 10^{-7}. The reconstructed weights and statistics are saved; the encoders remain fixed throughout the audit.

Let $R = H_0 - \mathbf{1}\mu_0^\top$ and $X = H_1 - \mathbf{1}\mu_1^\top$ denote the old and new fitting features after centering. The global scale estimate is $c_{\mathrm{RMS}} = \|X\|_F / \|R\|_F$. Mean-only correction feeds $h - \mu_1 + \mu_0$ to the original preprocessing; scale-only feeds h/c_{RMS}; their combination feeds $(h - \mu_1)/c_{\mathrm{RMS}} + \mu_0$.

Table 31: Paired checkpoint continuation with held-out calibration. Means ± population seed standard deviations. Each branch starts from the same healthy model and Adam state; the future minibatch order is shared. Trace is relative to the initial healthy checkpoint. HAR uses 60 healthy plus 30 continuation epochs; RadioML uses 25 plus 15. The real auxiliary head is trained on detached features during warm-up.

Case / dose	arm	frozen	input recal.	retrained	own	rel. trace
HAR variance, 10 seeds						
0	primary only	0.943 ± 0.007	0.943 ± 0.006	0.945 ± 0.005	0.944 ± 0.005	1.0897
0.1	joint	0.789 ± 0.161	0.951 ± 0.005	0.953 ± 0.002	0.951 ± 0.007	0.0213
0.1	STF-cal	0.943 ± 0.007	0.943 ± 0.006	0.945 ± 0.005	0.944 ± 0.005	1.0897
3	joint	0.453 ± 0.135	0.947 ± 0.005	0.952 ± 0.004	0.948 ± 0.006	0.0037
3	STF-cal	0.943 ± 0.007	0.943 ± 0.005	0.945 ± 0.005	0.941 ± 0.009	1.0868
100	joint	0.221 ± 0.067	0.829 ± 0.062	0.935 ± 0.016	0.796 ± 0.060	0.0003
100	STF-cal	0.941 ± 0.009	0.942 ± 0.005	0.944 ± 0.005	0.941 ± 0.006	1.0918
HAR projected, 5 seeds						
0	primary only	0.941 ± 0.007	0.940 ± 0.004	0.944 ± 0.004	0.943 ± 0.004	1.0636
0.1	joint	0.844 ± 0.128	0.932 ± 0.022	0.950 ± 0.005	0.945 ± 0.007	0.3460
0.1	STF-cal	0.941 ± 0.007	0.940 ± 0.004	0.944 ± 0.004	0.943 ± 0.004	1.0636
3	joint	0.652 ± 0.125	0.782 ± 0.077	0.946 ± 0.003	0.936 ± 0.011	0.3489
3	STF-cal	0.939 ± 0.011	0.940 ± 0.004	0.945 ± 0.004	0.943 ± 0.004	1.0737
100	joint	0.214 ± 0.056	0.398 ± 0.033	0.851 ± 0.083	0.724 ± 0.110	0.0640
100	STF-cal	0.939 ± 0.011	0.940 ± 0.003	0.944 ± 0.005	0.943 ± 0.004	1.0717
RadioML SNR, 5 seeds						
0	primary only	0.304 ± 0.011	0.306 ± 0.015	0.320 ± 0.010	0.319 ± 0.009	1.1045
3	joint	0.293 ± 0.011	0.300 ± 0.014	0.314 ± 0.012	0.312 ± 0.011	0.8280
3	STF-cal	0.296 ± 0.012	0.304 ± 0.015	0.318 ± 0.012	0.317 ± 0.012	0.9778
10	joint	0.269 ± 0.017	0.281 ± 0.016	0.302 ± 0.011	0.298 ± 0.013	0.7274
10	STF-cal	0.292 ± 0.015	0.301 ± 0.015	0.317 ± 0.011	0.315 ± 0.011	0.8520
30	joint	0.243 ± 0.020	0.253 ± 0.016	0.285 ± 0.012	0.277 ± 0.014	0.6394
30	STF-cal	0.286 ± 0.014	0.297 ± 0.013	0.312 ± 0.011	0.309 ± 0.010	0.7111

Table 32: Actual-step audit on a fixed 256-example fitting batch, using the first minibatch of predeclared continuation epochs. Disagree counts opposite signs for the gradient-flow derivative and actual Adam variance change. Aux. contracts compares the joint step with a primary-only Adam step from the same optimizer state. The final column counts negative flow auxiliary flux with a contracting auxiliary counterfactual. Checkpoints within a trajectory are dependent.

Case	α	updates	disagree	aux. contracts	$c < 0$, aux. contracts
HAR variance	0.1	80	28	80	0
HAR variance	3	80	22	79	2
HAR variance	100	80	23	76	3
HAR projected	0.1	40	14	31	0
HAR projected	3	40	11	25	1
HAR projected	100	40	8	29	2
RadioML SNR	3	25	10	12	1
RadioML SNR	10	25	13	10	0
RadioML SNR	30	25	10	10	1

Per-coordinate recalibration replaces each input mean and standard deviation. These estimators use marginal fitting statistics, with no new labels or test-statistic fitting.

The richer repairs are fixed post-hoc controls. Paired diagonal regression predicts each old coordinate from its new counterpart; scaled orthogonal Procrustes fits a rotation and global scale. Covariance alignment maps centered features by $C_1^{-1/2} C_0^{1/2}$, following CORAL (Sun et al., 2016), with eigenvalues floored at 10^{-6} of the largest. Paired ridge maps new standardized anchors to old standardized anchors with $\lambda = 10^{-3}$. These constants are fixed independently of test outcomes. Covariance and marginal-statistic repairs need only marginal features; paired regressions and Pro-

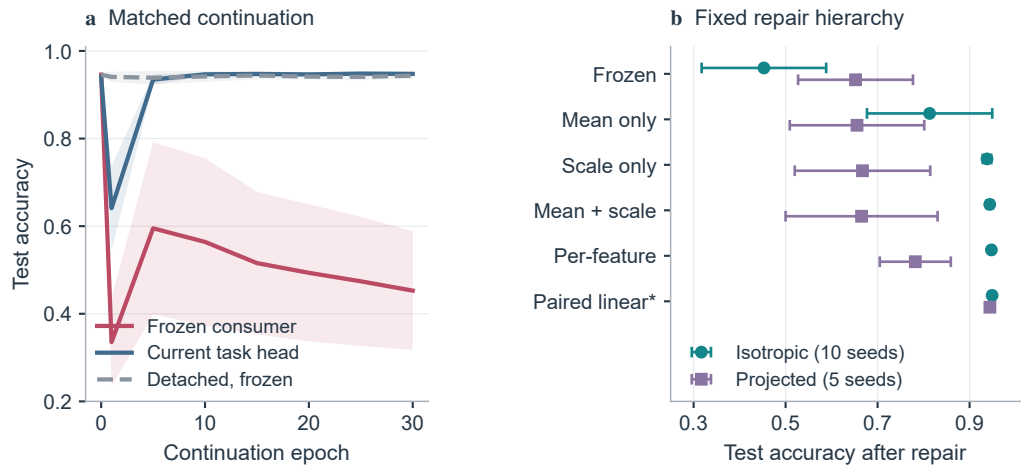


Figure 11: Input repair reveals the scale component of a learned compatibility failure. (a) Frozen-consumer and current-head trajectories at $\alpha = 3$, sharing healthy weights, Adam state, and future minibatches. (b) Fixed repairs fitted to training features without new labels. Global scale correction largely repairs isotropic contraction; projected contraction needs richer alignment. *Paired linear alignment requires old/new features of the same anchors. Shading and bars show population seed SD: 10 isotropic and 5 projected seeds.

Table 33: Fixed repair hierarchy on archived paired checkpoints. Means across 10 HAR isotropic, 5 HAR projected, and 5 RadioML SNR seeds. All maps use fitting features only, without labels or test-statistic fitting. PD, Proc., and ridge require paired old/new anchors; covariance alignment uses only marginal second moments. Ridge is fixed at 10^{-3} in standardized coordinates; covariance eigenvalue flooring is 10^{-6} of the largest eigenvalue. Full seed dispersions, NLL, ECE, prediction agreement, and fitting/held-out residuals accompany the records.

Case	α	frozen	mean	scale	mean +scale	diag.	PD	Proc.	cov.	ridge
HAR isotropic	0	0.943	0.941	0.943	0.941	0.943	0.943	0.941	0.945	0.946
HAR isotropic	0.1	0.789	0.941	0.946	0.950	0.951	0.952	0.949	0.930	0.950
HAR isotropic	3	0.453	0.813	0.938	0.944	0.947	0.949	0.948	0.912	0.949
HAR isotropic	100	0.221	0.280	0.603	0.782	0.829	0.841	0.855	0.774	0.915
HAR projected	0	0.941	0.938	0.941	0.938	0.940	0.940	0.940	0.944	0.944
HAR projected	0.1	0.844	0.896	0.850	0.898	0.932	0.924	0.931	0.553	0.950
HAR projected	3	0.652	0.655	0.667	0.665	0.782	0.711	0.862	0.493	0.944
HAR projected	100	0.214	0.362	0.333	0.354	0.398	0.354	0.431	0.383	0.769
RadioML SNR	0	0.304	0.306	0.303	0.306	0.306	0.302	0.310	0.312	0.310
RadioML SNR	3	0.293	0.296	0.294	0.299	0.300	0.291	0.302	0.302	0.303
RadioML SNR	10	0.269	0.275	0.271	0.283	0.281	0.267	0.280	0.285	0.287
RadioML SNR	30	0.243	0.246	0.247	0.254	0.253	0.233	0.248	0.257	0.266

crustes need old/new features for the same anchors. Jointly trained BC-Aligner (Hu et al., 2022b) has a different fitting procedure.

At isotropic dose 3, scale-only correction reaches 0.938, against 0.813 for mean correction, 0.944 for their combination, and 0.947 for per-coordinate statistics. A scale-sensitive component survives substantial feature drift. At projected dose 3, scale-only reaches 0.667, per-coordinate statistics 0.782, and paired ridge 0.944, close to the refitted probe. At isotropic dose 100, the corresponding scores are 0.603, 0.829, and 0.915. The effective repair therefore depends on the drift. Covariance matching also leaves covariance-preserving rotations unidentified. Numerical controls recover

Table 34: Paired continuation with real reconstruction auxiliaries (five seeds per domain). All branches share the healthy encoder, primary head, Adam state, and future batches. MSE ratio compares held-out reconstruction error with the dose-matched detached decoder; values below one indicate improved reconstruction. Trace is relative to the healthy encoder. All entries are seed means.

Case	α	arm	frozen	recal.	refit	own	rel. trace	MSE ratio
HAR recon.	0	primary	0.941	0.940	0.944	0.943	1.064	–
HAR recon.	1	joint	0.941	0.940	0.945	0.946	1.073	0.886
HAR recon.	1	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	1	STF-cal	0.942	0.942	0.944	0.945	1.027	3.779
HAR recon.	10	joint	0.930	0.932	0.943	0.943	0.658	0.784
HAR recon.	10	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	10	STF-cal	0.939	0.941	0.944	0.945	0.756	3.799
HAR recon.	100	joint	0.826	0.863	0.941	0.941	0.208	0.765
HAR recon.	100	detach	0.941	0.940	0.944	0.943	1.064	1.000
HAR recon.	100	STF-cal	0.914	0.932	0.941	0.937	0.288	3.778
RadioML I/Q	0	primary	0.304	0.306	0.320	0.319	1.105	–
RadioML I/Q	30	joint	0.303	0.305	0.319	0.319	1.106	0.995
RadioML I/Q	30	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	30	STF-cal	0.303	0.305	0.319	0.319	1.106	0.995
RadioML I/Q	300	joint	0.303	0.305	0.319	0.318	1.098	0.987
RadioML I/Q	300	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	300	STF-cal	0.303	0.305	0.319	0.318	1.098	0.987
RadioML I/Q	3000	joint	0.301	0.305	0.319	0.318	1.061	1.007
RadioML I/Q	3000	detach	0.304	0.306	0.320	0.319	1.105	1.000
RadioML I/Q	3000	STF-cal	0.301	0.305	0.319	0.318	1.061	1.007

known shifts, scales, positive diagonal changes, and scaled rotations, and include a covariance-identical sign flip.

All nine arms, all doses, and the primary-only, detached, and STF-cal endpoints are retained. Records include accuracy, NLL, ECE, agreement with healthy predictions, fitting/calibration/test alignment residuals, and map hashes. Fitting residuals and evaluation residuals are reported separately. Drivers: `checkpoint_repair_audit.py` and `readout_alignment.py`; records: `repro/results/CPUCONT/mechanism/d*.json`.

H.9 Real Reconstruction Under Matched Continuation

We extend the unchanged paired-continuation driver to a real reconstruction decoder, $64 \to 256 \to d_{\text{input}}$ with a ReLU hidden layer and mean squared error. All five seeds and the fixed grids are retained: HAR doses $1, 10, 100$, with 60 healthy and 30 continuation epochs; RadioML I/Q doses $30, 300, 3000$, with 25 healthy and 15 continuation epochs. During warm-up the decoder learns on detached features. Every continuation branch starts from identical encoder, primary head, decoder, and Adam states and reuses the same future minibatch order. All detached encoder and primary-head endpoints again match primary-only continuation bitwise.

The trained decoder is evaluated on fitting, calibration, and test data at the fixed endpoint. The reported ratio averages seed-wise test-MSE ratios to the dose-matched detached decoder. STF-cal evaluation uses its forward filter and frozen gate on ordered batches of 256; raw-input decoder error is also archived. Reconstruction progress, primary-task benefit, and frozen-consumer preservation are scored separately.

At HAR dose 100, reconstruction improves while frozen accuracy is 0.826 and current-head accuracy 0.941; the primary-only current head reaches 0.943. The paired frozen difference is -0.115, with descriptive seed-level 95% interval $[-0.216, -0.014]$. Relative scatter is 0.449 against primary-only continuation, with no seed below 0.3. Per-coordinate repair reaches 0.863 against 0.941 after refitting. Reconstruction thus improves while the consumer breaks, and input repair recovers most of it; primary-task benefit is measured in the low-label study. RadioML I/Q is a near-null control: at 3000, frozen accuracy is 0.301 versus 0.304 detached and trace is 1.061 of healthy. All grid cells and paired seed intervals are retained.

Table 35: Waveform CNN reconstruction continuation on the same prepared RadioML inputs and split as the MLP study. Five seeds; 25 healthy plus 15 continuation epochs; the original heads, optimizer, cloned-state and batch controls are retained. Entries are means, with population seed SD for frozen and own accuracy; all dispersions are archived. MSE ratio compares reconstruction error with the matched detached decoder. All metrics here use the full original test corpus, separately from the monitor/remainder audit.

α	arm	frozen	recal.	refit	own	rel. trace	MSE ratio
0	primary	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	–
30	joint	0.417 ± 0.018	0.427	0.443	0.437 ± 0.015	1.060	0.918
30	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000
300	joint	0.418 ± 0.018	0.428	0.443	0.437 ± 0.015	1.053	0.851
300	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000
3000	joint	0.415 ± 0.023	0.425	0.442	0.437 ± 0.015	1.032	0.917
3000	detach	0.418 ± 0.018	0.428	0.442	0.436 ± 0.015	1.058	1.000

The prepared RadioML cache is checked read-only against the original dictionary using the data loader's per-modulation 50/50 split, NumPy `RandomState(0)`, and original dictionary iteration order. Every input, modulation label, and SNR target tensor must match exactly; the record stores source/cache and index hashes. This verifies the seeded RadioML benchmark partition. Calibration is then reserved within its training portion before encoder updates. Drivers: `reconstruction_task_audit.py` and `radioml_split_audit.py`; full records: `repro/results/CPUCONT/{mechanism,task_validation}/`.

H.10 Waveform CNN Continuation

The paired real-head studies above use small MLPs. We therefore predeclare a waveform CNN on the identical prepared RadioML I/Q inputs, with five seeds and reconstruction doses $30, 300, 3000$. The flattened vector is reshaped to 2×128. The encoder uses a padded kernel-7 convolution ($2 \to 8$ channels), ReLU and max-pooling by two; a padded kernel-5 convolution ($8 \to 16$), ReLU and pooling by two; and a linear map from the resulting 16×32 features to width 64. There is no stochastic layer or normalization. The linear primary head and $64 \to 256 \to 256$ reconstruction decoder are unchanged.

We reuse the unchanged paired training engine: 25 healthy epochs with a detached auxiliary, 15 continuation epochs, batch 256, Adam at 10^{-3}, independent calibration reservation, complete model/optimizer cloning, matched future orders, and actual-step counterfactual audits. This study tests joint training, detachment, and primary-only continuation, for seven branches per seed; it does not test STF-cal on the CNN. The source records contain both the unchanged engine hash and the wrapper/architecture hash. Healthy heads are replayed deterministically and checked against their archived accuracy, NLL, and ECE before the consumer audit.

The healthy CNN readout reaches 0.424, compared with the MLP's 0.297 on the same split. At dose 3000, frozen accuracy is 0.415 versus 0.418 detached; recalibration gives 0.425, refitting 0.442, and the current head 0.437. Relative trace is 1.032 and the reconstruction MSE ratio to detachment is 0.917. The full dose grid and a pure-scale sweep for each healthy CNN readout are recorded. The small consumer loss contrasts with the MLP result on the same split. Driver: `cnn_reconstruction_continuation.py`; records: `repro/results/CPUCONT/frontier/c*.json`.

H.11 Conditional Auxiliary Head-State Factorial

To test whether auxiliary-head state explains the difference between historical SNR training and warm continuation, we branch from five archived healthy references. Encoder and primary weights, their Adam state, the frozen consumer, and future minibatches stay identical. At dose 30, warm versus initial pre-warm auxiliary weights are crossed with retained versus reset auxiliary Adam state, followed by 15 continuation epochs. The initial/retained cell isolates the interaction. The

Table 36: Conditional SNR head-state factorial at dose 30 (five seeds). Shared encoder/primary parameters and their Adam states, the frozen consumer, and future minibatches are identical at branching. Auxiliary weights and auxiliary Adam state form the four cells. Warm/retained replays the original model/Adam endpoint bitwise. Fresh probes share a fixed initialization seed across cells. Means with population seed SD for frozen accuracy; full paired intervals and early actual-step audits are archived.

Weights	aux. Adam	frozen	recal.	refit	own	rel. trace	SNR MSE
warm	retained	0.243 ± 0.020	0.253	0.284	0.277	0.639	0.460
warm	reset	0.241 ± 0.020	0.254	0.283	0.277	0.643	0.460
initial	retained	0.244 ± 0.025	0.245	0.281	0.275	0.726	0.459
initial	reset	0.226 ± 0.033	0.227	0.283	0.277	0.888	0.460

Table 37: Independent confirmation of the validation-selected SNR auxiliary: 20% of fitting primary labels, dose 1, ten new seeds. D/J denote detachment/joint training; G denotes the reference-anchored readout-nullspace constraint. Own accuracy uses the current task head. The same healthy consumer is retained for frozen accuracy. Means with population seed SD; bounds use a separate fixed 40-consumer bank and a five-point relative-error budget. The guarded representation requires both encoder evaluations at inference.

Arm	own accuracy	frozen accuracy	frozen NLL	ECE	certified
D	0.275 ± 0.010	0.259 ± 0.016	2.176	0.153	0/10
J	0.285 ± 0.009	0.258 ± 0.019	2.135	0.155	0/10
GD	0.279 ± 0.010	0.259 ± 0.013	2.031	0.048	10/10
GJ	0.288 ± 0.009	0.259 ± 0.013	2.031	0.048	10/10

design precedes outcome computation; seed 0 is the complete implementation check and remains in the five-seed analysis. The dose is fixed throughout.

Warm/retained reproduces the original model and Adam endpoint bitwise in all five seeds. Resetting auxiliary moments changes frozen accuracy from 0.243 to 0.241. Initial/reset reaches 0.226 with trace 0.888, versus 0.639 for warm/retained. More retained variance therefore need not give higher consumer accuracy. No factorial cell reproduces the historical severe scatter collapse: auxiliary state alone leaves the protocol difference unexplained at this reference and horizon.

Dense early audits and selected later audits record 620 dependent step observations, not 620 independent trials. First-step primary-only counterfactual increments match across all four cells; changing only auxiliary moments also leaves the first encoder increment unchanged, as expected from the parameterwise optimizer. Shared-state and batch hashes, CPU checkpoint validation, bitwise endpoint replays, and seed-paired intervals are archived in `head_state_verification.json` and `head_state_summary.json`. The 20 endpoint states include five exact replays and 15 new conditional trajectories; they are outside the fixed consumer-certificate bank below. Numerical driver: `auxiliary_state_factorial.py`.

H.12 Readout-Nullspace Protection with a Useful Real Auxiliary

The old consumer applies $z_0(x) = W_0 D_0^{-1}(h_0(x) - \mu_0) + b_0$, with fixed positive diagonal $D_0 = \mathrm{diag}(s_0)$. Let $A = W_0 D_0^{-1}$ and $P_\perp = I - A^\dagger A$, the orthogonal projector onto $\ker A$. We use established projection and backward-compatibility ideas (Farajtabar et al., 2020; Shen et al., 2020) as a constructive control for measuring retained benefit and compatibility.

Proposition 10 (Compatibility and remaining feature freedom)**.** *For any updated encoder h_θ and retained deterministic reference encoder h_0, define*

$$\widetilde{h}_\theta(x) = h_0(x) + P_\perp\big(h_\theta(x) - h_0(x)\big). \tag{28}$$

The frozen affine consumer has exactly the reference logits on every input in real arithmetic. It leaves $m - \mathrm{rank}(A)$ feature directions free. It is also the unique Euclidean-nearest feature vector to $h_\theta(x)$ among all vectors with the reference logits. Conversely, every feature displacement preserving these logits belongs to $\ker A$; if A has full column rank, no nonzero compatible displacement remains.

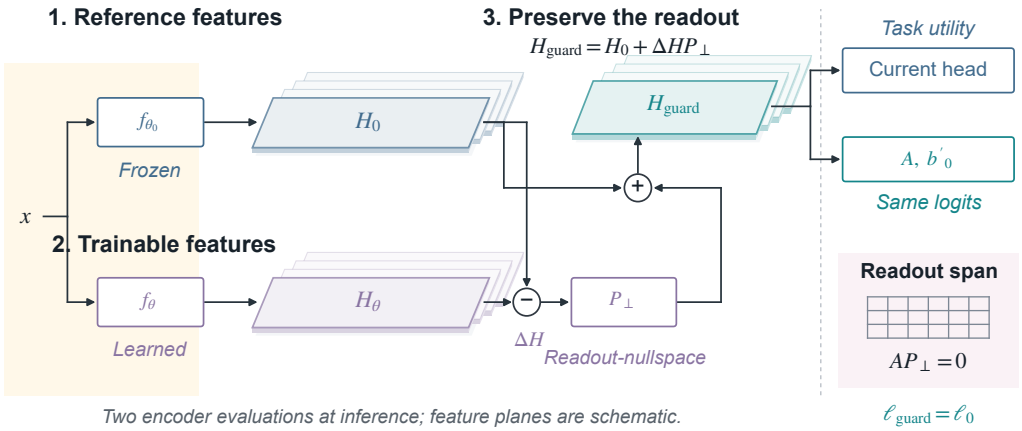


Figure 12: A fixed readout leaves a nullspace for learning. Projecting $\Delta H = H_\theta - H_0$ into $\ker A$ and adding H_0 gives one shared 64-vector with unchanged reference logits in real arithmetic. The normalized head is $A = W_0 D_0^{-1}$ and its effective bias is $b'_0 = b_0 - A\mu_0$. The inset shows the annihilation condition; planes are schematic. Both encoders are evaluated at inference.

Table 38: All predeclared primary-accuracy contrasts in percentage points, ten independent confirmation seeds. Paired two-sided 95% descriptive t intervals are conditional on the fixed dataset/split/recipe. GD controls the guarded architecture itself; GJ–GD isolates its auxiliary benefit. Noninferiority was declared at -0.5 point, not selected from the outcomes.

Contrast	mean (pp)	paired 95% interval	required lower limit
J–D	0.97	[0.63, 1.31]	0
GJ–GD	0.89	[0.60, 1.18]	0
GJ–D	1.25	[0.92, 1.58]	0
GJ–J	0.28	[0.08, 0.47]	−0.50

Proof. Since $AP_\perp = 0$, the logit difference is $AP_\perp(h_\theta - h_0) = 0$. Rank–nullity gives the free dimension and the converse. The construction is the orthogonal projection onto the affine set $h_0(x) + \ker A$, proving the minimum-distance statement. The fixed normalizer's mean and the head's bias cancel in the difference. □

We compute $P_\perp$ by float64 SVD before continuation and implement $\widetilde{H} = H_0 + (H_\theta - H_0)P_\perp$. Retaining H_0 preserves fixed post-processing as well as logits; corpus checks measure floating-point error.

Development seeds 200–202 compare HAR reconstruction at doses $1, 10, 100$ with 10% labels and RadioML SNR at $0.1, 1, 10$ with 20%. Validation is reserved before fitting. Each candidate shares the healthy state, Adam moments and minibatches. Selection maximizes the minimum validation gain among J–D, GJ–GD and GJ–D, where D/J denote detached/joint training and GD/GJ their guarded versions.

The selected SNR recipe uses dose 1, width 64, Adam 10^{-3}, batch 256 and $25 + 25$ warm-up/continuation epochs. Confirmation uses ten seeds 300–309 and all four branches. Logits and projection are float64; encoder, auxiliary parameters and Adam states are fp32. The 0.05 drift threshold is logged diagnostically at fixed epochs. All six development candidates and fixed-endpoint confirmation outcomes are archived.

All five confirmation conditions pass: positive paired lower 95% limits for J–D, GJ–GD and GJ–D; GJ–J above -0.5 points; and old-logit error below 10^{-8} with zero changed predictions. Table 38 reports the contrasts. SNR MSE is 0.766 of guarded detachment. Intervals describe seed variation on this fixed split and recipe.

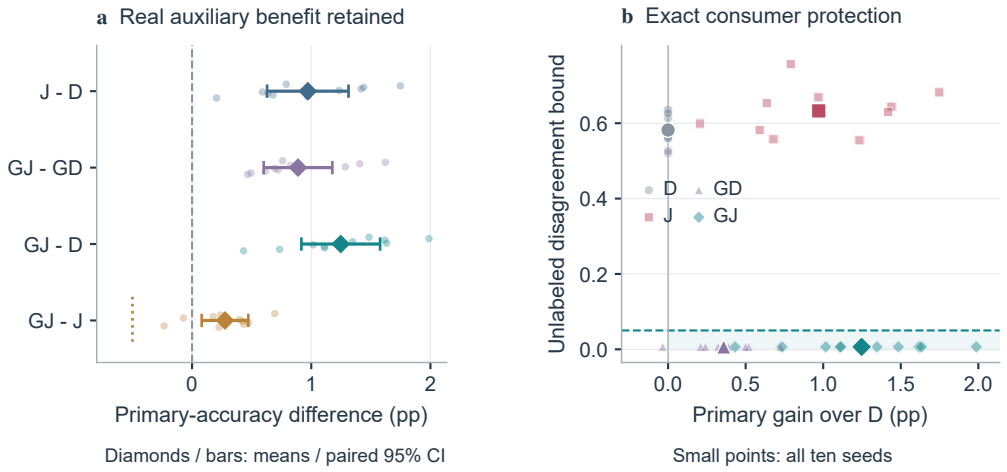


Figure 13: Independent confirmation retains auxiliary benefit and old outputs. (a) All predeclared primary-accuracy contrasts across ten new seeds, with paired 95% intervals. Zero marks the benefit threshold; -0.5 applies to the final noninferiority contrast. (b) All forty consumers: primary gain over D versus the unlabeled disagreement bound. The dashed line is the five-point relative-error budget; shading marks positive gain with a certified consumer. Ordinary consumers remain uncertified despite nearly unchanged aggregate frozen accuracy. Bounds cover the fixed corpus remainder.

The separate confirmation bank contains forty fixed consumers, total tail probability 0.05, an independent $1{,}024$-input monitor per seed, and a 0.05 relative classification-error budget. GD and GJ each certify in $10/10$ seeds with a bound near 0.0065; D and J each certify in $0/10$. Ordinary consumers change many predictions while aggregate frozen accuracy remains nearly unchanged. J's frozen ECE rises from 0.048 to 0.155; guarded predictions, NLL, and ECE match the reference, with maximum logit error 4.6×10^{-14}. This bank is distinct from the earlier $3{,}015$-candidate family.

All heads have rank 11, leaving 53 feature directions. Stacking affine heads reduces this nullspace; nonlinear consumers need a separate constraint. At batch $1{,}024$, nine post-warm-up repetitions on one Xeon Platinum 8570 thread give mean per-seed median cost $2.37\times$. The direct two-encoder baseline serves old outputs from h_0 and current outputs from J; projection additionally enforces one shared fixed-width representation. Training caches reference fitting features. Independent tensor replay checks all forty consumers; source and state hashes are in the artifact's verification records.

H.13 Folding Compatibility into a Single Encoder

Freezing the nonlinear trunk allows the compatibility constraint to fold into the final affine map. We test its learning cost under a separate fixed protocol.

Corollary 8 (Foldable affine compatibility)**.** *Let $h_0(x) = B_0\phi_0(x) + c_0$, with a retained deterministic trunk ϕ_0. For arbitrary trainable B, c, set*

$$\overline{B} = B_0 + P_\perp(B - B_0), \qquad \overline{c} = c_0 + P_\perp(c - c_0). \tag{29}$$

Then $\overline{h}(x) = \overline{B}\phi_0(x) + \overline{c}$ has the reference affine-consumer logits for every input in real arithmetic. Deploying $\overline{B}, \overline{c}$ needs one trunk and one final affine map of the original shapes, with no reference branch or projection. If the trunk has d outputs, this parameterization leaves $(m - \operatorname{rank} A)(d+1)$ affine degrees of freedom. Changing the trunk is outside this guarantee.

Proof. Both $A(\overline{B} - B_0)$ and $A(\overline{c} - c_0)$ vanish. Folding is a substitution of one weight and bias, not a second inference operation. Each of the $d+1$ weight/bias columns can vary in $\ker A$. □

This constrained affine adaptation (Farajtabar et al., 2020; Pham et al., 2026; Hu et al., 2022a) remains a linear predictor of fixed trunk features when its current head is linear.

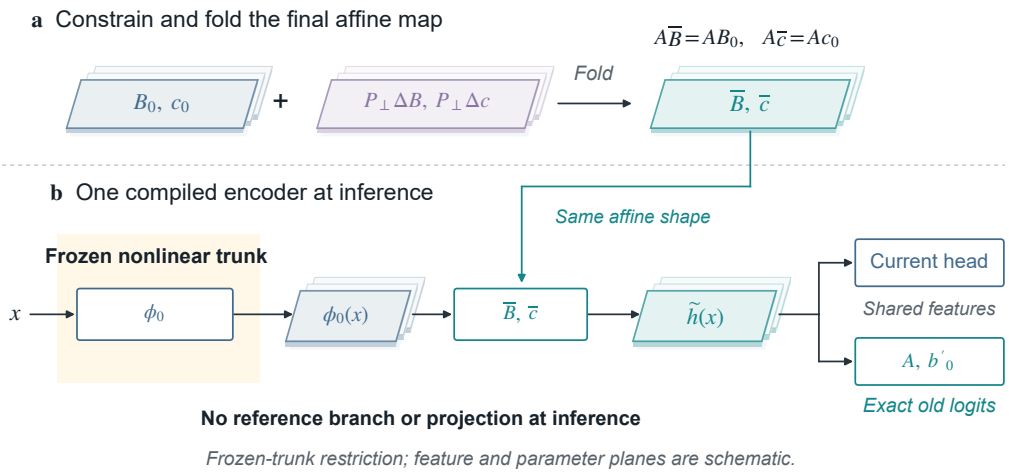


Figure 14: A frozen trunk allows the compatibility constraint to fold into one encoder. With $\Delta B = B - B_0$ and $\Delta c = c - c_0$, constrained weights and biases retain their native shapes (Equation 29); $b'_0 = b_0 - A\mu_0$ is the consumer's effective bias. Inference evaluates one encoder while preserving the reference logits. Freezing the nonlinear trunk restricts feature learning. Planes are schematic.

Table 39: Folded single-encoder continuation, ten new seeds per task. D/J update the full encoder; LD/LJ freeze the nonlinear trunk; CD/CJ additionally constrain the last affine map to preserve the normalized healthy consumer. All use the same native float64 final affine and float32 trunk. Means with population seed SD; the 120-consumer bank uses a five-point relative-error budget.

Task	arm	own accuracy	frozen accuracy	frozen NLL	ECE	certified
SNR	D	0.267 ± 0.008	0.243 ± 0.032	2.262	0.169	0/10
	J	0.278 ± 0.009	0.248 ± 0.026	2.198	0.166	0/10
	LD	0.242 ± 0.015	0.221 ± 0.025	2.261	0.139	0/10
	LJ	0.242 ± 0.015	0.219 ± 0.029	2.255	0.132	0/10
	CD	0.242 ± 0.015	0.244 ± 0.015	2.061	0.039	10/10
	CJ	0.242 ± 0.015	0.244 ± 0.015	2.061	0.039	10/10
HAR recon.	D	0.918 ± 0.008	0.913 ± 0.012	0.482	0.062	6/10
	J	0.919 ± 0.005	0.898 ± 0.019	0.540	0.069	2/10
	LD	0.918 ± 0.006	0.913 ± 0.011	0.440	0.059	6/10
	LJ	0.918 ± 0.006	0.906 ± 0.015	0.450	0.062	5/10
	CD	0.918 ± 0.006	0.918 ± 0.005	0.355	0.052	10/10
	CJ	0.918 ± 0.007	0.918 ± 0.005	0.355	0.052	10/10

RadioML uses SNR regression, dose 1 and 20% of fitting primary labels, inherited from the prior validation choice. HAR uses input reconstruction, dose 10 and 10% labels, the positive max-min candidate in the previously archived three-seed HAR validation grid. No new grid is searched. Each task uses ten new seeds 400–409, the same prepared test split, a stratified 20% training reservation before any model/head fit, width 64, Adam 10^{-3} and batch 256. HAR has 60 healthy plus 30 continuation epochs; RadioML has $25 + 25$. The last encoder affine map is natively float64 in every arm, as are classification logits; trunk and auxiliary parameters are fp32. The healthy consumer is fitted to that same mixed-precision interface. This explicit precision change prevents pooling with the earlier encoder-fp32 experiment.

Six branches clone each healthy model and full Adam state, with identical fitting-label masks and future minibatches: full-network D/J; frozen-trunk, unconstrained-last-layer LD/LJ; and frozen-trunk, constrained-last-layer CD/CJ. Heads continue learning in every branch. The constraint is fixed before continuation; the training-anchor diagnostic is recorded at fixed epochs. Validation/test inputs never train the encoder or healthy consumer. Frozen trunk parameters *and their Adam states* are checked unchanged. Every branch uses the declared endpoint and every seed is retained.

Table 40: All declared folded-control accuracy contrasts in percentage points, mean with paired descriptive 95% interval. Conditional on each fixed dataset/split/recipe. The first four rows have required lower limits $0, 0, 0, -0.5$ respectively; all other rows diagnose freezing and constraint costs regardless of sign.

Contrast	RadioML SNR	HAR reconstruction
J–D	$+1.11$ [0.90, 1.32]	$+0.04$ [-0.30, 0.37]
CJ–CD	$+0.01$ [-0.02, 0.04]	$+0.01$ [-0.08, 0.09]
CJ–D	-2.42 [-3.20, -1.64]	-0.03 [-0.43, 0.36]
CJ–J	-3.53 [-4.23, -2.83]	-0.07 [-0.47, 0.33]
LD–D	-2.45 [-3.21, -1.69]	-0.02 [-0.43, 0.39]
LJ–J	-3.53 [-4.21, -2.85]	-0.08 [-0.47, 0.30]
CJ–LJ	-0.003 [-0.039, 0.033]	$+0.010$ [-0.078, 0.099]
CD–LD	$+0.018$ [-0.025, 0.061]	-0.020 [-0.093, 0.052]
LJ–LD	$+0.028$ [0.001, 0.055]	-0.024 [-0.068, 0.021]

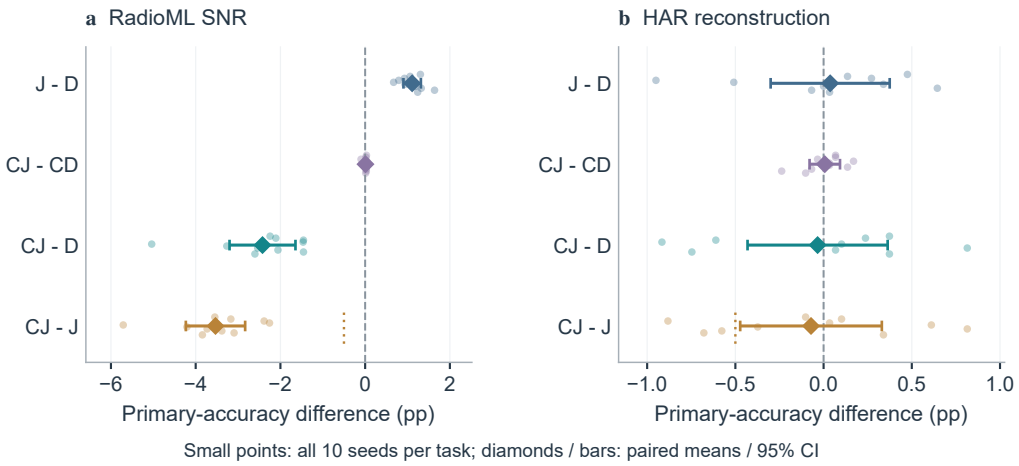


Figure 15: Folding retains compatibility but loses the SNR auxiliary gain. All four declared primary-accuracy contrasts, ten new seeds per task. Diamonds and bars show means and paired descriptive 95% intervals. D/J update the full encoder; CD/CJ freeze the trunk and constrain its last affine map. The -0.5 marker applies only to CJ–J. SNR retains an unrestricted gain; neither task passes the full five-criterion target. Panel scales differ.

The five criteria require positive lower paired 95% limits for J–D, CJ–CD, and CJ–D; CJ–J above -0.5 percentage point; and preserved old logits and predictions. Neither task passes all five. On SNR, unrestricted auxiliary gain is 1.11 points, $[0.90, 1.32]$, while compiled gain is 0.01, $[-0.02, 0.04]$. CJ–J is -3.53, $[-4.23, -2.83]$, below the noninferiority margin. The unconstrained frozen-trunk comparison LJ–J is -3.529 points, $[-4.213, -2.846]$; adding the constraint, CJ–LJ, changes accuracy by -0.003, $[-0.039, 0.033]$. Freezing accounts for the dominant measured cost in this recipe. LJ–LD retains a small 0.028-point gain, $[0.0006, 0.0554]$, far below unrestricted training.

HAR gives J–D of 0.04 points, $[-0.30, 0.37]$, and CJ–CD of 0.01, $[-0.08, 0.09]$. Both leave auxiliary benefit unresolved. CJ–J meets the half-point margin. Table 40 gives every planned contrast, including adverse and near-null outcomes. All intervals describe paired variation over ten seeds on each fixed dataset, split, and recipe.

All 40 CD/CJ consumers retain old predictions, NLL, and ECE, with maximum logit error 3.6×10^{-14}. Folded and training-time features are bitwise identical on the scored corpus. A separate fixed family of 120 consumers uses total $\delta = 0.05$, 1,024 uniformly sampled monitor inputs per seed,

and a five-point relative-error budget. All 40 compiled consumers certify on the corpus remainder; monitor information stays outside training.

Deployed encoders retain exactly three affine evaluations and the baseline parameter shapes, with no reference or projector parameters. At batch 1, mean ratios of per-seed median CPU latency are $1.00\times$ for SNR and $1.00\times$ for HAR. At batch 1,024, they are $1.01\times$ and $1.00\times$. Measurements cover the encoder and current head, using one Xeon Platinum 8570 thread, five warm-ups and 21 alternating timed repetitions per path; the baseline uses the same mixed precision. The corresponding two-encoder paths average $2.22\times$ and $2.16\times$ at batch 1,024. These deployment measurements establish the cost reduction, not useful adaptation.

State checks cover all 180 healthy, endpoint and compiled model–Adam states; independent replay covers 120 consumers. The compiled control reduces serving cost but loses the confirmed SNR auxiliary benefit.

H.14 A Compatible Nonlinear Channel: Development Gate and Limits

A narrow side network allows nonlinear learning while the base remains fixed. We combine side-tuning (Zhang et al., 2020) with established compatibility projections (Farajtabar et al., 2020; Shen et al., 2020):

$$\widetilde{h}(x) = f_0(x) + P_\perp g_\psi(x), \qquad A\widetilde{h}(x) = Af_0(x). \tag{30}$$

The healthy base f_0 stays fixed, and a two-ReLU side MLP g_ψ learns its nonlinear prefix. Both heads receive one shared 64-vector. Projection preserves affine-consumer logits; we test whether that flexibility retains useful learning.

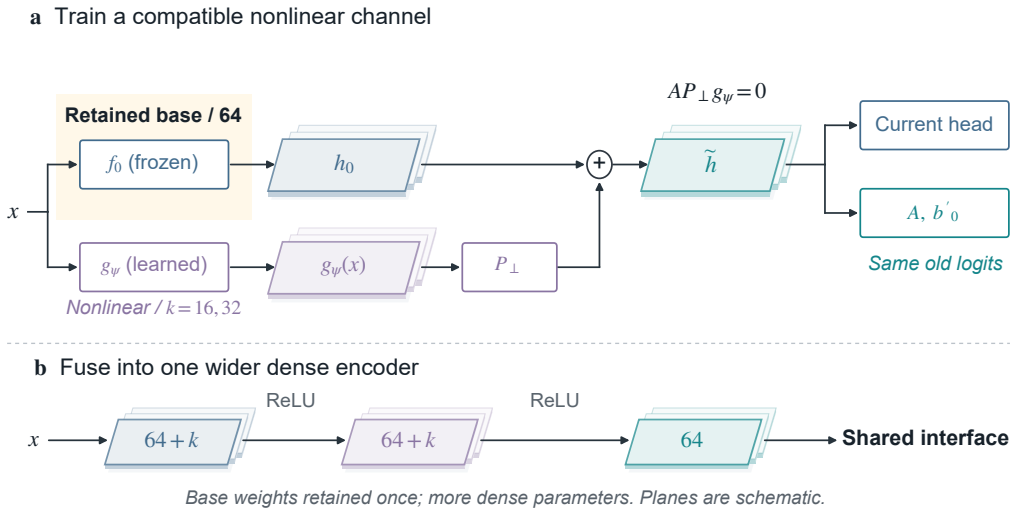


Figure 16: A compatible nonlinear channel adds trainable capacity. (a) The base stays fixed while the side learns, with $AP_\perp g_\psi = 0$. (b) Equal-depth paths fuse into a dense width-$64 + k$ MLP: the middle matrix is block diagonal and the final side block is projected. The base is embedded once, all dense zero blocks count toward storage and computation, and the output stays 64-dimensional. Planes are schematic; parameter ratios and the validation screen are reported separately.

For base weights (U_1, U_2, U_3) and side weights (V_1, V_2, V_3), the fused matrices are

$$\overline{U}_1 = \begin{bmatrix} U_1 \\ V_1 \end{bmatrix}, \quad \overline{U}_2 = \begin{bmatrix} U_2 & 0 \\ 0 & V_2 \end{bmatrix}, \quad \overline{U}_3 = \begin{bmatrix} U_3 & P_\perp V_3 \end{bmatrix}. \tag{31}$$

Stacking the first two biases and using final bias $b_3 + P_\perp c_3$ gives the same function with three affine layers and two ReLUs. The fused model embeds the base once and absorbs the separate projection. Its wider dense matrices still count all zero blocks: for side widths 16/32, encoder parameter ratios are 1.301/1.623 on SNR and 1.279/1.569 on HAR, at native float64. Full-data runtime evaluation

Table 41: Complete nonlinear-channel development grid: three seeds per task, validation only. D/J are full-network branches; TD/TJ are constrained side branches. All entries are percentage-point means. The score is $\min\{\overline{\mathrm{J}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{TD}}, \overline{\mathrm{TJ}-\mathrm{D}}, \overline{\mathrm{TJ}-\mathrm{J}}+0.5\}$. Confirmation requires strictly positive score, evaluated from integer correct counts. No candidate qualifies. HAR uses a new 1% fitting-label regime, SNR 20%.

Task	k	α	J–D	TJ–TD	TJ–D	TJ–J	score	confirm
SNR	16	1	$+0.34$	$+0.53$	-1.73	-2.07	-1.73	no
SNR	32	1	$+0.34$	$+0.32$	-0.69	-1.02	-0.69	no
HAR	16	1	$+0.00$	$+0.70$	$+0.34$	$+0.34$	0.00	no
HAR	32	1	$+0.00$	$+0.39$	$+0.25$	$+0.25$	0.00	no
HAR	16	10	$+1.50$	$+0.32$	-0.05	-1.54	-1.04	no
HAR	32	10	$+1.50$	-0.29	-0.43	-1.93	-1.43	no

was contingent on passing the development screen. The side's final weight and bias start at zero; prefix weights copy the first k healthy units and their $k \times k$ second-layer block.

The new plan was committed after the earlier studies and before numerical execution. Development uses seeds 500–502, the original width-64 base, Adam 10^{-3}, batch 256, and the same stratified 20% training reservation before model/head fitting. SNR uses 20% of fitting primary labels, dose 1, and $25 + 25$ healthy/continuation epochs. HAR uses a new 1% fitting-label reconstruction regime, doses $1, 10$, and $60 + 30$ epochs. Both test side widths $16, 32$. All feature-encoder arithmetic is float64, with fp32 head parameters and float64 classification logits. These changes prevent pooling with earlier confirmation results.

Each candidate compares full-network D/J, unconstrained side UD/UJ, and constrained side TD/TJ, with detached/joint auxiliary gradients. Healthy base/head parameters and their Adam states are retained; new side Adam states start empty. The common parameter ordering, paired side initialization, fitting-label masks and future minibatches are checked. The declared score is the minimum of three mean validation gains (J–D, TJ–TD, TJ–D) and the mean TJ–J contrast plus 0.5 point; ties prefer smaller width, then dose. A task enters ten-new-seed test confirmation only if its best score is strictly positive. No development test outcomes are computed.

The highest-scoring SNR candidate uses $k = 32$, dose 1. Its validation TJ–TD is 0.32 points, with descriptive paired 95% interval $[-0.17, 0.80]$, but TJ–D is -0.69, $[-1.93, 0.55]$, giving a negative score. The highest-scoring HAR candidate uses $k = 16$, dose 1: TJ–TD is 0.70, $[-1.76, 3.17]$, but J–D is exactly tied in aggregate correct counts, giving score zero. Neither qualifies for confirmation, and the artifact retains all six candidates.

Integer correct counts and exact rational means determine the strict-positive gate. Before confirmation, an arithmetic check corrected a floating mean near 3.7×10^{-17} for the tied HAR contrast to zero. Both serializations and the correction audit are retained; the grid and decision rule stay fixed. Because the screen fails, the planned test, auxiliary-MSE, and CPU-budget conjunction is not evaluated.

All 165 saved state files are locally checked: six healthy states, 87 endpoint model–Adam states and 72 deployed side models. A separate tensor-only process reconstructs all 87 validation consumers; accuracy, NLL and ECE replay exactly, with auxiliary errors also checked. All 36 constrained-side consumers retain their old validation predictions/NLL/ECE; maximum logit error is 2.5×10^{-14}. Fused/unfused feature error is at most 1.1×10^{-14}. The nonlinear prefix changes in every constrained branch, and frozen base weights and Adam states remain unchanged. Structural tests include arbitrary nonlinear changes, native-precision fusion, full-rank/no-nullspace and unconstrained/base-drift negative controls.

The construction retains affine-consumer outputs during nonlinear updates, but every candidate misses the joint utility/cost target. Isolating the prefix's contribution would need a matched side with its prefix frozen. A fixed-protocol vision screen did not reproduce the accuracy failure at the tested doses, although the frozen readout changed 42–56% of its decisions (Appendix H.21). Stronger vision auxiliaries and non-affine consumers are untested.

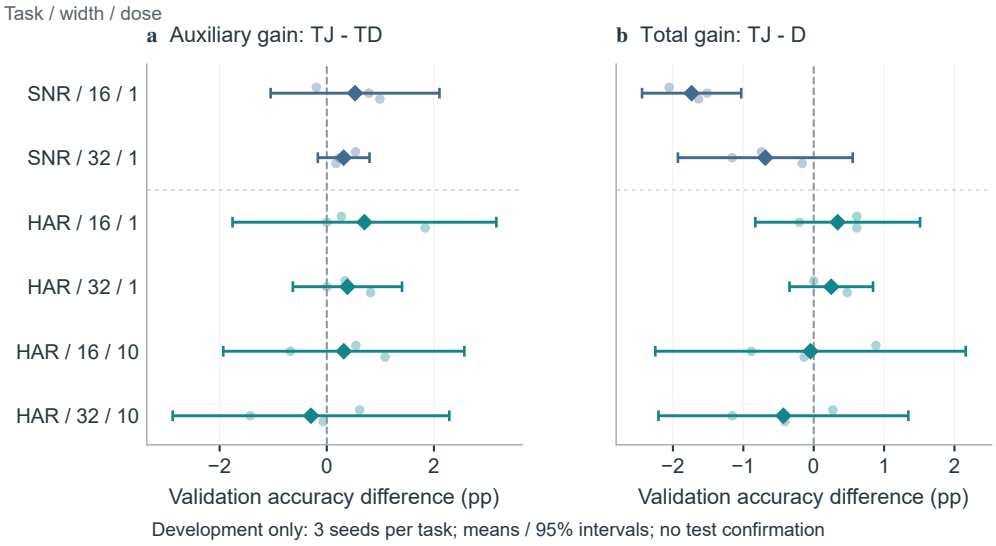


Figure 17: Neither task passes the nonlinear-channel development screen. Auxiliary and total gains for all six candidates, with three seed points and descriptive paired 95% intervals per row. Labels give task, side width, and dose. The screen additionally requires positive J–D and the TJ–J margin (Table 41). No candidate proceeds to test confirmation or full-data latency evaluation.

H.15 Post-hoc Coordinates after Nonlinear Learning

The frozen-trunk control removes reference inference but also restricts learning. Here both nonlinear layers remain trainable throughout continuation. Only after training do we fit an affine chart from new features into the old coordinates, using established linear alignment.

With row-vector features, write the current logits as $hW^\top+\beta$. An invertible chart emits $\widetilde{h} = hM+b$. Rebasing the current head by

$$W'^\top = M^{-1}W^\top, \qquad \beta' = \beta - bM^{-1}W^\top \tag{32}$$

preserves these logits in real arithmetic. Composing the chart with the encoder's final affine layer leaves its shapes unchanged. The old consumer benefits only to the extent that the chart matches its drift.

The committed protocol uses three development seeds 600–602 per task. HAR reconstructs inputs with 10% of fitting primary labels, dose 10 and $60 + 30$ epochs; RadioML predicts SNR with 20%, dose 1 and $25 + 25$ epochs. Both use width 64, Adam 10^{-3}, batch 256, fp32 nonlinear layers and a float64 final affine map. D/J clone the healthy model and complete optimizer state and share future minibatches. All nonlinear trunks change. Development evaluates validation only. Each chart uses 1,024 uniformly sampled fitting inputs, with no labels. Old anchor features are cached offline.

Two fixed candidates are checked in order: diagonal statistics, then paired ridge with standardized coefficient 10^{-3}. Singular values of the raw coordinate map are floored at 10^{-3} of the largest before rebasing. The development gate requires positive mean J–D, old accuracy within 0.5 point of healthy, NLL increase at most 0.05, ECE increase at most 0.02, unchanged current predictions with logit error below 10^{-8}, identical encoder shapes, and mean latency ratios at most $1.20/1.10$ for batches $1/1{,}024$. HAR fails the auxiliary-benefit gate. SNR diagonal repair fails ECE; paired ridge passes and alone opens confirmation.

Ten new seeds 700–709 receive the fixed SNR recipe. The execution ledger records one interrupted seed and its rerun. Current J–D is 1.16 points, paired descriptive 95% interval $[0.79, 1.54]$. Rebasing preserves every current prediction; maximum logit error is 1.44×10^{-12}. The old consumer improves from 0.251 healthy accuracy to 0.264, a difference of 1.36 points, $[0.93, 1.80]$. Mean NLL and ECE

Table 42: Post-hoc single-encoder chart, ten independent SNR confirmation seeds. Means on the full fixed test corpus. Current logits are algebraically preserved by head rebasing. Old-reader metrics can improve despite extensive prediction changes. The five-point disagreement budget certifies none of the ten repaired joint consumers.

Consumer	accuracy	NLL	ECE	changed predictions
Healthy old	0.251	2.044	0.043	0
Native J, old readout	0.260	2.168	0.166	56.69%
Chart J, old readout	0.264	1.984	0.019	44.07%
Native J, current head	0.281	1.980	0.053	—
Chart J, rebased current	0.281	1.980	0.053	0 (vs. native J)
Supervised refit	0.285	1.993	0.070	—

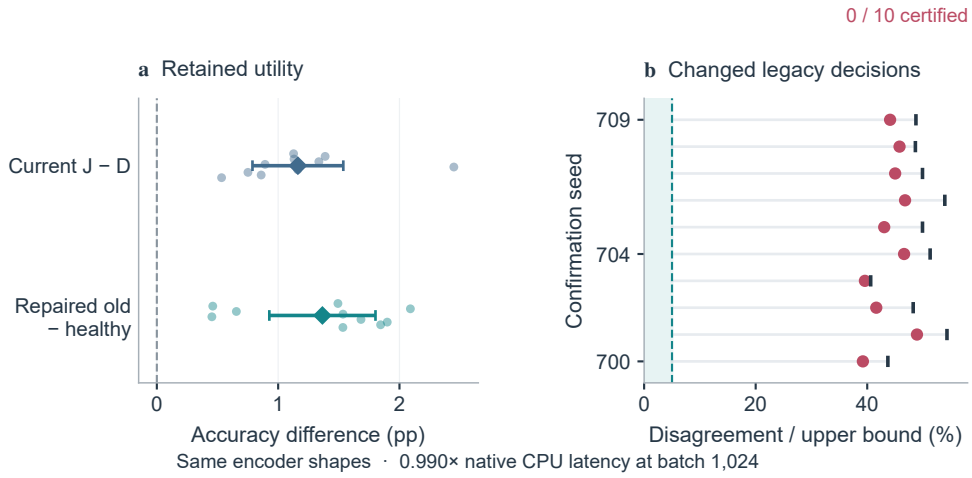


Figure 18: A folded chart retains utility at native encoder cost but changes many legacy decisions. (a) All ten seed differences and paired 95% intervals. (b) Repaired-J disagreement with healthy predictions and simultaneous finite-corpus upper bounds. Every bound exceeds the five-point budget. Healthy aggregate accuracy is 0.251; a higher aggregate score does not imply decision compatibility.

decrease by 0.0604 and 0.0242. Native J reaches 0.281 current accuracy; the supervised refit reaches 0.285. These are low-label results on one fixed modulation split.

The repaired consumers still change 44.07% of healthy predictions on average, missing the decision target. A separate forty-consumer family uses total $\delta = 0.05$, 1,024 uniform monitor inputs and a 0.05 error budget. None of the ten repaired joint consumers certifies; their upper remainder disagreement bounds range from 0.406 to 0.543. The monitor therefore abstains despite the positive paired accuracy differences.

On one CPU thread, alternating 31 timed repetitions after five warm-ups give mean median-latency ratios 0.900 at batch 1 and 0.990 at batch 1,024. Both branches evaluate final-affine features and primary logits in float64. The rebased head stores float64 parameters; the native fp32 head is cast for logit evaluation. The result is near-native encoder cost. Auxiliary heads are not included in these primary inference timings. The chart and current heads are composed offline; inference does not require the reference encoder or a separate projection. Offline fitting requires anchor feature extraction and matrix estimation.

All native model/Adam states, fitted charts, deployed models, source hashes and timing samples are retained locally. Drivers are `posthoc_chart_cpu.py` and its frozen confirmation wrapper; analysis is `summarize_posthoc_chart.py`.

Table 43: What each compatibility arm buys and costs, all on the same ten SNR confirmation seeds, splits and minibatch order. Gain is primary accuracy against the detached control, with paired descriptive 95% intervals; *changed* is the fraction of the healthy consumer's predictions that differ on the fixed test corpus; *certified* counts seeds whose exact finite-population bound is at most 0.05. Only the reference-anchored control certifies every seed, and it needs the reference encoder at inference. The decision-preserving arms share one encoder and still miss the budget; pinning the consumer's logits rather than its margin trades primary accuracy for lower disagreement. The post-hoc chart reaches 1.16 points at 44.07% on its own ten confirmation seeds.

Arm	gain (pp)	changed	certified	encoders
Joint training	0.97 [0.63, 1.31]	58.9%	0/10	1
Decision, $\gamma = 1$	0.72 [0.31, 1.12]	29.2%	0/10	1
Decision, $\gamma = 3$	-0.04 [-0.46, 0.38]	17.8%	0/10	1
Logit anchor, $\gamma = 10$	-1.59 [-2.18, -0.99]	5.0%	0/10	1
Reference-anchored	0.89 [0.60, 1.18]	0.0%	10/10	2

H.16 A Decision-Preserving Training Constraint

Each arm above fails one axis. A training-time constraint asks whether all three are reachable together. Continuation minimises $L_{\text{primary}} + \alpha L_{\text{aux}} + \gamma R$ with

$$R = \frac{1}{n} \sum_i \max\Big(0,\ \max_{j \neq p_{0i}} z_i[j] - z_i[p_{0i}]\Big), \tag{33}$$

where z_i is the frozen consumer's logit vector on the current features and p_{0i} its prediction on the healthy features. R vanishes at initialisation, so it penalises decisions that would flip, not drift. The healthy encoder and readout enter fitting only; inference is the single shared encoder.

The protocol is the ten-seed benefit confirmation unchanged: RadioML, 20% primary labels, width 64, Adam 10^{-3}, batch 256, dose 1, 25 warm-up and 25 continuation epochs, the same splits, minibatch order, healthy readout and monitor partition. On development seeds 800–802 the penalty trades disagreement for gain smoothly: at $\gamma = 0.1, 0.3, 1.0, 3.0$ the primary gain is $1.35, 1.26, 1.11, 0.08$ points and the validation disagreement is $38.4, 33.4, 24.3, 14.0\%$, against 1.06 points and 53.3% for joint training. The fixed rule selected the smallest γ whose development disagreement is at most 5%; none reaches it, so it selected $\gamma = 3$.

Ten confirmation seeds give 0.72 points $[0.31, 1.12]$ at $\gamma = 1$ with 29.2% disagreement and -0.04 $[-0.46, 0.38]$ at $\gamma = 3$ with 17.8%; both use one encoder at inference. Neither is certified: no seed meets the five-point budget and the largest exact upper bound is 0.26 at $\gamma = 3$. The frozen consumer's accuracy is 0.262 at $\gamma = 3$, above the healthy 0.258, while 17.8% of its decisions differ. The $\gamma = 1$ row is a post-hoc knee confirmation run after the fixed rule selected $\gamma = 3$ and the primary arm failed; it is reported separately and does not change the primary decision.

With the same inference graph, the constraint trades 0.25 of the 0.97-point gain for roughly half the decision changes, and still misses the budget. A second corpus tests whether the frontier belongs to the method or to the shift. An addendum, fixed before any HAR outcome, reruns the same grid, rule and seeds on HAR with its reconstruction auxiliary at dose 10, the highest-scoring HAR candidate in the earlier benefit development, where the auxiliary's primary gain is near zero. Every development value already meets the budget: disagreement is $1.6, 1.2, 1.1, 2.5\%$ at $\gamma = 0.1, 0.3, 1.0, 3.0$, against 2.6% for joint training, so the rule selected the smallest, $\gamma = 0.1$. Ten confirmation seeds at that value give $+0.22$ points $[-0.15, 0.60]$ over the detached control and $+0.31$ $[-0.68, 1.29]$ over joint training, with reconstruction error at 0.78 of the control against 0.81 for joint training; the gain axis is flat in both comparisons. Mean disagreement is 2.3%, between 1.5 and 3.1% per seed, and eight of ten seeds certify. The two remaining seeds have 2.7 and 3.0% point disagreement with exact upper bounds of 5.4 and 5.6%, so they abstain rather than contradict the budget. Unconstrained joint training on the same seeds changes 5.3% of legacy decisions, misses the budget on three seeds, certifies one, and loses 1.1 points of frozen accuracy, including a 5.6-point loss on one seed that the constrained arm avoids (Table 44). The budget is reachable where the shift is small and out of reach where it is large. Exact compatibility costs a second encoder at inference, approximate compatibility costs gain, and the five-point budget sits beyond the strength at which the gain vanishes.

Table 44: The same decision-preserving constraint on the second corpus: HAR with the reconstruction auxiliary at dose 10, ten confirmation seeds, and the same splits and minibatch order. Gain is primary accuracy against the detached control with paired descriptive 95% intervals; reconstruction is held-out MSE as a ratio to that control; *changed* and *certified* are as in Table 43. The fixed rule selected $\gamma = 0.1$ because every development value already sat below the budget. The constraint certifies more seeds than the detached control it is measured against, joint training is the only arm whose mean changes more than five percent, and the gain is flat.

Arm	gain (pp)	reconstruction	changed	certified	encoders
Detached control	—	1.00	2.2%	6/10	1
Joint training	−0.08 [−0.86, 0.69]	0.81	5.3%	1/10	1
Decision, $\gamma = 0.1$	0.22 [−0.15, 0.60]	0.78	2.3%	8/10	1

A second constraint tests whether the frontier belongs to the hinge. The anchor replaces R by $\frac{1}{n}\sum_i \|z_i - z_{0i}\|^2$, the squared distance between the frozen consumer's logits on the current features and on the healthy features of the same example, so it penalises every logit change rather than only changes that would flip a decision. A separate recorded plan fixed the grid $\gamma \in \{0.1, 1, 10, 100\}$ and reused the rule, splits, order and seeds. Development gives $+1.18, -0.16, -0.83, -1.10$ points with disagreement $29.4, 11.8, 4.6, 2.8\%$; the rule selects $\gamma = 10$. Ten confirmation seeds give -1.59 points $[-2.18, -0.99]$ against the detached control with no seed positive, and -2.56 against joint training. Mean disagreement is 5.0%, at most 6.4% per seed, and no seed certifies because the exact upper bounds reach 0.10 at that rate. Reconstruction error is 0.94 of the control against 0.77 for joint training, so the anchor also gives up most of the auxiliary benefit. In these runs, stronger logit constraints reduce primary and auxiliary progress together.

Exact preservation with fixed weights requires $A(h_\theta(x) - h_0(x)) = 0$ pointwise. This constrains feature functions; its runtime and utility depend on the implementation. The tested penalties lose SNR utility, while the reference-anchored construction retains it with a second encoder at $2.37\times$ cost. These outcomes describe the tested controls rather than a necessary cost law. Decision compatibility has a different tolerance: nonzero logit changes can leave every prediction unchanged. A five-point disagreement budget caps the fraction of examples that can newly become wrong relative to the old consumer. The reported curves allow another tolerance, and the learned-head studies below test a different parameterization of the legacy interface.

H.17 Learning a Legacy Readout

The constraints above keep the old weights on the new feature. A weaker requirement asks whether only the old *function* must survive. Let p be a linear head with n_c outputs, initialised to the healthy consumer map (A, B) and trained with the encoder on

$$\frac{1}{n}\sum_i \left\| p(h_\theta(x_i)) - z_0(x_i) \right\|^2, \tag{34}$$

where $z_0(x) = Ah_0(x) + B$ is the frozen consumer's logit vector, precomputed for the fitting set before continuation. Deployment applies p to $h_\theta(x)$: one encoder, one forward pass, $n_c m + n_c$ extra parameters (390 on HAR, 715 on SNR), and no reference forward. This replaces the legacy weights and aims to retain their predictions; it does not give an exact logit identity. Logit distillation (Hinton et al., 2015) and Learning without Forgetting (Li & Hoiem, 2016) provide the methodological precedents. The loss updates both the head and encoder, so this arm tests their joint ability to retain legacy predictions. Protocol, splits, minibatch order and monitor are those of the benefit confirmation; $\lambda \in \{0.1, 1, 10\}$ was selected on seeds 800–802 by the same rule that ignores accuracy and gain, and confirmation uses seeds 300–309.

On SNR the selected $\lambda = 10$ gives -1.29 points $[-1.78, -0.80]$ against detachment, 5.0% disagreement and no certificate, with SNR-regression MSE at 0.891 of the control against 0.763 for joint training. The learned head halves the change of the fixed weights on the same features, from 11.8% to 5.0%, under the selected setting.

HAR reconstruction is more favorable under the selected $\lambda = 0.1$. The arm gives $+0.44$ points $[-0.03, 0.91]$ on the primary task, reconstruction error 0.767 of detachment, close to joint training,

Table 45: Protected-head continuation on the two corpora. The arm trains a linear head, initialised to the healthy consumer map, to reproduce the frozen consumer's logits during continuation; inference is one encoder and one forward pass. *primary* is the absolute accuracy for the detached control and the change against it in points for the other arms; ***MSE*** is test auxiliary MSE relative to detachment (reconstruction on HAR, SNR regression on RadioML); *changed* is the fraction of healthy consumer predictions that differ; *certified* counts seeds whose exact finite-population bound is at most 0.05. HAR uses the dose-100 reconstruction auxiliary with the selected $\lambda = 0.1$; SNR uses the benefit protocol with the selected $\lambda = 10$. Each row aggregates ten confirmation seeds. All measured HAR disagreements are below budget, but one monitor abstains; neither corpus passes its complete specified conjunction.

Corpus	Arm	primary	MSE	changed	certified	forwards
HAR	Detach	0.912	1.000	2.2%	—	1
HAR	Joint	+0.20	0.767	21.8%	0/10	1
HAR	Protected, $\lambda = 0.1$	+0.44	0.767	1.8%	9/10	1
SNR	Detach	0.259	1.000	54.2%	—	1
SNR	Joint	+0.95	0.763	59.4%	0/10	1
SNR	Protected, $\lambda = 10$	−1.29	0.891	5.0%	0/10	1

and 1.8% disagreement, at most 2.2% per seed (Table 45). Nine of ten seeds meet the specified certificate criterion; the remaining seed has an observed disagreement of 1.8% and an exact bound of 5.1%, and abstains. Applying the original weights to the same features changes 3.3% of decisions, so the head absorbs roughly half the prediction changes on these same features. The primary-gain interval includes zero, and maximum healthy-logit error is 4.05, so neither a resolved primary gain nor exact function retention follows. Absolute accuracies differ from the full-label reconstruction study of Section 5 because this run keeps the 10% label protocol; all comparisons are within run.

The full specified conjunction requires every seed to certify. HAR therefore misses it by one conservative abstention, despite all measured full-corpus disagreements being below budget; SNR also fails the utility criteria. The cross-corpus difference does not identify information loss: teacher regularization, linear approximation and fitting can each affect the result. Appendix H.18 separates readout fitting on frozen features from jointly constrained learning.

H.18 Separating Readout Fitting from Encoder Regularization

The learned-head arm changes both the reader and its features. To separate these effects, we fixed a retrospective control before computing any new fits. It uses every saved confirmation encoder from Appendix H.17: detached (D), unconstrained joint (J), and jointly protected (P), with ten seeds per corpus. These retrospective fits use known outcomes; the original HAR $9/10$ and SNR $0/10$ confirmation decisions remain unchanged. Seeds 300–309 are held out from this experiment's development seeds, but were used in earlier studies. The released plans and hashes record the protocols; their combined plan/result commits do not establish an independent registration timestamp.

For each saved encoder, we fit an affine reader to the healthy teacher logits on all fitting inputs, without labels. We center the targets, standardize the features using fitting-set statistics, and solve either ordinary least squares (OLS) or ridge regression with coefficient 10^{-3} and an unpenalized intercept. OLS uses float64 SVD and tolerance $\max(n, m)\,\epsilon\, s_{\max}$; feature standard deviations are floored at 10^{-6}. Both fits were fixed in the control plan, with no validation-based choice. The encoder, current head and auxiliary head remain unchanged.

All six fitted readers and the original online head share a fresh uniform $1{,}024$-input monitor per endpoint. The family contains 140 consumers across the two corpora and ten seeds, with total $\delta = 0.05$ and the same five-point budget. Table 46 reports all fits, including the detached controls; Figure 19 compares the joint and protected features. The new bounds apply to this fixed family of saved candidates.

On HAR, fitting OLS to the protected encoder reduces mean disagreement from 1.77% to 1.44% and certifies all ten saved seeds. Reconstruction MSE and current-head accuracy are unchanged by construction. Applying the identical fit to the unconstrained joint encoder leaves 3.35% disagree-

Table 46: Fixed retrospective teacher-head fits on saved encoders. All fits use training features and healthy logits without labels. D/J/P denote detached, unconstrained joint, and jointly protected encoders. Certificates use the same fresh monitor and 140-consumer family for every row; original 9/10 and 0/10 decisions remain unchanged. Means over ten saved seeds per corpus.

Corpus	donor / legacy reader	accuracy	changed	certified
HAR	D / ols	0.911	0.97%	10/10
HAR	D / ridge	0.911	1.15%	10/10
HAR	J / ols	0.915	3.35%	1/10
HAR	J / ridge	0.916	3.56%	0/10
HAR	P / ols	0.914	1.44%	10/10
HAR	P / ridge	0.914	1.50%	9/10
HAR	P / online	0.915	1.77%	9/10
SNR	D / ols	0.269	38.58%	0/10
SNR	D / ridge	0.273	39.80%	0/10
SNR	J / ols	0.272	41.49%	0/10
SNR	J / ridge	0.275	42.61%	0/10
SNR	P / ols	0.259	4.04%	0/10
SNR	P / ridge	0.259	5.36%	0/10
SNR	P / online	0.259	5.00%	0/10

ment and certifies one seed; ridge certifies none. Thus a head-only fit on unrestricted features does not recover the result of joint distillation. Regularizing the encoder contributes to this comparison. OLS and ridge both recover the detached control in all ten seeds, providing a check that the fitting procedure can recover an accessible legacy function. On SNR, protected-feature OLS lowers mean disagreement to 4.04%, but no seed certifies, and the original primary accuracy loss remains. A richer reader or a different training recipe remains an open alternative.

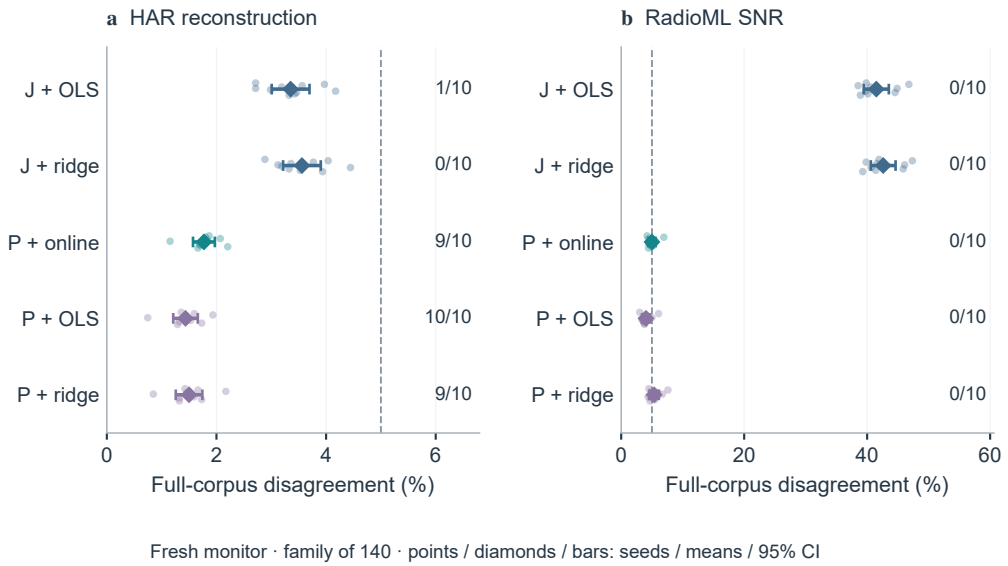


Figure 19: A reader fit and encoder regularization have different effects. J and P are the unconstrained joint and jointly protected encoders; online denotes the head trained during continuation. Points show all saved seeds, diamonds their means, and bars 95% intervals across seeds. Counts are certificates from the new 140-consumer audit, while the horizontal coordinate is full-corpus disagreement. The dashed line marks the 5% budget; a mean below it does not imply a certificate. All controls were specified after the original outcomes were known.

Independent numerical replay recovers all 60 original legacy-consumer metric sets exactly. A second process reloads the 120 new affine readers and 20 online heads without refitting, reproduces their metrics and bounds exactly, and checks the convex fits' stationarity conditions; the largest residual is 8.50×10^{-10}. Tensor states are retained locally; the supplementary archive contains the numerical replay records and drivers.

The original timing closures used identical current-head operations for the ordinary and protected paths, so they did not measure serving both outputs. An isolated corrected benchmark returns current and legacy logits from a single encoder call, using four CPU threads, five warm-ups and 51 alternating repetitions per path. At batch $1{,}024$, the online protected head costs $1.054\times$ current-only inference on HAR and $1.076\times$ on SNR. The float64 OLS head costs $1.181\times$ and $1.243\times$, respectively. At batch one, the corresponding ratios are $1.100/1.143$ on HAR and $1.103/1.156$ on SNR (online/OLS). Ratios average per-seed medians. Encoder features and online heads use float32; the current-head diagnostic evaluates logits in float64, and the OLS reader is stored in float64. These timings include both requested outputs and account for their numerical implementations.

H.19 Joint Alignment Baselines and Fresh Confirmation

We compare BC-Aligner-style feature alignment (Hu et al., 2022b) with logit alignment and direct-head distillation, then confirm the selected HAR candidates on fresh seeds.

Development reuses the healthy checkpoints, Adam moments, splits and future minibatches of seeds 800–802. Both new families learn a bias-free 64×64 map M, initialized to identity. Feature alignment minimizes $\|Mh - h_0\|^2/64$ per example; logit alignment minimizes $\|AMh + B - z_0\|^2/n_c$. The former is the single-update linear objective of BC-Aligner with an explicit width normalization, so its coefficient in the per-example squared-norm convention is $\lambda/64$. There is no historical composition in this experiment. The two maps share $4{,}096$ trainable parameters; serving composes AM into one legacy head. The direct head is the original logit-distillation arm, with 390 parameters on HAR.

Feature coefficients are $\{0.01, 0.1, 1, 10, 100\}$ and logit coefficients are $\{0.1, 1, 10\}$. Each family selects the smallest coefficient whose mean validation disagreement is at most 5%, or the largest if none qualifies. Utility is checked only after this choice. A third candidate applies the fixed training-only OLS fit to the original protected features and casts the reader to float32. Table 47 includes every candidate.

On HAR, feature alignment selects 0.01 and loses 0.63 primary points, missing the half-point utility gate. Stronger feature penalties have viable validation tradeoffs: at $\lambda = 1$, primary changes by -0.09 points, MSE is 0.715 of detachment, and disagreement is 1.29%. Thus the compatibility-only selection policy misses a viable feature-alignment tradeoff. Logit-map 0.1 and the direct-head OLS fit pass and open HAR confirmation. On SNR, feature-map 100 still changes 16.5% of decisions; logit-map 10 and direct-head OLS lose 0.62 and 0.65 primary points. None opens confirmation, and no coefficient is retuned.

We train ten new HAR seeds, 1300–1309, on the same official train/test benchmark, with new healthy checkpoints and the fixed low-label reconstruction recipe. Detached, joint and direct-head branches are paired with the selected logit map; OLS changes only the final reader on the direct-head features. The bank allocates $\delta = 0.05/160$ per consumer, covering up to two corpora, ten seeds, four protected readers and two monitor sizes. Unused allocations are retained. The primary monitor has $1{,}024$ uniform inputs. A nested $2{,}048$-input monitor reports additional observation cost and cannot replace a failed primary decision.

The direct head meets the complete confirmation criterion. Reconstruction MSE is 0.780 of detachment, with paired 95% interval $[0.756, 0.804]$, and primary accuracy differs by $+0.30$ points $[-0.01, 0.61]$. This interval clears the -0.5-point noninferiority margin. Mean legacy disagreement is 1.67%, and all ten primary-monitor bounds meet the five-point budget. OLS reduces disagreement to 1.46% with the same primary and reconstruction outcomes and ten certificates. The logit map has 1.64% mean disagreement but certifies nine seeds. Its ten secondary-monitor certificates leave that primary failure unchanged.

An isolated serving audit evaluates both heads in float32, using four CPU threads, five warm-ups and 51 alternating repetitions. At batch $1{,}024$, direct and OLS ratios are $1.064\times$ and $1.071\times$ current-

Table 47: Complete fixed development grid, three seeds per corpus. Primary differences are percentage points relative to detachment; MSE is auxiliary error relative to detachment. Means use validation only. A dagger marks the coefficient selected by mean disagreement before utility is checked; all other coefficients remain visible.

Corpus	target / reader	λ	primary	MSE / D	changed
HAR	Feature map	$0.01^{\dagger}$	-0.63	0.728	3.02%
HAR	Feature map	0.1	-0.45	0.733	2.38%
HAR	Feature map	1	-0.09	0.715	1.29%
HAR	Feature map	10	-0.16	0.798	0.79%
HAR	Feature map	100	-0.05	0.941	0.11%
HAR	Logit map	$0.1^{\dagger}$	-0.18	0.701	0.93%
HAR	Logit map	1	$+0.02$	0.783	0.45%
HAR	Logit map	10	$+0.09$	0.942	0.20%
HAR	Direct + OLS-fp32	$0.1^{\dagger}$	-0.09	0.703	0.95%
SNR	Feature map	0.01	$+1.30$	0.783	43.17%
SNR	Feature map	0.1	$+1.16$	0.765	43.72%
SNR	Feature map	1	$+1.28$	0.785	41.30%
SNR	Feature map	10	$+1.24$	0.787	31.41%
SNR	Feature map	$100^{\dagger}$	$+0.06$	0.807	16.48%
SNR	Logit map	0.1	$+1.30$	0.768	34.43%
SNR	Logit map	1	$+0.09$	0.814	15.53%
SNR	Logit map	$10^{\dagger}$	-0.62	0.880	4.98%
SNR	Direct + OLS-fp32	$10^{\dagger}$	-0.65	0.880	3.63%

Table 48: Fresh HAR confirmation, seeds 1300–1309. Primary values are absolute accuracy for detachment and paired percentage-point differences for other rows; brackets are 95% intervals. MSE is reconstruction error relative to detachment. Certified counts use the primary $1{,}024$-input monitor and fixed 160-consumer tail allocation. Native float32 CPU ratios return both current and legacy outputs at batch $1{,}024$. OLS changes only the reader.

Arm	primary	MSE / D	changed	certified	CPU
Detach	0.909	1.000	3.71%	—	—
Joint	$+0.02\ [-0.38, +0.43]$	0.767	27.66%	—	—
Direct	$+0.30\ [-0.01, +0.61]$	0.780	1.67%	$10/10$	$1.064\times$
Logit map	$+0.42\ [+0.03, +0.80]$	0.777	1.64%	$9/10$	$1.067\times$
OLS-fp32	$+0.30\ [-0.01, +0.61]$	0.780	1.46%	$10/10$	$1.071\times$

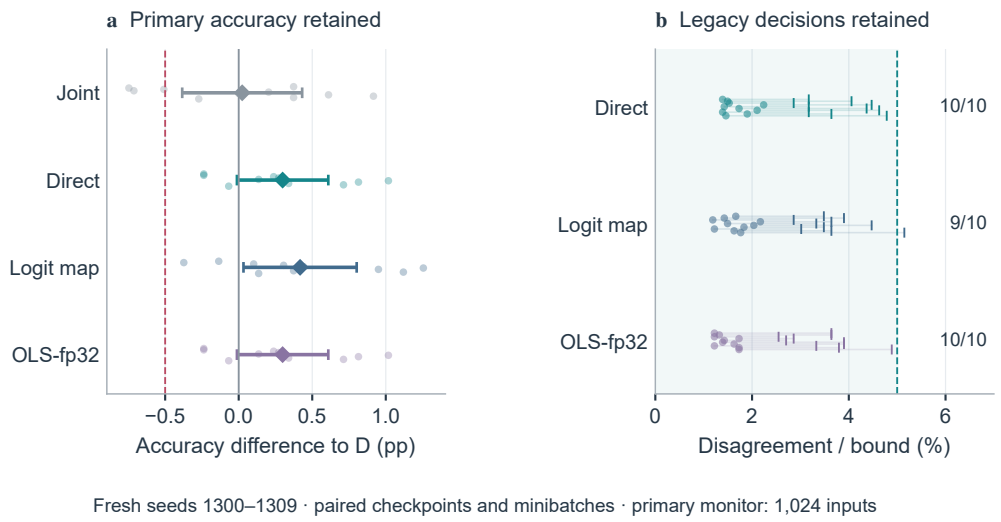


Figure 20: Fresh HAR confirmation separates aggregate utility from individual consumer acceptance. (a) All ten paired primary differences, their means and 95% intervals; the dashed line is the -0.5-point margin. (b) Each seed's observed legacy disagreement (dot) and finite-population upper bound (tick), with certificate counts at the fixed 5% budget. Direct and OLS-fp32 share the same encoder, current head and auxiliary head. These are new training seeds; the earlier $9/10$ HAR study and retrospective audit remain separate.

only inference; at batch one they are $1.121\times$ and $1.132\times$. Both outputs use one encoder call. The additional reader occupies 390 float32 parameters (1,560 bytes). Both arms also meet the original diagnostic timing limits, whose current head computes logits in float64. Native primary predictions match that diagnostic across all 50 saved branch endpoints.

Tensor replay reproduces all 50 metric sets and bounds. Casting OLS to float32 changes 11 of 29,470 seed–example predictions, with maximum logit difference 0.182; certificates use the actual float32 reader.

The fresh HAR confirmation therefore retains a bounded fraction of legacy decisions under the actual float32 implementation.

H.20 Matched Strong Controls and Versioned Consumer Contracts

This retrospective completion uses all ten saved fresh-confirmation checkpoints and a fixed plan published before the additional computations. It leaves the original development choice, confirmation bank and verdicts unchanged. From the earlier development grid, we choose the feature-map coefficient with the lowest reconstruction ratio among candidates with mean primary difference within 0.5 points of detachment, mean reconstruction ratio at most 0.95, and disagreement at most 5% on every development seed. This utility-aware rule selects $\lambda = 1$; the earlier compatibility-only rule selected 0.01 and failed its primary gate. Both decisions remain reported. The new branch starts from each saved healthy model and Adam state and uses the same 30 epochs, reconstruction dose 100 and future minibatches. Both nonlinear encoder layers update. The full-feature map is folded into the float32 legacy reader for serving.

We also fit the same standardized OLS reader on detached, joint, distilled and feature-aligned endpoints. Every fit uses fitting-set features and healthy teacher logits, without labels, test inputs or a coefficient search. The previously saved OLS reader and its identical new fit are counted once in comparisons: 100 recorded endpoints contain 90 distinct model/reader pairs. Learned and OLS readers use their actual float32 serving weights; raw detached and joint readers retain the original float64 diagnostic interface.

Table 49: Matched HAR completion on the ten saved fresh-confirmation seeds. This retrospective comparison uses development-selected full-feature alignment and training-only OLS; no test fitting or coefficient search. Primary values are paired percentage-point differences to detachment. Each changed/certified cell reports mean disagreement and the number of new $1{,}024$-monitor bounds below 5%, with tail $0.05/1{,}000$. Selective consumers use fixed healthy-training confidence thresholds targeting 80% or 50% coverage. These new bounds do not replace the earlier confirmation.

Arm	primary	MSE / D	class	selective 80	selective 50
Detach	$+0.00$ $[+0.00, +0.00]$	1.000	3.71%; $3/10$	9.64%; $0/10$	17.64%; $0/10$
Detach + OLS	$+0.00$ $[+0.00, +0.00]$	1.000	0.93%; $10/10$	2.20%; $4/10$	2.67%; $3/10$
Joint	$+0.02$ $[-0.38, +0.43]$	0.767	27.66%; $0/10$	52.71%; $0/10$	38.72%; $0/10$
Joint + OLS	$+0.02$ $[-0.38, +0.43]$	0.767	3.22%; $0/10$	8.31%; $0/10$	9.85%; $0/10$
Distilled head	$+0.30$ $[-0.01, +0.61]$	0.780	1.67%; $8/10$	4.71%; $0/10$	6.50%; $0/10$
Distilled + OLS	$+0.30$ $[-0.01, +0.61]$	0.780	1.46%; $9/10$	3.52%; $0/10$	4.82%; $0/10$
Feature align. ($\lambda = 1$)	$+0.56$ $[+0.24, +0.88]$	0.771	1.92%; $8/10$	5.42%; $0/10$	7.23%; $0/10$
Feature align. + OLS	$+0.56$ $[+0.24, +0.88]$	0.771	1.74%; $8/10$	4.15%; $0/10$	5.37%; $0/10$
Logit map	$+0.42$ $[+0.03, +0.80]$	0.777	1.64%; $9/10$	4.59%; $0/10$	6.43%; $0/10$

The stronger feature-alignment baseline retains reconstruction progress (MSE 0.771 of detachment) with primary difference $+0.56$ points $[0.24, 0.88]$. Full-feature alignment is therefore competitive with the distilled head. Unconstrained joint training followed by OLS reaches 3.22% mean class disagreement, versus 1.67% for joint distillation. These intervals describe a retrospective matched comparison on the saved benchmark.

Decision disagreement counts several kinds of change. With labels available, we report negative flips (healthy-correct to updated-wrong), positive flips, and changes between two wrong classes (Yan et al., 2021). Their rates sum to total disagreement, while the change in legacy accuracy equals positive minus negative flips. Joint training's 27.66% class disagreement contains 22.97% negative flips; OLS lowers those to 1.27%. Joint distillation has 0.58% negative flips and 0.91% positive flips. Total agreement is a conservative compatibility contract, rather than a measure of harmful changes alone.

For a probability-dependent consumer, define $q_\tau(x) = \arg\max_j p_j(x)$ when $\max_j p_j(x) \geq \tau$, and $q_\tau(x) = -1$ otherwise. We freeze two thresholds at the 20th and 50th percentiles of healthy fitting-set confidence. They target 80% and 50% fitting coverage; healthy test coverage is 71.32% and 42.65%. The new audit uses a fresh uniform $1{,}024$-input monitor and tail $0.05/1{,}000$, covering the declared new reader/contract bank. Its classification counts, including distilled $8/10$ and distilled-plus-OLS $9/10$, are separate from the earlier $10/10$ confirmation under its original monitor and 160-consumer allocation. Neither monitor is selected by outcome.

No tested reader certifies either selective consumer on all seeds. Among the auxiliary-learning readers, every selective count is $0/10$; detached-plus-OLS certifies $4/10$ and $3/10$, with no reconstruction gain. The distilled head's selective disagreement is 4.71% and 6.51%, versus 1.67% for classification; OLS reduces these to 3.51% and 4.82%, without certifying the threshold contracts. Distilled-head coverage is 69.14% and 39.86%, and OLS coverage is 70.61% and 41.64%. Per-seed coverage, accepted-set accuracy, NLL, ECE, flows and bounds are retained in the numerical records. Class agreement alone does not retain probability-dependent decisions. Independent saved-state replay checks predictions, flow decomposition, thresholded decisions and bounds. An isolated native float32 audit returns both outputs in one encoder call. At batch $1{,}024$, joint-plus-OLS, feature alignment and feature-plus-OLS cost $1.064\times$, $1.057\times$ and $1.060\times$ current-only inference; at batch one their ratios are $1.109\times$, $1.102\times$ and $1.110\times$. The folded feature-map reader has the same 390 serving parameters as the direct distilled head; its $4{,}096$ map weights are needed only during training.

H.21 A Vision Boundary Screen

We next test whether the scale-sensitive accuracy failure appears on a pretrained vision backbone. We ran two fixed-protocol development screens and did not open the official test split. The model is an ImageNet-pretrained ResNet-18 (He et al., 2016) on CIFAR-100 (Krizhevsky, 2009) at $64\times$

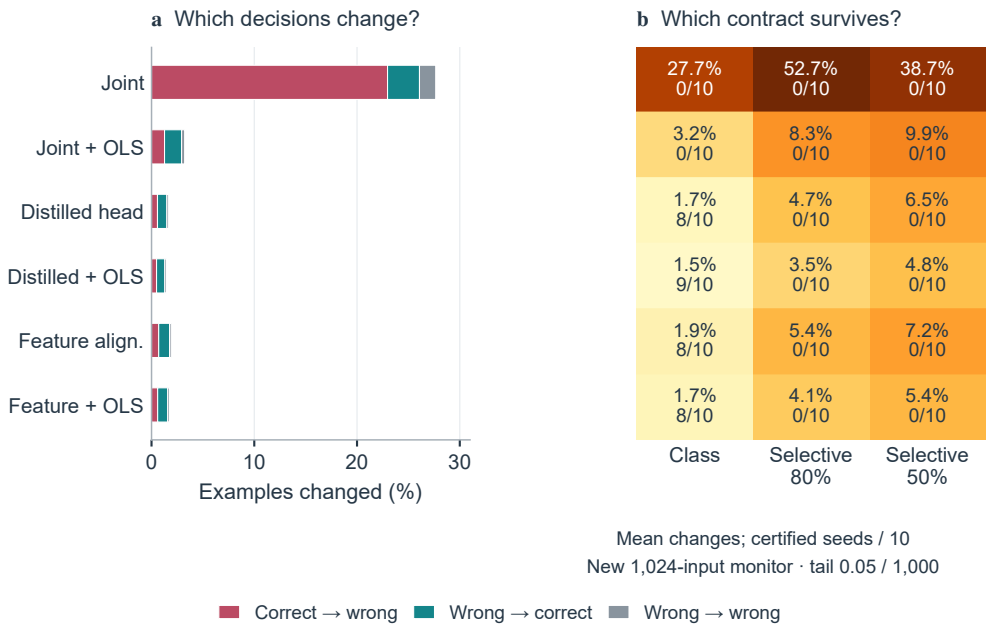


Figure 21: Consumer-specific retention on the ten saved HAR seeds. Left: the three disjoint contributions to class disagreement. Right: mean disagreement and certified seed counts for the class decision and two fixed selective consumers. Coverage labels name healthy-training targets, not test guarantees. The new monitor and family allocation leave earlier confirmation counts intact.

64 input, continued for three epochs with SGD at 10^{-3} for the encoder and 10^{-2} for the heads, momentum 0.9, weight decay 5×10^{-4}, batch 128, and frozen batch-normalization running statistics. A linear readout fitted on the healthy features reaches 0.521 validation accuracy across the three replication seeds (0.513–0.528). Each branch clones the pretrained model and shares minibatch order, and 2,048 paired anchors are isolated from fitting.

The first screen used bf16 continuation at seed 900, with a natural rotation auxiliary at doses 1 and 10. Against healthy accuracy 0.513, the frozen readout reached 0.598 in the primary-only branch and 0.588 at dose 1; the corresponding current heads reached 0.616 and 0.615. The paired primary difference was -0.06 points, so this screen did not establish an auxiliary benefit. The dose-10 branch produced a non-finite loss in its first epoch and was not retuned or rerun. The specified failure criterion was a loss of at least five points. The screen therefore cannot test repair: the fixed diagonal-statistics family stayed near 0.61–0.62, and the paired-ridge family was unstable across anchor budgets (0.17–0.58).

The second screen crossed the affine boundary deliberately. The dose was fixed before outcomes from the feature trace alone: of two one-epoch scale instruments, $\lambda = 0.2$ gave a trace ratio of 0.180 and $\lambda = 0.5$ gave 0.116, so the rule selected 0.5 for a three-epoch branch; readout accuracy was not used. Its endpoint trace ratio is 0.099 ± 0.003, a scale of 0.315 ± 0.005, below the median top-two crossing scale 0.411 of the healthy readout. The predeclared crossing criterion still fails. Frozen accuracy is 0.557, 3.6 points above healthy, and the standardized refit reaches 0.637. A rotation auxiliary at dose 3 reaches scale 0.367 and frozen 0.484 (-2.9 points), refit 0.594. At dose 5 the scale is 0.067; the frozen readout collapses to the constant predictor (0.013, equal to the largest healthy class frequency) and the current head collapses with it (0.023). No branch combines a crossed scale with a silent readout loss.

What the scale does change is the decisions. A post-hoc replay contracts the standardized features $z = (h-\mu_0)/s_0$ and evaluates logits $cWz+b$ over a fixed grid. This is contraction around the healthy feature mean; the raw-feature top-two crossing scale above describes a different path. Accuracy falls

Table 50: Vision boundary screens on ImageNet-pretrained ResNet-18/CIFAR-100 at 64×64. Within each seed, branches clone the same healthy model, share minibatch order, and train three epochs. Trace ratio and scale are measured on the fixed 5,000-example validation split against healthy features; *changed* is the fraction of the frozen readout's predictions that differ from healthy. The primary-only and $\lambda = 0.5$ rows are three-seed means (seeds 900–902, healthy 0.521); their largest SDs are 0.037 (trace ratio), 0.027 (scale), 0.006 (frozen), 0.013 (current), 0.018 (refit) and 1.1 points (changed). The dose rows are single runs at seed 900 (healthy 0.513); dose 1 uses bf16 and doses 3/5 use FP32. Across noncollapsed rows, frozen accuracy moves by -2.9 to $+8.5$ points while 42–56% of decisions change. The official test split is never loaded.

Branch	trace ratio	scale	frozen	current	refit	changed
Healthy	1.000	1.000	0.521	—	—	0
Primary only	0.474	0.688	0.606	0.620	0.637	42.1%
Rotation, dose 1	0.321	0.567	0.588	0.615	0.632	44.8%
Rotation, dose 3	0.134	0.367	0.484	0.583	0.594	55.8%
Scale, $\lambda = 0.5$	0.099	0.315	0.557	0.643	0.637	47.0%
Rotation, dose 5	0.005	0.067	0.013	0.023	0.079	constant

by only 3.5 points at the grid value $c = 0.432$, where 21.1% of predictions change, and by 7.0 points at $c = 0.318$, where 33.9% change. Only at $c = 0.15$, where accuracy falls to 0.169, do 77.7% of decisions change. At $c = 0.432$, 6.2% of the corpus moves from correct to wrong and 2.7% the other way, while 57.6% of all changes stay between two wrong classes. At $c = 0.318$, correct-to-wrong and wrong-to-correct changes account for 10.7% and 3.7% of the corpus, while 57.4% of changed predictions stay between wrong classes, giving a net 7.0-point loss. An incompatible interface hides behind stable accuracy because most of its traffic never touches the label.

The trained branches show the same separation. The primary-only branch raises frozen accuracy by 8.5 points, changes 42.1% of predictions, and turns 7.2% of the corpus from correct to wrong; dose 1 raises it by 7.4 points and changes 44.8%, reproduced within one point in an independent bf16 run (42.5% at 8.5 points); the selected scale branch gains 3.6 points and changes 47.0%; dose 3 loses 2.9 points, with 15.8% of the corpus moving from correct to wrong, 12.8% the other way, and 27.2% of the corpus changing between wrong classes (Table 50). The standardized refit ranks the two auxiliary branches above the primary-only branch, while the frozen readout ranks them below it; the largest accuracy gain of the four still arrives with a 42.1% change of legacy decisions.

The fixed unlabeled readout-input baselines also leave substantial disagreement. Recomputing per-feature input statistics at the readout input raises the old readout from 0.606 to 0.622 on the primary-only branch and from 0.557 to 0.630 on the selected scale branch, yet 40.4% and 41.7% of its decisions still differ. The paired new-to-old ridge map reaches 0.585 and 0.580 with 42.2% and 44.0% changed. Alignment recovers accuracy; it does not recover the interface.

Neither screen reproduces the HAR accuracy failure. Pure contraction has an exact affine law, whereas trained features also change direction; their trace ratio alone does not predict legacy accuracy. The HAR variance auxiliary reaches a scale of 0.015 against a median top-two crossing scale of 0.054 while remaining decodable. The strongly contracting vision branch instead loses both frozen and refitted accuracy. Decision compatibility fails across the tested vision continuations, including those that improve aggregate accuracy.

H.22 A Distinct Vision Consumer Under Primary-Task Updates

The final fixed screen separates the current task from the legacy task. An ImageNet-V1 ResNet-18 emits 512 features for CIFAR-100 at 64×64. The current head predicts 100 fine classes and the legacy head predicts 20 coarse classes (Krizhevsky, 2009). Both healthy heads fit the same 20% labeled subset. Coarse labels are excluded from continuation and OLS fitting; they remain available to an explicitly supervised refit. The coarse label is derivable from the fine label.

Seed 1800 partitions the official training split into 42,952 fitting, 2,048 isolated anchor and 5,000 validation inputs. Healthy features are cached once; running BN statistics remain frozen. The published one-epoch screen uses float32, batch 128, SGD momentum 0.9, weight decay $5 \cdot 10^{-4}$,

encoder learning rate 10^{-3} and head/map rate 10^{-2}. All arms clone the same healthy encoder and affine heads and share future minibatches. F freezes the encoder; U updates it without protection. P learns a coarse head by healthy-logit MSE at coefficients $0.1, 1, 10$; B learns a full feature map at the same coefficients. OLS uses all paired fitting features and healthy logits. The supervised refit uses the original coarse labeled subset. No auxiliary loss or augmentation is added: the question is transfer of the interface contract under a primary-task update.

Table 51: Completed validation endpoints in the distinct-consumer vision screen. Fine accuracy evaluates the current task; coarse accuracy, disagreement and negative flip rate (NFR) evaluate the legacy task. Healthy fine/coarse accuracies are $0.443/0.542$. All repair fits use the fitting pool.

Update	legacy reader	fine acc.	coarse acc.	changed	NFR
F	fixed	0.397	0.542	0.0%	0.0%
U	fixed	0.359	0.428	58.9%	22.0%
U	OLS	0.359	0.575	45.7%	12.1%
U	supervised refit	0.359	0.565	50.5%	14.0%
P (0.1)	fixed	0.429	0.555	45.2%	12.3%
P (0.1)	learned	0.429	0.568	44.6%	12.0%
P (0.1)	OLS	0.429	0.605	41.2%	9.2%

U fails the required two-point current-task gain over F: its difference is -3.78 points. P (0.1) improves over both continuation baselines but changes 44.56% of coarse decisions, above the 5% budget. OLS improves coarse accuracy while leaving 41.20% of decisions changed. The stronger P coefficients 1 and 10 produce nonfinite losses. The bounded queue stops after these two failures: B (0.1) is interrupted, and B $(1, 10)$ are never launched. The B branches remain unmeasured. The failed utility gate ends the screen before official test evaluation. The saved complete endpoints undergo independent checkpoint/reader replay; failure logs and queue states are retained separately. This initial full-backbone recipe yields no positive utility/compatibility result.

H.23 A Fixed-Corpus Unlabeled Consumer Check

For a present checkpoint, we complement the variance audit with unlabeled disagreement against the healthy consumer. The check combines the relative-loss inequality with exact finite-population sampling, extending unlabeled performance estimation (Garg et al., 2022) and update certificates (Usama & Qadir, 2026) with reference predictions and an option to abstain.

Equation (6) bounds error increase by disagreement for every label assignment. Using its fixed-corpus notation, we obtain a sampling bound for the unmonitored inputs.

Uniformly sample n monitor inputs without replacement, independently of the fixed candidate family. If c predictions disagree on this monitor and the unknown full-corpus disagreement count is K, then c has the hypergeometric distribution with parameters (N, K, n). With family size M and total tail probability δ, define

$$K^+(c) = \max\{K \in \{c, \dots, N - n + c\} : F_{N,K,n}(c) \geq \delta/M\}, \qquad U = \frac{K^+(c) - c}{N - n}. \tag{35}$$

Here F is the exact hypergeometric CDF. Inverting its lower tail bounds the full-corpus disagreement count, hence $D_T \leq U$ on the unmonitored remainder. For each candidate the failure probability is at most δ/M; a union bound gives simultaneous coverage for all M candidates, allowing selection from that fixed family. The statement needs no independence between adjacent windows, and it covers the fixed corpus with classification error, not a new population, future updates, NLL, ECE, or conformal coverage. Monitor-adaptive training and unconstrained construction of new candidates would need a different analysis.

We predeclare $n = 1{,}024$, $\delta = 0.05$, and budgets 0.05 and 0.10 for relative loss increase. The bank contains all 300 endpoints from the 30 earlier paired runs plus 35 CNN endpoints, each evaluated with the same nine fixed repair arms: $M = 3015$. All maps use the full fitting feature set; a restricted anchor budget would need its own evaluation. A seeded uniform monitor partition is shared across candidates in a run, and old/new reference predictions on those monitor inputs are required. Healthy reference predictions can be cached for a fixed corpus or obtained from the retained healthy

Algorithm 3 Auditing representation survival and probe validity

Require: Diagnostic batch, healthy reference, checkpoint θ, operating losses and dose α, recorded calibration value q.

1: Report task metrics, covariance trace $2V$, effective rank, standardized linear probe, and the declared operational scatter criterion.
2: Audit healthy-readout accuracy/NLL/ECE, mean/global/per-feature repairs, and any paired alignment; compare with a refitted probe.
3: With unlabeled monitor inputs and healthy reference predictions, bound consumer disagreement; select a certified candidate or abstain.
4: Report gradient-flow fluxes (r, c) from (4). For Adam, audit $v^\top s_\theta$ and the actual post-step V change; specify any auxiliary counterfactual.
5: Record $1-\alpha q$, the signed balance $\psi = d-\alpha a$, and the corresponding residuals. Form $q^\star = a/d$ only when $d \neq 0$.
6: **if** $|1 - \alpha q| > \Delta$ **then**
7: Mark the gradient-flow sign locally certified for this batch.
8: **else**
9: Use the signed audit; report restoration misalignment, geometry, and probe mismatch separately.
10: **end if**
11: Compare joint training, detachment, and any selective intervention using both survival and task utility at matched doses and seeds.

Table 52: Fixed-family consumer audit. Each corpus supplies 1,024 uniformly sampled unlabeled monitor inputs; remaining labels score the decisions afterward. The 3,015 fixed endpoint/repair candidates share total tail probability 0.05. *Scatter false safe* uses a scatter ratio ≥ 0.3 relative to healthy features on the same monitor, despite frozen-consumer loss increase above budget. Selection uses the first certified candidate, including identity/frozen; uncertified cases abstain. Counts describe dependent checkpoints, not independent trials.

Case	budget	joint	scatter false safe	frozen certified	candidate selected	selected false safe
HAR isotropic	0.05	30	0	0	6	0
HAR isotropic	0.10	30	0	3	24	0
HAR projected	0.05	15	10	0	3	0
HAR projected	0.10	15	8	0	10	0
HAR recon.	0.05	15	3	1	9	0
HAR recon.	0.10	15	3	9	15	0
RadioML SNR	0.05	15	3	0	0	0
RadioML SNR	0.10	15	0	0	0	0
RadioML I/Q MLP	0.05	15	0	0	0	0
RadioML I/Q MLP	0.10	15	0	0	0	0
RadioML I/Q CNN	0.05	15	0	0	0	0
RadioML I/Q CNN	0.10	15	0	0	0	0

encoder; this access is additional to paired training anchors for richer repairs. The first repair in the predeclared hierarchy satisfying $U \leq \varepsilon$ is selected, or the audit abstains. Monitor labels never enter this choice. Candidate remainder accuracy and loss increase are scored only after choices are fixed; archived healthy/frozen metrics are also checked independently for reproducibility.

Across 105 joint endpoints, the 5-point budget selects 18 candidates and abstains on 87; the 10-point budget selects 49 and abstains on 56. The latter includes 12 unchanged frozen consumers and 37 repaired consumers. Observed selected-consumer loss increases exceed budget in 0 and 0 cases, respectively. The healthy-reference scatter heuristic makes 16 and 11 false-safe declarations. This comparison uses scatter relative to healthy features on the same monitor, which is available without a continued primary-only counterfactual; it is distinct from retrospective comparison with a fully trained single-task endpoint. The fixed unsigned probe's safe declarations and primary-only controls are also retained in the numerical summaries.

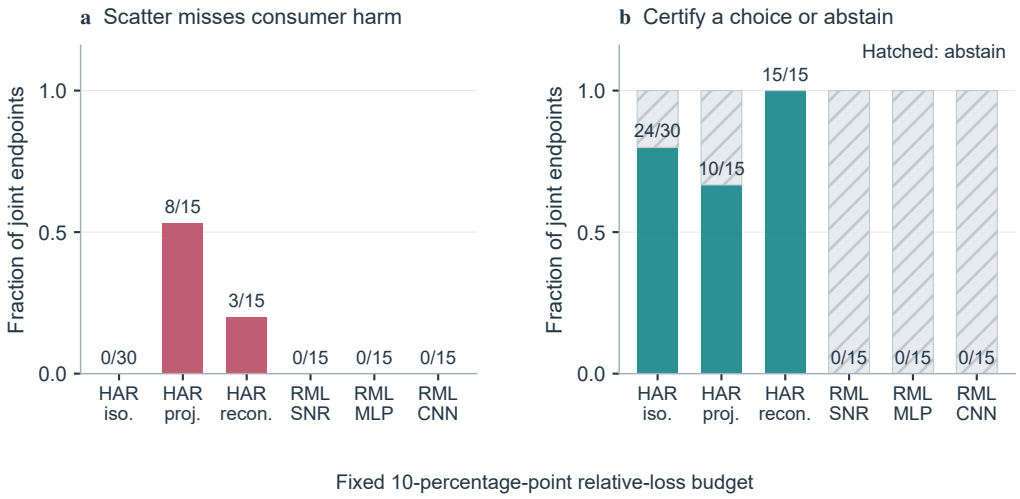


Figure 22: Unlabeled disagreement supports conservative repair selection. Fixed 10-percentage-point relative-error budget. Left: a monitor-scatter rule can declare a harmful frozen consumer safe. Right: the first certified repair is selected from the fixed hierarchy; otherwise the rule abstains. The final two groups use I/Q reconstruction. Bars show fractions of joint endpoints, with counts annotated. Within-seed branches are dependent. Simultaneous coverage applies to the fixed candidate family on the unmonitored corpus remainder.

At the 10-point budget, HAR supplies 49 selections; RadioML supplies 0 of 45, including its near-null continuation cases. Disagreement can leave an accuracy-preserving or improving candidate uncertified when many decisions change. The check bounds relative 0–1 error on the fixed corpus; absolute accuracy and warning lead time require other evidence. All candidates, decisions, bounds, remainder scores, map hashes, and source links are archived. Checks cover finite-population coverage, boundary cases, monotonicity, sharpness, selection, abstention, and temperature-only NLL drift. Drivers: `consumer_risk.py` and `checkpoint_consumer_audit.py`; analysis: `summarize_consumer_frontier.py`.

H.24 Dissociation Between Contraction and Utility

Three quantities are often conflated: variance contraction, scale-invariant information, and readout utility. Table 53 separates them in six cases, each using the protocol cited in its section.

The six cases show why no single scalar orders representation quality. In HAR, scatter contracts by a factor of 60 while task information remains linearly available and input recalibration repairs the readout. MATR has negative transfer without contraction. SNR combines relative collapse and readout failure, but mostly negative sampled flow flux leaves its Adam attribution unresolved. I/Q retains scatter 1.05 and refitted accuracy 0.351 versus 0.312 detached, while the frozen readout falls to 0.195. Subject-ID similarly gives joint accuracy 0.887 at scatter 0.93, yet frozen accuracy 0.384 versus 0.901 refitted. The last row is benign at its tested dose.

H.25 RadioML Dose–Response and Natural Scale

Table 54 varies the auxiliary dose for two real RadioML heads at $m = 64$, 25 epochs and three seeds, fully labeled: SNR regression (the head that collapses at the reported dose) and I/Q reconstruction (a benign head). The detached reference is 0.306 accuracy with $0/3$ collapses for SNR and 0.307 with $0/3$ for reconstruction; its relative scatter is 1.031 and 1.101, respectively.

SNR collapse begins between $\alpha = 3$ and 10, with all seeds below the criterion by 30; the historical 3000 comparison is deep within this observed regime. I/Q reconstruction stays near or above reference scatter through 1000 and reaches 0.641 at 10000, above 0.3. Thus nominal dose alone does not

Table 53: Six cases separating contraction, decodability, and readout utility. Relative S is the same-seed scatter ratio. The scale check is the standardized probe where available; PR measures spectral concentration in the battery row. Utility names the primary or frozen-readout metric. SNR and subject-ID readout studies use five seeds, I/Q three; the battery row uses the corrected MATR application.

Case	Rel. S	Scale check	Utility	Diagnosis
HAR toxifier (100)	0.016	probe 0.955 (healthy 0.947)	frozen 0.241; input recal 0.952; retrained 0.955	contraction; information intact; readout broken
MATR battery	3.55	PR 2.49	SOH 5.57 vs 2.10 detached	no contraction; negative transfer
RadioML SNR (3000)	0.137	n/a	own 0.205; frozen 0.099; retrained 0.229; detached readout 0.315	relative collapse; readout near chance; sampled flow flux mostly negative
RadioML I/Q (3000)	1.05	n/a	own 0.344; frozen 0.195; retrained 0.351; detached readout 0.312	no contraction; frozen readout broken by drift
HAR subject-ID (10)	0.93	n/a	own 0.887; frozen 0.384; retrained 0.901	no contraction; frozen readout broken by drift
HAR reconstruction	1.00	n/a	0.942	benign head

Table 54: RadioML dose–response, fully labeled. Joint training, three seeds, $m = 64$. "Coll." is the scatter-collapse count; $\alpha\rho$ is the nominal dose times the measured per-example radial ratio. The SNR onset lies between $\alpha = 3$ and 10, and the head collapses in every seed from $\alpha = 30$ onward; the reconstruction head never collapses at any tested dose.

α	SNR acc.	SNR coll.	SNR rel. S	SNR $\alpha\rho$	I/Q acc.	I/Q coll.	I/Q rel. S
1	0.329	0/3	0.932	0.28	0.305	0/3	1.054
3	0.318	0/3	0.600	1.28	0.308	0/3	1.038
10	0.277	2/3	0.291	5.84	0.312	0/3	1.096
30	0.248	3/3	0.211	22.6	0.303	0/3	1.005
100	0.229	3/3	0.163	70.3	0.312	0/3	1.057
300	0.214	3/3	0.146	187	0.321	0/3	1.252
1000	0.207	3/3	0.144	557	0.338	0/3	1.267
10000	0.204	3/3	0.140	4138	0.320	0/3	0.641

order the two heads. Their measured radial loadings differ sharply: at 30, SNR has $\alpha\rho = 22.6$, while I/Q loading is near zero. The signed-flux audit below checks what those unsigned measurements can explain.

Table 55 adds STF-cal at the three doses around and above the SNR onset. At $\alpha = 30$, calibrated filtering reaches 0.303 against joint 0.248 and detached 0.306, with no collapse; at $\alpha = 100$ and 300 the recovery shrinks to 0.278 and 0.236 even though the gate opens in every seed. The reconstruction gate stays shut (gate rate 0.00) at all three doses and STF-cal stays at 0.308. Filtering helps near the onset, not at every dose.

H.26 Readout Dose Scan and Signed Flux

Table 56 repeats the healthy-readout transfer across the SNR dose scan at $m = 64$, 25 epochs, and five seeds. The frozen readout is 0.176 at $\alpha = 3$ (relative scatter 0.558, above the 0.3 criterion), 0.121 at $\alpha = 10$ (relative scatter 0.274, at the criterion), and near chance from $\alpha = 30$ onward. Retraining recovers 0.322 at $\alpha = 3$ and 0.229 at $\alpha = 3000$, and ex-

Table 55: STF-cal at the SNR onset (RadioML, fully labeled, three seeds, $m = 64$). Gate rate and mean margin are for the calibrated arm; joint and STF-0 show the reference endpoints.

α	joint acc.	STF-0 acc.	STF-cal acc.	gate / margin
30	0.248	0.306	0.303	1.00 / 14.1
100	0.229	0.306	0.278	1.00 / 47.1
300	0.214	0.306	0.236	1.00 / 141.3

ceeds the joint encoder's own accuracy at every dose, so the representation retains linearly recoverable information. The readout therefore degrades before the operational scatter criterion fires; a scatter threshold alone is not a deployment certificate. Detachment preserves 0.315 at every dose. Driver: `repro/scripts/real_head_readout_transfer.py`; records: `results/READOUT/rml_snr_readout_a*.json`.

Table 56: Historical RadioML SNR readout dose scan (five seeds, $m = 64$, 25 epochs). Own uses the joint primary head; frozen and recalibrated reuse the healthy weights; retrained fits new weights; detached uses the healthy readout on the STF-0 encoder. Relative S is the same-seed scatter ratio.

α	rel. S	healthy	frozen	recal.	retrained	detached	own
3	0.558	0.315	0.176	0.146	0.322	0.315	0.315
10	0.274	0.315	0.121	0.114	0.283	0.315	0.273
30	0.198	0.315	0.102	0.099	0.256	0.315	0.247
100	0.160	0.315	0.106	0.090	0.240	0.315	0.229
300	0.145	0.315	0.103	0.097	0.234	0.315	0.214
3000	0.137	0.315	0.099	0.094	0.229	0.315	0.205

The signed flux audit uses the same driver with `--flux-trace`. On a fixed diagnostic batch of 384 test examples we compute $V = \|C_B H\|_F^2/(2B)$, $v = \nabla_\theta V$, $r = -v^\top G_p$, and $c = v^\top G_a$, and record them at every epoch end; inside the first two epochs we also sample every 10 minibatches. At $\alpha = 3000$ the seed-median c_t is -3.97 at epoch 1, -49.3 at epoch 5, -66.1 at epoch 10, and -107.5 at epoch 25; the aggregate signed cosine is below -0.90 at the endpoint. Of 75 epoch-end checkpoints, 74 have negative c_t; the exception is $+0.76$. Epoch-end sampling alone cannot rule out an early effect from that checkpoint. I/Q reconstruction gives $|c_t| < 10^{-5}$ at every epoch end, consistent with no variance flux. The SNR head thus shows relative collapse without consistently positive sampled gradient-flow flux. These observations alone do not identify its Adam-step contributions; Appendix H.7 audits actual updates separately. Records: `results/READOUT/rml_snr_flux*.json` and `results/READOUT/rml_iq_flux.json`.

At $\alpha = 3000$, every calibrated SNR gate opens and the selected deletion reaches its cap of 3 (calibrated $\rho = 0.44$–0.49, margins 1.3–1.5×10^3). Relative scatter remains 0.25–0.36; frozen accuracy rises only from 0.094 (joint seed median) to 0.136, with recalibration 0.124 and refitting 0.212. The selected rank and reliably open gate therefore fail to restore compatibility at this dose. Records: `results/READOUT/rml_snr_readout_stfcal.json`.

I Anti-Collapse Regularizers at Matched Doses

Table 25 evaluates two regularizer families at the doses and seeds of Table 4: VICReg's variance hinge plus off-diagonal covariance penalty, and Barlow Twins' redundancy-reduction penalty. Each is applied to the shared representation with weight λ. VICReg's invariance term is omitted because this experiment has no augmented-view pair. All rows use the same joint-training baseline; these are adaptations of the regularizers to this setting, not evaluations of the complete original pipelines.

Regularization can clear the operational collapse threshold without restoring the single-task reference or improving the task. In the coarse grid, VICReg first avoids collapse in all seeds at $\lambda = 10$ for HAR, 300 for battery, and 10000 for RadioML. Barlow Twins does not do so at any tested weight.

On HAR, VICReg scatter stays near 0.36 from $\lambda = 10$ to 10^4, while accuracy falls from 0.944 to 0.821. At the lowest clearing weight, accuracy is close to joint training (0.945) and detachment. On battery, scatter increases through the reference level, but the best tested RMSE is 0.339 at $\lambda = 300$ (relative scatter 0.59), compared with 0.297 for detachment. Larger weights worsen RMSE to 0.390 at 10^3 and 0.423 at 10^4; detachment is better in nine of ten seeds. On RadioML, $\lambda = 10000$ gives zero collapse and scatter 1.29, but accuracy drops to 0.215, below both collapsed joint training (0.256) and detachment (0.341). These cases separate the scatter criterion from predictive utility.

The Barlow Twins adaptation yields absolute scatter as low as 0.011, 0.012, and 0.013 for HAR, battery, and RadioML, respectively, or factors of 172, 6, and 24 below their references. It fails to clear the collapse threshold throughout the tested grid, with HAR accuracy reaching approximately 0.34–0.44 and RadioML approximately 0.11 in the low-accuracy configurations.

Additional ten-seed sweeps use $\lambda \in \{2, 5, 20, 50, 200, 500\}$ on HAR and $\lambda \in \{200, 500, 700, 1500, 2000, 5000\}$ on battery. The recorded battery transition is bracketed by 150 and 200, and the lowest measured battery RMSE remains 0.339 at $\lambda = 300$, above detachment. HAR scatter remains near 0.36 above its crossing. RadioML was not refined: the sampled collapse rate is 1.0 at 1000, 0.10 at 3000, and 0.0 at 10000. The experiments therefore locate an operational transition over a finite grid rather than a precise critical weight.

I.1 Shared-Reference Replay

Retaining the old function need not mean running two full encoders. Let $a_1^0 = \sigma(W_1^0 x + b_1^0)$ and $t_2^0 = W_2^0 a_1^0 + b_2^0$ denote the healthy trace. Change k rows S of the first affine map and use a rank-r second-map update UV. The current a_1 can cross nonlinear activation boundaries, but $\delta a_1 = a_1 - a_1^0$ is still supported on S.

Proposition 11 (Shared-reference replay)**.** *For any coordinatewise activation σ, the current second preactivation equals*

$$t_2 = t_2^0 + W_2^0[:, S]\, \delta a_1[S] + UV a_1 + \delta b_2.$$

Evaluating $\sigma(t_2)$ and $\sigma(t_2^0)$ produces the merged current and healthy functions in real arithmetic, including when activation gates change. Their extra MACs are $kd+km+2rm+Cm$, with separate affine current and old heads, rather than a second dense trunk.

Proof. Substitute $a_1 = a_1^0 + \delta a_1$ into $(W_2^0 + UV)a_1 + b_2^0 + \delta b_2$ and use the support of δa_1. □

The earlier one-encoder controls either restrict a common feature interface or learn a replacement readout. Shared-reference replay retains the healthy input-to-output function on a separate path, freeing the current features to change. The reference parameters remain available, but their dense computation is shared with the current model. This access condition differs from repairing terminal features alone. No old-label or distillation loss is added to continuation.

For a width-m two-nonlinearity trunk, first-map changes have row support S, $|S| = k$, and the second-map change is UV of rank at most r. Proposition 11 evaluates both nonlinearities without linearizing the update. The final affine encoder map and current task head remain trainable; their product is folded for native serving. The healthy final affine map can similarly be folded into the old head. One healthy dense first and second map is stored, along with sparse/low-rank updates and both heads. The serving package omits a second full encoder. Extra multiply–accumulates (MACs), excluding biases and elementwise work, are $kd + km + 2rm + Cm$ for input dimension d and C old outputs. For the selected HAR and SNR configurations these are 3,396 and 6,848, respectively: 8.4% and 32.3% above the merged current network. Measured latency differs substantially from these arithmetic ratios.

Low-rank parameters alone do not give a low-rank activation difference. For n scalar inputs $x_i = i$, take $W_1^0 = 0$, biases $b_j = -(j - \frac{1}{2})$, and the rank-one update $\Delta W_1 = \mathbf{1}$. The healthy ReLU activations vanish, whereas the change matrix has entries $\max\{i - j + \frac{1}{2}, 0\}$. It is lower triangular with diagonal $1/2$ and hence has rank n. Row support supplies the invariant used in our construction: after any coordinatewise nonlinearity, δa_1 is still supported on S. This invariant motivates the hybrid restriction.

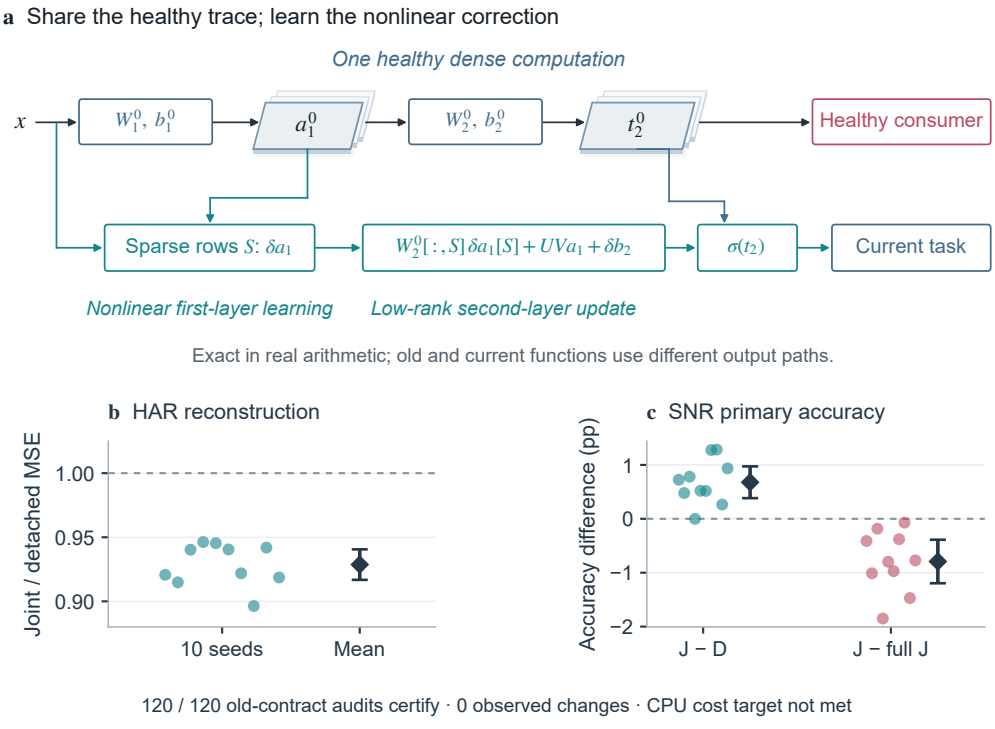


Figure 23: Shared products for the original sparse-row construction, followed by all ten fresh HAR and SNR outcomes. HAR reconstruction improves; sparse SNR gains over detachment but loses accuracy against unrestricted joint training. The dense follow-up below closes that accuracy gap. CPU cost remains above target.

The same identity applies to an identity residual block with fixed normalization. Updating k output channels of its first convolution confines the activation difference to those channels. A 3×3 rank-reducing convolution followed by a 1×1 expansion represents the second update. Adding the unchanged shortcut and applying ReLU gives the merged current block in real arithmetic. Before the visual study, fixed BN is folded into the healthy convolutions; this defines the numerical reference from the start.

I.2 Dense Replay, Support and Update Rank

Row support is one admissible structure, not the only one. Replace the first map by $W_1^0 + PQ$ with $P \in \mathbb{R}^{m \times s}$, $Q \in \mathbb{R}^{s \times d}$, the second by $W_2^0 + UV$ of rank r, and add biases $\delta b_1, \delta b_2$. With $a_1 = \sigma(W_1^0 x + b_1^0 + PQx + \delta b_1)$ and $a_1^0 = \sigma(W_1^0 x + b_1^0)$, substitution into $(W_2^0 + UV)a_1 + b_2^0 + \delta b_2$ gives

$$t_2 = t_2^0 + W_2^0(a_1 - a_1^0) + UVa_1 + \delta b_2 \tag{36}$$

exactly, for any coordinatewise σ. The correction is dense, so the extra MACs above the folded current-only graph are $sd + ms + m^2 + 2rm + n_c m$: the saving comes from reusing the healthy dense products, not from sparsity.

For each saved full-joint endpoint we truncate the first- and second-map deltas by SVD, keep everything else fixed, and score the official test split (Table 57). A truncated update is a member of the dense low-rank family, so a nonnegative gap proves that this structure can reproduce joint accuracy. Training and truncation give different results: the trained rank-$16/12$ SNR arm below reaches parity, whereas truncating the saved first-map update requires rank 32 for the same tolerance. Thus this retrospective screen characterizes a saved update, rather than the necessary rank of a trained family.

Development selected $(16, 12)$ for SNR, the smallest grid configuration clearing the predeclared gate on seeds 800–802. Ten fresh seeds then give a primary gain of 1.257 points $[1.037, 1.477]$ over de-

Table 57: Mean accuracy difference, in points, to the saved full-joint endpoint after truncating its first-map (s) or second-map (r) update and scoring the official test split. Ten endpoints per task. A truncated update is a member of the dense low-rank family, so a nonnegative gap is achievable accuracy, not a lower bound on the family.

Update kept	HAR	SNR
first map, $s = 16$	-0.13	-0.56
first map, $s = 32$	-0.07	-0.01
second map, $r = 16$	-0.18	-0.16
second map, $r = 24$	–	$+0.01$
$(s, r) = (16, 16)$	-0.28	–
$(s, r) = (32, 24)$	–	-0.02

tachment, a difference of -0.008 points $[-0.227, 0.210]$ to ordinary joint training, and an auxiliary MSE ratio of 0.788 $[0.754, 0.822]$. Accuracies are 27.43%, 28.69% and 28.70%. Every observed consumer disagreement is zero, every certificate meets the five-point budget with maximum upper bound 0.0092, the exact reference error is 0, and packed serving differs by at most 8.4×10^{-6} logits without changing a merged prediction.

The HAR gate rejects the same family: primary differences are -0.59 to -1.18 points on development while reconstruction improves, so no HAR configuration was confirmed. The larger update family helps SNR and reduces HAR accuracy under the specified recipe. The sparse HAR arm and its confirmed reconstruction result are unchanged.

The matched SNR current-only graph costs $21{,}184$ MACs after folding its final affine map into the primary head. Dense replay adds $11{,}456$ MACs (54.1%), versus $6{,}848$ (32.3%) for sparse replay. The earlier $12{,}288$ count mixed an extra head already present in the current baseline with 128 bias additions and divided by the unfused $24{,}576$-MAC encoder; we retain that historical record and use the matched graph here. Measured latency is reported in Appendix I.4.

Layerwise update rank also depends on the parameterization. The standard positive-rescaling symmetry of ReLU networks (Dinh et al., 2017) gives a simple check: $(W_1, b_1, W_2, b_2) \mapsto (2W_1, 2b_1, W_2/2, b_2)$ preserves the function, while both matrix differences can have full row rank. Our SVD results therefore concern saved updates in fixed coordinates, rather than an intrinsic rank of the learned function. The support/rank factorial in Appendix J.1 measures the distinction directly under matched future training orders.

I.3 Development, Selection and Fresh Confirmation

The fixed plan precedes development and is included with its source hashes. Development uses saved healthy checkpoints, optimizer states and fitting/ validation splits for seeds 800–802 on both tasks. The four configurations are $(k, r) \in \{(0, 4), (4, 4), (8, 8), (16, 8)\}$. On $1{,}024$ fitting inputs, the original joint-loss gradient ranks first-map rows and initializes the right second-map factors from its singular vectors. Update amplitudes start at zero. This training-only initialization rule is fixed before outcomes. The paired detached and joint arms share support, factors, inherited moments, and future minibatches. New update parameters start with zero Adam moments. We retain the original unclipped Adam, learning rate 10^{-3}, batch 256, $30/25$ HAR/SNR epochs, doses $100/1$, and $10\%/20\%$ primary labels.

Development requires every old-contract disagreement to be at most 1%. HAR additionally requires mean reconstruction ratio at most 0.95 and mean primary difference at least -0.5 points; SNR requires a positive paired primary-gain interval and mean accuracy within 0.5 points of unrestricted joint training. Eligible configurations are ranked by extra MACs. This selects HAR $(4, 4)$ and SNR $(16, 8)$. No test score enters selection.

The native development cost screen fails the fixed batch-$1/1024$ limits of $1.20/1.10$. This failure is recorded before confirmation; the fixed plan permits a partial utility/compatibility frontier. Twenty new warm checkpoints, seeds 2500–2509 on each task, then train the frozen structured pair and ordinary full-update detached/joint controls. These are new training seeds on the existing data splits, not new populations. Fresh $1{,}024$-input monitors use RNG $10700001 +$ seed, plus 100000 for HAR,

Table 58: Fresh shared-replay confirmation, ten seeds per task. J–D compares the matched structured arms. Intervals are paired 95% t intervals; accuracy differences use percentage points. Every old class/selective consumer certifies, but SNR misses noninferiority to full joint training.

Outcome	HAR $(k, r) = (4, 4)$	SNR $(16, 8)$
Primary J–D	-0.027	$+0.678$
95% interval	$[-0.468, 0.413]$	$[0.383, 0.973]$
Reconstruction/regression MSE J/D	0.929	0.817
95% interval	$[0.917, 0.941]$	$[0.782, 0.853]$
Primary J–full J	-0.024	-0.792
95% interval	$[-0.689, 0.641]$	$[-1.196, -0.388]$
Class / selective certificates	$20/20$ each	$20/20$ each
Maximum old decision disagreement	0	0

and tail probability $0.05/600$. Each evaluates classification and two fixed confidence thresholds calibrated on healthy fitting features. Certificates concern the finite test-corpus remainder with a five-point budget. Utility intervals are paired, two-sided 95% t intervals over all ten seeds; the criteria and cost limits are unchanged.

HAR confirms reconstruction progress and primary noninferiority relative to its structured detached arm. SNR confirms positive auxiliary transfer, but misses the half-point noninferiority criterion against full joint training. The unrestricted SNR pair gains 1.073 points; the structured pair gains 0.678. Both nonlinear layers change their parameters and observed ReLU gates. All 120 old-contract bounds pass, with no observed decision changes. The largest remaining-disagreement bound is 1.092% on HAR and 0.917% on SNR. These findings establish useful learning with class and selective compatibility, while leaving a cost failure and an SNR capacity cost.

The dense companion study was planned separately before its outcomes. Its grid is $(s, r) \in \{(16, 12), (24, 16), (32, 24)\}$; the same development seeds, warm checkpoints, orders, initialization rule and gate decide selection, and seeds 2520–2529 confirm the chosen pair. HAR fails the primary gate at all three configurations, so only SNR is confirmed, at $(16, 12)$. No criterion was changed after any outcome.

The full reference path reproduces stored healthy logits exactly under the declared evaluation arithmetic. The compressed native-float32 old heads have maximum errors 1.27×10^{-5} on HAR and 8.84×10^{-6} on SNR; their actual decisions are the ones monitored. An independent raw-state replay verifies all 40 structured endpoints and 120 contracts. Packed and independently merged current serving differ by at most 1.53×10^{-5}, with two near-tied current classifications changing across all replayed endpoints. Floating-point reassociation accounts for these near-tie differences. Reported utility uses the actual packed two-output implementation.

I.4 Native Serving and Strong Reuse Controls

An isolated four-thread CPU process measures both arms of all six selected development endpoints. Both outputs use float32; the merged current-only baseline folds the same final affine maps and emits the same current function in real arithmetic. We compare eager and TorchScript serving, two independently evaluated functions, and a stronger control sharing the healthy first map but recomputing both dense second maps. Packed replay combines small products before serving. Each measurement warms all paths, alternates operation order, and retains 71 repetitions. Table 59 reports medians across the six paired endpoint measurements per task.

Kernel launches, strided intermediates and small products dominate these small networks. At batch 1, sharing fewer arithmetic operations is slower than evaluating two independently folded functions. At batch 1024, HAR replay is modestly faster than that control; SNR replay is slower. A separate fixed-shape Inductor experiment on the joint seed-800 states yields matched packed/current ratios $1.44/1.46$ for HAR and $1.50/1.73$ for SNR at batch $1/1024$, still above budget. Its current-only batch-1 latencies are $41.8/38.9\,\mu$s, slower than TorchScript's $20.6/16.8\,\mu$s. Initial setup ranges from 0.10 to 13.88 seconds across compiled paths. Both absolute times and ratios are retained; the smaller compiled ratio does not establish a faster deployment. No universal latency claim follows from the exact identity or MAC count.

Table 59: Isolated float32 TorchScript inference, four CPU threads. Times are microseconds per batch; Replay/current is the median paired ratio. Prefix shares the healthy first map and recomputes both second maps. The registered limits are $1.20\times$ at batch 1 and $1.10\times$ at 1024.

Task	Batch	Current	Replay	Prefix	Independent	Replay/current
HAR	1	20.6	45.3	43.4	40.1	$2.20\times$
HAR	1024	276.1	508.3	508.9	543.9	$1.81\times$
SNR	1	16.8	42.7	39.5	32.4	$2.54\times$
SNR	1024	157.2	324.7	352.7	312.4	$2.07\times$

The same protocol on the ten dense SNR endpoints gives median packed/current ratios of 3.02 at batch 1 and 2.57 at batch 1024 under TorchScript ($3.22/2.63$ eager), against $2.54/2.07$ for the sparse arm. Denser corrections raise both the analytic MAC count and the measured cost, so the dense arm is reported as the utility endpoint of the frontier, not as a deployment proposal. For a fixed input dimension of 561, the same sparse implementation improves from $1.53\times$ at width 64 to $1.12\times$ at width 1024 for batch 1024, and remains between $1.56\times$ and $4.70\times$ at batch 1; the batch-1 target is not met by any measured two-output path at these widths.

I.5 ResNet-18 Capacity Boundary

The prospective ImageNet-V1 ResNet-18/CIFAR-100 screen uses 64×64 images, $42{,}952$ fitting, $5{,}000$ validation and $2{,}048$ isolated anchor images, and a fixed 20% fine-labeled fitting mask. Healthy fine-100 and coarse-20 heads use the same labeled subset. Coarse labels are not used in continuation; their derivability from fine labels is explicit. Earlier blocks and normalization are fixed; only the last identity-residual block can change. Current-head normalization remains fixed to healthy labeled features. Three AdamW epochs use identical original fitting-index orders, batch 128, head learning rate 10^{-3} and weight decay 10^{-4}. The ordinary dense-block control uses learning rate 10^{-4}.

The initial $(k, r) = (0, 8), (8, 8), (16, 16)$ screen at update rate 10^{-3} does not pass the prescribed two-point gain over both healthy and head-only accuracy, with a half-point tolerance against the dense control. A second fixed capacity screen, declared before execution, tests $(64, 32), (128, 64), (256, 128)$ at the dense block's update rate 10^{-4}. Figure 24 retains all six outcomes. The largest configuration reaches 45.30%, against 43.74% healthy, 43.48% head-only and 45.74% dense. Its 1.56-point net gain still misses the fixed two-point gate. Every old decision is unchanged, but neither screen proceeds to more development seeds or official test scoring. A folded-bias gradient initialization error before the first continuation is retained as a technical failure, separately from these completed scientific outcomes.

I.6 The Geometry of a Versioned Consumer

Let p_0 be a healthy categorical distribution with unique winning class k. In addition to the class reader, consider fixed consumers that return k above a confidence threshold and abstain below it. Their joint output fixes the winning class and one confidence bin $[l, u)$. We retain a buffered subset of that bin: $r = \exp(\eta \min\{\log p_0[k] - \max_{j \neq k} \log p_0[j], 1\})$, $\eta = 0.1$, and internal confidence bounds $l' = 0.9l + 0.1p_0[k]$, $u' = 0.9u + 0.1p_0[k]$. Missing outer bounds remain 0 or 1. The closed cell is

$$\mathcal{C}_0 = \{p \in \Delta^{C-1} : p[k] \geq rp[j] \ (j \neq k), \quad l' \leq p[k] \leq u'\}. \tag{37}$$

It contains p_0, is convex, and retains all specified consumer outputs when the healthy class is unique and rejected-threshold boundaries are buffered. This construction uses healthy predictions and fixed thresholds, not labels. Set-valued targets and KL projection are established ideas; label relaxation already replaces a point label by a probability set (Lienen & Hüllermeier, 2021). Here the set is induced by the particular deployed class and confidence-threshold consumers.

Proposition 12 (Projection onto a buffered consumer cell)**.** *For a positive new distribution q, the minimizer of $\mathrm{KL}(p\|q)$ over $\mathcal{C}_0$ is unique. Sort the non-k components as $q_{(1)} \geq \cdots \geq q_{(C-1)}$. In the*

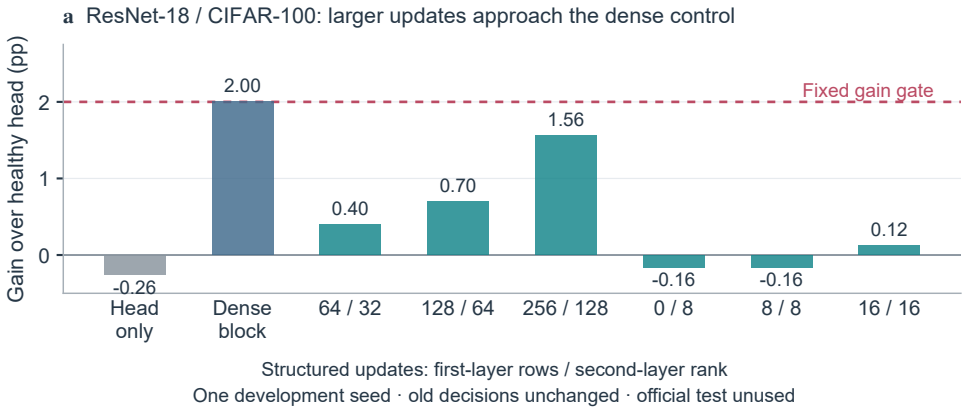


Figure 24: All six ResNet-18 shared-replay capacity candidates at seed 2600. Bars show validation accuracy gain over the healthy fine head. The dense last-block control reaches the fixed two-point gate; the structured updates do not. The two grids use their prospectively fixed update rates. Old coarse class and selective decisions are unchanged. This development screen uses no official test examples and is not an auxiliary-benefit study.

absence of confidence bounds, pool k and the violating prefix of length m, with

$$g_m = \left(\frac{q[k]^r \prod_{j=1}^{m} q_{(j)}}{r^r} \right)^{1/(r+m)}, \qquad t_0 = \frac{r g_m}{(r+m) g_m + \sum_{j>m} q_{(j)}}. \tag{38}$$

The prefix satisfies $q_{(m)} \geq g_m \geq q_{(m+1)}$, with the usual empty-prefix conventions. Set $t = \mathrm{clip}(t_0, [l', u'])$. The full projection is $p^*[k] = t$ and $p^*[j] = \min(t/r, a q[j])$ for $j \neq k$, where $a \geq 0$ makes the non-k mass $1 - t$. The latter scale follows from the same sorted prefix. For logits z and $q = \mathrm{softmax}(z)$, the envelope gradient of $L_{\mathrm{cell}}^{\mathrm{KL}} = \min_{p \in \mathcal{C}_0} \mathrm{KL}(p \| q)$ is $q - p^*$.

Proof. Strict convexity gives uniqueness. With m active ratio constraints write $p[k] = rs$, $p[j] = s$ for the active non-k classes and $p[j] = a q[j]$ elsewhere. Stationarity gives $s = a g_m$; normalization yields t_0. The value function at fixed $p[k] = t$ is convex, so imposing the interval clips its unconstrained minimizer. At fixed t, the remaining KKT conditions give capped proportional allocation. Differentiation of $-\sum_j p[j] \log q[j]$ at the unique optimizer gives $q - p^*$; differentiating through the sorting step is unnecessary. □

The probability metric can underweight an important consumer decision. Take binary $p_0 = (1 - \epsilon/2, \epsilon/2)$, $q = (1 - 2\epsilon, 2\epsilon)$ and threshold $\tau = 1 - \epsilon$. The class agrees but the threshold decision flips. The buffered lower confidence is $1 - 0.95\epsilon$. Thus $L_{\mathrm{cell}}^{\mathrm{KL}} = O(\epsilon)$ and its binary-logit gradient is $O(\epsilon)$, whereas the log-odds violation tends to $\log(2/0.95) > 0$. Small probability divergence need not mean small operational change for a high-confidence threshold.

Let $s_k = z[k] - \log \sum_{j \neq k} \exp z[j]$. Define the largest positive cell violation in logit/log-odds coordinates as

$$v(z) = \max\left\{ 0, \max_{j \neq k} (z[j] - z[k] + \log r), \mathrm{logit}(l') - s_k, s_k - \mathrm{logit}(u') \right\}, \qquad L_{\mathrm{cell}}^{\mathrm{odds}} = v(z)^2, \tag{39}$$

omitting missing outer confidence bounds. Both cell losses vanish inside the same protected set and ignore a common additive logit offset. The log-odds loss avoids the saturation in the example; its gradient is not bounded, so all matched training arms use the same explicit gradient-norm cap.

For each example let δ be the smallest protected distance to an original decision boundary: $\log r$ for class changes, $\mathrm{logit}(l') - \mathrm{logit}(l)$ for an accepted lower threshold, and $\mathrm{logit}(u) - \mathrm{logit}(u')$ for a rejected upper threshold. Missing boundaries impose no restriction. If $\delta > 0$, any change of

Table 60: Matched SNR support/rank factorial, three development seeds. Both first-map families have rank ceiling 16; dense updates also have more parameters and a larger bias support. Accuracy is in percent; the gain is paired J–D, in points. These are validation outcomes, not fresh confirmation.

First map / second rank	Update params	D	J	J–D
sparse/8	5,200	26.38	27.52	+1.14
sparse/12	5,712	26.58	27.70	+1.12
dense/8	6,272	26.75	27.89	+1.14
dense/12	6,784	26.95	27.85	+0.90

the specified consumers implies $L_{\text{cell}}^{\text{odds}} \geq \delta^2$: a changed class violates a ratio constraint, while a changed threshold crosses its buffered log-odds boundary. Hence the change indicator is at most $\min\{1, L_{\text{cell}}^{\text{odds}}/\delta^2\}$. Zero-margin cases make this bound uninformative and are counted explicitly. This pointwise surrogate relation complements the finite-corpus monitor, which tests actual served decisions under their floating-point implementation.

I.7 Consumer-Cell Development Study

The preceding geometry motivated a separately planned loss comparison before shared replay. Its purpose was to preserve specified decisions without forcing every logit to equal its healthy value. All choices use healthy predictions and fitting inputs; selection never uses new labels for the old consumer. The class and two fixed confidence-threshold outputs define the primary contract. A separately declared class-only ablation removes confidence constraints from training but still scores all consumers. Standard distillation and full-feature alignment are retained comparators.

Stage A freezes saved development encoders for seeds 800–802 on both tasks. It fits affine, diagonal-quadratic and width-32 residual readers with temperature-$1/2/4$ KL, centered logit MSE, consumer-cell KL and consumer-cell log-odds objectives, plus OLS: 114 endpoints. Adam uses learning rate 10^{-3}, weight decay 10^{-4}, batch 1024, 60 epochs and a common gradient-norm cap of 5. Neither task finds a candidate meeting the 2.5% development disagreement limit for all three consumers.

Stage B jointly continues the original healthy encoder and Adam state. Each of the six records includes detached/joint controls, conditioned raw logit distillation, full-feature alignment, and four objectives at four fixed coefficients with affine/residual readers: 216 endpoints. Primary label fractions, auxiliary doses, $30/25$ epochs and batch 256 remain fixed. All arms use the same gradient cap and future orders. The separate class-only study adds 96 endpoints. Complete grids and immutable plans are included in the artifact; no test scoring enters development.

Neither task passes the primary three-consumer gate. The class-only HAR cell log-odds candidate is formally eligible, but is development-dominated by a standard temperature-1, coefficient-1 KL control: reconstruction ratios are 0.633 versus 0.626, primary differences -0.272 versus -0.136 points, and worst class disagreements 2.109% versus 2.041%. We retain that eligibility and apply an additional, explicitly retrospective futility stop before any new test confirmation. SNR has no eligible cell candidate. The proposed cell losses therefore add no confirmed training advantage over the standard controls. Independent SLSQP, envelope-gradient and serving checks verify their numerical implementation.

J Support, Learned Scope and Serving Precision

J.1 Separating Support from Rank

The dense SNR result changes the first-map family as well as the second-map rank. We therefore cross sparse versus dense first-map updates, both with rank ceiling 16, with second-map ranks 8 and 12. The original SNR development references, seeds 800–802, retain their data splits, Adam moments, 25 continuation epochs, dose 1, and future masked fitting orders. Both calibrations use the same 1024 fitting inputs and the same second-map right factors. Detached and joint arms are paired within every cell. No test metrics are computed.

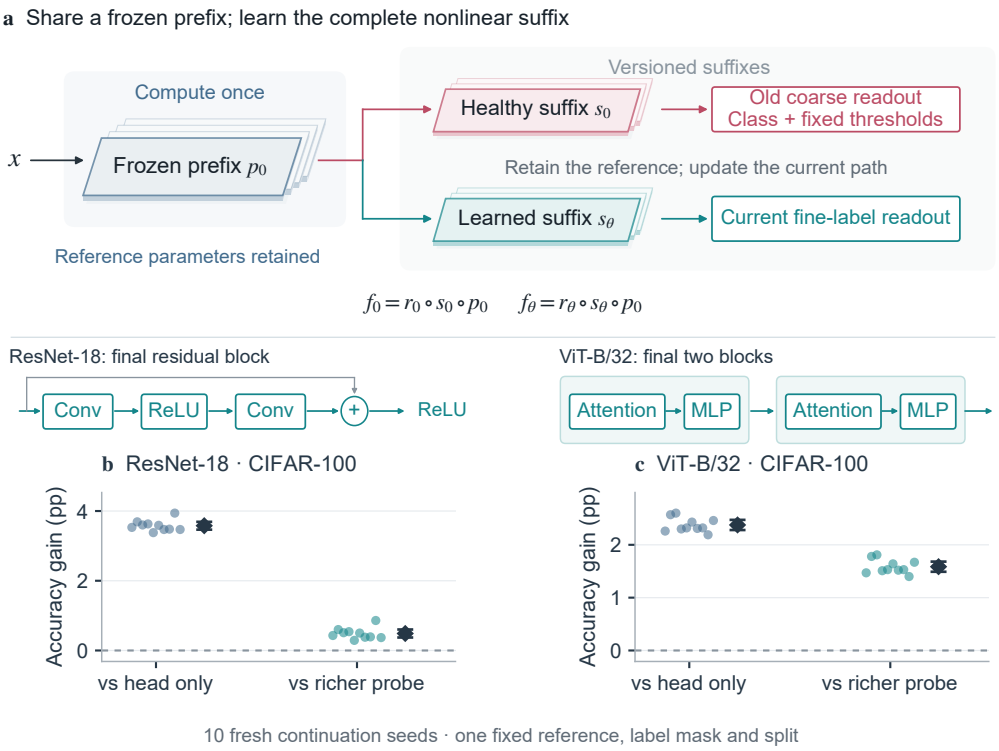


Figure 25: Shared computation with enough nonlinear learning capacity. (a) A frozen prefix feeds healthy and current suffixes; expanded views show the learned residual block or two attention blocks, with ViT normalization and residual paths omitted. The old coarse readout keeps its class and fixed-threshold rules; the current task uses fine labels. (b,c) Serving gains at batch 128 over head-only F and the richer frozen-feature probe H. Dots show all ten fresh continuation seeds; diamonds and bars show paired 95% intervals. One pretrained reference, label mask and split are shared within each backbone.

At second-map rank 8, dense minus sparse joint accuracy is $+0.37$ points with a paired three-seed 95% interval $[-0.63, 1.37]$. Raising the second rank from 8 to 12 changes sparse accuracy by $+0.18$ points and dense accuracy by -0.05. The dense first map also has 1072 more update parameters and a bias on all 64 channels rather than 16. The intervals cover zero, and the parameter-count difference prevents attributing the sparse-to-dense gain to rank alone. The SVD projection and fresh dense confirmation describe the respective saved and trained updates.

J.2 Opening a Complete Nonlinear Suffix

All three pretrained backbones use a fixed healthy reference and split: 42,952 CIFAR-100 fitting inputs, 5000 validation inputs and 2048 isolated anchors. A fixed Bernoulli mask provides 8705 fine-labeled fitting inputs. Healthy fine-100 and coarse-20 affine heads are fitted on that same subset. Coarse labels are derivable from the fine labels. Continuation does not use coarse labels or a consumer-preservation loss.

The ImageNet-V1 ResNet-18 and ImageNet-V2 ResNet-50 use 64×64 bilinear input resizing; ImageNet-V1 ViT-B/32 uses 224×224. The ImageNet input normalizer is fixed. Official torchvision weight URLs and SHA-256 fingerprints are retained. The pretrained parameter count describes the source backbone, separately from the subset updated during transfer.

The final-map screen tests unrestricted and rank-16/64/128 updates, head-only continuation F, a width-128 residual GELU head N, and a linear probe H on healthy features concatenated with pooled pre-projection features. None of its three-epoch pilots reaches the two-point gain over F. Refitting primary readers with the original standardized LBFGS/L2 recipe recovers much of the online-head

Table 61: ResNet-18/CIFAR-100 test utility after the publicly frozen three-seed development choice. Ten fresh continuation seeds share one healthy reference and split. Accuracies are in percent; U–control intervals are paired 95% intervals in points. H is a stronger frozen-feature readout, so its improvement does not establish encoder learning.

Arm	Test accuracy	U–control [95% interval]
Head only (F)	43.87	+3.58 [+3.47, +3.69]
Residual head (N)	44.49	+2.96 [+2.74, +3.17]
Richer frozen probe (H)	46.96	+0.49 [+0.37, +0.60]
Final map (L)	44.22	+3.23 [+3.10, +3.35]
Complete nonlinear suffix (U)	47.45	–

decline, without clearing that gate on any backbone. These negative screens remain separate from the following ten-epoch recipe.

The complete-suffix study opens both convolutions of ResNet-18's final identity block, all three maps of ResNet-50's final identity block, or both maps of ViT's final CLS MLP. Normalization parameters stay fixed. For CNNs, the suffix BN transforms are folded before defining its two versions; their deviation is measured against the pre-existing reference arithmetic. Earlier computation is frozen and may be cached during training. Inference evaluates the actual prefix.

Training uses ten AdamW epochs, weight decay 10^{-4}, current-head rate 10^{-4} and suffix/factor rate 3×10^{-5}. CE includes 10^{-4} times squared primary-head weights and bias; N includes its trainable linear layers in that penalty. This explicit head regularization differs from AdamW decay. Conditioning uses the two-view healthy fitting mean and SD, floored at 0.1 times median positive SD. The initial affine head is rebased to retain its function in real arithmetic. The old readout is unchanged.

Views are clean input and one fixed CIFAR training augmentation per labeled image: reflection padding 4, a uniform 32×32 crop, and horizontal flip with probability $1/2$, using RNG 3901 in fitting order. All learned arms use the same original fitting-index orders (seed$+20000$) and view choices (seed$+71000$). Final primary readers refit the same two views and labels, with no encoder changes. Online heads and final refits are retained as distinct objects. The final-map-only L control uses the same recipe; R64 permits rank-64 changes to every suffix map.

Development requires at least two accuracy points over healthy and F, within half a point of U, N and H, and at most 1% disagreement for each old class/selective consumer. ResNet-18 U passes all three seeds 3800–3802; the choice is publicly frozen before test loading. Ten fresh continuation seeds 3810–3819 then gain 3.58 points $[3.47, 3.69]$ over F and 0.49 $[0.37, 0.60]$ over H (Table 61). U exceeds L by 3.23 points. Intervals describe continuation variability conditional on the healthy reference, label mask and split.

H reaches 46.96% test accuracy without changing the encoder; U reaches 47.45%. H improves the readout on frozen features; U learns the nonlinear suffix and adds the smaller gain shown against H. The cached-prefix FP32 U path changes none of the three old decisions on any test input; all 30 specified U contracts certify the five-point remainder budget, with largest upper bound 0.0108. Full-prefix implementation checks are reported separately from this numerical representation comparison.

J.3 How Much of the Transformer Must Learn?

The complete final-MLP pilot does not reach the utility gate on ViT-B/32. Opening its complete final attention block raises validation accuracy to 75.64%, against 73.64% healthy, 74.70% F and 75.06% H. The 0.94-point gain over F still fails. Neither screen evaluates official test data. A final fixed scope check opens the last two blocks, freezing the first ten and retaining every LayerNorm parameter. The first suffix block processes all tokens; the last needs only its CLS query and MLP. Independent full-transformer output and input-gradient checks verify this pruning, including dynamic-batch TorchScript evaluation.

Table 62: ViT-B/32/CIFAR-100 test utility after the public two-block choice. Ten fresh continuation seeds share one reference and split. Accuracy is in percent; U–control intervals are paired 95% intervals in points.

Arm	Test accuracy	U–control [95% interval]
Head only (F)	75.07	+2.38 [+2.28, +2.47]
Residual head (N)	75.17	+2.27 [+2.16, +2.39]
Richer frozen probe (H)	75.86	+1.59 [+1.49, +1.68]
Final MLP (M)	75.79	+1.66 [+1.54, +1.78]
Two complete blocks (U)	77.45	–

Under the same views, rates, loss and final-refit recipe, U reaches 77.04%, 77.00% and 76.96% on development seeds 3800–3802. All three clear the fixed gate; M, which opens only the final MLP maps, remains near 75%. This choice is publicly fixed before official test loading. The scope comparison isolates the nonlinear learning capacity required by this transfer recipe.

Ten fresh continuations 3810–3819 reach 77.45% test accuracy. The paired gain over F is 2.38 points $[2.28, 2.47]$; over H it is 1.59 $[1.49, 1.68]$. U also exceeds the final-MLP M control by 1.66 points. All 30 class/selective contracts in this cached-prefix FP32 comparison certify, with zero observed changes and maximum remainder bound 0.0108. As in the ResNet-18 study, the reference, label mask and split are fixed.

J.4 Matched CPU Implementations

For the small MLPs, a fused float32 C++ kernel uses the same transposed weight layout, SIMD/compiler flags and preallocated buffers for protected and current-only execution. Empty binding cost and C-internal repetition timing are retained separately. Development chooses the fastest absolute implementation: native at batch 1, TorchScript at 1024. This choice is frozen before independent full-corpus serving checks on the ten saved HAR and dense SNR states. The original TorchScript and slower native large-batch paths remain in the evidence. Absolute runtime determines selection; each ratio uses the matched current-only implementation. Both selected implementations retain every old class/selective decision on the full small-model corpora. Independent current merging differs on two near-tied predictions in the optimized audit; real-arithmetic equality does not imply bitwise identity. Every primary metric uses its stated arithmetic.

For vision, monolithic, modular-prefix and grouped dense suffix programs compete against equivalently optimized one-version current programs. Grouped convolutions and batched projections apply to both one- and two-version implementations. FP32 ResNet-18 development serving is about $1.12\times$ at batch 1 and $1.20\times$ at 128; ViT is about $1.12/1.112\times$. The full-precision batch-128 results miss the 1.10 target. H reuses a healthy trace for its own richer features and old head, so it has a different, cheaper frontier.

Timings use an isolated twelve-thread CPU process, twelve warmups and 31 alternating repetitions. Current-only, protected and independent function evaluations receive the same preprocessing and pruning. Input decoding/preprocessing, compilation and warmup are outside forward latency. The Intel Xeon Platinum 8570 runs PyTorch $2.7.1$ and torchvision $0.22.1$ on CPU. AVX-512, BF16 and AMX capability are recorded; individual kernel dispatch is not profiled. The chosen ABI is fixed on development, not selected on test timing. Full chosen-code replay is separated from isolated timing; identical H functions are counted once, with each seed's monitor certificate recomputed from its actual saved output arrays. Untimed frozen-prefix reuse checks reference, source, corpus and state fingerprints.

Two export jobs fail before timing because a reference-only comparator registered the same readout tensor twice. A subsequent completed ResNet-18 record is retained despite a queue-envelope bookkeeping error; only the pending ViT job is reissued. These technical events, exact executed source versions and logs are retained. They do not change training or selection.

During the separate full-corpus replay, two ViT jobs exceed an author-set 8 GiB RSS ceiling; the queue then interrupts another job. The host is not exhausted: approximately 2.7 TB remains avail-

Table 63: Direct old coarse readout on updated visual features: descriptive means over the three fixed development seeds. All quantities are percentages on the same 5000 validation inputs. Accuracy can improve while formerly correct predictions regress. Labels diagnose flows here; they do not fit or select a repair.

Backbone	Healthy acc.	Direct acc.	Changed	Correct → wrong	Wrong → correct
ResNet-18	54.00	55.79	19.88	4.75	6.54
ViT-B/32	82.66	82.89	9.29	3.40	3.63

able at inspection. After the resource clarification, all completed ResNet-18 records are retained and the missing ViT endpoints run with the unchanged numerical programs, at up to four twelve-thread workers. Fixed RSS ceilings are replaced by a 256 GiB host-available reserve. This verifier holds full-corpus features and several programs; its RSS is distinct from single-program deployment storage or activation memory. Models, input counts, decisions, monitors and the already completed isolated timings do not change.

J.5 Precision Belongs to the Consumer Contract

The protected-only precision probe keeps the current function and shared prefix at FP32. Reference matrix products use BF16; their results return to FP32 before adding the original biases. Norms, attention, activations, residuals and the frozen affine consumer remain FP32. This changes the reference implementation, so its claim is a measured decision contract, not the exact-function identity of the FP32 construction.

On 5000 development inputs, ResNet-18 class/80/50 disagreements are $0.32/0.28/0.18\%$; ViT disagreements are $0.02/0.06/0.08\%$. Both pass the same one-percent screen, using only paired outputs for precision selection. No labels, fitted temperatures or output corrections select the precision. Maximum logit errors are 0.104 and 0.087, respectively; the class and selective decisions are checked directly.

Standard whole-encoder CPU AMP, retaining the FP32 affine consumer, is a stronger numerical control. It changes ResNet-18 class/80/50 decisions on $1.28/1.72/0.98\%$ of inputs, missing the screen even though legacy coarse accuracy rises from 54.00% to 54.10% and aggregate coverage remains close. ViT's $0.20/0.50/0.26\%$ passes the decision screen and proceeds to the fixed three-state utility control. Its matched lower-precision current-only baseline is reported separately from a fixed-current-FP32 ratio.

These numerical operations are standard. Low-bit language-model studies also show decision-dependent margin changes (Wu et al., 2026). Here the precision choice tests a named legacy interface alongside retained current utility and measured CPU cost. A small aggregate metric change supplies no replacement for the class and fixed-threshold audits.

J.6 Does an Unlabeled Affine Repair Suffice?

Before ViT test loading, we evaluate the existing nine-arm repair hierarchy on all three fixed U development states for each selected backbone. The original 2048 anchors are disjoint from model fitting and validation. Healthy and current features of these same images fit the maps, without labels. All estimators retain the original variance convention, ridge 10^{-3} and covariance eigenvalue floor. A raw-coordinate affine map is composed into the old readout, leaving the current fine-label function unchanged. This offers the cheaper one-encoder/two-head control, with paired-feature access distinguished from marginal statistics.

No arm passes all three consumers in every seed (Figure 26). The best worst-consumer arm is Procrustes on ResNet-18, with 24.0% disagreement; paired ridge reaches 9.16% on ViT. The direct old readout on new features reaches $25.86/15.44\%$ worst-consumer disagreement. With 2048 anchors and the fixed estimator family, affine repair leaves substantial residual disagreement. Retaining the healthy suffix avoids that residual in this transfer setting.

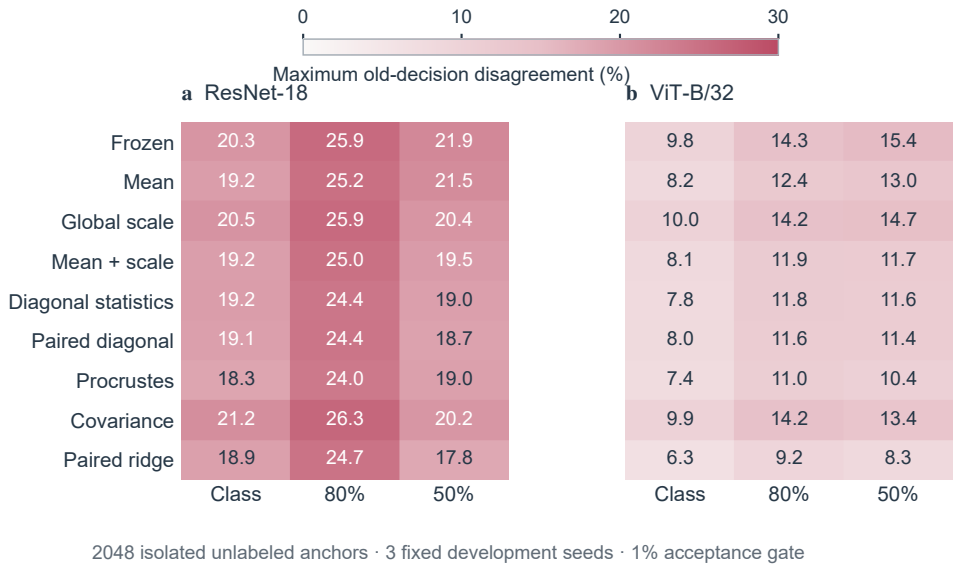


Figure 26: Complete development-only affine repair screen. Each cell is the largest disagreement across the three fixed continuation seeds, on all 5000 validation inputs. Class and the two original fixed confidence thresholds are distinct consumers; 80/50 denote their healthy fitting coverage targets. Every arm misses the one-percent acceptance gate. No official test outcome selects a repair, and no failed arm proceeds to test.

Table 64: Fixed one-forward distillation follow-up, development seed 3800 only. A learned coarse head is initialized to the healthy affine function. Classification and selective columns report disagreement in percent; fine accuracy is validation percent after the same final refit. None meets the fixed joint utility/consumer gate; no test evaluation follows.

Backbone	λ	Fine accuracy	Class	Selective 80	Selective 50
ResNet-18	1	46.80	14.78	18.58	13.84
ResNet-18	10	43.88	12.84	17.50	13.32
ResNet-18	100	43.24	12.80	18.08	13.40
ViT-B/32	1	76.50	8.60	12.88	12.24
ViT-B/32	10	76.30	8.10	11.24	11.34
ViT-B/32	100	76.10	8.26	11.20	12.70

Direct coarse accuracy need not fall when that interface changes. Across these three development seeds it rises from 54.00% to 55.79% on ResNet-18; ViT remains close at 82.89% versus 82.66% healthy. Yet 4.75% and 3.40% of inputs, respectively, move from correct to wrong. The corresponding beneficial corrections are 6.54% and 3.63%. Thus aggregate accuracy and preservation of specified decisions measure different effects of the update.

J.7 A Cheaper Learned Legacy Head

A subsequent control asks whether a learned affine coarse head can avoid the retained reference suffix. This follow-up is planned after opening the unconstrained test results and uses development only. It leaves the original models and serving choices fixed. Initialize the added head to the healthy coarse function, then train it and the same current suffix with fine-label CE/L2 plus $\lambda\,\mathrm{KL}(p_0\|p_\theta)$, temperature one. Targets are healthy coarse predictions on the same 8705 labeled fitting inputs and two views; coarse labels are not used. The current head/suffix rates, ten epochs, explicit L2, AdamW decay and final fine-head refit are unchanged. The added head uses rate 10^{-4} without decay or explicit L2. It adds 10,260 parameters on ResNet-18 and 15,380 on ViT.

Table 65: Actual selected CPU serving, ten fresh continuation seeds. Ratios use an independently optimized fixed FP32 current-only function. Intervals describe seed-paired timing ratios, not independent timing repeats. Mixed precision changes only the old suffix; full precision is a separate fixed control. Forward latency excludes preprocessing and compilation.

Backbone / batch	Reference	Current (ms)	Protected (ms)	Ratio [95%]
ResNet-18 / 1	FP32	1.467	1.652	1.126 [1.121, 1.131]
ResNet-18 / 1	Selected	1.467	1.652	1.126 [1.121, 1.131]
ResNet-18 / 128	FP32	34.227	41.994	1.228 [1.201, 1.256]
ResNet-18 / 128	Selected	34.227	38.231	1.118 [1.095, 1.140]
ViT-B/32 / 1	FP32	15.824	17.745	1.122 [1.114, 1.129]
ViT-B/32 / 1	Selected	15.824	18.080	1.143 [1.136, 1.150]
ViT-B/32 / 128	FP32	950.892	1035.193	1.089 [1.056, 1.122]
ViT-B/32 / 128	Selected	950.892	1022.693	1.076 [1.057, 1.095]

All fixed weights $\lambda \in \{1, 10, 100\}$ fail the joint gate on pilot seed 3800, ending this control at development (Table 64). A read-only initialization audit reconstructs both initial functions on all $17{,}410$ fitting views and 5000 validation inputs: initial KL is below 10^{-5}, and every validation class/selective decision agrees with the original reference. This confirms that the disagreement emerges during learning, rather than from inconsistent teacher targets or cached features.

J.8 Actual Serving Outputs and Isolated Cost

After the final precision screen, the fastest passing implementation is fixed separately at batches 1 and 128, with an unchanged FP32 current function. ResNet-18 selects its monolithic FP32 path at batch 1 and old- suffix autocast at 128. ViT selects manual BF16 reference products at 1 and old-suffix autocast at 128. The fastest full-precision reference implementation is also fixed as a separate control. No implementation, threshold, precision or training hyperparameter is selected on test results.

The final standard whole-AMP control remains important: at ViT batch 128 it takes 647 ms against its own 580 ms current-only baseline, a 1.115 ratio, versus approximately $1003/941$ ms for selected reference- only autocast at fixed-current FP32. Whole AMP is faster in absolute time but misses its matched 1.10 gate in the final development run. Its earlier timing run passed. Both records are retained, and it stops at development. The selected mixed implementation trades some of that absolute speed for FP32 current outputs.

Every selected endpoint is reconstructed from its saved state, then run from image tensors through the actual frozen TorchScript prefix and chosen suffix code at both fixed batch sizes. F/N/H/L or M controls use the same physically evaluated FP32 prefix. The original healthy coarse logits remain the reference. Full-precision replay and the approximate reference receive separate class, selective, NLL/ECE and logit-error checks. The fresh monitor uses RNG $14700001 + \text{seed}$, 1024 inputs and tail allocation $0.05/2000$; the one-percent empirical and five-percent remainder gates are unchanged. Full-corpus output arrays are recovered for local recalculation of accuracy, disagreements, decision flows and certificates. Local replay recalculates these metrics from the recovered outputs; the complete backbone forward is checked by the numerical driver. Serialized program sizes are collected after latency sampling, counting a modular prefix once. They measure exported code/constants, separately from runtime activation memory and process RSS; no size measurement selects an ABI.

ResNet-18's selected actual serving has maximum class/selective disagreement 0.39%, with $60/60$ checks certified and largest remainder bound 0.0233. Its full-precision control has maximum disagreement 0.00% and largest logit error $8.68e-05$. Each set contains three rules, two batch implementations and ten current endpoints.

At batch 1, current/selected/full-reference serialized programs occupy $42.8/60.9/60.9$ MiB. At batch 128, current/selected/full-reference serialized programs occupy $42.8/51.9/60.9$ MiB. These export sizes exclude activations and runtime libraries.

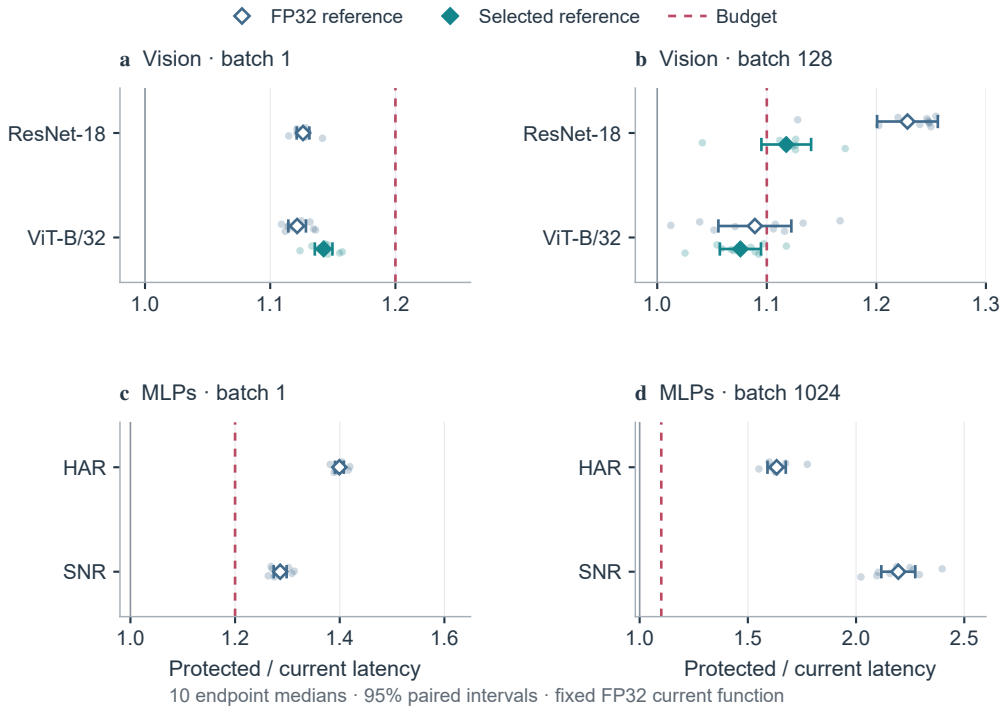


Figure 27: Matched actual serving cost on the same CPU. Points show ten endpoint medians per implementation; diamonds and bars show means and paired 95% intervals. Top panels resolve vision overhead at batches $1/128$; bottom panels show the larger MLP overhead at $1/1024$, on separate scales. Dashed lines mark the unchanged $1.20/1.10$ budgets. Identical selected and full-precision programs appear once. Each denominator is the development-selected current-only implementation with the same FP32 current function. MLP batch-1 uses a fused single-thread C++ kernel, large-batch MLP uses four-thread TorchScript, and vision uses twelve-thread TorchScript. The exact-function construction and approximate-reference decision contract remain separate claims.

ViT-B/32's selected actual serving has maximum class/selective disagreement 0.14%, with $60/60$ checks certified and largest remainder bound 0.0178. Its full-precision control has maximum disagreement 0.00% and largest logit error $7.25e-05$. The same three rules and two batch implementations are checked at every endpoint.

At batch 1, current/selected/full-reference serialized programs occupy $334.0/361.1/388.1$ MiB. At batch 128, current/selected/full-reference serialized programs occupy $334.0/361.0/388.1$ MiB. These export sizes exclude activations and runtime libraries.

J.9 Conditional Reference Routing

We test whether referring only difficult inputs to the healthy suffix can reduce preservation cost. Cascades and learned deferral are established methods (Jitkrittum et al., 2023); here routing affects only the old output, leaving the complete FP32 current function fixed.

For cheap logits z with winner k, define $s_k = z[k] - \log \sum_{j \neq k} e^{z[j]}$. The nearest decision margin is the minimum of the winning-class gap and $|s_k - \mathrm{logit}(\tau)|$ over the fixed thresholds. Let $\omega = \max_j (z_0[j] - z[j]) - \min_j (z_0[j] - z[j])$. If the margin exceeds ω, the class and all threshold outputs agree with the reference z_0: each class margin changes by at most ω, and log-sum-exp gives the same bound for the winning-class log-odds. Common logit offsets cancel. A fitted estimate of ω is a routing score, not a pointwise certificate.

Exact fallback removes errors only on the referred set. For each nonempty three-consumer error pattern S, let p_S be its corpus mass and f_S the mass referred. The fractional oracle minimizes

$\sum_S f_S$ subject to $0 \le f_S \le p_S$ and $\sum_{S:j\in S} f_S \ge e_j - \epsilon_j$ for each consumer's initial error e_j and budget ϵ_j. Its seven pattern variables account for overlapping errors. The optimum lower-bounds integral referral for that fixed reader under unit cost; it says nothing about measured batch latency or a different reader family.

The fixed screen uses all six saved visual development endpoints. An isolated $2{,}048$-anchor permutation provides $1{,}024$ reader-fitting, 512 gate-fitting and 512 cutoff-calibration examples. An affine ridge reader and its width-64 GELU residual extension predict healthy logits without new labels. Four gates use confidence, boundary distance, distance divided by predicted residual, or learned disagreement. Ten calibration-quantile cutoffs yield 80 policies per endpoint, 480 in total. The protocol advances a family only if every seed meets one-percent disagreement for all three consumers and its mean referral is at most 50%.

Table 66: Mean development reference-referral fractions (%). Every tested gate shown meets the one-percent class/selective disagreement limit on all three seeds, but exceeds the fixed 50% referral limit. The fractional oracle fixes the affine reader; the tested columns allow both affine and residual readers. It assumes unit fallback cost and is not a latency bound.

Backbone	Affine oracle	Confidence gate	Best tested gate
ResNet-18	43.67	77.62	76.98
ViT-B/32	29.39	78.40	78.40

Neither backbone qualifies. The least-referral ResNet family uses the residual reader and normalized boundary gate; ViT's confidence gate ties the best boundary family. Both need about four fifths of inputs referred. No test or latency confirmation follows. Local replay reconstructs all 480 policy outputs from the recovered cheap/reference logits and gate scores.

J.10 A Real Visual Auxiliary on New Fitted References

This follow-up targets the gap between supervised visual transfer and actual auxiliary benefit. It uses the same ImageNet weights and learned suffix scopes, but independently fits the healthy fine/coarse readers at reference seed 6200. A new permutation reserves $10{,}000$ fitting inputs, $5{,}000$ validation inputs and $2{,}048$ isolated anchors from the official training population. A fixed Bernoulli-0.2 mask selects fitting labels. Clean and fixed augmented views share an FP32 frozen-prefix cache. No official test input is loaded.

A 256-unit ReLU decoder reconstructs bilinear, antialiased 8×8 RGB images from terminal features. Fitting-target variance normalizes MSE. Five detached warm-up epochs initialize the decoder; ten continuation epochs use CE on labeled inputs and reconstruction on all fitting inputs. The fixed auxiliary weights are $0.01, 0.1, 1$, with identical initialization, orders and views. The CE-only encoder has a separate dose-matched detached decoder for each comparison. Final primary readers use the same two-view labeled pool; online-head and refitted scores are recorded separately.

The initial ResNet pilot was excluded after an optimizer-state alias check. The unchanged protocol was reissued with independent copied moments and warm-state hash checks. Executed sources and technical failures remain in the artifact. Separate saved-model processes replay the valid branches and healthy consumers.

The first-pilot rule requires at least 0.20 points of primary gain and an MSE ratio at most 0.95, with unchanged old decisions, before extending a weight to two further development seeds. Neither backbone passes. ResNet's CE-only branch also trails H by 1.02 points. ViT's CE-only branch exceeds H by 2.38 points, triggering the separately specified reference-replication study. These six auxiliary outcomes remain development results; no auxiliary test confirmation is run.

J.11 Confirmation Across Fitted References and Partitions

The predeclared fallback confirms CE-only ViT transfer at reference seeds 6201–6204, with continuation seeds 6310–6312 nested within each. Each reference independently permutes the official training population, selects its label mask and fits healthy fine/coarse readers. The ImageNet pretraining, two-view construction, suffix scope and training recipe remain fixed. Validation labels

Table 67: Real reconstruction auxiliary on the independently fitted development reference, one continuation seed per backbone. Gains compare the final refitted primary reader with CE-only detachment D; MSE ratios use its dose-matched detached decoder. Extension requires gain ≥ 0.20 points, ratio ≤ 0.95 and unchanged old decisions. No coefficient passes.

Backbone	Auxiliary weight	Primary gain (points)	MSE ratio	Extend
ResNet-18	0.01	+0.00	0.996	No
ResNet-18	0.1	+0.06	0.975	No
ResNet-18	1	−0.10	0.897	No
ViT-B/32	0.01	+0.12	0.919	No
ViT-B/32	0.1	+0.04	0.929	No
ViT-B/32	1	−0.52	0.864	No

score the frozen confirmation protocol; the official test split stays closed. These groups vary reference fitting, label masks and partitions under the same pretrained backbone.



Figure 28: Development auxiliary outcomes and independent-reference transfer. (a) All six real-auxiliary pilot candidates; labels give the auxiliary weight. Shading marks the gain/MSE region required for development expansion. Neither backbone enters it. (b) ViT transfer over the richer frozen-feature reader H. Circles show three new continuation seeds per fitted reference; teal diamonds show their means. The dark diamond and interval use the four reference means, not twelve pooled continuations. No official test inputs are used.

Table 68: ViT-B/32 transfer confirmation across four fitted references and train/validation partitions. Each entry averages three continuation seeds. F is an affine frozen-feature reader, N a learned nonlinear reader, H the richer frozen-feature reader, and D the CE-only learned suffix. All use the same fitting labels and final reader procedure. Accuracies are percentages; D−H is in percentage points. The ImageNet pretrained weights remain shared.

Reference	Labels	F	N	H	D	D−H
6201	2011	68.80	68.63	68.64	70.57	+1.93
6202	1996	69.06	69.00	69.08	70.36	+1.28
6203	1980	68.22	68.16	68.80	69.97	+1.17
6204	2002	68.84	68.91	69.30	70.45	+1.15

The primary contrast D−H averages 1.39 points, with 95% interval $[0.80, 1.97]$. The four reference means, each averaging three paired continuations, are the replication units (t, three degrees of freedom). Relative to the highest mean F/N/H score within each reference, the gain is 1.35 points

$[0.88, 1.81]$. All continuation seeds and reference means appear in Figure 28. Transfer remains positive across these reference and partition changes. Saved-model replay retains all three old decision rules on the new endpoints; the largest legacy-logit rounding difference is 1.05×10^{-5}. The pretrained backbone and training population remain shared, and serving costs come from the original paired study. The visual auxiliary screen remains a separate negative result.

J.12 Cross-View Auxiliary Learning on ViT-B/32

We continue the final two attention blocks on two fixed views of CIFAR-100. The frozen prefix, label mask, update schedule and final reader fitting match the supervised control D. Each reference uses $10{,}000$ fitting images with labels on 20%, $5{,}000$ held-out validation images, and a separate $2{,}048$-image anchor pool. The auxiliary applies NT-Xent with temperature 0.2 directly to the normalized terminal features. It adds no learned projection or annotations. The selected view supplies the primary cross-entropy and its counterpart supplies the paired contrastive view.

Weights 0.01, 0.1, and 1 yield pilot gains of 0.18, 0.50, and 0.90 points. The two qualifying weights extend to two further development seeds. Their three-seed mean gains are 0.56 and 0.99 points, and their mean loss ratios to D are 0.809 and 0.734. The fixed rule selects the smaller qualifying weight. Independent replay reproduces the pilot D weights and logits from the earlier supervised run. This check supports reuse of its matched D endpoints in confirmation.

Across four fitted references and three continuations per reference, the paired primary gain is 0.84 points $[0.66, 1.03]$ and the auxiliary loss ratio is 0.805 $[0.794, 0.816]$. Intervals use the four reference means. The references share one ImageNet-pretrained checkpoint. All comparisons use the held-out training-population splits.

Two supervised controls match the training suffix-forward budget. D2 averages cross-entropy on both views for ten epochs. D20 continues the original one-view objective for twenty epochs. Each arm processes $200{,}000$ image views through the trainable suffix in $1{,}580$ minibatch forwards. D2 and joint take 790 optimizer steps, and D20 takes $1{,}580$. The contrastive objective additionally computes paired-feature similarities. Across the full four-reference, three-continuation grid, joint gains 0.73 points $[0.29, 1.16]$ over D2 and 0.53 points $[0.20, 0.86]$ over D20. Their auxiliary loss ratios are 0.791 $[0.781, 0.800]$ and 0.785 $[0.777, 0.793]$. All 24 control accuracies are independently recomputed from the saved validation outputs. Elapsed times are recorded for their separate numerical executions.

We repeat the two arms with AugReg ViT-B/32 weights pretrained on ImageNet-21k and fine-tuned on ImageNet-1k (Steiner et al., 2022). The four pairs reuse the fitting examples, label masks, views, input normalization and continuation seed 6310. Both arms start from the same newly fitted reference in each pair. We audit the retained FP32 suffix against the original unsplit AugReg encoder on every validation image. The gain is 0.60 points $[0.15, 1.05]$ and the contrastive loss ratio is 0.915 $[0.910, 0.921]$. Restricting the TorchVision comparison to seed 6310 gives 0.88 points $[0.65, 1.11]$ and a loss ratio of 0.806 $[0.797, 0.815]$. The unchanged coarse head applied directly to the updated features changes 6.92% of class decisions on average. The retained reference changes zero of the $60{,}000$ audited decisions across the three rules, with a maximum FP32 logit difference of 1.15×10^{-5}.

On the pilot split, using both views for cross-entropy reaches 70.62% and extending one-view training to twenty epochs reaches 70.92%. The ten-epoch D and selected joint runs reach 70.94% and 71.44%. Widening a frozen-feature residual decoder to $1{,}024$, $2{,}048$ and $4{,}096$ hidden units gives 68.46%, 67.90% and 66.96%. These comparisons use the same labeled examples and reader-fitting recipe.

We rerun the image prefix and evaluate both readers on its physical outputs. The unchanged coarse head applied directly to the updated features changes 7.35% of class decisions on average. With the retained reference suffix, full-precision execution changes none of the audited old decisions. The selected lower-precision reference has at most 0.18% disagreement. Both variants preserve all audited current classifications. Each variant passes the five-point remainder bound in all 72 checks, covering twelve endpoints, two serving implementations and three old decision rules. The full-corpus output audit uses chunks of 128. An additional 128 individual batch-one inputs per

endpoint check the batch-one implementation. Instance retrieval between the paired validation views improves by 0.69 points $[0.60, 0.77]$ over D in the secondary analysis.

Isolated CPU timing uses one TorchVision endpoint per reference, twelve warmups, and thirty-one interleaved forward passes on preprocessed inputs. At batch one the original selected program costs $1.139\times$ $[1.124, 1.154]$. Its joint trace at batch 128 gives $1.098\times$ $[1.020, 1.176]$. We repeat batch 128 with the same compiled prefix and FP32 current suffix in the one-output and two-output programs. The added BF16 reference costs $1.078\times$ $[1.069, 1.088]$ in total, with unchanged current classifications and at most 0.18% old disagreement. The FP32 reference costs $1.105\times$ $[1.092, 1.119]$ and preserves every audited decision. The BF16 interval fits the 1.10 budget on this CPU (Figure 29).

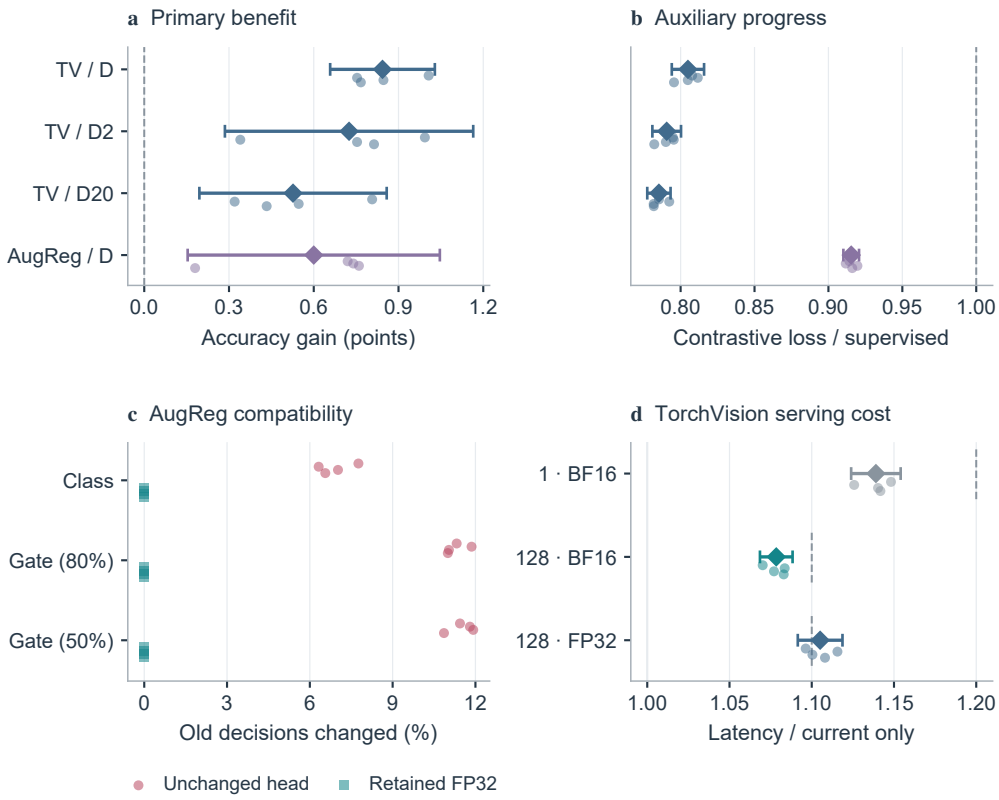


Figure 29: Cross-view utility and compatibility. (a,b) Accuracy gains and loss ratios against one-view (D), two-view (D2) and twenty-epoch (D20) supervision. Points average three seeds per TorchVision reference (TV) and use seed 6310 for AugReg. Diamonds show means and paired 95% intervals. (c) Old decisions on AugReg. Gates target 80% and 50% coverage on fitting data. (d) Native latency on TorchVision endpoints. Batch 128 holds the FP32 current graph fixed. Dashed segments mark the latency limits.

J.13 Small-Budget Estimation of Consumer Disagreement

The finite-corpus check answers whether a candidate meets a disagreement budget. A small sample can also estimate the amount of disagreement, although direct binary counts have substantial sampling variation. We test whether statistical propagation through the cheap reader improves this estimate. Conditional-expectation variance reduction and model-assisted estimation are established tools; prediction-powered inference (Angelopoulos et al., 2023) provides related context for combining inexpensive predictions with exact samples. Here we evaluate point-estimation error, separately from the earlier certificate.

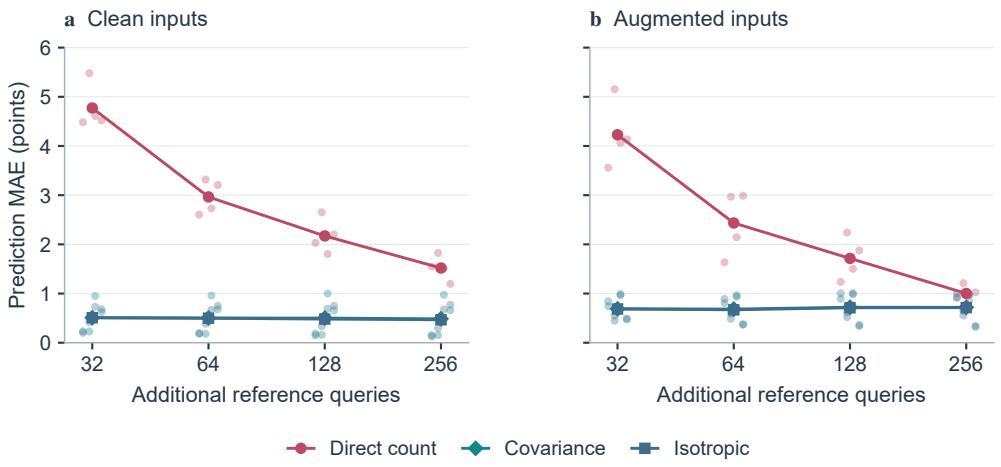


Figure 30: Reference-query budget and disagreement-estimation error. Solid markers average four fitted references; pale points show every reference mean, each averaging three continuations and twenty monitor subsets. Budgets count additional reference calls after the common $1{,}024$ map-fitting anchors. The two propagation models additionally use 512 cheap current outputs. (a) Clean validation. (b) Fixed augmented views with clean monitor anchors. The isotropic control ties full covariance within uncertainty.

Let Z be repaired current logits and E the healthy-minus-current logit residual. A known conditional law gives $m_j(Z) = \Pr\{C_j(Z+E) \neq C_j(Z) \mid Z\}$ for consumer j. Then $\mathbb{E}m_j(Z) = \Pr\{C_j(Z+E) \neq C_j(Z)\}$ and the conditional-variance identity gives $\mathrm{Var}(m_j(Z)) \leq \mathrm{Var}(\mathbf{1}\{C_j(Z+E) \neq C_j(Z)\})$. We approximate the residual law by a shared Gaussian mean/covariance estimated from a small monitor, with 10% shrinkage toward an isotropic covariance. An isotropic comparator retains the same mean and covariance trace.

An exploratory screen on all six earlier visual development endpoints fixes the recipe before confirmation. Each repair uses $1{,}024$ isolated anchors to fit a standardized centered-logit ridge reader with coefficient 10^{-3}. A separate $1{,}024$-anchor pool supplies monitor subsets of size $32/64/128/256$. Both estimators use the same subsets. Propagation averages 64 fixed Gaussian draws over 512 inexpensive current outputs; direct counting uses the observed old/current consumer differences. No consumer labels enter either estimator. The additional monitor-query budget is separate from the initial $1{,}024$ paired map-fitting outputs, which can be cached across updates.

Confirmation uses the twelve saved CE-only ViT endpoints from four independently fitted references and partitions, with three continuation seeds per reference. Their training accuracies were previously observed, but the new repair and risk outcomes were computed after freezing this analysis. The primary target is the maximum class/selective disagreement over $5{,}000$ clean validation inputs. Twenty monitor subsets measure each endpoint's sampling variation. Intervals use four reference means, each averaging three continuations and their subsets (t, three degrees of freedom). The official test split remains closed.

At 32 additional queries, covariance propagation reduces MAE from 4.773 to 0.509 points; the paired reduction is 4.264 points, 95% interval $[3.148, 5.380]$. At 64 queries, MAE falls from 2.964 to 0.502, a reduction of 2.462 points $[1.573, 3.352]$. Both pass the fixed 20% improvement and positive-lower-interval criteria. With augmented validation inputs and clean monitor anchors, the corresponding MAEs are $0.690/0.683$ against $4.228/2.434$, with positive improvement intervals at both primary budgets. The augmented-view intervals cross zero at $128/256$ queries.

At $32/64$ queries the isotropic model obtains $0.508/0.499$ points, close to full covariance. Averaging errors across consumers gives clean MAEs of $0.772/0.779$ versus $3.488/2.514$ for direct counting. Independent rate-level replay reproduces the forecasts.

The estimate depends on the joint law. For equally likely binary logits $Z = \pm a$, the residuals $E = 2Z$ and $E = -2Z$ share a marginal law and yield zero and complete sign disagreement, respectively. The observed confirmation disagreements are 6.20–7.96%, above the 1% repair limit. The supplementary archive supplies the arrays and replay.

J.14 Reusing the Fitting Anchors

The ridge reader already uses $1{,}024$ paired anchors. Its mean leverage is 0.741 on the supervised continuation endpoints, making fitted residuals substantially smaller than leave-one-out residuals. With the feature normalizer and ridge penalty held fixed, the standard leave-one-out identity rescales residual e_i to $e_i/(1 - h_i)$, where h_i is the diagonal of the ridge hat matrix including the intercept. The identity agrees with independent row-deleted refits.

On these endpoints, clean-corpus MAE falls from 4.85 points for in-sample decision counts to 1.09 for leave-one-out counts and 0.49 for isotropic propagation of the leave-one-out residuals. The latter uses the original fitting outputs and no additional reference queries. The original 32-query isotropic estimate has MAE 0.51 points. The leave-one-out result on these endpoints is a retrospective control.

We fix the leave-one-out estimator before examining repair risk on the twelve cross-view continuations. The clean-corpus MAE is 0.47 points with zero additional reference queries and 0.49 with 32 queries. The paired difference is -0.02 points with 95% interval $[-0.27, 0.22]$, within the fixed 0.25-point noninferiority margin. On augmented inputs the MAEs are 1.48 and 1.28 points, with a paired difference of 0.20 points $[0.10, 0.31]$. These fits reuse the same reference models and the same initial $1{,}024$ paired accesses. The clean-corpus disagreement of the repaired outputs ranges from 7.92% to 9.30%.

On the clean saved-model corpus, exact stratified audits take 129.60 queries to reject a 1% budget, compared with 78.67 under uniform sampling. Holding the residual marginal fixed and reassigning residuals to current logits yields constructed disagreement ranges spanning 39.49 points on average.

J.15 Early Decisions for a Planned Encoder Update

On CIFAR-10 (Krizhevsky, 2009), we freeze the ResNet-18 prefix and continue the final block for eight epochs. The deployed consumer is the initial ten-class reader and its two confidence gates. We fit readers with labels on 20% of the $4{,}096$ fitting examples. The repair and warning use $1{,}024$ unlabeled anchors and a separate 512-example monitor. Six confirmation references each have a separate $2{,}000$-image evaluation partition.

The early monitor estimates the largest disagreement across the three consumer rules. We compare forecasts obtained by scaling the change from healthy features to epoch-two features. Across two development references, the selected multiplier is one. Each confirmation run saves its risk estimates and chosen repair family at epoch two, then resumes the same optimizer and data-order state. At epoch eight, we fit the chosen family on the original anchors and evaluate it alongside the unchanged head.

Among the 18 planned updates, the unchanged head exceeds the 5% budget in 17. The epoch-two prediction has MAE 1.99 points, compared with 1.08 for a predictor that uses only the learning rate and the development mean for that schedule. With one safe update and the persistence multiplier, the original forecasting gate fails. The original early repair choices keep every held-out disagreement within budget, using the full reference suffix for $12/18$ updates. Five of the 17 unsafe cases use a repair with no reference suffix, below the prespecified half-coverage target.

Before inspecting confirmation outcomes, we fix a separate stability control and a limited-reference replay. The six lower-rate stability runs all stay below budget. Across the 24 updates, the early warning detects $15/17$ exceedances and clears $7/7$ safe cases. Both misses are mild updates, whose risk rises from 4.30% to 6.85% and from 4.49% to 7.90%. These updates use moment repair in the limited-reference replay.

The replay selects direct, moment, or ridge repair on the sealed epoch-two monitor. It can also reserve exact reference calls for a fixed fraction of low-margin examples. The policy uses a 2.5% monitor target and a 5% held-out budget. It meets the latter on all 24 updates, with a maximum of 4.10%. Its median reference fraction on unsafe updates is 25%, compared with 75% for uniform

referral under the same rule. At matched query fractions, margin ranking reduces disagreement by 0.93 points $[0.72, 1.13]$. On the original 18 updates, mean reference use falls from 66.7% to 38.3%.

Both old-output programs use the same current predictions. Completing the update improves current accuracy by 1.45 points $[1.06, 1.84]$ over stopping at epoch two. At the nominal learning rate, joint continuation and the matched supervised control have the same mean accuracy, with difference 0.00 points $[-0.12, 0.12]$. The auxiliary loss ratio is 0.980 $[0.977, 0.982]$. Intervals use the six reference means. Reference calls measure access in the batched replay.

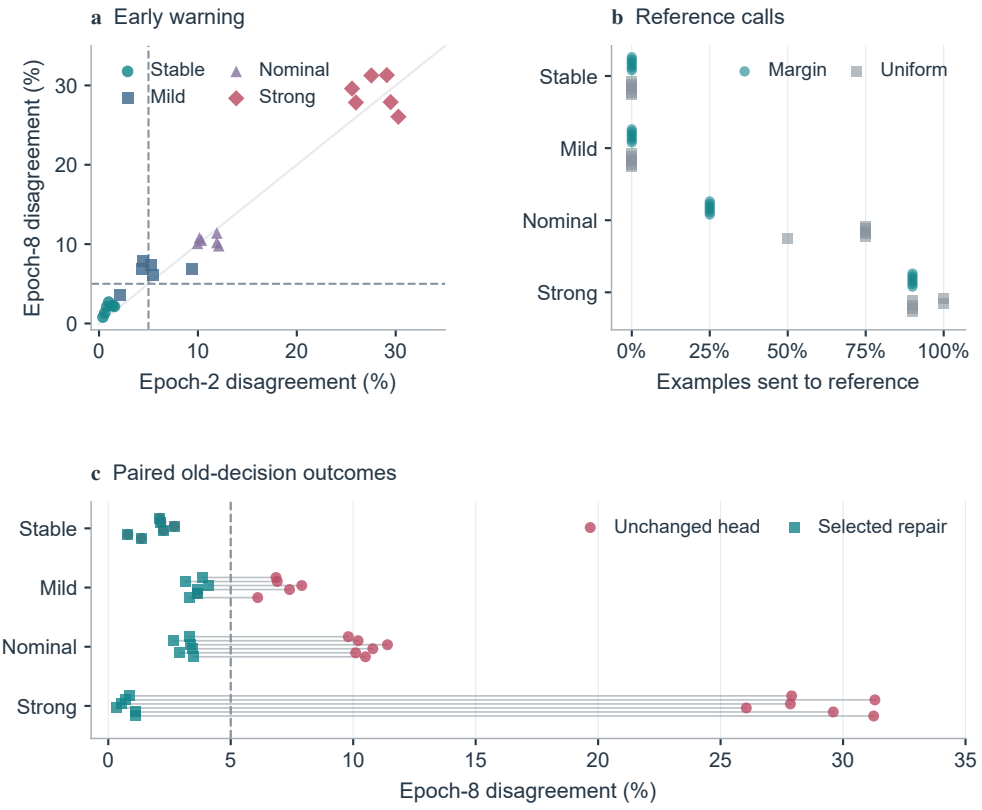


Figure 31: Early decisions on CIFAR-10 with ResNet-18. Each point is one planned update of a fitted reference. (a) Maximum direct disagreement at epochs two and eight. The dashed lines mark the 5% limit. (b) Reference fractions fixed by the epoch-two replay for margin and uniform referral. (c) Paired held-out disagreement for the unchanged head and the selected margin policy.

J.16 Legacy Precision Under Strong Updates

The strong-update policy in Appendix J.15 sends 90% of examples to the exact old suffix. We first test narrowing the old block while retaining its skip path and output gate. A head-independent student distils the teacher's activations and pooled features on $12{,}000$ unlabeled image views. Intermediate widths 64, 128, and 256 require 80%, 70%, and 50% referral on both development monitors. The width-256 path reaches the query target with an old-block MAC proxy of one. All widths miss the joint development criterion.

Precision gives a separate option with the original channel widths. We calibrate static x86 INT8 observers on $1{,}024$ unlabeled prefix tensors. The selected program evaluates the two old convolutions in INT8 and retains FP32 residual addition, the output gate, pooling and the old affine reader. Its maximum monitor disagreements are 0.98% and 1.17%. Quantizing the skip and output gate gives 0.59% and 1.37%. The BF16 convolution control gives 0.20% on both. The fixed priority selects INT8 with the FP32 skip.

We reconstruct the six strong continuations and seal both old-output programs before accessing the official test batch. Each reference uses a disjoint $1{,}500$-image partition. The previous terminal-feature policy calls the exact old block on 90.6% of test inputs after batch rounding. The selected INT8 program calls it on none. Its largest class or threshold disagreement across all six references is 1.73%, within the 5% budget. Both programs use the same FP32 current predictions. The INT8 program evaluates the complete low-precision old block on each input. It takes only the frozen prefix and fitted legacy reader, so replacing the trainable suffix leaves this approximate legacy function unchanged for a fixed batch.

We time four programs on the same CPU batches with a shared FP32 current graph. Each reference uses 12 warmups and 31 interleaved passes at batch 128. The INT8 program costs $1.043\times$ $[1.036, 1.050]$ relative to current-only, compared with $1.187\times$ $[1.175, 1.200]$ for the prior referral program and $1.185\times$ $[1.171, 1.199]$ for the exact two-output program. The paired INT8 ratio is 0.878 $[0.874, 0.883]$ relative to prior referral and 0.880 $[0.875, 0.886]$ relative to exact two-output serving. Intervals use six reference ratios.

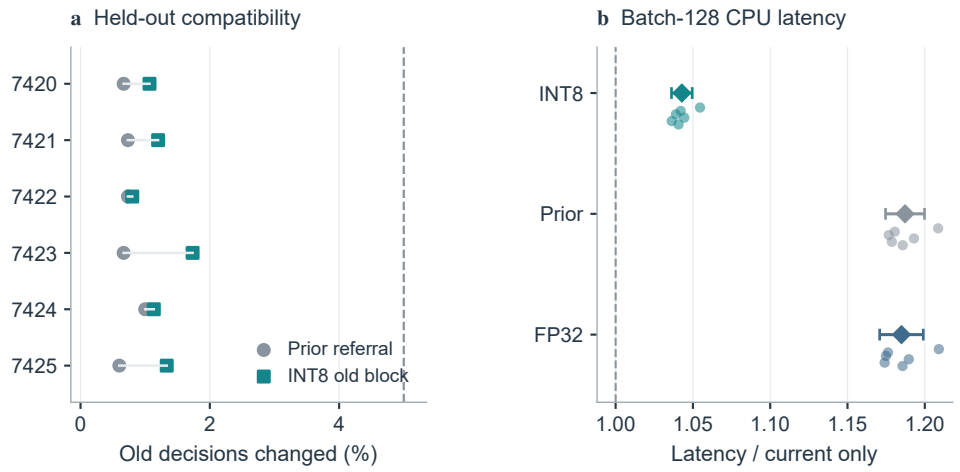


Figure 32: A low-precision legacy block under strong continuation. (a) Maximum class or threshold disagreement on six disjoint CIFAR-10 test partitions. The prior program refers 90.6% of inputs to the exact old block. INT8 uses zero exact calls. (b) Physical CPU latency with a shared FP32 current graph. Points are reference ratios. Diamonds show means and paired 95% intervals.

J.17 Forecasts and auxiliary gains on held-out subjects

We partitioned the UCI-HAR training subjects by reference and used 3% of each fitting activity's windows as labels. All arms shared an eight-epoch warm start, Adam state, minibatches and encoder forwards. During continuation, both arms applied supervised loss to clean and masked views. Reconstruction trained the same decoder in both arms. Joint training also passed its gradient to the encoder. Each mask replaced 25% of the coordinates by their fitting means.

Across eight development splits, none of eight reconstruction, contrastive or neighborhood-consistency settings reached both the 0.5-point activity gain and the 0.98 auxiliary loss ratio set in advance. Reconstruction at weight 3 came closest, with a 0.36-point gain and a loss ratio of 0.815. We used the fixed weight-0.3 fallback for the forecast experiment. Across 16 new references on nine official test subjects, its reconstruction loss ratio was 0.897 ($[0.886, 0.908]$) and its activity accuracy changed by 0.06 points (95% CI $[-0.17, 0.28]$) at the nominal learning rate.

We fit two frozen readers to the healthy features, for activity and movement versus rest. For each reader we recorded class decisions and fixed confidence rules at 80% and 50% healthy coverage. The forecast uses unlabeled logits at epochs 0, 1 and 3. For each rule it fits terminal disagreement with a ridge logistic model of the signed preservation margins, their two changes through epoch 3 and the two observed early disagreement indicators. For an accepted reference output, the margin is

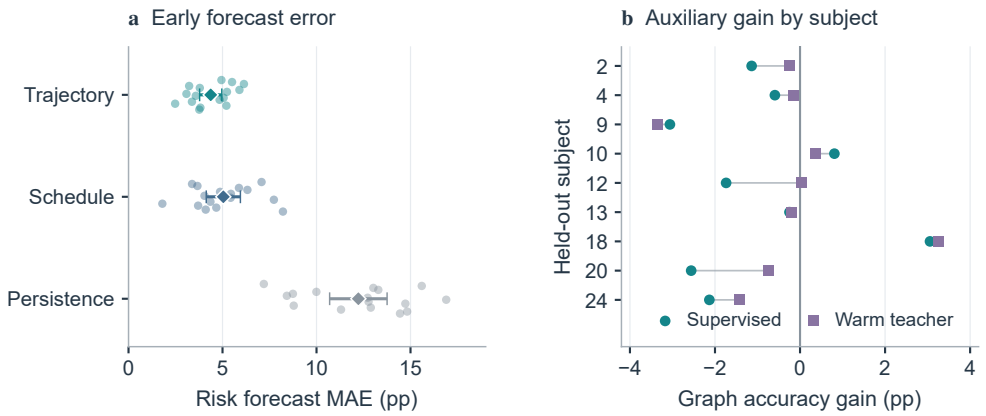


Figure 33: Subject-disjoint continuation on UCI-HAR. (a) Error in forecasting the maximum old-rule disagreement at epoch 24. Each small point averages eight forecasts from one reference. Diamonds and bars show means and 95% intervals across references for each estimator. (b) Accuracy gains from graph-derived targets over the two matched controls, averaged across references for each of the nine test subjects.

the smaller of the class margin and the reference class log odds minus the threshold log odds. For an abstained reference output, the margin is the threshold log odds minus the current winning class log odds. The class-only rule uses the class margin.

The schedule estimator knows the objective, learning rate, reader and rule. It uses the cell's mean terminal disagreement in development. The trajectory estimator adds the change in mean predicted risk between that cell and the new monitor. A held-reference calibration selected the scale of that correction from $\{0, .25, .5, .75, 1\}$, choosing 1. We refit on all 12 calibration references, then sealed the forecasts before epoch 4 in each confirmation run. Monitoring and evaluation subjects were disjoint. The trajectory, schedule and epoch-3 persistence estimators had mean absolute errors of 4.37, 5.04 and 12.23 points (Figure 33a). The paired trajectory improvement over persistence was 7.85 points ($[6.10, 9.61]$); its improvement over schedule was 0.67 ($[-0.38, 1.72]$). All 128 reader–update pairs exceeded the 5% budget. The trajectory and schedule estimators flagged all of them, and persistence missed 14.

For a separate auxiliary, we propagated fitting activity labels over a 30-nearest-neighbor cosine graph (Zhu et al., 2003) and matched its soft targets on masked unlabeled windows. Controls use supervision alone or healthy-model soft targets. Graph setup averages 0.51 CPU seconds. Weight 0.1 gains $1.00/2.59$ points over these controls on eight development splits. On 16 new references across nine official test subjects, it instead loses 0.82 points $[-1.52, -0.13]$ versus supervision and 0.22 $[-0.93, 0.49]$ versus the healthy teacher (Figure 33b). Intervals vary references on this fixed cohort. Retained healthy forwards reproduce both old readers' logits bit for bit.

J.18 Early warning under matched RadioML continuations

We trained an eleven-class modulation reader with 3% of fitting modulation labels and used signal-to-noise ratio as the auxiliary target. The detached and joint arms shared their warm model, Adam state and minibatches at each of three learning rates. We used the same signed-margin forecast with snapshots at epochs 0, 2 and 6 to predict epoch-24 risk. Calibration used 12 references. Across 16 new references and six arms per reference, we sealed forecasts before epoch 7 and evaluated 4,400 held-out signals per reference.

Trajectory, schedule-only and epoch-6 persistence gave errors of 2.25, 2.87 and 8.27 points. The paired improvement was 0.62 points (95% CI $[0.28, 0.96]$) over schedule and 6.02 ($[4.79, 7.26]$) over persistence. At a 5% budget, the trajectory and schedule rules kept the reference for all 96 episodes. Five would have met the budget with the unchanged reader. Persistence retained 69 references and violated the budget on 22 episodes. Keeping the healthy branch exactly reproduced the old logits.

Joint SNR training changed modulation accuracy by 0.05 ($[-0.02, 0.11]$), 0.35 ($[0.20, 0.50]$), and 0.13 ($[-0.18, 0.43]$) points at rates 10^{-5}, 10^{-4}, and 10^{-3}. The interval for the middle rate is $[0.16, 0.54]$ with a Bonferroni correction across the three rate-specific contrasts.

J.19 Choosing a readout repair before continuation ends

In a separate RadioML experiment, we varied three learning rates from 3×10^{-6} to 10^{-4} and added an affine repair. We estimated the residual between the unchanged reader and the healthy logits on 512 paired unlabeled anchors. Ridge regression predicts that residual from the current features. The alignment uses a separate anchor set and adds an affine map to the old output. The current prediction is shared.

For each of 16 new references, we sealed an epoch-six choice for each of six continuations. The policy takes the first path whose forecast plus 0.5 points fits within 5%. It tries the unchanged reader, then alignment, then the exact reference. All policies were evaluated after epoch 24 on $4{,}400$ held-out signals per reference.

Table 69: Early release decisions on 96 RadioML continuations. Counts give the number of updates assigned to each path. Violations count updates exceeding the 5% old-decision budget on held-out signals. Max is the largest disagreement across updates and rules, in percent.

Policy	Direct	Aligned	Reference	Violations	Max
Schedule, direct only	0	0	96	0	0.00
Trajectory, direct only	6	0	90	0	3.84
Persistence with alignment	59	16	21	39	9.77
Schedule with alignment	0	64	32	3	7.57
Trajectory with alignment	6	57	33	2	6.70

Alignment reduced trajectory-based reference retention from 90 to 33 of 96 updates, with 2 budget violations. The schedule policy with alignment retained 32 references and had 3 violations. At rate 10^{-4}, the joint SNR arm changed modulation accuracy by 0.24 points ($[0.05, 0.42]$). Both violations were class changes. Without alignment, early monitoring released six updates with no violations, while schedule retained all 96. At the middle rate, all 32 continuations retained the reference under the three-path trajectory policy, including the joint runs with a confirmed accuracy gain.

J.20 Exact one-pass serving at a useful SNR update rate

RadioML continuation uses rate 10^{-4}. Each reference used $12{,}000$ fitting signals, 3% primary labels, a width-64 encoder and a healthy warm start. We trained joint and detached teachers with the same 24 continuation epochs, data and optimizer state. In confirmation we evaluated $4{,}400$ test signals per reference after freezing every serving program.

We first compared post-hoc repair with training a shared student. A fixed-feature affine repair used all fitting windows, alongside the 512-anchor ridge control. Nonlinear repairs added 8, 16 or 32 units to the second hidden layer while keeping the current encoder fixed. They still exceeded the four-point development budget. A width-64 student learned both outputs from the two teachers. The best student started at the joint endpoint. In 16 fresh references it gained 0.32 points (95% CI $[0.21, 0.42]$) over detachment, but class disagreement reached 5.68% and exceeded the budget on four references (Figure 34).

We then kept the old nonlinear activations inside the student. Write $a = \sigma(W_1^0 x + b_1^0)$ for the healthy first layer and $b = \sigma(W_2^0 a + b_2^0)$ for the healthy second layer. Add $u = \sigma(Ua + c)$ and fit the current head on $[b; u]$ to the joint teacher. If the folded old head is $C_0 b + d_0$, the two outputs are

$$\begin{pmatrix} z_{\text{current}} \\ z_{\text{old}} \end{pmatrix} = \begin{pmatrix} C_b & C_u \\ C_0 & 0 \end{pmatrix} \begin{pmatrix} b \\ u \end{pmatrix} + \begin{pmatrix} d_{\text{current}} \\ d_0 \end{pmatrix}. \tag{40}$$

Stacking W_2^0 with U and both heads gives one graph. Zero old-head weights on the added units preserve the old function in real arithmetic. Every saved program retains the reference affine coefficients; the current student determines how much teacher utility survives.

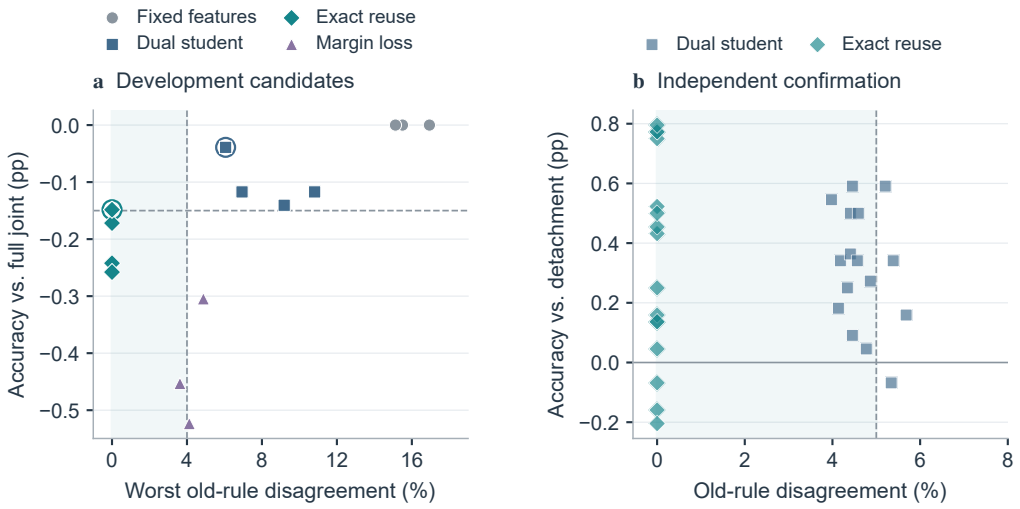


Figure 34: Useful RadioML continuations with a single serving graph. (a) The largest old-rule disagreement across eight development references and the mean accuracy change from unrestricted joint. The shaded band and horizontal line mark the 4% and -0.15-point development limits. Rings mark the two frozen choices. (b) Each point is one of 16 fresh references for its method. The band marks the 5% decision budget. The accuracy baseline is matched detached continuation.

On fresh development references we compared 0, 4, 8 and 16 added units with a same-width student initialized at the healthy checkpoint and trained with a preservation-margin loss. The width-16 expansion was the cheapest candidate meeting both limits. It used 36 offline distillation epochs on unlabeled fitting signals. On 16 fresh confirmation references its accuracy gain over detachment was 0.33 points ($[0.15, 0.51]$) and its change from unrestricted joint was -0.05 ($[-0.19, 0.09]$). All old class and confidence decisions agreed. The largest FP32 old-logit difference after folding was 1.72×10^{-5}. The corresponding worst class disagreements were 17.61% for the 512-anchor ridge control and 16.32% for full-fit affine repair.

We timed the folded programs in isolation on the same CPU with one thread. The current-only baseline used the same affine folding and the same input shape. Exact reuse cost $1.016\times$ ($[1.014, 1.017]$) at batch 1 and $1.070\times$ ($[1.068, 1.072]$) at batch 128. The ratios to two separately executed encoders were $0.557\times$ ($[0.554, 0.559]$) and $0.545\times$ ($[0.544, 0.546]$). No serving input calls a separate healthy encoder. These timings cover model forward on standardized signals.

J.21 Equal distillation work

To isolate the auxiliary contribution, we repeated exact reuse on 16 new references with both teachers. Both students used the same healthy hidden units, 16 added ReLU units, initialization, minibatches and 36 distillation epochs. Their teachers differed only in whether the SNR gradient reached the encoder. Both students were frozen before test evaluation.

The joint student's accuracy exceeded the detached student's by 0.28 points (95% CI $[0.11, 0.45]$). It differed from the unrestricted joint teacher by -0.12 points ($[-0.24, 0.002]$), within the prespecified -0.25 noninferiority margin. Descriptive gain intervals were 0.30 points ($[0.11, 0.49]$) below 0 dB and 0.27 ($[0.06, 0.48]$) at or above 0 dB. Every class and confidence decision agreed across all 32 student programs. All saved programs contained the healthy subgraph; maximum FP32 old-logit difference was 1.91×10^{-5}.

J.22 Exact old vision interfaces with selective current computation

We transferred the construction to ImageNet-pretrained ViT-B/32 on CIFAR-100. The first ten transformer blocks form a shared frozen prefix. The healthy two-block suffix produces the old represen-

tation h_0 and its last MLP activation u_0. We fit a student fine-class readout on their concatenation. Every old reader still receives h_0. The student learns the current output of a saved teacher, and a gate can send selected prefix tokens through that teacher's updated two-block suffix.

Contrastive, two-view supervised and doubled-epoch supervised teachers share a 20% fitting-label budget. Students use both fixed views of 10,000 fitting images and their own teacher targets, without further labels. An affine residual with ridge coefficients $0.001, 0.01, 0.1, 1$ retains the contrastive advantage but loses 0.76–1.04 teacher-accuracy points. We add a 128- or 512-unit GELU logit residual and train for 30 epochs with normalized centered MSE, unit-temperature KL, batch 256 and Adam at 0.001; the ridge-0.001 affine term stays fixed. Width 128 loses 0.32 pilot points against a 0.25-point retention limit.

Routing uses the student's top-two logit gap, with cutoffs at the 10%, 25% or 50% fitting-score quantile. The single-reference pilot selects 10%; four-reference training-only validation selects 25%, the smallest point meeting gain and retention limits after averaging three continuations per reference. This final choice precedes official-test predictions. Healthy test-feature preparation supplies no student or teacher accuracy. All student fitting and width selection use the training partition.

Proposition 13 (A retained interface and a selective current output)**.** *Let $h_0(x)$ be the retained healthy representation, C any fixed consumer, S a student classifier, T its teacher and $r(x) \in \{0, 1\}$ a routing rule. Define*

$$F_r(x) = \begin{cases} T(x), & r(x) = 1, \\ S(x), & r(x) = 0. \end{cases}$$

Returning $C(h_0(x))$ preserves that consumer for every x in real arithmetic, independently of S, T, r. On any finite audit corpus, for any assignment of task labels,

$$|\mathrm{err}(F_r) - \mathrm{err}(T)| \le \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{r(x_i) = 0,\ S(x_i) \neq T(x_i)\}.$$

Proof. The old path computes the original composition. On routed inputs the current output equals the teacher; on unrouted inputs, its error indicator can differ only when student and teacher decisions differ. Summing that pointwise inequality gives the bound. □

The gate uses confidence as in adaptive inference (Bolukbasi et al., 2017). Here, extra computation serves the new output while the old interface always follows its retained path. The proof concerns complete consumer functions, including their normalization and thresholds. The current-task bound can be measured without semantic labels on the audited inputs.

On all 10,000 official test images, the routed joint program gains $0.86[0.68, 1.05]$ points over the routed two-view student and $0.60[0.41, 0.79]$ over the routed doubled-epoch student. Its change from the unrestricted joint teacher is $-0.023[-0.044, -0.003]$ points. It uses the current teacher on 32.2% of inputs. The pure nonlinear student gains $0.52[0.18, 0.87]$ and $0.31[0.06, 0.56]$ points, respectively, but loses $-0.97[-1.14, -0.79]$ points from its teacher. Thus the test distinguishes a useful one-pass student from the point that also retains teacher accuracy.

A direct supervised control uses the same retained features, 128-unit head, fitting views, labels, 30 epochs and optimizer. Its affine term can also learn. The routed joint program exceeds this control by $2.23[1.93, 2.54]$ points. The twelve programs use four historical fitted references and the same TorchVision pretrained weights. The official test images were reserved from every fit and model-selection phase; intervals describe variation across the four reference means.

To separate routing quality from extra computation, we performed a retrospective equal-call control on the saved predictions. For a uniform subset of exactly k of n inputs, each teacher prediction is used with probability $p = k/n$. Its expected accuracy is therefore $(1 - p)\,\mathrm{acc}(S) + p\,\mathrm{acc}(T)$, and its expected teacher disagreement is $(1 - p)D(S, T)$. We set k to each fixed program's actual call count. The confidence route exceeds this exact random expectation by 0.63 points $[0.51, 0.75]$. It captures 99.33% $[99.06, 99.59]$ of student–teacher class differences. Remaining teacher disagreement is 0.041% $[0.023, 0.058]$, versus 4.095% $[3.848, 4.343]$ under random routing. Thus the chosen inputs matter, even at the same additional suffix-evaluation count. This comparison uses already inspected test outputs; it introduces no new fitted policy or independent confirmation. Call counts do not equate measured latency.

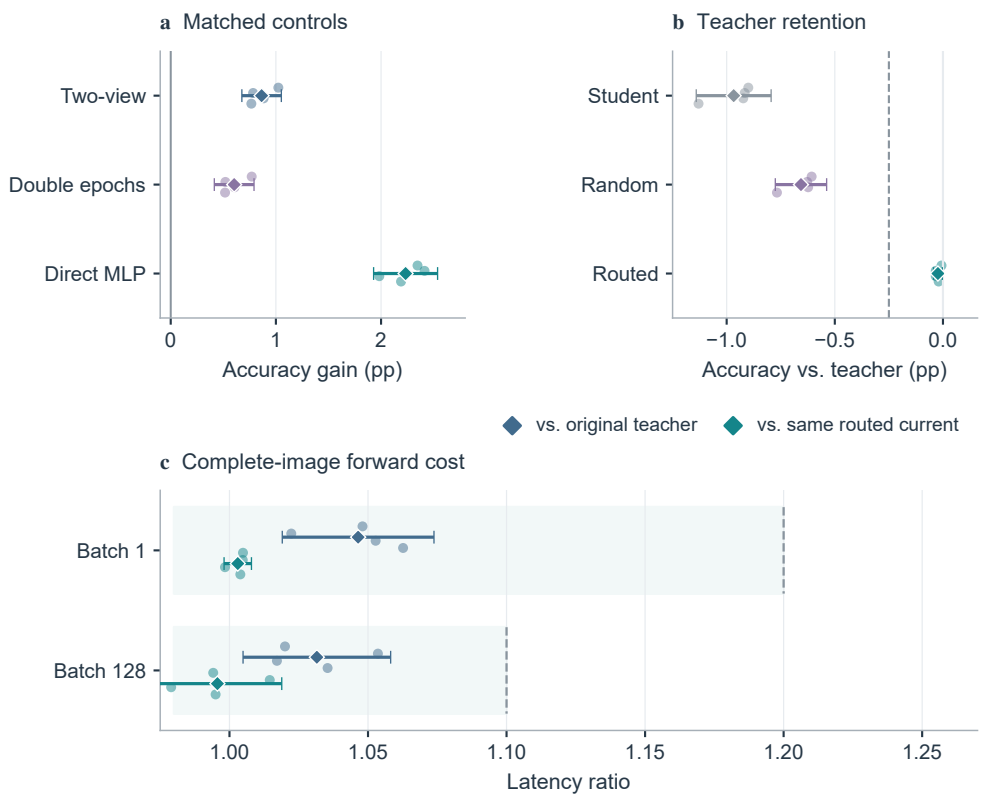


Figure 35: Official CIFAR-100 evaluation of the fixed visual program. Small points are the four fitted-reference means, each averaging three continuations; diamonds and bars show means and paired 95% intervals. (a) Routed joint-student gains over three supervised controls. (b) Accuracy change from the unrestricted joint teacher for the student, the exact expectation of uniform random routing with equal calls, and the fixed confidence route. The dashed line is the -0.25-point margin. (c) Full-image CPU cost against the original teacher and the same routed current function. Row-specific marks show the $1.20/1.10$ cost limits.

The old fine and coarse consumers each have a class rule and two fixed confidence rules. Every audited rule agrees. The largest current-class disagreement from the full joint teacher is 0.06% across the twelve programs, so their accuracy gaps on this corpus are bounded by 0.06 points for any labels (Proposition 13).

We measured complete FP32 image forwards with twelve physical CPU cores on one NUMA node. A first traced implementation evaluated one padded suffix item per batch. Its ratios were $1.162[1.104, 1.220]$ at batch 1 and $1.047[0.984, 1.110]$ at batch 128, missing the confidence-limit gates. Scripted branches skip empty teacher sets, yielding $1.046[1.019, 1.074]\times$ and $1.032[1.005, 1.058]\times$. Batch-one blocks average 32 fixed images; batch 128 uses the first 128. Nine alternating blocks yield one paired ratio per reference. The implementation preserves all 120,000 current predictions, 720,000 old-rule decisions and 384 individual-input executions. The archive retains the padded failure and excluded overlapping timing.